

GenLayer Full Documentation - StudioNet Builder Research

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The Adjudication Layer for the Agentic Economy

Source: <https://docs.genlayer.com/>

GenLayer uses decentralized AI-validator consensus to resolve contracts that require judgment, not just code.

Intelligent Contracts interpret language, process unstructured data, and pull live web inputs. No oracles, no intermediaries.

- Bitcoin — Trustless Money
- Ethereum — Trustless Computation
- GenLayer — Trustless Adjudication

The agentic-commerce stack is being built in public — Coinbase's x402 (<https://www.x402.org/>) for payments, Ethereum's ERC-8004 (<https://eips.ethereum.org/EIPS/eip-8004>) for trustless agent identity, the Linux Foundation's A2A (<https://a2a-protocol.org>) for agent interoperability, plus Stripe/OpenAI's ACP, Visa's Trusted Agent Protocol, Google's AP2, and Mastercard's Agent Pay. Every layer engineers the happy path. None ships dispute resolution. GenLayer fills that gap.

Explore use cases (</understand-genlayer-protocol/typical-use-cases>) or jump straight into building.

Note:

Start with Skills  — The fastest way to build a contract or run a validator is the GenLayer Skills (<https://skills.genlayer.com/>) plugin for Claude Code. One command scaffolds, deploys, and operates — no manual setup.

bash

claude /plugin marketplace add genlayerlabs/skills

then: genlayer-dev (build contracts) or genlayernode (run a validator)

Get Started

- Discover the Protocol (/understand-genlayer-protocol) - What GenLayer is, how Optimistic Democracy works, and what makes it different.
- For Developers (/developers) - Build Intelligent Contracts in Python. Deploy to testnet or Studio.
- For Validators (/validators/setup-guide) - Run a validator node and participate in the GenLayer network.

Community and Careers

Join the community on Discord (<https://discord.gg/8Jm4v89VAu>) or Telegram (<https://t.me/genlayer>) to ask questions, share feedback, and showcase your builds.

We're hiring — see open roles (<https://x.com/GenLayer/jobs>).

What is GenLayer?

GenLayer is the first AI-native blockchain built for AI-powered smart contracts—called Intelligent Contracts—capable of reasoning and adapting to real-world data. Its foundation is the Optimistic Democracy consensus mechanism, an enhanced Delegated Proof of Stake (dPoS) model where validators connect directly to Large Language Models (LLMs). This setup allows for non-deterministic operations—such as processing text prompts, fetching live web data, and executing AI-based decision-making—while preserving the reliability and security of a traditional blockchain.

Core Technology

At the heart of GenLayer lies Optimistic Democracy—an enhanced Delegated Proof of Stake (dPoS) consensus mechanism that integrates AI models directly into validator operations. This synergy delivers three capabilities traditional blockchains cannot match:

1. On-Chain AI Processing

Validators connect to leading AI models (GPT, LLaMA, Meta, etc.) to execute complex reasoning on-chain, from natural language comprehension to data-driven predictions.

2. Consensus-Backed Security

Multiple validators vote on outcomes, ensuring collective agreement and robust reliability for every transaction—even those involving non-deterministic AI outputs.

3. Intelligent Contracts

Smart contracts in GenLayer gain reasoning abilities, allowing them to understand natural language, process real-world data, and adapt to evolving conditions.

Technical Implementation

To integrate AI seamlessly with the blockchain, GenLayer employs a distributed neural consensus network, wherein validators run specialized software connected via API to advanced AI models. This approach unifies:

- Delegated Proof of Stake (dPoS) for efficient block production and governance.
- Neural Consensus for non-deterministic transactions requiring advanced AI reasoning.

This architecture supports autonomous DAOs, self-executing prediction markets, and dynamic DeFi protocols that react to real-world data in real time.

Optimistic Democracy: How Consensus Works

Source: <https://docs.genlayer.com/understand-genlayer-protocol>

Inspired by Condorcet's Jury Theorem (<https://jury-theorem.genlayer.com/>) (click the link to check out our interactive model), Optimistic Democracy merges probabilistic AI systems with deterministic blockchain rules, ensuring secure and accurate consensus at scale.

1. User Submits a Transaction

A user sends a transaction request to the network (see the diagram's Step 1).

2. Leader (Validator) Proposes Result

The network selects a Leader, who processes the request and proposes an outcome (Step 2).

3. Validators Recompute

A group of Validators independently re-compute the transaction (Step 3). If the output aligns with the Leader's proposal, they approve; otherwise, they deny.

This multi-layer validation ensures majority agreement, adding a safety net for AI-driven computations.

Validator Selection Mechanism

Token holders bolster network security by delegating tokens to validator candidates. A deterministic function $f(x)$ then randomly designates Leader-Validator and Validators for each transaction. This process not only promotes fairness but also helps decentralize validation power, strengthening GenLayer's security and trustlessness.

Validator Operational Framework

Each GenLayer validator node integrates:

- Validator Software

Handles core blockchain functions: networking, block production, and transaction management.

- AI Model Integration

Connects to Large Language Models (LLMs) or other AI services for complex reasoning, natural language processing, and real-time data retrieval.

Validators seamlessly manage both:

- 1. Deterministic Transactions typical of traditional blockchains.
- 2. Non-Deterministic Transactions that leverage AI-driven logic (e.g., searching the internet, analyzing data, making probabilistic inferences).

By splitting tasks between standard deterministic and advanced AI-powered transactions, GenLayer ensures high performance without compromising on security.

Putting It All Together

With Optimistic Democracy guiding consensus and validators empowered by AI, GenLayer enables a new class of blockchain applications. From DAOs that self-govern based on real-time data to DeFi protocols that dynamically adjust parameters in response to market changes, developers can now build truly intelligent decentralized solutions.

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Source: <https://docs.genlayer.com/understand-genlayer-protocol/what-is-genlayer>

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- Bitcoin — Trustless Money
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- GenLayer — Trustless Adjudication

Where Bitcoin reached consensus on the *order* of transactions and Ethereum on the *execution* of code, GenLayer reaches consensus on the meaning of transactions. The technical primitive that makes this possible is the Equivalence Principle (/developers/intelligent-contracts/features/non-determinism): two answers can be different in form yet equivalent in meaning, and a network of independent AI validators can agree on that.

The Missing Layer

The agentic-commerce stack is being built in the open — and every layer of it stops at the happy path.

Layer	Standard	What it handles	Dispute resolution
Payments	[x402](https://www.x402.org/)	(Coinbase)	Internet-native, agent-friendly payments
Identity & reputation	[ERC-8004](https://eips.ethereum.org/EIPS/eip-8004)	(Ethereum)	Trustless agent identity
Agent interoperability	[A2A](https://a2a-protocol.org)	(Linux Foundation / Google)	How agents discover each other and exchange

Other players are stacking in alongside them — Stripe + OpenAI's ACP (already powering ChatGPT checkout), Visa's Trusted Agent Protocol, Google's AP2, Mastercard's Agent Pay + BVNK, and Tempo (Stripe / Paradigm L1). The shape is the same in every case: the payment goes through, the job is accepted, the reputation is updated — and the moment a single material dispute appears, the stack reaches for a function it does not have.

That function is adjudication. Not a sovereign court. A credible, machine-speed mechanism for evaluating contested commitments, weighing evidence, interpreting language, reaching a verdict, and attaching consequences to it.

GenLayer is that layer.

What Intelligent Contracts Can Do

Subjective Decisions

Evaluate context and nuance. Turn judgment calls into enforceable on-chain outcomes — content moderation, claim assessment, quality evaluation.

Internet Access

Fetch live web data directly on-chain. Contracts can read websites, call APIs, and verify real-world information without oracles or intermediaries.

Natural Language Processing

Interpret human-readable inputs via LLMs. Contracts can analyze text, extract meaning, and make decisions based on qualitative criteria.

Image & Visual Processing

Pass images to LLMs for analysis — screenshot a webpage and verify its content, check visual evidence for claims, analyze receipts or documents. Contracts can capture screenshots via `gl.nondet.web.render()` and send them to LLMs via `gl.nondet.exec_prompt(images=[...])`.

Unstructured Data

Process text, images, audio transcripts, and qualitative evidence. Handle real-world complexity that traditional smart contracts cannot.

How It Compares

Feature	Traditional Smart Contracts	Intelligent Contracts
Language	Solidity, Rust	Python
Data sources	On-chain only (or oracles)	On-chain + live web data
Decision logic	Deterministic only	Deterministic + subjective
AI integration	Not possible	Native LLM access (text + images)
Consensus	All nodes must agree on exact output	Validators assess equivalence of results

Architecture: Two Layers

GenLayer operates as two integrated layers:

GenLayer Chain — an EVM-compatible L2 (zkSync Elastic Chain). Holds account balances via ghost contracts, handles standard Ethereum operations (eth_* methods), and anchors to Ethereum's security model.

GenVM — the execution environment for Intelligent Contracts. A WebAssembly-based VM (built on Wasmtime (<https://wasmtime.dev>)) that runs a Python interpreter with native access to LLMs, web data, and non-deterministic operations. Can also execute compiled native code.

Every Intelligent Contract has a corresponding ghost contract on the chain layer at the same address. Ghost contracts hold the contract's GEN balance, relay transactions to consensus, and execute external messages. See Messages (/developers/intelligent-contracts/features/messages#ghost-contracts) for details.

Transactions enter via addTransaction on the chain layer. GenVM executes the contract logic. Results settle back on-chain.

Develop in Python

Intelligent Contracts are Python classes extending gl.Contract:

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }
from genlayer import *
import json

class WizardOfCoin(gl.Contract):
    has_coin: bool

    def __init__(self):
        self.has_coin = True

    @gl.public.write
    def ask_for_coin(self, request: str) -> None:
        if not self.has_coin:
            raise gl.vm.UserError("I don't have a coin!")

        prompt = f"""
        You are a wizard guarding a gold coin.
        An adventurer says: {request}
        Should you give them the coin?
        Respond as JSON: {{"give_coin": true/false}}
        """

        def leader_fn():
            return gl.nondet.exec_prompt(prompt, response_format="json")

        def validator_fn(leaders_res) -> bool:
            if not isinstance(leaders_res, gl.vm.Return):
                return False
            my_result = leader_fn()
            return my_result["give_coin"] == leaders_res.calldata["give_coin"]

        result = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)
        if result["give_coin"]:
            self.has_coin = False

    @gl.public.view
    def get_has_coin(self) -> bool:
        return self.has_coin
```

Full SDK available: genlayer-js (/api-references/genlayer-js) (TypeScript), genlayer-py (/api-references/genlayer-py) (Python), CLI (/api-references/genlayer-cli).

Get started → (/developers/intelligent-contracts/first-contract)

How GenLayer Works

Source: <https://docs.genlayer.com/understand-genlayer-protocol/optimistic-democracy-how-genlayer-works>

Optimistic Democracy

GenLayer uses Optimistic Democracy for consensus — a mechanism where validators running diverse AI models independently evaluate transactions and vote on outcomes. It applies Condorcet's Jury Theorem (https://en.wikipedia.org/wiki/Condorcet%27s_jury_theorem): a group of independent reasoners is more likely to reach the correct answer than any individual. This is what lets GenLayer act as an adjudication layer for the agentic economy — judgments emerge from a diverse validator set rather than from any one model, operator, or jurisdiction.

Transactions are accepted if a majority of validators agree. Anyone can appeal an accepted result, triggering re-evaluation by a new, larger validator set. This process can escalate through multiple rounds until a final decision is reached.

Transaction Lifecycle

Every transaction moves through these stages:

1. Pending — queued, waiting to be picked up
2. Proposing — a leader validator executes the contract and proposes a result
3. Committing — other validators execute independently and submit encrypted votes
4. Leader Revealing — the leader reveals execution data and decryption keys
5. Revealing — validators reveal their votes
6. Accepted — majority consensus reached; transaction enters the appeal window
7. Finalized — appeal window closed, result is permanent and irreversible

If consensus is not reached, the transaction may be marked Undetermined or rotate to a new leader.

See Transaction Execution (</understand-genlayer-protocol/core-concepts/transactions/transaction-execution>) for the full state machine.

Non-Determinism and Consensus

Because Intelligent Contracts use LLMs and web data, validators may produce different outputs for the same input. GenLayer provides several strategies for reaching consensus on non-deterministic results:

- Strict equality — all validators must produce the exact same output (for deterministic operations)
- LLM-based comparison — an LLM compares validator outputs against developer-defined criteria
- Custom validation — developers write explicit leader/validator function pairs with full control over consensus logic

See Non-determinism (</developers/intelligent-contracts/features/non-determinism>) for implementation details.

Appeals and Finality

After a transaction is accepted, it enters a finality window during which anyone can appeal the result.

- An appeal triggers a new round with a fresh, larger validator set
- Appeals can escalate through multiple rounds
- The final round's decision is binding

Once the finality window closes without appeal (or after the final appeal round), the transaction is finalized — permanent and irreversible.

See Appeal Process (/understand-genlayer-protocol/core-concepts/optimistic-democracy/appeal-process) and Finality (/understand-genlayer-protocol/core-concepts/optimistic-democracy/finality) for details.

Use Cases

Source: <https://docs.genlayer.com/understand-genlayer-protocol/typical-use-cases>

GenLayer is built for commitments where outcomes depend on judgment — evaluating evidence, interpreting language, or assessing quality — and where a deterministic smart contract alone cannot resolve them.

If you are deciding whether a feature belongs on GenLayer or in a normal backend, start with the builder fit checklist (/developers/intelligent-contracts/when-to-use-genlayer).

The use cases below are grouped by where the need for adjudication is most acute today.

1. Performance & Milestone Adjudication

Payments, rewards, or recognition that depend on whether some obligation was actually fulfilled under criteria that are partly measurable and partly interpretive. The money is already on-chain. The commitment is already written down. The dispute already happens — and today it is resolved by human bottlenecks that do not scale.

- Bounties where payout depends on whether the work met the spec
- Grant milestones where tranche release turns on contested deliverables
- Retroactive funding rounds where allocation depends on impact assessment
- Creator economies where AI-scored rewards generate disputes with no resolution layer
- Prediction markets with subjective outcomes (example contract (/developers/intelligent-contracts/examples/prediction))
- Contributor performance tied to vesting or continued compensation
- Freelance and gig work — was the deliverable satisfactory? AI consensus replaces subjective back-and-forth
- Chargebacks — buyer/seller disputes resolved by analyzing shipping records, communication logs, and transaction history

GenLayer can support agreed dispute-resolution workflows, but it is not a court and does not automatically make a result legally binding. For legal or contractual use cases, parties still need the appropriate agreements, jurisdiction, and escalation process around the Intelligent Contract.

2. Adjudication Inside the Agentic-Commerce Stack

GenLayer plugs into the infrastructure the industry is building right now: payment rails (x402 (<https://www.x402.org/>)), agent identity and reputation (ERC-8004 (<https://eips.ethereum.org/EIPS/eip-8004>)), agent-to-agent task exchange (A2A (<https://a2a-protocol.org/>)), plus Stripe/OpenAI's ACP, Visa's Trusted Agent Protocol, Google's AP2, and Mastercard's Agent Pay. Each standard ships the happy path and carves the moment of disagreement out as someone else's problem. GenLayer is that someone else.

- Agent-executed job disputes — was the task delivered to spec?
- Escrow release on ambiguous completion — agent counterparties needing neutral resolution
- SLA enforcement on agent work — quality, latency, or scope claims
- Reputation claims contested between counterparties in ERC-8004-style identity systems
- Coverage claims on agent-to-agent commitments — parametric and evidence-based
- Multi-agent workflows where responsibility for failure has to be assigned across participants

3. Rule & Constitution Verification

Check whether something meets a set of criteria defined in natural language — a foundational primitive that many applications reduce to.

- Does a new prediction market meet listing guidelines?
- Does a DAO proposal comply with the organization's charter?
- Does a content submission follow community standards?
- Does a transaction comply with regulatory requirements?

4. Adjacent Surfaces

The same primitive shows up across digital commerce:

- Insurance — parametric and evidence-based claims evaluated by AI validators. Contracts fetch weather data, flight statuses, or photographic evidence to assess claims and trigger payouts automatically — no adjusters, no weeks of waiting.
- Social content verification — AI validators assess quality, detect plagiarism, and distribute rewards based on originality and engagement. Replaces centralized moderation with consensus-driven evaluation.
- Code & work quality assurance — staked submissions where reviewers are economically incentivized to find issues; AI validators assess deliverable completeness or compliance with specifications.
- AI-governed organizations — DAOs where proposals are written in natural language, evaluated against real-time data, and executed automatically when conditions align.
- Compliance automation — real-time screening against sanctions lists, KYC/AML requirements, and changing regulations. Contracts read authoritative sources directly — no manual updates needed.
- Argumentation and debate markets — structured debates where participants stake positions and AI consensus determines outcomes, a new primitive for information markets.

What Makes These Possible

All of these share a common pattern: they require judgment that traditional smart contracts can't perform. GenLayer's Intelligent Contracts can:

- Fetch and interpret live web data
- Process natural language and unstructured inputs
- Make subjective decisions through multi-validator AI consensus
- Execute outcomes on-chain with full finality — justice in minutes, not months

See projects building on GenLayer (<https://portal.genlayer.foundation/#/>) for live examples.

Start building → (</developers/intelligent-contracts/first-contract>)

Core Concepts

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts>

Dive into GenLayer's fundamental building blocks. These core concepts elucidate how Intelligent Contracts remain secure, efficient, and reliable in a non-deterministic environment:

- [GenVM \(/core-concepts/genvm\)](/core-concepts/genvm)
- [Transactions \(/core-concepts/transactions\)](/core-concepts/transactions)
- [Optimistic Democracy \(/core-concepts/optimistic-democracy\)](/core-concepts/optimistic-democracy)
- [⚖ Equivalence Principle \(/core-concepts/optimistic-democracy/equivalence-principle\)](/core-concepts/optimistic-democracy/equivalence-principle)
- [Appeal Process \(/core-concepts/optimistic-democracy/appeal-process\)](/core-concepts/optimistic-democracy/appeal-process)
- [Finality \(/core-concepts/optimistic-democracy/finality\)](/core-concepts/optimistic-democracy/finality)
- [Staking \(/core-concepts/optimistic-democracy/staking\)](/core-concepts/optimistic-democracy/staking)
- [⚡ Slashing \(/core-concepts/optimistic-democracy/slashing\)](/core-concepts/optimistic-democracy/slashing)
- [Unstaking \(/core-concepts/optimistic-democracy/unstaking\)](/core-concepts/optimistic-democracy/unstaking)

Validators and Validator Roles

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/validators-and-validator-roles>

Overview

Validators are essential participants in the GenLayer network. They are responsible for validating transactions and maintaining the integrity and security of the blockchain. Validators play a crucial role in the Optimistic Democracy consensus mechanism, ensuring that both deterministic and non-deterministic transactions are processed correctly.

Key Responsibilities

- Transaction Validation: Validators verify the correctness of transactions proposed by the leader, using mechanisms like the Equivalence Principle for non-deterministic operations.
- Leader Selection: Validators participate in the process of randomly selecting a leader for each transaction, ensuring fairness and decentralization.
- Consensus Participation: Validators cast votes on proposed transaction outcomes, contributing to the consensus process.
- Staking and Incentives: Validators stake tokens to earn the right to validate transactions and receive rewards based on their participation and correctness.

Validator Selection and Roles

- Leader Validator: For each transaction, a leader is randomly selected among the validators. The leader is responsible for executing the transaction and proposing the result to other validators.
- Consensus Validators: Other validators assess the leader's proposed result and vote to accept or reject it based on predefined criteria.

Becoming a Validator

- Staking Requirement: Participants must stake a certain amount of tokens to become validators.
- Validator Configuration: Validators must configure their nodes with the appropriate LLM providers and models, depending on the network's requirements.
- Reputation and Slashing: Validators must act honestly to avoid penalties such as slashing of their staked tokens.

GenVM (GenLayer Virtual Machine)

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/genvm>

The GenVM is the execution environment for Intelligent Contracts in the GenLayer protocol. It serves as the backbone for processing and managing contract operations within the GenLayer ecosystem.

Source code at GitHub (<https://github.com/genlayerlabs/genvm>)

Purpose of GenVM

The only purpose of the GenVM is to execute Intelligent Contracts, which can have non-deterministic code while maintaining blockchain security and consistency.

In summary, the GenVM plays a crucial role in enabling GenLayer's unique features, bridging the gap between traditional smart contracts and AI-powered, web-connected Intelligent Contracts.

Key Features That Make the GenVM Different

Unlike traditional blockchain virtual machines such as Ethereum Virtual Machine (EVM), the GenVM has some advanced features.

- Integration with LLMs: the GenVM facilitates seamless interaction between Intelligent Contracts and Large Language Models
- Web access: the GenVM provides access to the Internet
- User friendliness: Intelligent Contracts can be written in Python, which makes the learning curve much more shallow

How the GenVM Works

1. Contract Deployment: When an Intelligent Contract is deployed, the GenVM compiles and executes the contract code.
2. Transaction Processing: As transactions are submitted to the network, the GenVM executes the relevant contract functions and produces the contract's next state.

Developer Considerations

When developing Intelligent Contracts for the GenVM:

- Utilize Python's robust libraries and features
- Consider potential non-deterministic outcomes when integrating LLMs
- Implement proper error handling for web data access
- Optimize code for efficient execution within the rollup environment

Optimistic Democracy

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/optimistic-democracy>

Optimistic Democracy is the consensus method used by GenLayer to validate transactions and operations of Intelligent Contracts. This approach is especially good at handling unpredictable outcomes from transactions involving web data or AI models, which is important for keeping the network reliable and secure.

Key Components

- Validators: Participants who stake tokens to earn the right to validate transactions. They play a crucial role in both the initial validation and any appeals process if needed.
- Leader Selection: A process that randomly picks one validator to propose the outcome for each transaction, ensuring fairness and reducing potential biases.

How It Works

Optimistic Democracy relies on a mix of trust and verification to ensure transaction integrity:

[Image:]

1. Initial Validation: When a transaction is submitted, a small group of randomly selected validators checks its validity. One is chosen as the leader. The leader executes the transaction, and the other validators assess the leader's proposal using the Equivalence Principle.
2. Majority Consensus: If most validators accept the leader's proposal, the transaction is provisionally accepted. However, this decision is not final yet, allowing for possible appeals during a limited window of time, known as the Finality Window.

Note:

If any validator fails to vote within the specified timeframe, they are replaced, and a new validator is selected to cast a vote.

3. Initiating an Appeal: If a participant disagrees with the initial validation (if it's incorrect or fraudulent), they can appeal during the Finality Window. They must submit a request and provide a bond. After the appeal starts, a new group of validators joins the original ones. This group first votes on whether the transaction should be re-evaluated. If they agree, a new leader is chosen to reassess the transaction, and all validators then review this new evaluation.
4. Appeal Evaluation: The new leader re-evaluates the transaction, while the other validators assess the leader's proposal using the Equivalence Principle. This step involves more validators, increasing the chances of an accurate decision.
5. Escalating Appeals: If the appealing party is still not satisfied, the process can escalate, with each round involving more validators. Each round doubles the number of validators. A new leader is only chosen if the transaction is overturned.
6. Final Decision: The appeals process continues until a majority consensus is reached or until all validators have participated. The final decision is recorded, and the transaction's state is updated accordingly. If the appealing party is correct, they receive a reward for their efforts, while incorrect appellants lose their bond.

Equivalence Principle Mechanism

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/optimistic-democracy/equivalence-principle>

The Equivalence Principle mechanism is a cornerstone in ensuring that Intelligent Contracts function consistently across various validators when handling non-deterministic outputs like responses from Large Language Models (LLMs) or data retrieved through web browsing. It plays a crucial role in how validators assess and agree on the

outcomes proposed by the Leader.

The Equivalence Principle protects the network from manipulations or errors by ensuring that only suitable, equivalent outcomes influence the blockchain state.

Key Features of the Equivalence Principle

The Equivalence Principle is fundamental to how Intelligent Contracts operate, ensuring they work reliably across different network validators.

- **Consistency in Decentralized Outputs:** The Equivalence Principle allows outputs from various sources, such as LLMs or web data, to be different yet still considered valid as long as they meet predefined standards. This is essential to maintain fairness and uniform decision-making across the blockchain, despite the natural differences in AI-generated responses or web-sourced information.
- **Security Enhancement:** To protect the integrity of transactions, the Equivalence Principle requires that all validators check each other's work. This mutual verification helps prevent errors and manipulation, ensuring that only accurate and agreed-upon data affects the blockchain.
- **Output Validation Flexibility:** Intelligent Contracts often need to handle complex and varied data. This part of the principle allows developers to set specific rules for what counts as "equivalent" or acceptable outputs. This flexibility helps developers tailor the validation process to suit different needs, optimizing either for accuracy or efficiency depending on the contract's requirements.

Types of Equivalence Principles

Validators work to reach a consensus on whether the result set by the Leader is acceptable, which might involve direct comparison or qualitative evaluation, depending on the contract's design. If the validators do not reach a consensus due to differing data interpretations or an error in data processing, the result might be challenged or an appeal process might be initiated.

Comparative Equivalence Principle

In the Comparative Equivalence Principle, both the Leader and the validators perform identical tasks and then directly compare their respective results with the predefined criteria in the Equivalence Principle to ensure consistency and accuracy. This method uses an acceptable margin of error to handle slight variations in results between validators and is suitable for quantifiable outputs. However, since multiple validators perform the same tasks as the Leader, it increases computational demands and associated costs.

For example, if an Intelligent Contract is tasked with calculating the average rating of a product based on user reviews, the Equivalence Principle specifies that the average ratings should not differ by more than 0.1 points. Here's how it works:

1. **Leader Calculation:** The Leader validator calculates the average rating from the user reviews and arrives at a rating of 4.5.
2. **Validators' Calculations:** Each validator independently calculates the average rating using the same set of user reviews. Suppose one validator calculates an average rating of 4.6.
3. **Comparison:** The validators compare their calculated average (4.6) with the Leader's average (4.5). According to the Equivalence Principle, the ratings should not differ by more than 0.1 points.
4. **Decision:** Since the difference (0.1) is within the acceptable margin of error, the validators accept the Leader's result as valid.

Non-Comparative Equivalence Principle

In contrast, the Non-Comparative Equivalence Principle does not require validators to replicate the Leader's output, which makes the validation process faster and less costly. Instead, validators assess the accuracy of the Leader's result against the criteria defined in the Equivalence Principle. This method is particularly useful for qualitative outputs like text summaries.

For example, in an Intelligent Contract designed to summarize news articles, the process works as follows:

1. **Leader Summary:** The Leader validator generates a summary of a news article.
2. **Evaluation Criteria:** The Equivalence Principle defines criteria for an acceptable summary, such as accuracy, relevance, and length.
3. **Validators' Assessment:** Instead of generating their own summaries, validators review the Leader's summary and check if it meets the predefined criteria.
 - **Accuracy:** Does the summary accurately reflect the main points of the article?
 - **Relevance:** Is the summary relevant to the content of the article?
 - **Length:** Is the summary within the acceptable length?
4. **Decision:** If the Leader's summary meets all the criteria, it is accepted by the validators.

Key Points for Developers

- **Setting Equivalence Criteria:** Developers must define what 'equivalent' means for each non-deterministic operation in their Intelligent Contract. This guideline helps validators judge if different outcomes are close enough to be treated as the same.
- **Ensuring Contract Reliability:** By clearly defining equivalence, developers help maintain the reliability and predictability of their contracts, even when those contracts interact with the unpredictable web or complex AI models.

Appeals Process

Source:

<https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/optimistic-democracy/appeal-process>

The appeals process in GenLayer is a critical component of the Optimistic Democracy consensus mechanism. It provides a means for correcting errors or disagreements in the validation of Intelligent Contracts. This process ensures that non-deterministic transactions are accurately evaluated, contributing to the robustness and fairness of the platform.

How It Works

- **Initiating an Appeal:** Participants can appeal the initial decision by submitting a request and a required bond during the Finality Window. A new set of validators is then added to the original group to reassess the transaction.

- Appeal Evaluation: The new validators first review the existing transaction to decide if it needs to be overturned. If they agree it should be re-evaluated, a new leader re-evaluates the transaction. The combined group of original and new validators then review this new evaluation to ensure accuracy.
- Escalating Appeals: If unresolved, the appeal can escalate, doubling the number of validators each round until a majority consensus is reached or all validators have participated.

Once a consensus is reached, the final decision is recorded, and the transaction's state is updated. Correct appellants receive a reward, while those who are incorrect may lose their bond.

Gas Costs for Appeals

The gas costs for an appeal can be covered by the original user, the appellant, or any third party. When submitting a transaction, users can include an optional tip to cover potential appeal costs. If insufficient gas is provided, the appeal may fail to be processed, but any party can supply additional gas to ensure the appeal proceeds.

Finality

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/optimistic-democracy/finality>

Finality refers to the state in which a transaction is considered settled and unchangeable. In GenLayer, once a transaction achieves finality, it cannot be appealed or altered, providing certainty to all participants in the system. This is particularly important for applications that rely on accurate and definitive outcomes, such as financial contracts or decentralized autonomous organizations (DAOs).

Finality Window

The Finality Window is a time frame during which a transaction can be challenged or appealed before it becomes final. This window serves several purposes:

1. Appeals: During the Finality Window, any participant can appeal a transaction if they believe the validation was incorrect. This allows for a process of checks and balances, ensuring that non-deterministic transactions are evaluated properly.
2. Re-computation: If a transaction is appealed, the system can re-evaluate the transaction with a new set of validators. The Finality Window provides the time necessary for this process to occur.
3. Security: The window also acts as a security feature, allowing the network to correct potential errors or malicious activity before finalizing a transaction.

Deterministic vs. Non-Deterministic Transactions

In GenLayer, Intelligent Contracts are classified as either deterministic or non-deterministic.

Deterministic Contracts

These contracts have a shorter Finality Window because their validation process is straightforward and not subject to appeals. However, it is essential that all interactions with the contract remain deterministic to maintain this efficiency.

Non-Deterministic Contracts

Non-deterministic contracts involve Large Language Model (LLM) calls or web data retrieval, which introduce variability in their outcomes. These contracts require a longer Finality Window to account for potential appeals and re-computation.

Note:

If a specific transaction within the contract is deterministic but interacts with a non-deterministic part of the contract, it will be treated as non-deterministic. This ensures that any appeals or re-computations of previous transactions are handled consistently, maintaining the integrity of the contract's overall state.

Fast Finality

For scenarios requiring immediate finality, such as emergency decisions in a DAO, it is possible to pay for all validators to validate the transaction immediately. This approach, though more costly, allows for fast finality, bypassing the typical Finality Window.

Note:

Fast finality only works if there are no previous non-deterministic transactions still within their Finality Window. Even if your transaction is considered final, if a previous transaction is reverted, your transaction will have to be recomputed as it might depend on the same state.

Appealability and Gas

When submitting a transaction, users can include additional gas to cover potential appeals. If a transaction lacks sufficient gas for appeals, third parties can supply additional gas during the Finality Window. Developers of Intelligent Contracts can also set minimum gas requirements for appealability, ensuring that critical transactions have adequate coverage.

Staking in GenLayer

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/optimistic-democracy/staking>

Participants, known as validators, commit a specified amount of tokens to the network by locking them up on the rollup layer. This commitment supports the network's consensus mechanism and enables validators to actively participate in processing transactions and managing the network.

Validators vs Delegators

Role	Minimum Stake	Infrastructure	Rewards
Validator	42,000 GEN	Must run a node	10% operational fee + stake rewards
Delegator	42 GEN	None required	Passive stake rewards

Validators run the consensus infrastructure and are responsible for executing intelligent contracts and validating transactions. They receive a 10% operational fee from rewards before distribution.

Delegators stake their tokens with validators without running infrastructure. They earn passive rewards proportional to their stake, minus the validator's operational fee.

How Staking Works

- **Stake Deposit:** To become a validator on GenLayer, participants must deposit GEN tokens on the rollup layer. This deposit acts as a security bond and qualifies them to join the pool of active validators.
- **Validator Participation:** Only a maximum of 1000 validators with the highest stakes can be part of the active validator set. Once staked, validators take on the responsibility of validating transactions and executing Intelligent Contracts. Their role is crucial for ensuring the network's reliability and achieving consensus on transaction outcomes.
- **Delegated Proof of Stake (DPoS):** GenLayer enhances accessibility and network security through a Delegated Proof of Stake system. This allows token holders who are not active validators themselves to delegate their tokens to trusted validators. By delegating their tokens, users increase the total stake of the validator and share in the rewards. Typically, the validator takes a configurable fee (around 10%), with the remaining rewards (90%) going to the delegating user.
- **Earning Rewards:** Validators, and those who delegate their tokens to them, earn rewards for their contributions to validating transactions, paid in GEN tokens. These rewards are proportional to the amount of tokens staked and the transaction volume processed.
- **Risk of Slashing:** Validators, and by extension their delegators, face the risk of having a portion of their staked tokens slashed (</understand-genlayer-protocol/core-concepts/optimistic-democracy/slashing>) if they fail to comply with network rules or if the validator supports fraudulent transactions.

Owner, Operator, and ValidatorWallet

When a validator joins, the system creates three distinct entities:

ValidatorWallet: A separate smart contract wallet created automatically on `validatorJoin()`. This is the primary validator identifier and holds staked GEN tokens.

Owner Address: The address that creates the validator (`msg.sender`). It controls staking operations and can change the operator address. Should use a cold wallet for security.

Operator Address: Used for consensus operations. Can differ from the owner (hot wallet recommended). It can be changed by the owner but cannot be the zero address or reused across validators.

Epoch System

The network operates in epochs (1 day):

- **Epoch +2 Activation Rule:** All deposits become active 2 epochs after they are made
- Epoch finalization requires all transactions to be finalized
- Cannot advance to epoch N+1 until epoch N-1 is finalized

- Validators are "primed" via `validatorPrime()` each epoch (permissionless - anyone can call it)

Critical: If `validatorPrime()` isn't called, the validator is excluded from the next epoch's selection.

Genesis Epoch 0

Epoch 0 is the genesis bootstrapping period with special rules designed to facilitate network launch. The normal staking rules are relaxed to allow rapid network bootstrapping.

What is Epoch 0?

Epoch 0 is the bootstrapping period before the network becomes operational. During epoch 0:

- No transactions are processed - the network is not yet active
- No consensus occurs - validators are not yet participating
- Stakes are registered and prepared for activation in epoch 2

Important: The network transitions directly from epoch 0 to epoch 2 (epoch 1 is skipped). Validators and delegators who stake in epoch 0 become active in epoch 2, but only if they meet the minimum stake requirements.

Special Rules for Epoch 0

Rule	Normal Epochs (2+)	Epoch 0
Validator minimum stake	42,000 GEN	No minimum to join
Delegator minimum stake	42 GEN	No minimum to join
Activation delay	+2 epochs	Active in epoch 2
<code>validatorPrime</code> required	Yes, each epoch	Not required
Share calculation	Based on existing ratio	1:1 (shares = input)
Transaction processing	Yes	No (bootstrapping only)

Activation requires meeting minimums: While you can join with any amount during epoch 0, your stake will only be activated in epoch 2 if it meets the minimum requirements (42,000 GEN for validators, 42 GEN for delegators). Stakes below the minimum remain registered but inactive.

Validators in Epoch 0

Key behaviors:

1. No minimum stake to join: Validators can join with any non-zero amount during epoch 0
2. Registered for epoch 2: Stakes are recorded and will become active when epoch 2 begins
3. No priming required: `validatorPrime()` is not needed during epoch 0
4. No consensus participation: Validators do not process transactions in epoch 0

Do validators need to take any action in epoch 0 to be active in epoch 2?

No. Validators who join in epoch 0:

- Have their stake registered during epoch 0
- Become active automatically in epoch 2 (epoch 1 is skipped) only if they have at least 42,000 GEN staked
- Must start calling `validatorPrime()` in epoch 2 for continued participation in epoch 4+

Important: Validators who joined in epoch 0 with less than 42,000 GEN will not be active in epoch 2. They must deposit additional funds to meet the minimum requirement before epoch 2 begins.

Delegators in Epoch 0

Key behaviors:

1. No minimum delegation: Any non-zero amount accepted during epoch 0
2. Registered for epoch 2: Delegation is recorded and will become active when epoch 2 begins
3. No rewards in epoch 0: Since no transactions are processed, no rewards are earned during epoch 0

Is a delegation made in epoch 0 active in epoch 2?

Yes. Delegations made in epoch 0 become active in epoch 2 (epoch 1 is skipped). Unlike normal epochs where you wait +2 epochs, epoch 0 delegations activate as soon as the network becomes operational.

Activation Timeline Comparison

Normal Epochs (2+):

```
Epoch N:  validatorJoin() or delegatorJoin() called
Epoch N+1: validatorPrime() stages the deposit
Epoch N+2: validatorPrime() activates the deposit → NOW ACTIVE
```

Epoch 0 (Bootstrapping):

```
Epoch 0: validatorJoin() or delegatorJoin() called → stake registered (not yet active)
          No transactions processed, no consensus
Epoch 2: Stakes become active (if minimum met), network operational, validatorPrime() required
```

Share Calculation in Epoch 0

In epoch 0, shares are calculated at a 1:1 ratio with the input amount:

```
Shares = Input Amount
Example: Deposit 1,000 GEN → Receive 1,000 shares
```

This is because there's no existing stake pool to calculate a ratio against. Starting from epoch 2, shares are calculated based on the current stake-to-share ratio.

Transitioning from Epoch 0 to Epoch 2

When the network advances from epoch 0 to epoch 2 (epoch 1 is skipped):

1. Epoch 0 stakes that meet minimums become active - validators need 42,000 GEN, delegators need 42 GEN
2. Normal minimum requirements apply for new joins/deposits
3. +2 epoch activation delay applies to all new deposits
4. validatorPrime() becomes mandatory for validators to remain in the selection pool
5. Existing validators must ensure their nodes begin calling validatorPrime() in epoch 2

FAQ: Epoch 0 Special Cases

Q: Can I join as a validator with less than 42,000 GEN in epoch 0?

A: Yes, any non-zero amount is accepted during epoch 0. However, you will not be active in epoch 2 unless you have at least 42,000 GEN staked by then.

Q: If I delegate in epoch 0, when does it become active?

A: In epoch 2. Unlike normal epochs with a +2 delay, epoch 0 delegations activate when the network becomes operational.

Q: Do I need to call `validatorPrime()` in epoch 0?

A: No. Priming is not required during epoch 0. Your node should start calling it automatically when epoch 2 begins.

Q: Will my epoch 0 stake still be active after epoch 0 ends?

A: Yes, if you meet the minimum requirements. Stakes from epoch 0 carry forward and remain active in all subsequent epochs.

Q: What happens to my stake if I joined in epoch 0 but my node doesn't call `validatorPrime()` in epoch 2?

A: You'll be excluded from validator selection in epoch 4, but your stake remains. Once priming resumes, you'll be eligible for selection again.

Shares vs Stake

The staking system uses shares to track ownership:

Shares are fixed quantities that never change. You receive shares when depositing and exit by burning shares. They represent immutable claims on the stake pool.

Stake is the dynamic GEN token amount. It increases with rewards/fees and decreases with slashing. The exchange rate is calculated as:

```
stake_per_share = total_stake / total_shares
```

Example: 100 shares representing 1,000 GEN (10 GEN per share). After rewards are distributed, the same 100 shares might represent 1,050 GEN (10.5 GEN per share). Rewards automatically compound without user action.

Validator Selection and Weight

Validators are selected for consensus based on their weight, calculated using:

```
Weight = (ALPHA × Self_Stake + (1-ALPHA) × Delegated_Stake)^BETA
```

Parameters:

- ALPHA = 0.6: Self-stake counts 50% more than delegated stake
- BETA = 0.5: Square-root damping prevents whale dominance

Effects:

- Higher stake leads to higher weight and higher selection probability
- Doubling stake only increases weight by approximately 41%
- Encourages distribution across validators

- Smaller validators often provide higher returns per GEN staked

Reward Distribution

Sources:

1. Transaction Fees
2. Inflation (starting at 15% APR, decreasing to 4% APR over time)

Distribution Pattern:

- 10% → Validator owners (operational fee)
- 75% → Total validator stake (validators + delegators)
- 10% → Developers
- 5% → Locked future allocation for the DeepThought AI-DAO

Within the 75% stake allocation:

- Self-stake receives a portion based on the validator's own staked amount
- Delegated stake is split among delegators proportionally to their shares

Rewards automatically increase the stake-per-share ratio without requiring user action.

Unbonding Period

Both validators and delegators face a 7-epoch unbonding period when withdrawing:

- Prevents rapid stake movements that could destabilize the network
- Exit is not processed immediately - validator remains active until next `validatorPrime()` call at next epoch
- Exited tokens stop earning rewards only after the exit is processed in the next epoch
- Countdown starts from the exit epoch
- Funds become claimable when: `current_epoch >= exit_epoch + 7`

Detailed Exit Flow:

```
Epoch N:  validatorExit(all_shares) called
          → Exit scheduled in contract state
          → Validator STILL ACTIVE and earning rewards
          → Can still be selected for consensus participation

Epoch N+1: validatorPrime() called (by anyone)
          → Exit processed: stake reduced to 0
          → Removed from validator tree
          → No longer active or earning rewards
          → Cannot be selected for new transactions

Epoch N+2: epochAdvance() called
          → Validator officially not in active validator set
          → Fully excluded from consensus operations

Epoch N+7: validatorClaim() callable
          → 7 epochs have passed since exit call (epoch N)
          → Tokens released to validator owner
```

Validator Priming

validatorPrime(address validator) is a critical function that:

- Activates pending deposits
- Processes pending withdrawals
- Distributes previous epoch rewards
- Applies pending slashing penalties
- Sorts the validator into the selection tree

Key Properties:

- Monitoring Required: Ensure correct execution
- Permissionless: Anyone can call it
- Incentivized: Caller receives 1% of any slashed amount
- Critical: If the node fails to prime, the validator is excluded from the next epoch
- No Loss: Missing priming doesn't lose rewards, but the validator can't be selected

Unstaking and Withdrawing

Both validators and delegators can withdraw their staked tokens, but must follow the unbonding process.

For Validators

To stop validating or retrieve staked tokens, validators must:

1. Calculate shares to exit: Determine how many shares to withdraw (partial or full)
2. Call validatorExit(shares): Initiate the unbonding process
3. Wait 7 epochs: Tokens are locked during the unbonding period
4. Call validatorClaim(): Retrieve tokens after unbonding completes

Validators can perform partial exits while remaining active, as long as their stake stays above the 42,000 GEN minimum.

For Delegators

Delegators follow a similar process:

1. Calculate shares to exit: Use sharesOf(delegator, validator) to check current shares
2. Call delegatorExit(validator, shares): Initiate unbonding for a specific validator
3. Wait 7 epochs: Tokens are locked during the unbonding period
4. Call delegatorClaim(delegator, validator): Retrieve tokens after unbonding

Important for delegators:

- Exit each validator separately if delegating to multiple validators
- Claims are permissionless—anyone can trigger them on your behalf

- Tokens stop earning rewards immediately upon calling exit
- Multiple exits create separate withdrawals that can be claimed together

For detailed step-by-step instructions and code examples, see the Staking Guide (/developers/staking-guide).

Governance and Safeguards

- 24-Hour Delay: All slashing actions have a governance delay period
- Parameters like ALPHA, BETA, minimum stakes, and unbonding periods are adjustable through governance
- Maximum 1,000 active validators per epoch (adjustable)

Next Steps

- Staking Guide (/developers/staking-guide) - Practical guide for staking operations
- Unstaking (/understand-genlayer-protocol/core-concepts/optimistic-democracy/unstaking) - Detailed unstaking process
- Slashing (/understand-genlayer-protocol/core-concepts/optimistic-democracy/slashing) - Slashing conditions and penalties

Slashing in GenLayer

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/optimistic-democracy/slashing>

Slashing is a mechanism used in GenLayer to penalize validators who engage in behavior detrimental to the network. This ensures that validators act honestly and effectively, maintaining the integrity of the platform and the Intelligent Contracts executed within it.

By penalizing undesirable behavior, slashing helps align validators' incentives with those of the network and its users.

Slashing Process

1. Violation Detection: The network identifies a violation, such as missing an execution window.
2. Slash Calculation: The amount to be slashed is calculated based on the specific violation and platform rules.
3. Stake Reduction: The slashed amount is deducted from the validator's stake.
4. Finality: The slashing becomes final after the Finality Window closes, ensuring that the validator's balance is finalized and accounts for any potential appeals.

When Slashing Occurs

Validators in GenLayer can be slashed for several reasons:

1. **Missing Transaction Execution Window:** Validators are expected to execute transactions within a specified time frame. If a validator misses this window, they are penalized, ensuring that validators remain active and responsive.
2. **Missing Appeal Execution Window:** During the appeals process, validators must respond within a set time frame. If they fail to do so, they are slashed, which motivates validators to participate in the appeals process.

Amount Slashed

The amount slashed varies based on the severity of the violation and the specific rules set by the GenLayer platform. The slashing amount is designed to be substantial enough to deter malicious or negligent behavior while not being excessively punitive for honest mistakes.

Unstaking in GenLayer

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/optimistic-democracy/unstaking>

Unstaking in GenLayer is the process by which validators disengage their staked tokens from the network, ending their active participation as validators. They must initiate an unstaking process, which includes a cooldown period to finalize all pending transactions. This procedure ensures that all obligations are fulfilled and pending issues resolved, maintaining the network's integrity and securing the platform's operations.

How Unstaking Works

The unstaking process includes several key steps:

1. **Initiating Unstaking:** Validators initiate their exit from active duties by submitting an unstaking transaction, signaling their intention to cease participation in validating transactions.
2. **Validator Removal:** Once the unstaking request is made, the validator is promptly removed from the pool of active validators, meaning they will no longer receive new transactions or be called upon for appeal validations.
3. **Finality Period:** During this period, validators must wait for all transactions they have participated in to reach full finality. This is crucial to ensure that validators do not exit while still having potential influence over unresolved transactions. This cooldown period helps prevent the situation where new transactions with new finality windows could prevent them from ever achieving full finality on all transactions they were involved in.
4. **Withdrawing Stake:** After all transactions have achieved finality and no outstanding issues remain, validators and their delegators can safely withdraw their staked tokens and any accrued rewards.

Purpose of Unstaking

The unstaking process is designed to:

- **Ensure Accountability:** By enforcing a Finality Window, validators are held accountable for their actions until all transactions they influenced are fully resolved. This prevents premature exit from the network and ensures that

all potential disputes are settled.

- **Align Incentives:** The requirement for validators to wait through the Finality Window aligns their incentives with the long-term security and reliability of the network, promoting responsible participation.
- **Maintain Network Security:** The unstaking process discourages abrupt departures and ensures that validators address any possible security concerns related to their past validations before leaving.

Implications for Validators and Delegators

For validators, this process mandates careful planning regarding their exit strategy from the network, considering the need to wait out the Finality Window. Delegators must also be patient, understanding that their assets will remain locked until their validator has cleared all responsibilities, safeguarding their investments from potential liabilities caused by unresolved validations.

Rollup Integration

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/rollup-integration>

GenLayer leverages Ethereum rollups, such as ZKSync or Polygon CDK, to ensure scalability and compatibility with existing Ethereum infrastructure. This integration is crucial for optimizing transaction throughput and reducing fees while maintaining the security guarantees of the Ethereum mainnet.

Key Aspects of Rollup Integration

Scalability

- **High Transaction Throughput:** Rollups allow GenLayer to process a much higher number of transactions per second compared to Layer 1 solutions.
- **Reduced Congestion:** By moving computation off-chain, GenLayer helps alleviate congestion on the Ethereum mainnet.

Cost Efficiency

- **Lower Transaction Fees:** Users benefit from significantly reduced gas fees compared to direct Layer 1 transactions.
- **Batched Submissions:** Transactions are batched and submitted to the Ethereum mainnet, distributing costs across multiple operations.

Security

- **Ethereum Security Inheritance:** While execution happens off-chain, the security of assets and final state is guaranteed by Ethereum's robust consensus mechanism.
- **Fraud Proofs/Validity Proofs:** Depending on the specific rollup solution (Optimistic or ZK), security is ensured through either fraud proofs or validity proofs.

How Rollup Integration Works with GenLayer

1. Transaction Submission: Users submit transactions to the rollup.
2. Transaction Execution: Transactions are executed within the GenVM environment.
3. Consensus: The rollup layer implements the Optimistic Democracy mechanism to reach consensus on the state updates.
4. State Updates: The rollup layer maintains an up-to-date state of all accounts and contracts.
5. Batch Submission: Periodically, batches of transactions and state updates are submitted to the Ethereum mainnet.
6. Verification: The Ethereum network verifies the integrity of the submitted data, ensuring its validity.

Benefits for Developers and Users

- Ethereum Compatibility: Developers can leverage existing Ethereum tools and infrastructure.
- Improved User Experience: Lower fees and faster transactions lead to a better overall user experience.

Considerations

- Withdrawal Periods: Depending on the rollup solution, there might be waiting periods for withdrawing assets back to the Ethereum mainnet.
- Rollup-Specific Features: Different rollup solutions may offer unique features or limitations that developers should be aware of.

By integrating with Ethereum rollups, GenLayer combines the innovative capabilities of Intelligent Contracts with the scalability and efficiency of Layer 2 solutions, creating a powerful platform for next-generation decentralized applications.

Non-Deterministic Operations Handling

Source:

<https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/non-deterministic-operations-handling>

Overview

GenLayer extends traditional smart contracts by allowing Intelligent Contracts to perform non-deterministic operations, such as interacting with Large Language Models (LLMs) and accessing web data. Handling the variability inherent in these operations is crucial for maintaining consensus across the network.

Challenges

- Variability of Outputs: Non-deterministic operations can produce different outputs when executed by different validators.
- Consensus Difficulty: Achieving consensus on varying outputs requires specialized mechanisms.

Solutions in GenLayer

- Equivalence Principle: Validators assess whether different outputs are equivalent based on predefined criteria, allowing for consensus despite variability.
- Optimistic Democracy: The consensus mechanism accommodates non-deterministic operations by allowing provisional acceptance of transactions and providing an appeals process.

Developer Considerations

- Defining Equivalence Criteria: Developers must specify what constitutes equivalent outputs in their Intelligent Contracts.
- Testing and Validation: Thorough testing is essential to ensure that non-deterministic operations behave as expected in the consensus process.

Large Language Model (LLM) Integration

Source:

<https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/large-language-model-llm-integration>

Overview

Intelligent Contracts in GenLayer can interact directly with Large Language Models (LLMs), enabling natural language processing and more complex decision-making capabilities within blockchain applications.

Key Features

- Natural Language Understanding: Contracts can process and interpret instructions written in natural language.
- Dynamic Decision Making: Utilizing LLMs allows contracts to make context-aware decisions based on complex inputs.

Implementation

1. LLM Providers: Validators are configured with LLM providers (e.g., OpenAI, Ollama) to process LLM requests.
2. Equivalence Principle: LLM outputs are validated using the Equivalence Principle to ensure consensus among validators.
3. Prompt Design: Developers craft prompts to interact effectively with LLMs, specifying expected formats and constraints.

Considerations

- Cost and Performance: LLM interactions may incur additional computational costs and latency.
- Security: Care must be taken to prevent prompt injections and ensure the reliability of LLM responses.

Web Data Access in Intelligent Contracts

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/web-data-access>

Overview

GenLayer enables Intelligent Contracts to directly access and interact with web data, removing the need for oracles and allowing for real-time data integration into blockchain applications.

Key Features

- Direct Web Access: Contracts can retrieve data from web sources.
- Dynamic Applications: Access to web data allows for applications that respond to external events and real-world data.
- Equivalence Validation: Retrieved data is validated across validators to ensure consistency.

Implementation

1. Data Retrieval Functions: GenLayer provides mechanisms for fetching web data within contracts.
2. Data Parsing and Validation: Contracts must parse web data and validate it according to defined equivalence criteria.
3. Security Measures: Contracts should handle potential security risks such as untrusted data sources and ensure data integrity.

Considerations

- Network Dependencies: Reliance on external web sources introduces dependencies that may affect contract execution.
- Performance Impact: Web data retrieval may introduce latency and affect transaction processing times.

Transactions

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/transactions>

Transactions are the fundamental operations that drive the GenLayer protocol. Whether it's deploying a new contract, sending value between accounts, or invoking a function within an existing contract, transactions are the means by which state changes occur on the network.

Here is the general structure of a transaction:

```
{
  "consensus_data": {
    "leader_receipt": {
      "args": [
        [
          2,
          "0x793Ae2CfF17462cc9f9D68e194b7b949d2080Ea2"
        ]
      ],
      "class_name": "LlmErc20",
      "contract_state": "gASVnAAAAAAAAACMF2JhY2t1bmQubm9kZS5nZW52bS5iYXNlIiwITGxtRXJjMjMjCuk5QpgZR9lIwIYmFsYW5jZX0UfZQojCoweEQyNzFjNzRBNzgwODNGMzU3YTlmOGQzMWQ1YWRDNTlCMzk1Y2YxNmKUS2KMKjB4NzkzQWUyQ2ZGMTc0NjJjYzlmOUQ2OGUxOTRiN2I5NDlkmjA4MEVhMpRLAnVzYi4=",
      "eq_outputs": {
        "leader": {
          "0": "{\"transaction_success\": true, \"transaction_error\": \"\", \"updated_balances\": {\"0xD271c74A78083F357a9f8d31d5adC59B395cf16b\": 98, \"0x793Ae2CfF17462cc9f9D68e194b7b949d2080Ea2\": 2}}"}
        }
      },
      "error": null,
      "execution_result": "SUCCESS",
      "gas_used": 0,
      "method": "transfer",
      "mode": "leader",
      "node_config": {
        "address": "0x185D2108D9dE15ccf6beEb31774CA96a4f19E62B",
        "config": {},
        "model": "gpt-4o",
        "plugin": "openai",
        "plugin_config": {
          "api_key_env_var": "OPENAIKEY",
          "api_url": null
        }
      },
      "provider": "openai",
      "stake": 1
    },
    "vote": "agree"
  },
  "validators": [
    {
      "args": [
        [
          2,
          "0x793Ae2CfF17462cc9f9D68e194b7b949d2080Ea2"
        ]
      ],
      "class_name": "LlmErc20",
      "contract_state": "gASVnAAAAAAAAACMF2JhY2t1bmQubm9kZS5nZW52bS5iYXNlIiwITGxtRXJjMjMjCuk5QpgZR9lIwIYmFsYW5jZX0UfZQojCoweEQyNzFjNzRBNzgwODNGMzU3YTlmOGQzMWQ1YWRDNTlCMzk1Y2YxNmKUS2KMKjB4NzkzQWUyQ2ZGMTc0NjJjYzlmOUQ2OGUxOTRiN2I5NDlkmjA4MEVhMpRLAnVzYi4=",
      "eq_outputs": {
        "leader": {
          "0": "{\"transaction_success\": true, \"transaction_error\": \"\", \"updated_balances\": {\"0xD271c74A78083F357a9f8d31d5adC59B395cf16b\": 98, \"0x793Ae2CfF17462cc9f9D68e194b7b949d2080Ea2\": 2}}"}
        }
      },
      "error": null,
      "execution_result": "SUCCESS",
      "gas_used": 0,
      "method": "transfer",
      "mode": "validator",
      "node_config": {
        "address": "0x31bc9380eCbF487EF5919eBa7457F457B5196FCD",
        "config": {},
        "model": "gpt-4o",
        "plugin": "openai",
        "plugin_config": {

```

```

        "api_key_env_var": "OPENAIKEY".
        "api_url": null
    },
    "provider": "openai",
    "stake": 1
},
"pending_transactions": [],
"vote": "agree"
},
...
],
"votes": {
    "0x185D2108D9dE15ccf6beEb31774CA96a4f19E62B": "agree",
    "0x2F04Fb1e5daf7DCbf170E4CB0e427d9b11aB96cA": "agree",
    "0x31bc9380eCbF487EF5919eBa7457F457B5196FCD": "agree"
}
},
"created_at": "2024-10-02T20:32:50.469443+00:00",
"data": {
    "function_args": "[2, \"0x793Ae2CfF17462cc9f9D68e194b7b949d2080Ea2\"]",
    "function_name": "transfer"
},
"from_address": "0xD271c74A78083F357a9f8d31d5adC59B395cf16b",
"gas_limit": 66,
"hash": "0xb7486f70a3fec00af5f929fc1cf1078af9ff3a063afe8b6f370a44a96635505d",
"leader_only": false,
"nonce": 66,
"r": null,
"s": null,
"status": "FINALIZED",
"to_address": "0x5929bB548a2Fd7E9Ea2577DaC9c67A08BbC2F356",
"type": 2,
"v": null,
"value": 0
}

```

Explanation of fields:

- consensus_data: Object containing information about the consensus process
- leader_receipt: Object containing details about the leader's execution of the transaction
- args: Arguments passed to the contract function
- class_name: Name of the contract class
- contract_state: Encoded state of the contract
- eq_outputs: Outputs from every equivalence principle in the execution of the contract method
- error: Any error that occurred during execution (null if no error)
- execution_result: Result of the execution (e.g., "SUCCESS" or "ERROR")
- gas_used: Amount of gas used in the transaction
- method: Name of the method called on the contract
- mode: Execution mode (e.g., "leader" or "validator")
- node_config: Configuration of the node executing the transaction
- address: Address of the node
- config: Configuration of the node
- model: Model of the node
- plugin: Plugin used for the LLM provider connection
- plugin_config: Configuration of the plugin
- api_key_env_var: Environment variable containing the API key for the given provider
- api_url: API URL for the given provider

- provider: Provider of the node
- stake: Stake of the validator
- vote: The leader's vote on the transaction (e.g., "agree")
- validators: Array of objects containing similar information for each validator
- votes: Object mapping validator addresses to their votes
- created_at: Timestamp of when the transaction was created
- data: Object containing details about the function call in the transaction
- from_address: Address of the account initiating the transaction
- gaslimit: Maximum amount of gas the transaction is allowed to consume
- hash: Unique identifier (hash) of the transaction
- leader_only: Boolean indicating whether the transaction is to be executed by the leader node only
- nonce: Number of transactions sent from the from_address (used to prevent double-spending)
- r: Part of the transaction signature (null if not yet signed)
- s: Part of the transaction signature (null if not yet signed)
- status: Current status of the transaction (e.g., "FINALIZED")
- to_address: Address of the contract or account receiving the transaction
- type: Internal type of the transaction (2 indicates a contract write call)
- v: Part of the transaction signature (null if not yet signed)
- value: Amount of native currency (GEN) being transferred in the transaction

Types of Transactions

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/transactions/types-of-transactions>

There are three different types of transactions that users can send in GenLayer. All three types are sent through the same RPC method, but they differ in the data they contain and the actions they perform.

1. Deploy a Contract

Deploying a contract involves creating a new Intelligent Contract on the GenLayer network. This transaction initializes the contract's state and assigns it a unique address on the blockchain. The deployment process ensures that the contract code is properly validated and stored, making it ready to be called.

Example

```
{
  "consensus_data": {
    "leader_receipt": {
      "args": [
        {
          "total_supply": 100
        }
      ],
      "class_name": "LlMERC20",
      "contract_state": "gASVnAAAAAAAAACMF2JhY2t1bmQubm9kZS5nZW52bS5iYXNl1IwITGxtRXJjMjCuk5QpgZR9liwIYmFsYW5jZX0UfZQojCoweEQyNzFjNzRBNzgWODNGMzU3Yt1m0GQzMWQ1YWRDNTlCMzk1Y2YxNmKUS2KMKjB4NzkzQWUyQ2ZGMTc0NjJjYzlmOUQ2OGUxOTRiN2I5NDlkMjA4MEV"
```

```
hMpRLANvZyI4=",
    ...
  },
  "validators": [
    ...
  ],
  ...
},
"data": {
  "constructor_args": "{\\"total_supply\\":100}",
  "contract_address": "0x5929bb548a2Fd7E9Ea2577DaC9c67A08BbC2F356",
  "contract_code": "import json\nfrom backend.node.genvm.icontract import IContract\nfrom\nbackend.node.genvm.equivalence_principle import EquivalencePrinciple\n\n\nclass LlmErc20(IContract):\n    def\n    _init__(self, total_supply: int) -> None:\n        self.balances = {}\n\n    self.balances[contract_runner.from_address] = total_supply\n...",
  },
  ...
}
```

2. Send Value

Sending value refers to transferring the native GEN token from one account to another. This is one of the most common types of transactions. Each transfer updates the balance of the involved accounts, and the transaction is recorded on the blockchain to ensure transparency and security.

Example

```
{
  "consensus_data": null,
  "created_at": "2024-10-02T21:21:04.192995+00:00",
  "data": {},
  "from_address": "0x0Bd6441CB92a64fA667254BCa1e102468fffB3f3",
  "gaslimit": 0,
  "hash": "0x6357ec1e86f003b20964ef3b2e9e072c7c9521f92989b08e04459b871b69de89",
  "leader_only": false,
  "nonce": 2,
  "r": null,
  "s": null,
  "status": "FINALIZED",
  "to_address": "0xf739FDe22E0C0CB6DFD8f3F8D170bFC07329489E",
  "type": 0,
  "v": null,
  "value": 200
}
```

3. Call Contract Function

Calling a contract function is the process of invoking a specific method within an existing Intelligent Contract. This could involve anything from querying data stored within the contract to executing more complex operations like transferring tokens or interacting with other contracts. Each function call is a transaction that modifies the contract's state based on the inputs provided.

Example

```
{
  "consensus_data": {
    "leader_receipt": {
      "args": [
        [
          2,
          "0x793Ae2CfF17462cc9f9D68e194b7b949d2080Ea2"
        ]
      ],
      "class_name": "LlmErc20",
      "contract_state": "gASVnAAAAAAAAACMF2JhY2t1bmcubm9kZS5nZW52bS5iYXNlllIwITGxtRXJjMjCuk5QpgZR9lIwIYmFsYW5jZXOUfZQojCoweEQyNzFjNzRBNzgwODNGMzU3YTlmOGQzMWw1YWRDNTlCMzk1Y2YxNmKUS2MKKjB4NzgzQWUyQ2ZGMTc0NjJjYzlmOUQ2OGUxOTRiN2I5NDlkMjA4MEVhMjRlANvZyI4=",
      "eq_outputs": {
        "leader": {
          "0": "{\"transaction success\": true, \"transaction error\": \"\", \"updated balances\":
```

```
{\ "0xD271c74A78083F357a9f8d31d5adC59B395cf16b\": 98, \ "0x793Ae2CfF17462cc9f9D68e194b7b949d2080Ea2\": 2}}"
  },
  ...
},
"validators": [
  ...
],
...
},
"data": {
  "function_args": "[2,\ "0x793Ae2CfF17462cc9f9D68e194b7b949d2080Ea2\ "]",
  "function_name": "transfer"
},
...
}
```

- For a list of all the fields in a transaction, see here (</core-concepts/transactions>)

Transaction Processing

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/transactions/transaction-statuses>

In GenLayer, transactions are processed through an account-based queue system that ensures orderliness. Here's how transactions transition through different statuses:

1. Pending

When a transaction is first submitted, it enters the pending state. This means it has been received by the network but is waiting to be processed. Transactions are queued per account, ensuring that each account's transactions are processed in the order they were submitted.

2. Proposing

In this stage, the transaction is moved from the pending queue to the proposing stage. A leader and a set of voters are selected from the validator set via a weighted random selection based on total stake. The leader proposes a receipt for the transaction, which is then committed to by the validators.

3. Committing

The transaction enters the committing stage, where validators commit their votes and cost estimates for processing the transaction. This stage is crucial for reaching consensus on the transaction's execution.

4. Revealing

After the committing stage, validators reveal their votes and cost estimates, allowing the network to finalize the transaction's execution cost and validate the consensus.

5. Accepted

Once the majority of validators agree on the transaction's validity and cost, the transaction is marked as accepted. This status indicates that the transaction has passed through the initial validation process successfully.

6. Finalized

After all validations are completed and any potential appeals have been resolved, the transaction is finalized. In this state, the transaction is considered irreversible and is permanently recorded in the blockchain.

7. Undetermined

If the transaction fails to reach consensus after all voting rounds, it enters the undetermined state. This status indicates that the transaction's outcome is unresolved, and it may require further validation or be subject to an appeal process.

8. Canceled

A transaction can be canceled by the user or by the system if it fails to meet certain criteria (e.g., insufficient funds). Once canceled, the transaction is removed from the processing queue and will not be executed.

Transaction execution

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/transactions/transaction-execution>

Once a transaction is received and properly verified by the `eth_sendRawTransaction` method on the RPC server, it is stored with a `PENDING` status and its hash is returned as the RPC method response. This means that the transaction has been validated for authenticity and format, but it has not yet been executed. From this point, the transaction enters the GenLayer consensus mechanism, where it is picked up for execution by the network's validators according to the consensus rules.

As the transaction progresses through various stages—such as proposing, committing, and revealing—its status is updated accordingly. Throughout this process, the current status and output of the transaction can be queried by the user. This is done by calling the `eth_getTransactionByHash` method on the RPC server, which retrieves the transaction's details based on its unique hash. This method allows users to track the transaction's journey from submission to finalization, providing transparency and ensuring that they can monitor the outcome of their transactions in real-time.

Transaction Status Transitions

In the journey of a transaction within the GenLayer protocol, it begins its life when an Externally Owned Account (EOA) submits it, entering the Pending state. Here, it awaits further processing unless it encounters an `OutOfFee` state due to insufficient fees. If the fees are topped up, it returns to Pending. Alternatively, the user can cancel the transaction, moving it to the Canceled state.

From Pending, the transaction progresses to the Proposing stage, where a leader is selected to propose a receipt. Upon successful proposal, it advances to the Committing stage, where all validators must commit to the transaction. If all validators commit, the transaction moves to the Revealing stage.

In the Revealing stage, the transaction's fate is determined. If a majority agrees, it is Accepted. However, if there is no majority agreement, it returns to Proposing. If all leaders are rotated without agreement, it becomes Undetermined. In cases of disagreement, a successful appeal can revert it to Pending, while a failed appeal results in Accepted.

Once Accepted, the transaction awaits the passing of the appeal window to become Finalized. An Undetermined transaction also becomes Finalized after the appeal window passes. However, an appeal can initiate a return to Committing from Accepted, or automatically revert an Undetermined transaction to Pending.

```
graph TD;
    Start(( )) -->|EOA submits| Pending(Pending)
    Start -->|Insufficient fee| OutOfFee(OutOfFee)
    OutOfFee -->|Fee topped up| Pending
    Pending -->|User cancels| Canceled(Canceled)

    Pending -->|Select leader| Proposing(Proposing)
    Proposing -->|Leader proposes receipt| Committing(Committing)
    Committing -->|All validators commit| Revealing(Revealing)

    Revealing -->|Majority agrees| Accepted(Accepted)
    Revealing -->|No majority agreement| Proposing
    Revealing -->|All leaders rotated| Undetermined(Undetermined)
    Revealing -->|Disagreement, appeal successful| Pending
    Revealing -->|Agreement, appeal failed| Accepted

    Accepted -->|Appeal window passes| Finalized(Finalized)
    Undetermined -->|Appeal window passes| Finalized
    Accepted -->|Appeal initiated| Committing
    Undetermined -->|Appeal automatically| Pending
```

Transaction encoding, serialization, and signing

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/transactions/transaction-encoding-serialization-and-signing>

In GenLayer, all three types of transactions needs to be properly encoded, serialized, and signed on the client-side. This process ensures that the transaction data is packaged into a reliable cross-platform efficient format, and securely signed using the sender's private key to verify the sender's identity.

Once prepared, the transaction is sent to the network via the `eth_sendRawTransaction` method on the RPC Server. This method performs the inverse process: it decodes and deserializes the transaction data, and then verifies the signature to ensure its authenticity. By handling all transaction types through `eth_sendRawTransaction`, GenLayer ensures that transactions are processed securely and efficiently while maintaining compatibility with Ethereum's specification.

Economic Model

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/economic-model>

Overview

GenLayer's economic model is designed to incentivize participants to maintain the network's security and functionality. It involves staking, rewards, transaction fees, and penalties.

Key Components

- Staking: Validators must stake tokens to participate in the validation process, aligning their interests with the network's health.
- Rewards: Validators receive rewards for correctly validating transactions.
- Transaction Fees: Users pay fees for transaction processing, which are partly used to reward validators.
- Slashing: Validators acting maliciously or incompetently can have their staked tokens slashed as a penalty.

Incentive Mechanisms

- Positive Incentives: Rewards and fees motivate validators to act in the network's best interest.
- Negative Incentives: Slashing and penalties deter malicious behavior.

Economic Security

- Stake-Based Security: The amount staked by validators serves as a deterrent against attacks, as they risk losing their stake.
- Balancing Supply and Demand: The economic model aims to balance the supply of validation services with demand from users.

Accounts and Addressing

Source: <https://docs.genlayer.com/understand-genlayer-protocol/core-concepts/accounts-and-addresses>

Overview

Accounts are fundamental to interacting with the GenLayer network. They represent users or entities that can hold tokens, deploy Intelligent Contracts, and initiate transactions.

Types of Accounts

1. Externally Owned Accounts (EOAs):
 - Controlled by private keys
 - Can initiate transactions and hold tokens
2. Contract Accounts:
 - Associated with deployed Intelligent Contracts
 - Have their own addresses and code

Account Addresses

- Address Format: GenLayer uses a specific address format, typically represented as a hexadecimal string prefixed with 0x.
- Public and Private Keys: Addresses are derived from public keys, which in turn are generated from private keys kept securely by the account owner.

Interacting with Intelligent Contracts

- Transaction Sending: Accounts initiate transactions to call functions on Intelligent Contracts or transfer tokens.
- Gas Fees: Transactions require gas fees to be processed.

Account Management

- Creating Accounts: Users can create new accounts using wallets or development tools provided by GenLayer.
- Security Practices: Users must securely manage their private keys, as losing them can result in loss of access to their funds.

Getting Started with GenLayer

Source: <https://docs.genlayer.com/developers>

This guide helps developers build, deploy, and integrate Intelligent Contracts on GenLayer.

Start with Skills (Recommended)

The fastest path from zero to deployed contract is the GenLayer Skills (<https://skills.genlayer.com/>) plugin for Claude Code. The genlayer-dev skill scaffolds a project, runs the linter and direct-mode tests, deploys to testnet, and walks you through the SDKs interactively.

```
claude /plugin marketplace add genlayerlabs/skills
claude /plugin install genlayer-dev@genlayerlabs
```

Then run `claude /genlayer-dev` (or just ask Claude what to do next). The skill bundles the same workflows documented on this site — it just runs them for you.

Note:

Prefer to set things up manually or read the references first? The cards below cover every topic in depth, and the Tooling Setup (</developers/intelligent-contracts/tooling-setup>) page describes the manual install path.

Reference

- 📖 Introduction (</developers/intelligent-contracts/introduction>)
- ⚙️ Tooling Setup (</developers/intelligent-contracts/tooling-setup>)
- 📄 First Contract (</developers/intelligent-contracts/your-first-contract>)

- [Types \(/developers/intelligent-contracts/types\)](/developers/intelligent-contracts/types)
- [Storage \(/developers/intelligent-contracts/storage\)](/developers/intelligent-contracts/storage)
- [First Intelligent Contract \(/developers/intelligent-contracts/your-first-intelligent-contract\)](/developers/intelligent-contracts/your-first-intelligent-contract)
- [Equivalence Principle \(/developers/intelligent-contracts/equivalence-principle\)](/developers/intelligent-contracts/equivalence-principle)
- [Debugging \(/developers/intelligent-contracts/debugging\)](/developers/intelligent-contracts/debugging)
- [Error Handling \(/developers/intelligent-contracts/error-handling\)](/developers/intelligent-contracts/error-handling)
- [Crafting Prompts \(/developers/intelligent-contracts/crafting-prompts\)](/developers/intelligent-contracts/crafting-prompts)
- [Advanced Features \(/developers/intelligent-contracts/advanced-features/contract-to-contract-interaction\)](/developers/intelligent-contracts/advanced-features/contract-to-contract-interaction)
- [Security and Best Practices \(/developers/intelligent-contracts/security-and-best-practices/prompt-injection\)](/developers/intelligent-contracts/security-and-best-practices/prompt-injection)
- [Examples \(/developers/intelligent-contracts/examples/storage\)](/developers/intelligent-contracts/examples/storage)
- [Tools \(/developers/intelligent-contracts/tools/genlayer-cli\)](/developers/intelligent-contracts/tools/genlayer-cli)
- [Ideas \(/developers/intelligent-contracts/ideas\)](/developers/intelligent-contracts/ideas)
- [Architecture \(/developers/decentralized-applications/architecture-overview\)](/developers/decentralized-applications/architecture-overview)
- [DApp Development \(/developers/decentralized-applications/dapp-development-workflow\)](/developers/decentralized-applications/dapp-development-workflow)
- [GenLayer JS \(/developers/decentralized-applications/genlayer-js\)](/developers/decentralized-applications/genlayer-js)
- [Reading Data \(/developers/decentralized-applications/reading-data\)](/developers/decentralized-applications/reading-data)
- [Writing Data \(/developers/decentralized-applications/writing-data\)](/developers/decentralized-applications/writing-data)
- [Querying \(/developers/decentralized-applications/queries-a-transaction\)](/developers/decentralized-applications/queries-a-transaction)
- [Testing \(/developers/decentralized-applications/testing\)](/developers/decentralized-applications/testing)
- [Project Boilerplate \(/developers/decentralized-applications/project-boilerplate\)](/developers/decentralized-applications/project-boilerplate)

Networks

Source: <https://docs.genlayer.com/developers/networks>

GenLayer operates as two layers: the GenLayer RPC handles intelligent contract operations (gen_* methods), while the GenLayer Chain is the underlying L2 (zkSync Elastic Chain) that handles standard Ethereum operations (eth_* methods). The GenLayer RPC also passes through all eth_* and zks_* calls to the underlying chain, so you can use either endpoint for standard wallet operations.

For the full RPC reference, see GenLayer Node API (/api-references/genlayer-node).

Your wallet connects to the GenLayer RPC — it handles both intelligent contract calls and standard Ethereum methods.

Testnet Bradbury

Production-like testnet with real AI/LLM workloads.

```
| Setting | Value |
|:--|:--|
| **GenLayer RPC** | `https://rpc-bradbury.genlayer.com` |
| **GenLayer Chain RPC** | `https://rpc.testnet-chain.genlayer.com` |
| **Chain ID** | 4221 |
| **Currency** | GEN |
| **Explorer** | [explorer-bradbury.genlayer.com](https://explorer-bradbury.genlayer.com) |
| **Chain Explorer** | [explorer.testnet-chain.genlayer.com](https://explorer.testnet-chain.genlayer.com/) |
| **Faucet** | [testnet-faucet.genlayer.foundation](https://testnet-faucet.genlayer.foundation) |
```

Testnet Asimov

Infrastructure and stress testing.

```
| Setting | Value |
|:--|:--|
| **GenLayer RPC** | `https://rpc-asimov.genlayer.com` |
| **GenLayer Chain RPC** | `https://rpc.testnet-chain.genlayer.com` |
| **Chain ID** | 4221 |
| **Currency** | GEN |
| **Explorer** | [explorer-asimov.genlayer.com](https://explorer-asimov.genlayer.com) |
| **Chain Explorer** | [explorer.testnet-chain.genlayer.com](https://explorer.testnet-chain.genlayer.com/) |
| **Faucet** | [testnet-faucet.genlayer.foundation](https://testnet-faucet.genlayer.foundation) |
```

Stunionet

Hosted development environment — no local setup required.

```
| Setting | Value |
|:--|:--|
| **GenLayer RPC** | `https://studio.genlayer.com/api` |
| **Chain ID** | 61999 |
| **Currency** | GEN |
| **Explorer** | [explorer-studio.genlayer.com](https://explorer-studio.genlayer.com) |
| **Faucet** | Built-in – use the  button in the account selector |
```

Localnet

Local development with full control. Requires GenLayer Studio
(/developers/intelligent-contracts/tools/genlayer-studio) or GLSim (/api-references/genlayer-test/glsim).

```
| Setting | Value |
|:--|:--|
| **GenLayer RPC** | `http://localhost:4000/api` |
| **Chain ID** | 61127 |
| **Currency** | GEN |
| **Explorer** | Bundled with Studio at `http://localhost:8080` |
| **Faucet** | Built-in – use the  button in the account selector |
```

GenLayer Chain (L2)

The underlying zkSync Elastic Chain that GenLayer runs on. You typically don't need to interact with this directly — the GenLayer RPC passes through all eth_* calls. But if you need direct L2 access (e.g., token transfers, contract debugging at the EVM level), you can add this chain to your wallet.

```
| Setting | Value |
|:--|:--|
| **RPC** | `https://rpc.testnet-chain.genlayer.com` |
| **Chain ID** | 4221 |
| **Currency** | GEN |
| **Explorer** | [explorer.testnet-chain.genlayer.com](https://explorer.testnet-chain.genlayer.com/) |
```

Network Comparison

```
| | Bradbury | Asimov | Studionet | Localnet |
|:--|:--|:--|:--|:--|
| **Purpose** | Production-like testing | Infrastructure testing | Hosted dev | Local dev |
| **Setup** | None – connect and go | None – connect and go | None – browser only | Docker or GLSim |
| **Persistence** | Persistent | Persistent | Temporary | Local only |
| **LLM execution** | Real models | Real models | Real models | Configurable |
| **Faucet** | [Available](https://testnet-faucet.genlayer.foundation) | [Available](https://testnet-faucet.genlayer.foundation) | Bu
```

Recommended Flow

1. Start on Studionet — zero setup, open studio.genlayer.com (<https://studio.genlayer.com>)
2. Move to Localnet — when you need full control and fast iteration
3. Deploy to Bradbury — when ready for production-like testing with real AI workloads

Introduction to Intelligent Contracts

Source: <https://docs.genlayer.com/developers/intelligent-contracts/introduction>

What are Intelligent Contracts?

Intelligent Contracts are an advanced evolution of smart contracts that combine traditional blockchain capabilities with natural language processing and web connectivity. Built in Python, they enable developers to create contracts that can understand human language, process external data, and make complex decisions based on real-world information.

Key Features of Intelligent Contracts

Natural Language Processing (NLP)

Intelligent Contracts leverage Large Language Models (LLMs) to process and understand human language inputs. This integration enables the contracts to interpret complex text-based instructions and requirements, moving beyond simple conditional logic.

Through NLP capabilities, these contracts can analyze qualitative criteria and make nuanced decisions based on contextual understanding, bringing a new level of intelligence to contract execution.

Web Connectivity

These contracts can actively interact with web APIs to fetch real-time information, enabling dynamic decision-making based on current data. By incorporating external services for data verification, they maintain a reliable connection to real-world events and conditions, bridging the gap between on-chain and off-chain environments.

Non-Deterministic Operations

Intelligent Contracts introduce a sophisticated approach to handling operations with unpredictable outputs, a significant advancement over traditional deterministic smart contracts. Through the implementation of a built-in equivalence principle, multiple validators can reach consensus even when processing non-deterministic results. This system supports both comparative validation, where outputs are directly matched, and non-comparative validation, where validators assess the reasonableness of results within defined parameters.

How Do Intelligent Contracts Work?

Contract Structure

Intelligent Contracts are written in Python using the GenVM SDK library. The basic structure consists of:

1. Dependencies Declaration: Specify required GenVM SDK modules
2. Declare Contract: Extend `gl.Contract` to define a contract class
3. State Variables: Declare with type annotations for strong typing
4. Methods:
 - `@gl.public.view`: Read-only methods that don't modify state
 - `@gl.public.write`: Methods that can modify contract state
 - `@gl.public.write.payable`: Methods that can modify contract state *and* receive value

Here's an example:

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }
from genlayer import *

class MyContract(gl.Contract):
    # State variables with type annotations
    variable: str

    def __init__(self):
        self.variable = "initial value"

    @gl.public.view
    def read_method(self) -> str:
        return self.variable

    @gl.public.write
    def write_method(self, new_value: str):
        self.variable = new_value
```

Validation Process

- When transactions are submitted to Intelligent Contracts, they are automatically queued in a contract-specific order and marked with a "pending" status

- A randomly selected group of validators is assigned to process the transaction, with one designated as the leader to propose the outcome
- Once all validators evaluate the proposal and reach consensus using the equivalence principle, the transaction is accepted and enters the Finality Window

Learn more about the validation process (</about-genlayer/core-concepts/optimistic-democracy>)

Advantages over Traditional Smart Contracts

Enhanced Decision Making

Intelligent Contracts can process complex and qualitative criteria that traditional smart contracts cannot handle. Through their natural language understanding capabilities, they can interpret and act on human-readable inputs without requiring strict formatting or coding syntax.

This flexibility allows the contracts to dynamically adapt to changing conditions, making them more responsive and intelligent in their decision-making processes.

External Data Integration

Intelligent Contracts can seamlessly integrate with external data sources, providing direct access to real-world information without intermediate layers. Their real-time data processing capabilities ensure that contract decisions are based on current and accurate information.

This direct connectivity significantly reduces the traditional reliance on oracle services, making the contracts more efficient and cost-effective.

Flexible Programming

Development of Intelligent Contracts leverages Python's robust ecosystem, providing developers with a familiar and powerful programming environment.

The platform supports the data structures needed to handle complex business logic and requirements.

Trade-offs

Intelligent Contracts introduce non-determinism and external dependencies that traditional smart contracts avoid. Key considerations:

- Non-deterministic outputs — LLM responses and web data vary across validators. The Equivalence Principle (</developers/intelligent-contracts/features/non-determinism>) lets you define how validators compare results to reach consensus.
- External data reliability — Web sources can be inconsistent or unavailable. Multiple validators independently fetch and cross-validate data as part of Optimistic Democracy (</understand-genlayer-protocol/optimistic-democracy-how-genlayer-works>).
- Compute cost — LLM calls and web requests add latency and cost compared to purely deterministic contracts. Design contracts to minimize non-deterministic calls where possible.

When to Use GenLayer

Source: <https://docs.genlayer.com/developers/intelligent-contracts/when-to-use-genlayer>

GenLayer is most useful when your application needs a shared, on-chain decision about something a normal deterministic smart contract cannot evaluate by itself.

Use GenLayer when the decision depends on evidence, language, or judgment — and when different parties need to trust the result without relying on a single backend, oracle, or operator.

Quick Fit Checklist

A feature is usually a good fit for GenLayer if most of these are true:

- There is a real on-chain consequence: payout, escrow release, market resolution, reputation update, rule enforcement, or another state change.
- The outcome requires judgment: interpreting evidence, applying natural-language rules, evaluating quality, or resolving ambiguity.
- The evidence can be independently checked: validators can fetch the same source material or enough comparable evidence to verify the leader's result.
- The decision benefits from neutral consensus: counterparties should not have to trust your centralized backend or one AI provider.
- The result can be made explicit: your contract can return a structured decision such as accepted, winner, score, amount, or reason.

If your feature is mostly personalization, summarization, chat, analytics, routing, or UX assistance with no consensus-critical state change, it is usually better handled by a normal backend or frontend LLM.

Architecture Pattern

A common GenLayer app has three layers:

1. Frontend or backend: collects user intent, displays data, handles indexing, and may prepare URLs or evidence.
2. Intelligent Contract: owns the consensus-critical decision and any on-chain state transition.
3. GenLayer validators: independently verify the leader's result using the evidence and equivalence principle you define.

The key question is:

What decision must not depend on my server alone?

That decision is the part that belongs in the Intelligent Contract.

Good Fits

Prediction or resolution markets

Good fit when the contract resolves a market by checking source evidence and applying listing or resolution rules.

- Frontend: shows the market, lets users trade, links evidence.
- Intelligent Contract: determines the outcome from trusted sources and settles the market.
- Validators: verify that the outcome follows the evidence and rules.

Dispute or milestone workflows

Good fit when parties agree up front that a contract should evaluate whether a deliverable, task, or condition was satisfied.

- Frontend/backend: stores submissions, attachments, and communication context.
- Intelligent Contract: evaluates the evidence against agreed criteria and releases funds or updates status.
- Validators: verify that the accepted result is reasonable given the evidence.

Rule or policy verification

Good fit when a decision depends on natural-language criteria.

Examples:

- Does a proposal comply with a DAO constitution?
- Does a submitted work item meet bounty rules?
- Does a listing meet market guidelines?
- Does a claim satisfy policy conditions?

Weak Fits and Anti-Patterns

Frontend computes the answer, GenLayer only stores it

If your frontend fetches data, asks an LLM for the answer, and sends the finished result to GenLayer for storage, GenLayer is not adding much trust.

Better pattern: send the contract stable evidence references or inputs, then let the Intelligent Contract perform the consensus-critical evaluation.

Generic AI brain

If GenLayer is only being used as an AI assistant, recommender, chatbot, or analytics engine, a normal backend is usually simpler and cheaper.

Better pattern: use off-chain AI for UX and exploration, then use GenLayer only for the part where users need neutral settlement or verifiable judgment.

Private data that validators cannot verify

If validators cannot access enough evidence to independently check the leader result, consensus becomes weak or impossible.

Better pattern: commit to public evidence, signed attestations, hashes, URLs, or other verifiable inputs that validators can inspect.

Unbounded subjective output

If the contract asks for open-ended text with no structured decision fields, validators may disagree even when they are all reasonable.

Better pattern: return structured fields and compare only the parts that matter. See the Equivalence Principle (</developers/intelligent-contracts/equivalence-principle>).

What Belongs Where

- Frontend: UX, forms, wallets, previews, charts, indexing, user guidance.
- Backend: caching, notifications, search, off-chain analytics, non-critical LLM assistance.
- Intelligent Contract: the decision that moves money, resolves a market, updates reputation, enforces rules, or creates another shared state transition.
- Evidence sources: stable URLs, APIs, signed data, user submissions, or other material validators can fetch or evaluate.

Dispute Resolution Is Not a Court

GenLayer can support agreed dispute-resolution workflows, but it does not automatically make a decision legally binding or replace a court.

Use safer framing such as:

- evidence-based settlement workflow
- agreed arbitration primitive
- contractual dispute-resolution mechanism
- milestone or deliverable adjudication

Avoid implying that an Intelligent Contract is a legal judge unless your application has the legal agreements, jurisdiction, and review process to support that claim.

Design Prompt

Before building, answer these questions:

1. What exact decision should GenLayer make?
2. What state changes if the decision is accepted?
3. What evidence will validators use to verify it?
4. Which fields must validators agree on exactly, and which can vary?

5. What happens if validators reject the leader result or the transaction becomes undetermined?

If the answers are clear, the feature is probably a good candidate for GenLayer. If not, start by moving more logic off-chain or narrowing the on-chain decision.

Intelligent Contract Features

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features>

Deterministic Features

- [Types \(types\)](#)
- [Storage \(features/storage\)](#)
- [Error Handling \(features/error-handling\)](#)
- [Upgradability \(features/upgradability\)](#)
- [Value Transfers \(features/value-transfers\)](#)
- [Messages \(features/messages\)](#)
- [Interacting with Intelligent Contracts \(features/interacting-with-intelligent-contracts\)](#)
- [Interacting with EVM Contracts \(features/interacting-with-evm-contracts\)](#)
- [Special Methods \(features/special-methods\)](#)
- [Vector Storage \(features/vector-storage\)](#)
- [Debugging \(features/debugging\)](#)
- [Random \(features/random\)](#)

Non-Deterministic Features

- [Non-determinism \(features/non-determinism\)](#)
- [Calling LLMs \(features/calling-llms\)](#)
- [Image Processing \(features/image-processing\)](#)
- [Web Access \(features/web-access\)](#)

Storage

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/storage>

Basic Types

Store persistent data using class attributes:

```
class Contract(gl.Contract):
    counter: u32
    name: str
    active: bool

    @gl.public.write
    def set_data(self, count: u32, new_name: str):
        self.counter = count
        self.name = new_name
```

DynArray

Use DynArray[T] instead of list[T] for persistent arrays:

```
class Contract(gl.Contract):
    items: DynArray[str]
    scores: DynArray[u32]

    @gl.public.write
    def add_item(self, item: str):
        self.items.append(item)

    @gl.public.view
    def get_all_items(self):
        return [item for item in self.items]
```

TreeMap

Use TreeMap[K, V] instead of dict[K, V] for persistent mappings:

```
class Contract(gl.Contract):
    balances: TreeMap[str, u32]

    @gl.public.write
    def update_balance(self, user: str, amount: u32):
        self.balances[user] = amount

    @gl.public.view
    def get_balance(self, user: str):
        return self.balances.get(user, u32(0))
```

Storage Classes

Create custom storage types with @allow_storage:

```
@allow_storage
@dataclass
class UserData:
    scores: DynArray[u32]
    username: str

class Contract(gl.Contract):
    users: DynArray[UserData]

    @gl.public.write
    def add_user(self, name: str):
        user = UserData(scores=DynArray[u32](), username=name)
        self.users.append(user)
```

Memory Operations

Copy storage objects to memory for non-deterministic operations:

```
@gl.public.write
def process_user(self):
    storage_user = self.users[0]
    memory_user = gl.storage.copy_to_memory(storage_user)

    def nondet_operation():
        return f"User: {memory_user.username}"

    result = gl.eq_principle.strict_eq(nondet_operation)
```

Note:

In future reading from storage directly in non-deterministic blocks will be supported.

Type Restrictions

- Use `DynArray[T]` instead of `list[T]`
- Use `TreeMap[K, V]` instead of `dict[K, V]`
- Use sized integers (`u32`, `i64`) instead of `int`
- Use `bigint` only for arbitrary precision needs
- All generic types must be fully specified

Default Values

Storage is zero-initialized:

Type	Default value
<code>`u*`, `i*`</code>	<code>`0`</code>
<code>`bigint`</code>	<code>`0`</code>
<code>`bool`</code>	<code>`False`</code>
<code>`float`</code>	<code>`+0.0`</code>
<code>`str`</code>	<code>`""`</code>
<code>`bytes`</code>	<code>`b""`</code>
<code>`Address`</code>	<code>`0x0...`</code>
<code>`DynArray`</code>	<code>`[]`</code>
<code>`TreeMap`</code>	<code>`{}`</code>

Error Handling

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/error-handling>

Unrecoverable Errors

Exit codes terminate execution immediately:

```
if invalid_condition:
    exit(1)
```

Unhandled exceptions also become unrecoverable:

```
raise Exception("Critical error") # Becomes exit(1)
```

UserError

User-generated errors with UTF-8 encoded messages:

```
# Can be caught in current sub-vm
raise gl.vm.UserError("Invalid input")

# Immediate user error, more efficient but can't be caught
gl.advanced.user_error_immediate("Insufficient funds")
```

VMError

VM-generated errors with predefined string codes:

```
# Non-zero exit codes become VMError
exit(1) # Results in VMError with specific code

# Resource limit violations also trigger VMError
# (handled automatically by the VM)
```

Catching UserError

Handle user errors from sub-VMs:

```
def risky_operation():
    raise gl.vm.UserError("Operation failed")

try:
    result = gl.eq_principle.strict_eq(risky_operation)
except gl.vm.UserError as e:
    print(f"Caught user error: {e.message}")
```

Error Propagation

Errors flow from non-deterministic to deterministic code:

```
def nondet_block():
    if some_condition:
        raise gl.vm.UserError("INVALID_STATE")
    return "success"

try:
    gl.eq_principle.strict_eq(nondet_block)
except gl.vm.UserError as e:
    if e.message == "INVALID_STATE":
        # Handle specific error condition
        pass
```

VM Result Types

GenVM produces four result types:

- Return - Successful execution with encoded result
- VMError - VM errors (exit codes, resource limits)
- UserError - User-generated errors with UTF-8 messages
- InternalError - Critical VM failures (not visible to contracts)

Error Patterns for Consensus

When using `run_nondet_unsafe`, errors affect how validators compare results. If the leader errors, the validator needs to decide: do I agree or disagree?

One approach is to classify errors by their nature, so validators can handle each type appropriately:

- Deterministic errors (business logic, client errors): should match exactly between leader and validator
- Transient errors (network issues, server errors): both hitting a transient failure is expected
- LLM errors (malformed output): disagreeing forces a rotation to a new leader, which is usually what you want

Example: Error Classification

```
def validator_fn(leaders_res: gl.vm.Result) -> bool:
    if not isinstance(leaders_res, gl.vm.Return):
        # Leader errored – run the same logic to see if we error too
        leader_msg = leaders_res.message if hasattr(leaders_res, 'message') else ''
        try:
            leader_fn()
            return False # Leader errored but we succeeded – disagree
        except gl.UserError as e:
            validator_msg = str(e)
            # Both hit the same business logic error – agree
            if validator_msg == leader_msg:
                return True
            return False
        except Exception:
            return False

    # Leader succeeded – validate the result
    validator_result = leader_fn()
    return compare_results(leaders_res.calldata, validator_result)
```

This pattern ensures that consensus failures on broken LLM output or transient network issues lead to retries rather than locking in bad state.

Upgradability

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/upgradability>

GenVM provides a native contract upgradability system that allows contracts to be modified after deployment while maintaining security guarantees and clear access controls.

The system is built around the Root Slot (`gl.storage.Root`), which stores:

Field	Description
code	The contract's source code
locked_slots	Storage slots that non-upgraders cannot write to
upgraders	Addresses authorized to modify locked slots (including code)

How It Works

1. At the start of a write transaction, GenVM reads the upgraders list

2. If the sender is in the upgraders list, no slot restrictions apply — the sender can modify any slot, including code
3. If the sender is not in the upgraders list, GenVM reads `locked_slots` and prevents writes to them

During deployment (init), after the constructor completes, the runtime automatically calls `root.lock_default()`, which locks four critical slots: the root slot, the code slot, the `locked_slots` slot, and the upgraders slot.

Making a Contract Upgradable

To make a contract upgradable, you need to:

1. Add authorized upgrader addresses in init
2. Expose a method that replaces the contract code

```
# v0.1.0
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *

class UpgradableStorage(gl.Contract):
    storage: str

    def __init__(self, initial_storage: str):
        self.storage = initial_storage

        # Set the deployer as an upgrader
        root = gl.storage.Root.get()
        root.upgraders.get().append(gl.message.sender_address)

        # lock_default() is called automatically after __init__
        # it locks: root slot, code, locked_slots, upgraders

    @gl.public.view
    def get_storage(self) -> str:
        return self.storage

    @gl.public.write
    def update_storage(self, new_storage: str) -> None:
        self.storage = new_storage

    @gl.public.write
    def upgrade(self, new_code: bytes) -> None:
        root = gl.storage.Root.get()
        code = root.code.get()
        # If sender is not in upgraders, this will raise a VMError
        code.truncate()
        code.extend(new_code)
```

Upgrading to a New Version

The upgraded contract code must maintain the same storage layout for compatibility. Only the code changes — all storage data, the upgraders list, and locked slots persist across upgrades.

```
# v0.1.0
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *

class UpgradableStorage(gl.Contract):
    # Storage layout must remain compatible with v1
    storage: str

    def __init__(self):
        pass

    @gl.public.view
    def get_storage(self) -> str:
```

```

    return self.storage

    @gl.public.write
    def update_storage(self, new_storage: str) -> None:
        self.storage = new_storage

    # New method added in v2
    @gl.public.view
    def get_storage_length(self) -> int:
        return len(self.storage)

    # Keep the upgrade method for future upgrades
    @gl.public.write
    def upgrade(self, new_code: bytes) -> None:
        root = gl.storage.Root.get()
        code = root.code.get()
        code.truncate()
        code.extend(new_code)

```

What Happens During an Upgrade

When an authorized upgrader calls the upgrade method with new code:

Component	After upgrade
code	**Replaced** with new code
Contract storage data (e.g. <code>storage: str</code>)	**Persists** unchanged
locked_slots	**Persists** unchanged
upgraders	**Persists** unchanged

This means:

- Upgrades are not one-shot — since the upgraders list persists, the same addresses can push another upgrade later
- Storage must be compatible — the new code must understand the existing storage layout. There is no automatic migration mechanism
- New methods can be added — but existing storage field positions must not change
- Upgraders can be modified — an upgrader can add or remove addresses from the upgraders list

Freezing a Contract

To make a contract permanently non-upgradable, either:

- Call `root.lock_default()` without adding any addresses to upgraders — the code and critical slots are locked and nobody can unlock them
- Remove all addresses from the upgraders list after locking

Note:

Once a contract is frozen (locked slots with no upgraders), it cannot be upgraded. This is irreversible.

Testing Upgrades

Using the GenLayer Testing Suite (</developers/intelligent-contracts/tools/genlayer-testing-suite>), you can test the full upgrade lifecycle:

```

from pathlib import Path
from gltest import get_contract_factory
from gltest.assertions import tx_execution_succeeded

```

```

CONTRACTS_DIR = Path(__file__).parent.parent / "contracts"

def test_upgradable_storage():
    # Deploy v1
    factory = get_contract_factory(
        contract_file_path=CONTRACTS_DIR / "upgradable_storage.py"
    )
    contract = factory.deploy(args=["hello"])

    # Use v1 methods
    assert contract.get_storage(args=[]).call() == "hello"

    # Read v2 code and upgrade the contract
    v2_code = (CONTRACTS_DIR / "upgradable_storage_v2.py").read_bytes()
    tx = contract.upgrade(args=[v2_code]).transact()
    assert tx_execution_succeeded(tx)

    # Rebuild the contract proxy from the v2 schema
    v2_factory = get_contract_factory(
        contract_file_path=CONTRACTS_DIR / "upgradable_storage_v2.py"
    )
    contract_v2 = v2_factory.build_contract(contract_address=contract.address)

    # Storage persists across upgrades
    assert contract_v2.get_storage(args=[]).call() == "hello"

    # New v2 method works
    assert contract_v2.get_storage_length(args=[]).call() == 5

```

See GenVM specification (<https://sdk.genlayer.com/v0.2.7/spec/04-contract-interface/04-upgradability.html>) for the full technical details.

Value Transfers

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/value-transfers>

Native Token (GEN)

GenLayer uses GEN as its native token. Values are denominated in wei (1 GEN = 10^{18} wei). Use the u256 type for value amounts in contract code.

Receiving Value

Mark a method as payable with `@gl.public.write.payable` to accept GEN:

```

from genlayer import *

class TipJar(gl.Contract):
    def __init__(self):
        self.total_tips = u256(0)

    @gl.public.write.payable
    def tip(self) -> None:
        v = gl.message.value
        if v == u256(0):
            raise gl.vm.UserError("send some value")
        self.total_tips = self.total_tips + v

    @gl.public.view
    def get_tips(self) -> u256:

```

```
return self.total_tips
```

gl.message.value is a u256 containing the GEN sent with the call. It is only available in methods decorated with @gl.public.write.payable.

Sending Value to Another Intelligent Contract

Send value to another IC using internal messages

(/developers/intelligent-contracts/features/messages#internal-messages-ic--ic):

```
other = gl.get_contract_at(recipient_address)

# Pure value transfer (triggers __receive__ on recipient)
other.emit_transfer(value=u256(amount), on='finalized')

# Value + method call (recipient method must be payable)
other.emit(value=u256(amount), on='finalized').deposit()
```

How Value Flows

When a message with value is emitted, the value is immediately deducted from the sending contract's balance and held in the message. It is only credited to the recipient when the message's child transaction is activated. The flow is:

Sender balance → message → recipient balance

If the child transaction fails, the value is not automatically returned to the sender.

See Messages (/developers/intelligent-contracts/features/messages) for details on timing (on='accepted' vs on='finalized').

Sending Value to an EOA or EVM Contract

Sending to an address on the GenLayer Chain (EOA or EVM contract) is an external message (/developers/intelligent-contracts/features/messages#external-messages-ic--chain-layer). It goes through the IC's ghost contract (/developers/intelligent-contracts/features/messages#ghost-contracts) and always executes on finalization.

```
@gl.evm.contract_interface
class _Recipient:
    class View:
        pass
    class Write:
        pass

class Faucet(gl.Contract):
    @gl.public.write.payable
    def send(self, recipient: str) -> None:
        v = gl.message.value
        if v == u256(0):
            raise gl.vm.UserError("send some value")
        _Recipient(Address(recipient)).emit_transfer(value=v)
```

Note:

The syntax for sending to an EOA uses the EVM contract interface even though the recipient is not a contract. This is because EOAs live on the chain layer, making this an external message — the same mechanism used for calling EVM contracts. This will be simplified in a future version.

Reading Balances

```
# Own contract balance
my_balance = self.balance

# Another IC's balance
other_balance = gl.get_contract_at(addr).balance

# EVM contract balance
evm_balance = EthContract(addr).balance
```

In write methods, `self.balance` reflects the current state including any value received in the current call. In view methods, it shows a snapshot without modifications.

Where Balance Lives

On the GenLayer network, an IC's GEN balance is held by its ghost contract (`/developers/intelligent-contracts/features/messages#ghost-contracts`) on the chain layer. `self.balance` reads from this. The balance is visible on both layers.

Note:

Studio: Balances are simulated in a local database. There is no EVM layer or ghost contracts in Studio.

Receiving Value Without a Method Call

When value is sent without specifying a method, the contract can handle it with special methods:

```
class Contract(gl.Contract):
    @gl.public.write.payable
    def __receive__(self):
        """Called for value-only transfers with no method name."""
        pass

    @gl.public.write.payable
    def __handle_undefined_method__(
        self, method_name: str, args: list, kwargs: dict
    ):
        """Fallback for calls to undefined methods."""
        pass
```

The dispatch logic:

```
graph TD
    msg[>incoming call] --> has_method[method exists?]
    has_method -->|yes| call_method[call_method{{call method}}]
    has_method -->|no| has_value[has value?]

    has_value -->|yes| has_method_name[has method name?]
    has_method_name -->|no| has_receive[has_receive[receive__ defined?]]
    has_receive -->|yes| call_receive[call_receive{{call __receive__}}]
    has_receive -->|no| has_fallback1[has_fallback1[handle_undefined_method__ defined and payable?]]
    has_fallback1 -->|yes| call_fallback1[call_fallback1{{call __handle_undefined_method__}}]
    has_fallback1 -->|no| error1[error1{{error}}]

    has_method_name -->|yes| has_fallback2[has_fallback2[handle_undefined_method__ defined and payable?]]
    has_fallback2 -->|yes| call_fallback2[call_fallback2{{call __handle_undefined_method__}}]
    has_fallback2 -->|no| error2[error2{{error}}]

    has_value -->|no| has_fallback3[has_fallback3[handle_undefined_method__ defined?]]
    has_fallback3 -->|yes| call_fallback3[call_fallback3{{call __handle_undefined_method__}}]
    has_fallback3 -->|no| error3[error3{{error}}]
```

Funding Accounts

Environment	How to fund
Studio (Localnet)	Built-in faucet – button in the account selector
Studio (Testnet)	Built-in faucet – button in the account selector
Testnet Asimov	testnet-faucet.genlayer.foundation
Testnet Bradbury	testnet-faucet.genlayer.foundation

See [Networks \(/developers/networks/\)](/developers/networks/) for full details on each environment.

Calling Payable Methods from JavaScript

Use the value parameter in `writeContract()` to send GEN:

```
import { createClient, createAccount } from 'genlayer-js';
import { testnetAsimov } from 'genlayer-js/chains';

const client = createClient({
  chain: testnetAsimov,
  account: createAccount(),
});

const txHash = await client.writeContract({
  address: contractAddress,
  functionName: 'tip',
  args: [],
  value: BigInt(5) * BigInt(10 ** 18), // 5 GEN in wei
});
```

Transaction Context

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/transaction-context>

Every contract execution has access to information about the transaction that triggered it — caller addresses, value, chain ID, and the transaction timestamp (datetime). This page covers how to read the current time, get a Unix timestamp, access caller and tx metadata, and what is *not* exposed (block number, gas, wall-clock).

gl.message — typed accessors

The most common fields are exposed as a typed `NamedTuple`:

Field	Type	Description
<code>`sender_address`</code>	<code>`Address`</code>	Immediate caller — EOA, EVM contract, or IC depending on how the call was initiated
<code>`origin_address`</code>	<code>`Address`</code>	Original transaction submitter — preserved through internal message chains
<code>`contract_address`</code>	<code>`Address`</code>	The current contract's own address
<code>`value`</code>	<code>`u256`</code>	GEN sent with the call (only in <code>@gl.public.write.payable`</code> methods)
<code>`chain_id`</code>	<code>`u256`</code>	Current chain ID

```
@gl.public.write
def deposit(self):
    self.balances[gl.message.sender_address] += gl.message.value
```

For the difference between `sender_address` and `origin_address` across internal message chains, see [Messages \(/developers/intelligent-contracts/features/messages/\)](/developers/intelligent-contracts/features/messages/).

gl.message_raw — full message dict

gl.message_raw is a TypedDict with everything the bootloader receives. It includes all gl.message fields plus:

```
| Field | Type | Description |
|---|---|---|
| `datetime` | `str` | Transaction datetime as an ISO 8601 string |
| `is_init` | `bool` | `True` iff this execution is a deployment |
| `stack` | `list[Address]` | Stack of view-method callers, excluding the current contract |
| `entry_kind` | `int` | `0` = MAIN, `1` = SANDBOX, `2` = CONSENSUS_STAGE |
| `entry_data` | `bytes` | Raw entry payload |
| `entry_stage_data` | - | Consensus-stage-specific data |

# Detect whether the current execution is a deploy
if gl.message_raw['is_init']:
    self.deployer = gl.message.sender_address
```

Note:

Prefer gl.message for the common fields — it gives you autocompletion and type checking. Reach for gl.message_raw only when you need one of the extra fields.

Time and Timestamps

Time inside the GenVM is deterministic and pinned to the transaction's timestamp. Every validator re-executing the transaction sees the same value, so you can use it for storage, comparisons, and prompt context without breaking equivalence.

The Python standard library's clock is wired to the transaction datetime — datetime.now(), datetime.now(tz), and time.time() all return the same value across validators.

Transaction Timestamp

The transaction timestamp is exposed three ways. Use whichever fits:

```
| Form | How | Use for |
|---|---|---|
| Unix seconds | `int(datetime.now(timezone.utc).timestamp())` | Arithmetic, expiries, deltas |
| Unix seconds (alt) | `int(time.time())` | Same — shorter |
| `datetime` object | `datetime.now(timezone.utc)` | Comparisons, formatting |
| ISO 8601 string | `datetime.now(timezone.utc).isoformat()` *or* `gl.message_raw['datetime']` | Storage, audit trails, prompts |
```

Common patterns

Unix timestamp for arithmetic:

```
from datetime import datetime, timezone

@gl.public.write
def mint(self):
    now = int(datetime.now(timezone.utc).timestamp())
    self.tokens[gl.message.sender_address] = Token(
        expiry=u256(now + 86400), # 24h from now
    )
```

Expiry check:

```
@gl.public.view
def is_expired(self, token_id: u256) -> bool:
    now = int(datetime.now(timezone.utc).timestamp())
    return now > int(self.tokens[token_id].expiry)
```

ISO string for audit trails or events:

```
@gl.public.write
def submit(self, payload: str):
    self.submissions.append(Submission(
        sender=gl.message.sender_address,
        payload=payload,
        timestamp=datetime.now(timezone.utc).isoformat(),
    ))
```

Raw ISO string from the message (skip the clock):

```
ts: str = gl.message_raw['datetime']
```

Note:

The clock returns the transaction datetime, not host wall-clock time. Do not call `datetime.now()` expecting current real-world time — by the time validators re-execute, the transaction may be minutes or hours old. Use it for relative arithmetic ("expires 24h after the tx") and timestamping, not for "is it past 5pm right now".

What is **not** in the context

- No block number, no block hash. The transaction context does not carry block-level metadata. If your contract needs block height, fetch it via Web Access (/developers/intelligent-contracts/features/web-access) against an RPC inside a non-deterministic block.
- No gas price, no nonce. Transaction metadata beyond what is listed above is not exposed to the contract.
- No host wall-clock time. As above — the clock is the transaction datetime, not the validator's current time.

Messages

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/messages>

GenLayer has three types of interaction, each operating at a different layer of the architecture.

Transactions (→ Intelligent Contract)

A transaction is the entry point for all IC execution. The caller signs an EVM transaction that calls `addTransaction()` on `ConsensusMain` or on the IC's ghost contract (which relays to `ConsensusMain`). This is submitted via `eth_sendRawTransaction`.

The caller can be an EOA or an EVM contract on GenLayer Chain.

- `gl.message.sender_address` = caller address
- `gl.message.origin_address` = caller address

The transaction enters the consensus pipeline: activation → proposal → commit → reveal → finalization.

Note:

Studio: The SDK sends the same signed addTransaction EVM transaction via `eth_sendRawTransaction`, but Studio intercepts and simulates the consensus pipeline rather than executing on a real EVM layer.

Internal Messages (IC → IC)

An internal message is an asynchronous call from one Intelligent Contract to another. It stays within the GenVM layer. During execution, `emit()` records the message as part of the execution result — the message is not sent immediately. The actual child transaction is only created when the parent transaction reaches the specified state (`on='accepted'` or `on='finalized'`). The child transaction then goes through consensus independently.

```
other = gl.get_contract_at(addr)

# Call a write method on another IC
other.emit(on='finalized').update_status("active")

# Call with value (recipient must be payable)
other.emit(value=u256(100), on='finalized').deposit()

# Pure value transfer (triggers __receive__ on recipient)
other.emit_transfer(value=u256(100), on='finalized')

# Deploy a child contract
addr = gl.deploy_contract(
    code=contract_code,
    args=[],
    salt_nonce=u256(1),
    on='finalized',
)
```

In the child transaction's execution context:

- `gl.message.sender_address` = calling contract's address
- `gl.message.origin_address` = original caller (preserved through the chain)

See [Interacting with Intelligent Contracts](#)

([/developers/intelligent-contracts/features/interacting-with-intelligent-contracts](#)) for full syntax reference.

External Messages (IC → Chain Layer)

An external message crosses from the GenVM layer back to the GenLayer Chain (EVM layer). This is how an IC sends value or calls an EOA or EVM contract.

External messages are executed by the IC's ghost contract via `handleOp()`, so the `msg.sender` seen by the recipient is the ghost contract address (which is the same as the IC's address).

```
@gl.evm.contract_interface
class ERC20:
    class View:
        def balance_of(self, owner: Address) -> u256: ...
    class Write:
        def transfer(self, to: Address, amount: u256) -> None: ...

# Read from an EVM contract (gl.message.contract_address is your own address)
balance = ERC20(token_address).view().balance_of(gl.message.contract_address)

# Write to an EVM contract (value is optional, defaults to 0)
ERC20(token_address).emit().transfer(recipient, amount)

# Send value to an EOA
@gl.evm.contract_interface
class _Recipient:
    class View:
        pass
    class Write:
```

```
pass
```

```
_Recipient(Address(eoa_address)).emit_transfer(value=u256(amount))
```

Note:

External messages can only be emitted on='finalized'. Using on='accepted' for external messages is not supported.

Note:

Studio: EVM contract interaction beyond value transfers to EOAs is not implemented. The @gl.evm.contract_interface calls are not functional in Studio.

Ghost Contracts

Every Intelligent Contract has a corresponding ghost contract on GenLayer Chain (the EVM layer). The ghost and IC share the same address.

What Ghost Contracts Do

- Hold the IC's GEN balance — the native balance on the ghost contract on-chain is the source of truth for self.balance
- Relay transactions — addTransaction() on the ghost forwards to ConsensusMain
- Execute external messages — on finalization, handleOp() forwards calls to recipients so msg.sender is the IC's address
- Bridge the two layers — ghost on EVM ↔ IC on GenVM, same address

Lifecycle

1. A deploy transaction is submitted
2. GhostFactory deploys a ghost contract at address 0xABC on GenLayer Chain
3. Consensus processes the deployment
4. If successful: IC deploys at the same address 0xABC on GenVM
5. If deployment is appealed and reverted: the ghost remains but has no IC behind it — it has no purpose or effect, but may be reused by consensus if deployment succeeds later

Note:

Ghost contract existence does not guarantee the IC is deployed. A reverted deployment leaves an empty ghost.

Note:

Studio: Ghost contracts are not implemented. The IC address is used directly, and balance is stored in a database table.

Message Context

Caller and execution metadata (sender_address, origin_address, contract_address, value, chain_id, transaction datetime, etc.) is available via gl.message and gl.message_raw. See Transaction Context (/developers/intelligent-contracts/features/transaction-context).

Timing: Accepted vs Finalized

Internal messages and contract deployments can execute at two different points:

on='finalized' (default)

The message executes after the parent transaction is fully finalized (appeal window has closed). This is the safe default.

on='accepted'

The message executes as soon as the parent transaction is accepted by initial consensus, before the appeal window closes.

Note:

Appeal risks with on='accepted':

- The re-execution may emit the message again — potentially multiple times across appeal rounds.
- If the appeal changes the outcome, the message may be "invalid" (would not have been emitted with the final outcome), but it cannot be taken back — it was already sent and executed.
- The receiving contract must be idempotent and must handle duplicate or unexpected messages gracefully.
- Contract design must account for the possibility that accepted messages may not reflect the final state.

External messages always use on='finalized' and cannot be emitted on acceptance.

Interfaces

IC Interfaces (@gl.contract_interface)

Define typed stubs for calling other Intelligent Contracts:

```
@gl.contract_interface
class Token:
    class View:
        def balance_of(self, owner: Address) -> u256: ...
    class Write:
        def transfer(self, to: Address, amount: u256) -> None: ...

token = Token(token_address)
balance = token.view().balance_of(user)          # typed view call
token.emit(on='finalized').transfer(to, amount)   # typed write call
```

This is purely for type safety and IDE autocompletion — at runtime it behaves identically to `gl.get_contract_at()`. No overhead.

EVM Interfaces (@gl.evm.contract_interface)

Define typed stubs matching Solidity function signatures:

```
@gl.evm.contract_interface
class IERC20:
    class View:
        def balance_of(self, owner: Address) -> u256: ...
```

```

def total_supply(self) -> u256: ...
class Write:
    def transfer(self, to: Address, amount: u256) -> bool: ...
    def approve(self, spender: Address, amount: u256) -> bool: ...

```

Parameters are automatically ABI-encoded. Type mapping:

```

| GenLayer | Solidity |
|---|---|
| `u256`, `u128`, `u64`, ... | `uint256`, `uint128`, `uint64`, ... |
| `i256`, `i128`, `i64`, ... | `int256`, `int128`, `int64`, ... |
| `Address` | `address` |
| `bool` | `bool` |
| `str` | `string` |
| `bytes` | `bytes` |

```

Interacting with Intelligent Contracts

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/interacting-with-intelligent-contracts>

This page covers syntax for internal messages (IC → IC). See Messages (/developers/intelligent-contracts/features/messages) for the conceptual model and Value Transfers (/developers/intelligent-contracts/features/value-transfers) for sending GEN.

Getting Contract References

Access other contracts by their address:

```

contract_address = Address("0x03FB09251eC05ee9Ca36c98644070B89111D4b3F")

dynamically_typed_contract = gl.get_contract_at(contract_address)

@gl.contract_interface
class GenLayerContractIface:
    class View:
        def method_name(self, a: int, b: str): ...

    class Write:
        pass

statically_typed_contract = GenLayerContractIface(contract_address)

```

Both approaches result in the same runtime value, however the statically typed approach provides type checking and autocompletion in IDEs.

Calling View Methods

Call read-only methods on other contracts:

```

addr: Address = ...
other = gl.get_contract_at(addr)
result = other.view().get_token_balance()

```

Emitting Messages

Send asynchronous messages to other contracts:

```
other = gl.get_contract_at(addr)
other.emit(on='accepted').update_status("active")
other.emit(on='finalized').update_status("active")

# With value (recipient method must be @gl.public.write.payable)
other.emit(value=u256(100), on='finalized').deposit()
```

Deploying New Contracts

```
gl.deploy_contract(code=contract_code)
salt: u256 = u256(1) # not zero
child_address = gl.deploy_contract(code=contract_code, salt=salt)
```

View vs Emit

- view() is synchronous — it reads state from another contract and returns the result immediately. The target contract's state is read as of the current block.
- emit() is asynchronous — it queues a write call that executes *after* the current transaction completes. The call is not blocking.

Accepted vs Finalized

```
# Fast – executes after the transaction is accepted by initial consensus
other.emit(on='accepted').do_something()

# Safe – waits until the transaction is fully finalized (after appeal window)
other.emit(on='finalized').do_something()
```

on='accepted' has important implications. If the emitting transaction is appealed, two things can happen:

1. The appeal changes the contract state such that the message should not have been emitted — but it was already sent and cannot be recalled.
2. The transaction is re-executed during the appeal, and the message (or a similar one) is emitted again. This can repeat across multiple appeal rounds — up to ~6 times depending on the validator set size.

The receiving contract must be idempotent — it must handle duplicate or unexpected messages gracefully. If the receiving logic cannot tolerate duplicates or messages that "shouldn't have been sent" based on the final state, use on='finalized' instead.

Factory Pattern

Deploy child contracts from a parent:

```
def __init__(self, num_workers: int):
    with open("/contract/Worker.py", "rt") as f:
        worker_code = f.read()

    for i in range(num_workers):
        addr = gl.deploy_contract(
            code=worker_code.encode("utf-8"),
            args=[i, gl.message.contract_address],
            salt_nonce=i + 1,
            on="accepted",
        )
        self.worker_addresses.append(addr)
```

Child contracts are immutable after deployment. To update worker logic, redeploy through the factory.

Interacting with EVM Contracts

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/interacting-with-evm-contracts>

These are external messages — they cross from the GenVM layer back to the GenLayer Chain via ghost contracts (/developers/intelligent-contracts/features/messages#ghost-contracts). See Messages (/developers/intelligent-contracts/features/messages) for the conceptual model.

Contract Interface Definition

Define EVM contract interfaces using decorators:

```
@gl.evm.contract_interface
class TokenContract:
    class View:
        def balance_of(self, owner: Address) -> u256: ...
        def total_supply(self) -> u256: ...

    class Write:
        def transfer(self, to: Address, amount: u256) -> bool: ...
        def approve(self, spender: Address, amount: u256) -> bool: ...
```

Calling EVM Contracts

Interact with EVM contracts through defined interfaces:

```
token_address: Address = ...
# Read from EVM contract
token = TokenContract(token_address)
supply = token.view().total_supply()

# Write to EVM contract
token.emit().transfer(receiver_address, u256(100))
```

Balance Access

Access EVM contract balances directly:

```
evm_contract = TokenContract(address)
balance = evm_contract.balance # Get contract's ETH balance
```

Message Emission

Send messages to EVM contracts:

```
TokenContract(token_address).emit().approve(spender, amount)
```

Note:

Messages to EVM contract can be emitted only on finality

Special Methods

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/special-methods>

For how these methods fit into value transfer flows, see Value Transfers ([/developers/intelligent-contracts/features/value-transfers#receiving-value-without-a-method-call](https://docs.genlayer.com/developers/intelligent-contracts/features/value-transfers#receiving-value-without-a-method-call)).

There are two special methods allowed in each contract definition:

```
class Contract(gl.Contract):
    @gl.public.write # .payable?
    def __handle_undefined_method__(
        self, method_name: str, args: list[typing.Any], kwargs: dict[str, typing.Any]
    ):
        """
        Method that is called for undefined method calls,
        must be either ``@gl.public.write`` or ``@gl.public.write.payable``
        """
        ...

    @gl.public.write.payable
    def __receive__(self):
        """
        Method that is called for no-method transfers,
        must be ``@gl.public.write.payable``
        """
        ...
```

Below is a diagram that shows how GenVM decides which method to pick, in case any regular method did not match:

```
graph TD
    unhandled_message[unhandled_message] -->|unhandled message| has_value[has_value]
    has_value -->|yes| has_method_name[has_method_name]
    has_value -->|no| fallback_defined1[fallback_defined1]
    has_method_name -->|yes| fallback_defined1_is_defined[is __handle_undefined_method__ defined?]
    has_method_name -->|no| receive_defined[is __receive__ defined?]
    receive_defined -->|no| fallback_defined1_is_defined
    receive_defined -->|yes| receive_call[call __receive__]
    fallback_defined1_is_defined -->|yes| fallback_is_payable[is __handle_undefined_method__ payable?]
    fallback_defined1_is_defined -->|no| error3[error3]
    fallback_is_payable -->|yes| fallback1_call[call __handle_undefined_method__]
    fallback_is_payable -->|no| error1[error1]
    has_value -->|no| fallback_defined2[fallback_defined2]
    fallback_defined2 -->|yes| fallback2_call[call __handle_undefined_method__]
    fallback_defined2 -->|no| error2[error2]
```

Vector Store

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/vector-storage>

The Vector Store feature in GenLayer allows developers to enhance their Intelligent Contracts by efficiently storing, retrieving, and calculating similarities between texts using vector embeddings. This feature is particularly useful

for tasks that require natural language processing (NLP), such as creating context-aware applications or indexing text data for semantic search.

Key Features of Vector Store

The Vector Store provides several powerful features for managing text data:

1. Text Embedding Storage

You can store text data as vector embeddings, which are mathematical representations of the text, allowing for efficient similarity comparisons. Each stored text is associated with a vector and metadata.

2. Similarity Calculation

The Vector Store allows you to calculate the similarity between a given text and stored vectors using cosine similarity. This is useful for finding the most semantically similar texts, enabling applications like recommendation systems or text-based search.

3. Metadata Management

Along with the text and vectors, you can store additional metadata (any data type) associated with each text entry. This allows developers to link additional information (e.g., IDs or tags) to the text for retrieval.

4. CRUD Operations

The Vector Store provides standard CRUD (Create, Read, Update, Delete) operations, allowing developers to add, update, retrieve, and delete text and vector entries efficiently.

How to Use Vector Store in Your Contracts

To use the Vector Store in your Intelligent Contracts, you will interact with its methods to add and retrieve text data, calculate similarities, and manage vector storage. Below are the details of how to use this feature.

Importing Vector Store

First, import the VectorStore class from the standard library in your contract:

```
from backend.node.genvm.std.vector_store import VectorStore
```

Creating a Contract with Vector Store

Here's an example of a contract using the Vector Store for indexing and searching text logs:

```
# {
#   "Seq": [
#     { "Depends": "py-lib-genlayermodelwrappers:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" },
#     { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }
#   ]
# }

from genlayer import *
import genlayermodelwrappers
import numpy as np
from dataclasses import dataclass

@allow_storage
@dataclass
class StoreValue:
    log_id: u256
```



```

text: str

# contract class
class LogIndexer(gl.Contract):
    vector_store: VecDB[np.float32, typing.Literal[384], StoreValue]

    def __init__(self):
        pass

    def get_embedding_generator(self):
        return genlayermodelwrappers.SentenceTransformer("all-MiniLM-L6-v2")

    def get_embedding(
        self, txt: str
    ) -> np.ndarray[tuple[typing.Literal[384]], np.dtypes.Float32DType]:
        return self.get_embedding_generator()(txt)

    @gl.public.view
    def get_closest_vector(self, text: str) -> dict | None:
        emb = self.get_embedding(text)
        result = list(self.vector_store.knn(emb, 1))
        if len(result) == 0:
            return None
        result = result[0]
        return {
            "vector": list(str(x) for x in result.key),
            "similarity": str(1 - result.distance),
            "id": result.value.log_id,
            "text": result.value.text,
        }

    @gl.public.write
    def add_log(self, log: str, log_id: int) -> None:
        emb = self.get_embedding(log)
        self.vector_store.insert(emb, StoreValue(text=log, log_id=u256(log_id)))

    @gl.public.write
    def update_log(self, log_id: int, log: str) -> None:
        emb = self.get_embedding(log)
        for elem in self.vector_store.knn(emb, 2):
            if elem.value.text == log:
                elem.value.log_id = u256(log_id)

    @gl.public.write
    def remove_log(self, id: int) -> None:
        for el in self.vector_store:
            if el.value.log_id == id:
                el.remove()

```

Debugging

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/debugging>

Print Statements

Use print calls for basic debugging, which will be present in node or studio output.

Use gl.trace to include time stamps in the output (it will be written to genvm_log instead of stdout)

Profiling

There is also `gl.trace_time_micro` which returns timestamp in the GenVM debug mode. Default bootloader includes possibility to enable profiling by setting environment variable `GENLAYER_ENABLE_PROFILER=true`.

```
# {
#   "Seq": [
#     { "SetEnv": { "name": "GENLAYER_ENABLE_PROFILER", "value": "true" } },
#     { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }
#   ]
# }
```

This will output base-64 encoded gzip-compressed profiling data to `stderr` after every single VM execution

Attaching a Debugger

Note:

Unfortunately, attaching a debugger to a running Intelligent Contract is not supported yet

Random

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/random>

Getting random in deterministic part may be difficult. Trying to delegate it to non-deterministic block will cause your contract to either never agree on that block or to trust the leader. For that reason it is advised to use seeded random.

Seed acquisition methods

1. Use some field of message
2. Use current time (transaction time)
3. Use stdin, as shown below

```
def get_random_seed() -> bytes:
    import os
    import hashlib
    f = os.fdopen(0, 'rb', buffering=0, closefd=False)
    f.seek(0)
    hash_obj = hashlib.sha256()
    while True:
        chunk = f.read(8192)
        if not chunk:
            return hash_obj.digest()
        hash_obj.update(chunk)
```

Non-determinism

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/non-determinism>

When to Use

Non-deterministic operations are needed for:

- External API calls
- LLM and AI model calls
- Random number generation
- Any operation that might vary between nodes

What Goes Inside vs Outside

Non-deterministic blocks (leader_fn, validator_fn, functions passed to strict_eq) run in a special execution context. The GenVM enforces strict rules about what can and cannot happen inside these blocks.

Must be INSIDE nondet blocks

All `gl.nondet.*` calls — web requests, LLM prompts — must be inside a nondet block. They cannot run in regular contract code.

```
@gl.public.write
def fetch_price(self):
    def leader_fn():
        response = gl.nondet.web.get(api_url)          # ✓ inside nondet block
        result = gl.nondet.exec_prompt(prompt)         # ✓ inside nondet block
        return parse_price(response)

    # gl.nondet.web.get(api_url)                        # ✗ would fail here
    self.price = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)
```

Must be OUTSIDE nondet blocks

Several operations must happen in the deterministic context — after the nondet block returns:

```
| Operation | Why |
|-----|-----|
| **Storage writes** (`self.x = ...`) | Storage must only change based on consensus-agreed values |
| **Contract calls** (`gl.get_contract_at()`) | Cross-contract calls must use deterministic state |
| **Message emission** (`.emit()`) | Messages to other contracts/chains must be deterministic |
| **Nested nondet blocks** | Nondet blocks cannot contain other nondet blocks |

@gl.public.write
def update_price(self, pair: str):
    def leader_fn():
        response = gl.nondet.web.get(api_url)
        return json.loads(response.body.decode("utf-8"))["price"]
        # self.price = price                # ✗ no storage writes here
        # other = gl.get_contract_at(addr)   # ✗ no contract calls here
        # other.emit().notify(price)         # ✗ no message emission here

    def validator_fn(leaders_res) -> bool:
        if not isinstance(leaders_res, gl.vm.Return):
            return False
        my_price = leader_fn()
        return abs(leaders_res.calldata - my_price) / leaders_res.calldata <= 0.02

    price = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)

    # ✓ All side effects happen AFTER consensus, in deterministic context
    self.prices[pair] = price
    oracle = gl.get_contract_at(self.oracle_address)
    oracle.emit().price_updated(pair, price)
```

Note:

The GenVM linter (/developers/intelligent-contracts/tooling-setup) catches all of these violations statically — run `genvm-lint` check before deploying to avoid runtime errors.

Why these rules exist

The leader and validators execute nondet blocks independently — each node runs its own `leader_fn` or `validator_fn`. If you wrote to storage inside a nondet block, each node would write a different value before consensus decides which one is correct. The same applies to contract calls and message emission: these must happen once, after consensus, using the agreed-upon result.

Equivalence Principle

GenLayer provides `strict_eq` for exact-match consensus and custom validator functions (`run_nondet_unsafe`) for everything else. Convenience wrappers like `prompt_comparative` and `prompt_non_comparative` exist for common patterns. For detailed information, see [Equivalence Principle \(/developers/intelligent-contracts/equivalence-principle\)](/developers/intelligent-contracts/equivalence-principle).

Strict Equality

Requires exact matches between validator outputs. Use when all nodes can converge on the same normalized value — e.g., fetching objective data from an API and extracting a structured result:

```
def fetch_current_block():
    response = gl.nondet.web.request("https://api.example.com/block/latest")
    data = json.loads(response)
    return json.dumps({"height": data["height"], "hash": data["hash"]}, sort_keys=True)

# All validators must return the exact same string
result = gl.eq_principle.strict_eq(fetch_current_block)
```

Note: `strict_eq` is not suitable for random number generation or LLM calls, since those inherently produce different results on each node. Use a custom validator function or one of the convenience wrappers below for those cases.

Comparative (Convenience Shortcut)

A convenience wrapper where both leader and validators perform the same task, then an LLM compares results using your criteria:

```
def comparative_example():
    return gl.nondet.web.request("https://api.example.com/count")

# Results are compared with acceptable margin of error
result = gl.eq_principle.prompt_comparative(
    comparative_example,
    "Results should not differ by more than 5%"
)
```

Non-Comparative (Convenience Shortcut)

A convenience wrapper where validators evaluate the leader's output against criteria without repeating the task:

```
result = gl.eq_principle.prompt_non_comparative(
    input="This product is amazing!",
    task="Classify the sentiment as positive, negative, or neutral",
    criteria=""
)
```

```

        Output must be one of: positive, negative, neutral
        Consider context and tone
    """
)

```

Custom Validator Functions

For full control over consensus logic, write a custom leader/validator pair with `run_nondet_unsafe`. This is the recommended approach for most contracts:

```

def custom_consensus_example(self, data: str):
    def leader_fn():
        # Leader performs the operation
        response = gl.nondet.exec_prompt(f"Rate this sentiment 1-10: <data>{data}</data>. Answer only with integer, without reasoning")
        return int(response.strip())

    def validator_fn(leader_result):
        own_score = leader_fn()

        if isinstance(leader_result, Exception):
            return False

        # Accept if within acceptable range
        return abs(own_score - leader_result) <= 2

    return gl.vm.run_nondet_unsafe(leader_fn, validator_fn)

```

Operations

Accessing External Data

Use non-deterministic blocks for external API calls. For more web access examples, see [Web Access \(/developers/intelligent-contracts/features/web-access/\)](/developers/intelligent-contracts/features/web-access/):

```

@gl.public.write
def fetch_external_data(self):
    def fetch_data():
        # External API call - inherently non-deterministic
        response = gl.nondet.web.request("https://example.com/data")
        return response

    # Consensus ensures all validators agree on the result
    data = gl.eq_principle.strict_eq(fetch_data)
    return data

```

LLM Integration

Execute AI prompts using comparative principle. For more detailed examples, see [Calling LLMs \(/developers/intelligent-contracts/features/calling-llms/\)](/developers/intelligent-contracts/features/calling-llms/):

```

@gl.public.write
def ai_decision(self, prompt: str):
    def call_llm():
        response = gl.nondet.exec_prompt(prompt)
        return response.strip()

    # Use comparative principle for LLM response consensus
    decision = gl.eq_principle.prompt_comparative(
        call_llm,
        principle="Responses should be semantically equivalent in meaning"
    )
    return decision

```

Calling LLMs

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/calling-llms>

Basic LLM Calls

Execute prompts using large language models:

```
def leader_fn():
    response = gl.nondet.exec_prompt("Answer this question")
    return response.strip().lower()

def validator_fn(leader_result) -> bool:
    if not isinstance(leader_result, gl.vm.Return):
        return False
    my_result = leader_fn()
    return my_result == leader_result.calldata

answer = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)
```

JSON Response Format

Request structured responses from LLMs:

```
def structured_llm_call():
    prompt = """
    Return a JSON object with these keys:
    - "score": random integer from 1 to 100
    - "status": either "active" or "inactive"
    """
    return gl.nondet.exec_prompt(prompt, response_format='json')

def validate_structured(leader_result) -> bool:
    if not isinstance(leader_result, gl.vm.Return):
        return False
    data = leader_result.calldata
    # Validate structure rather than exact match - LLM outputs are non-deterministic
    return (
        isinstance(data, dict)
        and isinstance(data.get("score"), int)
        and 1 <= data["score"] <= 100
        and data.get("status") in ("active", "inactive")
    )

result = gl.vm.run_nondet_unsafe(structured_llm_call, validate_structured)
score = result['score'] # Access JSON fields
```

Note:

Using `strict_eq` with LLM calls will fail consensus because LLM outputs are non-deterministic.

Always use `run_nondet_unsafe` with a custom validator that checks the *structure* and *validity* of the response rather than requiring an exact match.

This approach guarantees that `exec_prompt` returns a valid JSON object, however correspondence to the specified format depends on the underlying LLM.

Image Processing

Process images with vision models:

```
def vision_analysis():
    prompt = "Describe what you see in this image"
    image_data = b"\x89PNG..."
    return gl.nondet.exec_prompt(
        prompt,
        images=[image_data]
    )

def validate_description(leader_result) -> bool:
    if not isinstance(leader_result, gl.vm.Return):
        return False
    # Accept any non-empty string description
    return isinstance(leader_result.calldata, str) and len(leader_result.calldata.strip()) > 0

description = gl.vm.run_nondet_unsafe(vision_analysis, validate_description)
```

Note:

Limit of images is two

Response Validation

Validate and process LLM responses:

```
def generate_number():
    response = gl.nondet.exec_prompt(
        "Generate a number between 1 and 100",
        response_format='json'
    )

    # Validate the response
    if response['number'] < 1 or response['number'] > 100:
        raise Exception(f"Invalid number: {response['number']}")

    return response['number']

def validate_number(leader_result) -> bool:
    if not isinstance(leader_result, gl.vm.Return):
        return False
    num = leader_result.calldata
    return isinstance(num, int) and 1 <= num <= 100

result = gl.vm.run_nondet_unsafe(generate_number, validate_number)
```

Defensive Response Parsing

LLMs return unpredictable formats. Even with `response_format='json'`, the response may not match your expected schema. Here are some patterns for handling this:

Key Aliasing

LLMs sometimes use alternate key names. One approach is to check for common variations:

```
def parse_score(response: dict) -> int:
    """Extract a numeric score, handling common LLM variations."""
    raw = response.get("score")
    if raw is None:
        for alt in ("rating", "points", "value", "result"):
            if alt in response:
                raw = response[alt]
                break

    if raw is None:
        raise gl.UserError(f"Missing 'score' key. Got keys: {list(response.keys())}")
```

```
# Coerce to int – handles "3", "3.5", floats, whitespace
try:
    return max(0, int(round(float(str(raw).strip()))))
except (ValueError, TypeError):
    raise gl.UserError(f"Non-numeric score: {raw}")
```

JSON Cleanup

LLMs sometimes wrap JSON in markdown fences or add trailing commas:

```
import re

def clean_llm_json(text: str) -> dict:
    """Extract and clean JSON from LLM output."""
    import json
    first = text.find("{")
    last = text.rfind("}")
    if first == -1 or last == -1:
        raise gl.UserError(f"No JSON object found in response")
    text = text[first:last + 1]
    text = re.sub(r"(?!\\s*?[\\{\\[\\\"\\'\\w])", "", text) # Remove trailing commas
    return json.loads(text)
```

Always Use response_format='json'

```
result = gl.nondet.exec_prompt(task, response_format="json")
```

This instructs the LLM to return JSON. It significantly improves reliability, but you should still validate and clean the output — LLMs don't always comply.

Error Handling in LLM Calls

When using LLMs inside `run_nondet_unsafe`, consider how errors affect consensus. If the LLM returns garbage, you generally want the validator to **disagree** (return False) rather than agree on broken output — this forces a rotation to a new leader:

```
def leader_fn():
    result = gl.nondet.exec_prompt(prompt, response_format="json")
    if not isinstance(result, dict):
        raise gl.UserError(f"LLM returned non-dict: {type(result)}")
    return result
```

See Error Handling (/developers/intelligent-contracts/features/error-handling) for more on error classification patterns.

Image Processing

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/image-processing>

Intelligent Contracts can process images through LLMs — pass screenshots, photos, or any visual data alongside a prompt for analysis.

Sending Images to LLMs

Use the images parameter in `gl.nondet.exec_prompt()`:

```
from genlayer import *

class ReceiptVerifier(gl.Contract):
    verified: bool

    def __init__(self):
        self.verified = False

    @gl.public.write
    def verify_receipt(self, image_data: bytes, expected_amount: str) -> None:
        def leader_fn():
            return gl.nondet.exec_prompt(
                f"Does this receipt show a payment of {expected_amount}? "
                "Respond as JSON: {{\"matches\": true/false, \"actual_amount\": \"...\"}}",
                images=[image_data], # accepts raw bytes directly
                response_format="json",
            )

        def validator_fn(leaders_res) -> bool:
            if not isinstance(leaders_res, gl.vm.Return):
                return False
            my_result = leader_fn()
            return my_result["matches"] == leaders_res.calldata["matches"]

        result = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)
        self.verified = result["matches"]
```

The images parameter accepts a sequence of raw bytes (e.g., PNG/JPEG data) or `gl.nondet.Image` objects.

Capturing Screenshots from the Web

Combine web access (</developers/intelligent-contracts/features/web-access>) with image processing to screenshot a webpage and analyze it:

```
def check_website_status():
    url = "https://example.com/status-page"
    screenshot = gl.nondet.web.render(url, mode='screenshot')

    return gl.nondet.exec_prompt(
        "Is this status page showing all systems operational? "
        "Respond as JSON: {{\"all_operational\": true/false}}",
        images=[screenshot],
        response_format="json",
    )

result = gl.eq_principle.strict_eq(check_website_status)
```

Use Cases

- Visual evidence verification — insurance claims with photo proof, damage assessment
- Document analysis — receipts, invoices, certificates
- Web monitoring — screenshot a page and verify its content matches expectations
- Brand & content compliance — check if visual content meets guidelines
- UI verification — screenshot an app and verify it renders correctly

Note:

Image processing requires vision-capable LLM models. On the GenLayer network, validators handle model selection — your contract just sends images. In Studio running locally, ensure your configured validators use a model that supports image inputs (e.g., GPT-5, Claude Sonnet).

Web Access

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/web-access>

HTTP Requests

Send data to external services:

```
def post_request():
    url = "https://test-server.genlayer.com/body/echo"
    response = gl.nondet.web.request(
        url,
        method='POST',
        body={}
    )
    return response.status_code

status_code = gl.eq_principle.strict_eq(post_request)
```

Web Rendering

Render web page and extract content:

```
def render_page():
    url = "https://test-server.genlayer.com/static/genvm/hello.html"
    # Render HTML content
    html_content = gl.nondet.web.render(url, mode='html')
    return html_content

page_html = gl.eq_principle.strict_eq(render_page)
```

Screenshot Capture

Take screenshots of web pages:

```
def take_screenshot():
    url = "https://test-server.genlayer.com/static/genvm/hello.html"
    # Capture page as image
    screenshot = gl.nondet.web.render(url, mode='screenshot')
    return screenshot

image_data = gl.eq_principle.strict_eq(take_screenshot)
```

Handling HTTP Errors

External APIs can return error responses. Consider checking the status code:

```
def fetch_data():
    response = gl.nondet.web.request(api_url, method='GET')
    if response.status_code >= 400 and response.status_code < 500:
        raise gl.UserError(f"API returned client error: {response.status_code}")
    elif response.status_code >= 500:
        raise gl.UserError(f"API temporarily unavailable: {response.status_code}")
    return json.loads(response.body.decode("utf-8"))
```

Consensus-Friendly Web Requests

When using web data in non-deterministic blocks, remember that the leader and validators make independent requests. External APIs may return different data between calls — timestamps change, counts update, caches vary.

Extract Stable Fields

One approach is to return only the fields that won't change between calls:

```
def leader_fn():
    res = gl.nondet.web.get(api_url)
    data = json.loads(res.body.decode("utf-8"))
    # Only return fields that are stable across requests
    return {"id": data["id"], "login": data["login"], "status": data["status"]}
    # NOT: follower_count, updated_at, online_status
```

Derive Status from Variable Data

When raw data may differ between leader and validator (e.g., CI check counts change), consider comparing a derived summary instead of the raw values:

```
def validator_fn(leaders_res: gl.vm.Result) -> bool:
    validator_data = leader_fn()

    def derive_status(checks):
        if not checks:
            return "pending"
        for c in checks:
            if c.get("conclusion") != "success":
                return "failing"
        return "success"

    return derive_status(leaders_res.calldata) == derive_status(validator_data)
```

See the Equivalence Principle (</developers/intelligent-contracts/equivalence-principle>) page for more validation patterns.

Development Setup

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tooling-setup>

This guide covers everything you need to develop, test, and deploy Intelligent Contracts — from zero-setup exploration to a full local development environment with linting, testing, and agent-assisted workflows.

Recommended: GenLayer Skills

The GenLayer Skills (<https://skills.genlayer.com/>) plugin for Claude Code is the fastest path to a working development environment. The genlayer-dev skill scaffolds a project, installs prerequisites, runs the linter and direct-mode tests, and walks you through deployment — interactively.

```
claude /plugin marketplace add genlayerlabs/skills
claude /plugin install genlayer-dev@genlayerlabs
```

Then run `claude /genlayer-dev` and follow the prompts. The skill bundles the same workflows documented in the rest of this page — it just runs them for you.

The rest of this guide describes the manual setup, the reference path for anyone who wants fine-grained control or is integrating with existing tooling.

Quick Start: GenLayer Studio

The fastest way to start is studio.genlayer.com (<https://studio.genlayer.com>) — a web-based IDE where you can write, deploy, and interact with Intelligent Contracts with zero setup.

You can also import any deployed contract by address to interact with it directly in the Studio — useful for debugging or testing contracts you've deployed from your local environment.

For anything beyond exploration — local testing, CI, agent-assisted development — set up a local environment below.

Local Environment Setup

Prerequisites

- Python 3.12+ (<https://www.python.org/downloads/>) - For contracts, linting, and testing
- Node.js 18+ (<https://nodejs.org/en/download/>) - For GenLayer CLI and frontend
- Docker 26+ (<https://docs.docker.com/get-docker/>) - For GenLayer Studio (local)

Note:

Docker is only required if you want to run GenLayer Studio locally. Direct mode tests and GLSim work without Docker.

Clone the Boilerplate

The GenLayer Project Boilerplate (<https://github.com/genlayerlabs/genlayer-project-boilerplate>) is the recommended starting point. It includes contract templates, testing infrastructure, and a frontend scaffold.

```
git clone https://github.com/genlayerlabs/genlayer-project-boilerplate.git
cd genlayer-project-boilerplate
```

Install Dependencies

```
pip install -r requirements.txt
```

This installs:

- `genlayer-test` — testing framework with direct mode, GLSim, and integration test support
- `genvm-linter` — static analysis for contract safety and correctness

Project Structure

```
contracts/      # Your Intelligent Contracts
tests/
  direct/      # Fast in-memory tests (no server needed)
```

```

integration/      # Full tests against GenLayer Studio
frontend/        # Next.js frontend with GenLayerJS
deploy/          # Deployment scripts
gltest.config.yaml # Test network configuration

```

Development Workflow

The recommended development loop:

1. Write Your Contract

Create or edit a contract in `contracts/`:

```

from genlayer import *

class MyContract(gl.Contract):
    data: TreeMap[Address, str]

    def __init__(self):
        self.data = TreeMap()

    @gl.public.view
    def get_data(self, addr: Address) -> str:
        return self.data.get(addr, "")

    @gl.public.write
    def set_data(self, value: str):
        self.data[gl.message.sender_address] = value

```

2. Lint

Run the linter to catch issues before testing:

```
genvm-lint check contracts/my_contract.py
```

check runs both fast AST-based safety checks and SDK-based semantic validation. It catches:

- Forbidden imports — `os`, `sys`, `subprocess`, etc.
- Non-deterministic calls outside equivalence principle blocks
- Invalid storage types — must use `TreeMap/DynArray`, not `dict/list`
- Missing decorators and return type annotations

Use `--json` for machine-readable output (useful for CI and agent workflows).

3. Direct Mode Tests

The fastest feedback loop — runs contracts in-memory without any server:

```
pytest tests/direct/ -v
```

Direct mode provides Foundry-style test fixtures:

```

import json

def test_create_and_read(direct_vm, direct_deploy, direct_alice):
    contract = direct_deploy("contracts/my_contract.py")

    # Set the transaction sender
    direct_vm.sender = direct_alice

```

```
# Call write methods directly
contract.set_data("hello")

# Call view methods
result = contract.get_data(direct_alice)
assert result == "hello"
```

Mocking Web and LLM Calls

For contracts that use `gl.nondet.web` or `gl.nondet.exec_prompt`, mock the responses:

```
def test_with_mock(direct_vm, direct_deploy, direct_alice):
    contract = direct_deploy("contracts/my_contract.py")
    direct_vm.sender = direct_alice

    # Mock web requests (regex matches on URL)
    direct_vm.mock_web(
        r".*api\.example\.com/prices.*",
        {"status": 200, "body": '{"price": 42.5}'},
    )

    # Mock LLM responses (regex matches on prompt)
    direct_vm.mock_llm(
        r".*Extract the match result.*",
        json.dumps({"score": "2:1", "winner": 1}),
    )

    contract.update_price("ETH/USD")
    assert contract.get_price("ETH/USD") == 42.5

    # Reset mocks between scenarios
    direct_vm.clear_mock()
```

Testing Failures

```
def test_expected_failure(direct_vm, direct_deploy, direct_alice):
    contract = direct_deploy("contracts/my_contract.py")
    direct_vm.sender = direct_alice

    with direct_vm.expect_revert("Insufficient balance"):
        contract.withdraw(1000)
```

Available fixtures: `direct_vm`, `direct_deploy`, `direct_alice`, `direct_bob`, `direct_charlie`, `direct_owner`, `direct_accounts`

4. Integration Tests

Once direct tests pass, run against a GenLayer environment (see Environments (#environments) below) for full end-to-end validation:

```
gltest tests/integration/ -v -s
```

Integration tests use the `genlayer-test` SDK to deploy and interact with contracts through the JSON-RPC API:

```
from gltest import get_contract_factory
from gltest.assertions import tx_execution_succeeded

def test_full_flow():
    factory = get_contract_factory("MyContract")
    contract = factory.deploy(args=[])

    # Write methods return transaction receipts
    tx_receipt = contract.set_data(args=["hello"]).transact()
    assert tx_execution_succeeded(tx_receipt)

    # Read methods return values directly
    result = contract.get_data(args=[contract.address]).call()
    assert result == "hello"
```

See the GenLayer Test SDK reference (</api-references/genlayer-test>) for the full API.

Environments

These are the backends you can develop against. Your frontend, integration tests, and deployment scripts all connect to one of these via JSON-RPC.

GLSim

Setup: `pip install genlayer-test[sim]`

A lightweight simulator that implements the GenLayer JSON-RPC protocol. Starts in ~1 second, no Docker required. You can point your frontend and integration tests at it for fast iteration.

GLSim runs the Python runner natively — not inside GenVM — so there can be minor incompatibilities with the full runtime. Use it for fast development cycles, then validate against Studio before deploying.

```
# Start the simulator
glsim --port 4000 --validators 5

# Your frontend connects to http://localhost:4000/api
# Integration tests run against it
gltest tests/integration/ -v -s
```

GenLayer Studio (Local)

Setup: Docker 26+ and GenLayer CLI

Full GenVM execution with real consensus. You get complete validator logs, transaction inspection, and the Studio UI for interactive debugging. This is where you validate that everything works in the real runtime.

```
npm install -g genlayer
genlayer init
genlayer up
```

Once running, access Studio at <http://localhost:8080/>. Your frontend and tests connect to <http://localhost:4000/api>.

```
gltest tests/integration/ -v -s --network localnet
```

CLI options for `genlayer init` and `genlayer up`

`genlayer init` options:

- `--numValidators` — number of validators (default: 5)
- `--headless` — run without Studio UI
- `--reset-db` — start with a clean database
- `--localnet-version` — pin a specific localnet version

`genlayer up` options:

- `--reset-validators` — recreate all validators
- `--numValidators` — number of validators
- `--headless` — run without Studio UI
- `--reset-db` — clean database

studio.genlayer.com

Setup: none

Hosted Studio — convenient when you don't want to run Docker locally. Rate limited compared to a local instance. You can also import any deployed contract by address to interact with it directly in the browser.

```
gltest tests/integration/ -v -s --network studionet
```

Testnet Bradbury

Setup: account with private keys

The most realistic environment — the actual GenLayer testnet with real validators. You don't get full validator logs like you do with local Studio, so this is best for final pre-production validation rather than active debugging.

```
gltest tests/integration/ -v -s --network testnet_bradbury
```

Configure accounts in `gltest.config.yaml`:

```
networks:
  testnet_bradbury:
    accounts:
      - "${ACCOUNT_PRIVATE_KEY_1}"
      - "${ACCOUNT_PRIVATE_KEY_2}"
```

IDE Support

The GenLayer VS Code Extension (<https://marketplace.visualstudio.com/items?itemName=genlayer-labs.genlayer>) provides syntax highlighting, snippets, and language support for Intelligent Contracts. Works with VS Code, Cursor, and other VS Code-compatible editors.

Agent-Assisted Development

The boilerplate is designed to work well with AI coding agents. The linter and direct mode tests provide a fast feedback loop that agents can use without spinning up infrastructure.

For Claude Code, install the GenLayer Skills (<https://skills.genlayer.com/>) marketplace (covered at the top of this page) — it bundles the workflows below behind interactive commands.

For other IDEs and clients, the GenLayer MCP servers provide the same capabilities via the Model Context Protocol:

The GenLayer MCP Server (<https://www.npmjs.com/package/genlayer-mcp>) provides contract generation and GenLayer-specific guidance directly in your IDE:

```
# Add to Claude Code
claude mcp add genlayer npx -- -y genlayer-mcp
```

The GenLayer Docs MCP Server gives your AI agent searchable access to the full GenLayer documentation and SDK reference — so it can look up APIs, patterns, and examples while writing code:

```
# Add to Claude Code
claude mcp add genlayer-docs --transport sse https://docs-mcp.genlayer.com/sse
```


This is a hosted service — no local setup required. It provides a `search_docs` tool that searches across both the GenLayer documentation (<https://docs.genlayer.com>) and the GenLayer SDK reference (<https://sdk.genlayer.com>). Compatible with any MCP client (Claude Code, Cursor, Windsurf, etc.).

The boilerplate also includes a `CLAUDE.md` file pre-configured with commands, architecture context, and testing patterns — so agents understand the project structure immediately.

Frontend Development

The boilerplate includes a Next.js frontend with GenLayerJS integration. See the GenLayerJS SDK documentation ([/api-references/genlayer-js](#)) for the full API reference.

```
cd frontend
npm install
npm run dev
```

For building a frontend from scratch:

```
npm install genlayer-js

import { createClient } from 'genlayer-js';

const client = createClient({ endpoint: 'http://localhost:4000/api' });

// Read from a contract
const value = await client.readContract({
  address: contractAddress,
  functionName: "get_data",
  args: [],
});

// Write to a contract
const txHash = await client.writeContract({
  address: contractAddress,
  functionName: "set_data",
  args: [newValue],
});

const receipt = await client.waitForTransactionReceipt({
  hash: txHash,
  status: "FINALIZED",
});
```

Your First Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/first-contract>

While a GenLayer Intelligent Contract is a pure Python program, there are few things that must be present in your file.

Version Comment

First of all, you need to place a magic comment with the version of GenVM you wish to use on the first line of your file

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }
```

It is similar to Solidity's `pragma solidity`. The hash after the colon identifies the exact version of the Python runtime to use, ensuring your contract always runs against the same environment.

Importing the Standard Library

After the version comment, you can import the GenLayer standard library:

```
from genlayer import *
```

This will import all necessary types into the global scope, and everything else under gl namespace.

The Contract Class

An Intelligent Contract is a regular Python class, with a regular constructor and methods, which allows you to use your favorite IDE for type checking and auto completion. However, there are some additional requirements to make a class an Intelligent Contract.

Decorators

The GenVM makes use of decorators to identify the contract class and its methods.

The contract class must extend `gl.Contract` class so that GenVM knows that it's an Intelligent Contract. There can be only one contract class in a file.

Furthermore, public methods must be decorated with either `@gl.public.view` for read-only methods or `@gl.public.write` (or `@gl.public.write.payable`) for methods that modify storage. Constructor (`init`) must be private (not decorated)

Persistent data

The GenVM enables Intelligent Contracts to maintain persistent data. This data is stored on the blockchain and can be accessed and modified via transactions.

This is done by declaring fields in the contract class.

Note:

All persistent fields must be declared in the class body and annotated with types.

Types

In your contracts, you can use any Python types, but for persisted fields, there are some restrictions:

- `list[T]` needs to be replaced with `DynArray[T]`
- `dict[K, V]` needs to be replaced with `TreeMap[K, V]`
- `int` type isn't supported on purpose. You most likely wish to use some fixed-size integer type, such as `i32` or `u256`. If this is not the case and you are sure that you need big integers, you can annotate your field with `bigint`, which is just an alias for python `int`

Note:

Only fully instantiated generic types can be used, so `TreeMap` is forbidden, while `TreeMap[str, u256]` is not

Example

Here is a simple example of an Intelligent Contract that stores a name and allows changing it:

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *

class Hello(gl.Contract):
    name: str

    def __init__(self, name: str):
        self.name = name

    @gl.public.view
    def run(self) -> str:
        return f'Hello, {self.name}'

    @gl.public.write
    def set_name(self, name: str):
        print(f'debug old name: {self.name}') # <- you can use prints for debugging
        # they will be included in the GenVM execution log
        self.name = name
```

Note:

The GenLayer Studio automatically detects the constructor parameters from your code. When you run your contract in the Studio, it provides the UI for all parameters for both constructor and methods, making it easy to manage and modify these inputs.

Address Type

Source: <https://docs.genlayer.com/developers/intelligent-contracts/types/address>

The Address type represents a 20-byte blockchain address, similar to Ethereum addresses. It's one of the most important types in Intelligent Contracts for handling user accounts, contract addresses, and cross-contract interactions.

Creating Address Instances

Use these patterns to instantiate an Address from different encodings. Choose the one that matches your source data. Construction validates size and normalizes the value so later comparisons and serializations are consistent.

```
# Constructing Address values from common encodings
# From hex string (most common)
address1 = Address("0x742d35Cc6634C0532925a3b8D4C9db96C4b4d8b6")
address2 = Address("0x000000000000000000000000000000000000000000000000") # Zero address (useful as a sentinel/null)

# From base64 encoded string (handy when integrating with systems that emit base64)
address3 = Address("WzjaanAcVoVF3PywP8uHX1a+3cQ=")

# From bytes (20 bytes, useful for low-level calls or pre-parsed data)
address4 = Address(b'\x5b\x38\xda\x6a\x70\x1c\x56\x85\x45\xdc\xfc\xb0\x3f\xcb\x87\x5f\x56\xbe\xdd\xc4')
```

Address Properties and Conversions

Address exposes multiple read-only views for common encodings. Use these when you need a specific serialization for UI, storage, hashing, or interop with external systems.

```
class AddressConverter(gl.Contract):
    # Demonstrates converting a stored Address into several representations and when to use them
    stored_address: Address
```

```

def __init__(self, initial_address: str):
    # Store as Address (not string) so the value is validated and normalized once
    self.stored_address = Address(initial_address)

@gl.public.view
def get_address_hex(self) -> str:
    # Get EIP-55 checksum hex; ideal for UI display, logs, and user input round-trips
    return self.stored_address.as_hex

@gl.public.view
def get_address_bytes(self) -> bytes:
    # Get raw 20-byte representation; efficient for hashing and compact storage
    return self.stored_address.as_bytes

@gl.public.view
def get_address_b64(self) -> str:
    # Get base64 representation; useful for URLs and systems that prefer base64
    return self.stored_address.as_b64

@gl.public.view
def get_address_string(self) -> str:
    # Default string representation (same as as_hex)
    return str(self.stored_address)

@gl.public.view
def format_address(self, fmt: str) -> str:
    # Format with a specifier for consistent serialization across callers
    if fmt == "x":
        return f"{self.stored_address:x}" # hex (checksummed)
    elif fmt == "b64":
        return f"{self.stored_address:b64}" # base64 string
    elif fmt == "cd":
        return f"{self.stored_address:cd}" # calldata/ABI-friendly format
    else:
        return str(self.stored_address)

```

Address in Contract State

Store Address values directly in contract state for role management and registries. Collections like `DynArray[Address]` and `TreeMap[key, Address]` help model allowlists and name-to-address lookups safely.

```

class AddressStorage(gl.Contract):
    # Store owner/admin roles and registries using Address-typed fields and collections
    # Single address storage
    owner: Address
    admin: Address

    # Address collections
    authorized_users: DynArray[Address]
    contract_registry: TreeMap[str, Address]

    def __init__(self):
        # Initialize roles to the transaction sender to establish initial permissions
        self.owner = gl.message.sender_address
        self.admin = gl.message.sender_address

    @gl.public.write
    def add_authorized_user(self, user_address: str):
        # Convert input to Address so membership checks use a normalized value
        address = Address(user_address)
        self.authorized_users.append(address)

    @gl.public.write
    def register_contract(self, name: str, contract_address: str):
        # Keep a normalized Address in the registry for consistent lookups
        address = Address(contract_address)
        self.contract_registry[name] = address

    @gl.public.view
    def is_authorized(self, user_address: str) -> bool:
        # Membership check with Address ensures exact, normalized equality semantics
        address = Address(user_address)

```

```
return address in self.authorized_users
```

Address in Method Parameters and Returns

Methods often accept human-friendly strings and return normalized Address data. Convert inputs at the boundary to keep state consistent and make API responses easy for clients to consume.

```
class AddressOperations(gl.Contract):
    # Use Address as the key to guarantee consistent equality and map behavior
    balances: TreeMap[Address, u256]

    def __init__(self):
        # Start with an empty map; balances default to zero for missing keys
        pass

    @gl.public.write
    def set_balance_from_hex(self, account_hex: str, amount: int):
        # Convert caller-provided hex into Address to validate and normalize before use
        account = Address(account_hex)
        self.balances[account] = amount

    @gl.public.view
    def get_my_balance(self) -> int:
        # Look up the balance associated with the message sender's Address
        return self.balances.get(gl.message.sender_address)

    @gl.public.view
    def get_balance(self, account_hex: str) -> int:
        # Convert external input to Address so lookups match normalized keys
        account = Address(account_hex)
        return self.balances.get(account)

    @gl.public.view
    def get_my_address(self) -> str:
        # Return the sender's address as a string; convenient for clients/UI
        return str(gl.message.sender_address)
```

Primitive Types

Source: <https://docs.genlayer.com/developers/intelligent-contracts/types/primitive>

GenVM provides comprehensive sized integer types for efficient storage. These types are primarily used for type annotations in storage fields and enforce range constraints only when assigned to storage.

Note:

Important: Sized integer types (u8, u256, etc.) are typing.NewType aliases that behave like regular Python int outside of storage context. Range checking and overflow protection only occur when assigning values to storage fields.

Integer Types

Standard Integer Types (Most Commonly Used)

- u8, u16, u32, u64, u128, u256 - Unsigned integers of various sizes
- i8, i16, i32, i64, i128, i256 - Signed integers of various sizes

- bigint - Arbitrary precision integer (use with caution, prefer sized integers)

Full Range Available

GenVM also supports all intermediate sizes from 8-bit to 256-bit (e.g., u24, u40, u48, u56, u72, u80, u88, u96, u104, u112, u120, u136, u144, u152, u160, u168, u176, u184, u192, u200, u208, u216, u224, u232, u240, u248 and their signed counterparts).

```
class IntegerTypes(gl.Contract):

    uint8_val: u8
    uint256_val: u256
    uint24_val: u24    # 3-byte integers
    int64_val: i64

    def __init__(self):
        # Storage assignment enforces type constraints
        self.uint8_val = 255          # Range checked: must be 0-255
        self.uint256_val = 2**256 - 1 # Range checked: must fit in u256
        self.int64_val = -9223372036854775808 # Range checked: must fit in i64
        self.uint24_val = 16777215        # Range checked: must be 0-16777215

    @gl.public.view
    def get_uint8(self) -> int:
        return self.uint8_val

    @gl.public.write
    def set_uint256(self, value: int):
        self.uint256_val = value

    @gl.public.view
    def calculate_sum(self, a: int, b: int) -> int:
        # Note: u256(a) + u256(b) behaves exactly like a + b in Python
        # The u256() calls don't actually enforce range constraints here
        return a + b

    @gl.public.view
    def get_int64(self) -> int:
        return self.int64_val

    @gl.public.view
    def get_uint24(self) -> int:
        return self.uint24_val
```

String and Bytes Types

```
class StringBytesTypes(gl.Contract):
    text_data: str
    binary_data: bytes

    def __init__(self):
        self.text_data = "Hello, GenLayer!"
        self.binary_data = b"binary_data_here"

    @gl.public.write
    def store_text(self, text: str):
        self.text_data = text

    @gl.public.view
    def get_text_length(self) -> int:
        return len(self.text_data) # Return type annotation handles the conversion

    @gl.public.view
    def get_bytes_length(self) -> int:
        return len(self.binary_data) # Return type annotation handles the conversion

    # String operations
    @gl.public.view
    def concatenate_strings(self, str1: str, str2: str) -> str:
        return str1 + str2
```

Boolean Type

```
class BooleanTypes(gl.Contract):
    flag: bool
    flags: DynArray[bool]
    flag_map: TreeMap[str, bool]

    def __init__(self):
        self.flag = True

    @gl.public.write
    def toggle_flag(self):
        self.flag = not self.flag

    @gl.public.write
    def set_flag(self, flag_value: bool):
        self.flag = flag_value

    @gl.public.write
    def set_named_flag(self, name: str, value: bool):
        self.flag_map[name] = value

    @gl.public.view
    def get_flag_status(self) -> bool:
        return self.flag

    @gl.public.view
    def check_named_flag(self, name: str) -> bool:
        return self.flag_map.get(name, False)
```

Type Conversion Utilities

```
class TypeConversionUtils(gl.Contract):
    def __init__(self):
        pass

    @gl.public.view
    def string_to_bytes(self, text: str) -> bytes:
        return text.encode('utf-8')

    @gl.public.view
    def bytes_to_string(self) -> str:
        return b"binary_data_here".decode('utf-8')

    @gl.public.view
    def int_to_string(self, value: int) -> str:
        return str(value)

    @gl.public.view
    def string_to_int(self, value_str: str) -> int:
        return int(value_str)
```

Type Ranges

Type	Range	Use Case
`u8`	0 to 255	Small counters, flags
`u16`	0 to 65,535	Medium counters
`u32`	0 to 4,294,967,295	Large counters, timestamps
`u64`	0 to 18,446,744,073,709,551,615	Very large numbers
`u128`	0 to $2^{128} - 1$	Cryptographic values
`u256`	0 to $2^{256} - 1$	Token amounts, hashes
`u160`	0 to $2^{160} - 1$	Address integers

Type	Range	Use Case
`i8`	-128 to 127	Small signed values
`i16`	-32,768 to 32,767	Medium signed values
`i32`	-2,147,483,648 to 2,147,483,647	Large signed values
`i64`	-9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	Very large signed values
`i128`	-2^{127} to $2^{127} - 1$	Large signed calculations

Collection Types

Source: <https://docs.genlayer.com/developers/intelligent-contracts/types/collections>

GenVM provides storage-compatible collection types that replace Python's built-in collections. These are essential for managing dynamic data structures in Intelligent Contracts.

DynArray (Dynamic Arrays)

DynArray[T] replaces Python's list[T] and provides dynamic array functionality with storage compatibility.

Basic DynArray Usage

```
class ArrayOperations(gl.Contract):
    # Simple arrays
    numbers: DynArray[u256]
    names: DynArray[str]
    addresses: DynArray[Address]

    # Nested arrays
    matrix: DynArray[DynArray[u256]]

    def __init__(self):
        pass

    @gl.public.write
    def add_number(self, num: int):
        self.numbers.append(num)

    @gl.public.write
    def add_name(self, name: str):
        self.names.append(name)

    @gl.public.write
    def add_address(self, addr: str):
        address = Address(addr)
        self.addresses.append(address)

    @gl.public.view
    def get_number_at(self, index: int) -> int:
        if index < len(self.numbers):
            return self.numbers[index]
        return u256(0)

    @gl.public.view
    def get_array_length(self) -> int:
        return u256(len(self.numbers))

    @gl.public.write
    def remove_last(self):
        if len(self.numbers) > 0:
            self.numbers.pop()

    # Matrix operations
    @gl.public.write
    def add_row(self, row: DynArray[int]):
        self.matrix.append(row)

    @gl.public.view
    def get_matrix_element(self, row: int, col: int) -> int:
        if row < len(self.matrix) and col < len(self.matrix[row]):
```



```

        return self.matrix[row][col]
    return u256(0)

```

DynArray with Custom Types

```

from genlayer import *
from dataclasses import dataclass

import json
import typing

@allow_storage
@dataclass
class User:
    name: str
    age: u8
    balance: u256

class UserArrayOperations(gl.Contract):
    users: DynArray[User]
    user_ids: DynArray[str]

    def __init__(self):
        pass

    @gl.public.write
    def add_user(self, name: str, age: int, balance: int):
        user = User(name=name, age=age, balance=balance)
        self.users.append(user)
        self.user_ids.append(f"user_{len(self.users)}")

    @gl.public.view
    def find_user_by_name(self, name: str) -> TreeMap[str, typing.Any]:
        for user in self.users:
            if user.name == name:
                return user
        return User("", u8(0), u256(0))

```

TreeMap (Ordered Maps)

TreeMap[K, V] replaces Python's dict[K, V] and provides ordered map functionality with storage compatibility.

Basic TreeMap Usage

```

class MapOperations(gl.Contract):
    # Simple maps
    balances: TreeMap[Address, u256]
    names: TreeMap[str, str]
    flags: TreeMap[str, bool]

    def __init__(self):
        pass

    @gl.public.write
    def set_balance(self, address: str, amount: int):
        account = Address(address)
        self.balances[account] = amount

    @gl.public.write
    def set_name(self, key: str, name: str):
        self.names[key] = name

    @gl.public.view
    def get_balance(self, address: str) -> int:
        account = Address(address)
        return self.balances.get(account, u256(0))

    @gl.public.view
    def get_name(self, key: str) -> str:
        return self.names.get(key, "")

```

```

@gl.public.view
def has_balance(self, address: str) -> bool:
    account = Address(address)
    return account in self.balances

```

TreeMap with Complex Values

```

@allow_storage
@dataclass
class UserProfile:
    name: str
    email: str
    balance: u256
    is_active: bool

class UserProfileMap(gl.Contract):
    profiles: TreeMap[Address, UserProfile]
    permissions: DynArray[str]
    user_permissions: TreeMap[Address, DynArray[str]]

    def __init__(self):
        pass

    @gl.public.write
    def create_profile(self, name: str, email: str):
        address = gl.message.sender_address
        profile = UserProfile(
            name=name,
            email=email,
            balance=u256(0),
            is_active=True
        )
        self.profiles[address] = profile
        self.user_permissions[address] = self.permissions

    @gl.public.write
    def add_permission(self, user: str, permission: str):
        account = Address(user)
        if account in self.user_permissions:
            self.user_permissions[account].append(permission)

    @gl.public.view
    def get_profile(self, address: str) -> UserProfile:
        account = Address(address)
        return self.profiles.get(account, UserProfile("", "", u256(0), False))

    @gl.public.view
    def has_permission(self, user: str, permission: str) -> bool:
        account = Address(user)
        if account in self.user_permissions:
            return permission in self.user_permissions[account]
        return False

```

Dataclasses

Source: <https://docs.genlayer.com/developers/intelligent-contracts/types/dataclasses>

Dataclasses provide structured data types for Intelligent Contracts. Use `@allow_storage` decorator for storage compatibility.

Method Parameters and Returns

```

import typing

```

```

@allow_storage
@dataclass
class UserData:
    name: str
    balance: u256

class SimpleContract(gl.Contract):
    users: TreeMap[Address, UserData]

    def __init__(self):
        pass

    @gl.public.write
    def create_user(self, name: str, balance: int):
        self.users[gl.message.sender_address] = UserData(name, balance)

    @gl.public.view
    def get_user(self, user_address: str) -> TreeMap[str, typing.Any]:
        address = Address(user_address)
        return self.users.get(address, UserData("", u256(0)))

```

Generic Types

```

import typing

@allow_storage
@dataclass
class Item[T]:
    data: T
    label: str

class GenericContract(gl.Contract):
    string_items: DynArray[Item[str]]

    def __init__(self):
        pass

    @gl.public.write
    def add_item(self, data: str, label: str):
        item = gl.storage.inmem_allocate(Item[str], data, label)
        self.string_items.append(item)

    @gl.public.view
    def get_items(self) -> DynArray[TreeMap[str, typing.Any]]:
        return self.string_items

```

Persisting data on the blockchain

Source: <https://docs.genlayer.com/developers/intelligent-contracts/storage>

Usual data structures aren't suitable for representing blockchain persistent storage:

1. Allocated addresses (id in python terms) are not persistent
2. Allocation requires knowledge about all allocated addresses, which takes a lot of space and would cost a lot of reads at start time
3. Serialization works poorly as it will rewrite entire storage (consider rehash)

Intelligent Contracts store data publicly on chain, attached to their account's address. The storage starts zero-initialized until a contract is deployed and initializes a state.

For storage declaration GenLayer uses contract class fields.

Note:

All persistent fields must be declared in the class body and annotated with types.

Example:

```
class PersistentContract(gl.Contract):
    minter: Address

    def __init__(self):
        self.minter = gl.message.sender_address
```

In your contracts, you can use any Python types, but for persisted fields, there are some restrictions:

- list[T] needs to be replaced with DynArray[T]
- dict[K, V] needs to be replaced with TreeMap[K, V]
- int type isn't supported on purpose. You most likely wish to use some fixed-size integer type, such as i32 or u256. If this is not the case and you are sure that you need big integers, you can annotate your field with bigint, which is just an alias for python int

Note:

Only fully instantiated generic types can be used, so TreeMap is forbidden, while TreeMap[str, u256] is not

Simple examples:

```
class PersistentContract(gl.Contract):
    a: str
    b: bytes
    # c: list[str]          # [] `list` is forbidden!
    c: DynArray[str]
    # b: dict[Address, u256] # [] `dict` is forbidden!
    # b: TreeMap           # [] only fully specialized generic types are allowed!
    b: TreeMap[Address, u256]
    # d: int               # [] `int` is forbidden
    d: bigint             # [] most likely you don't need an arbitrary big integer
    d_sized: i256
```

Few words about DynArray and TreeMap

These types implement `python collections.abc.MutableSequence` and `collections.abc.MutableMapping` which makes them compatible with most of the python code

They can be encoded into calldata as-is as well, which means that following code is correct:

```
class PersistentContract(gl.Contract):
    storage: DynArray[str]

    @gl.public.view
    def get_complete_storage(self) -> collections.abc.Sequence[str]:
        return self.storage
```

Note:

Calldata format supports mappings only with str keys, like JSON does.

Using custom data types

You can use other python classes in storage, for example:

```
@allow_storage
@dataclass
class User:
    name: str
    birthday: datetime.datetime

class Contract(gl.Contract):
    users: DynArray[User]
```

Note that you must decorate them with `@allow_storage`. This is done to prevent confusion (#differences-from-regular-python-types)

However, there is a tricky case: allocating storage generics in-memory. It is different from regular python syntax because storage types don't have type erasure due to fixed memory layout.

```
@allow_storage
@dataclass
class User:
    data: TreeMap[str, str]

User() # error: data is absent (from dataclass)
User(gl.storage.inmem_allocate(TreeMap[str, str])) # works fine, this function takes a type and `*args, **kwargs` for
corresponding `__init__`. Also note that type must be fully instantiated and have no type variables.
```

Default values

By default storage is zero-initialized:

Type	Default value
<code>`u*`, `i*`</code>	<code>`0`</code>
<code>`bool`</code>	<code>`false`</code>
<code>`float`</code>	<code>`+0`</code>
<code>`str`</code>	<code>`""`</code>
<code>`DynArray`</code>	<code>`[]`</code>
<code>`TreeMap`</code>	<code>`{}`</code>

Struct types are zero-initialized "recursively"

Memory Management

gl.storage.inmem_allocate

Required for generic storage classes only. Regular dataclasses use normal initialization.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *
from dataclasses import dataclass
import datetime

@allow_storage
@dataclass
class User:
    name: str
    birthday: datetime.datetime

@allow_storage
@dataclass
class Gen[T]:
    name: T
    birthday: datetime.datetime

class Contract(gl.Contract):
```

```

user: User
gen_user: Gen[bytes]

def __init__(self):
    pass

@gl.public.write
def plain(self):
    user = User('Ada', datetime.datetime.now())
    self.user = user

    read_user = self.user
    copied_out = gl.storage.copy_to_memory(read_user)

    def nd():
        print('inmem: ok', user)
        try:
            print('storage: not ok', str(read_user))
        except Exception as e:
            print('storage: not ok', e)
        print('copied out: ok', copied_out)

    gl.eq_principle.strict_eq(nd)

@gl.public.write
def generic(self):
    user = gl.storage.inmem_allocate(Gen[bytes], b'Ada', datetime.datetime.now())
    self.gen_user = user

    read_user = self.gen_user
    copied_out = gl.storage.copy_to_memory(read_user)

    def nd():
        print('inmem: ok', user)
        try:
            print('storage: not ok', str(read_user))
        except Exception as e:
            print('storage: not ok', e)
        print('copied out: ok', copied_out)

    gl.eq_principle.strict_eq(nd)

```

gl.storage.copy_to_memory

Storage objects need to be copied to memory when you want to use them in non-deterministic blocks, since non-deterministic blocks cannot access storage directly:

```

# Read from storage
storage_user = self.user

# Copy to memory for use in non-deterministic blocks or other operations
memory_user = gl.storage.copy_to_memory(storage_user)

def my_nondet_function():
    # Storage objects cannot be used directly in nondet blocks
    # print('storage: not accessible', storage_user) # Error - storage not accessible!

    # But memory objects work fine
    print('copied out: ok', memory_user) # Works!
    return str(memory_user) # String conversion also works with memory objects

gl.eq_principle.strict_eq(my_nondet_function)

```

Differences from regular python types

Even though storage classes mimic python types, remember that they provide you only with a view on memory, not actual data that is "here". For example, consider the above example

```

self.users.append(User("Ada"))
user = self.users[-1]

```

```
self.users[-1] = User("Definitely not Ada", datetime.datetime.now())
assert user.name == "Definitely not Ada" # this is true!
```

Your First Intelligent Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/first-intelligent-contract>

Now is the time to utilize all the power of GenLayer!

For blockchain integrity reasons, non-determinism must be contained within special non-deterministic blocks. Such blocks are regular Python functions with no arguments, which can return arbitrary values. However, there are some limitations:

- Storage is inaccessible from non-deterministic blocks
- State of the Python interpreter is not passed back to the deterministic code (for instance, you won't see changes in global variables)

Simple Case

To illustrate how it works, let's get a webpage as plain HTML, and verify that it has a link to the owner:

```
example_web_address = 'https://example.org'
def my_non_deterministic_block():
    web_data = gl.nondet.web.render(example_web_address, mode='html')
    return 'iana' in web_data

print(gl.eq_principle.strict_eq(my_non_deterministic_block))
```

Here are a few important parts:

1. It is mandatory to call `gl.nondet.web.render` (or `gl.nondet.web.get`) from a function invoked via `gl.eq_principle.*`, otherwise it will give an error
2. Type annotations are not required
3. `example_web_address` gets captured without the need to specify it explicitly
4. We are getting the page in plain HTML, because we want text from a link (HTML tag), which is not visible in the text representation
5. We are using `gl.eq_principle.strict_eq` because we return a bool (True or False), so there is no need to run LLMs or other complex computations: validators will agree *if* they both get the exactly the same result, which makes sense for our case. However, if you wish to return more complex data (such as a text summary), you should use other `gl.eq_principle` methods. More on this topic on the next page

As a Full Contract

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *

class Contract(gl.Contract):
    had_iana: bool

    def __init__(self):
        example_web_address = 'https://example.org'
        def my_non_deterministic_block():
            web_data = gl.nondet.web.render(example_web_address, mode='html')
            return 'iana' in web_data
        self.had_iana = gl.eq_principle.strict_eq(my_non_deterministic_block)
```

The Equivalence Principle

Source: <https://docs.genlayer.com/developers/intelligent-contracts/equivalence-principle>

The Equivalence Principle is how GenLayer achieves consensus on non-deterministic operations — things like web requests, LLM calls, or any computation that might produce different results on different nodes.

The core idea: a leader executes the operation and proposes a result, then validators independently verify whether that result is acceptable.

Quick Reference: Which Pattern to Use

```
Can validators reproduce the exact same normalized output?
├── YES → strict_eq
│   │   Exact match. Use when outputs are deterministic or can be
│   │   canonicalized (e.g., JSON with sort_keys=True).
│   │   Examples: blockchain RPC, stable REST APIs.
└── NO  → Write a custom validator function (run_nondet_unsafe)
    │   You control the full logic: rerun and compare with tolerances,
    │   derive status, extract stable fields, or evaluate the leader's
    │   output directly without rerunning – whatever your contract needs.
```

For most contracts, you'll write a custom validator function. It gives you full control over comparison logic and error handling.

Note:

GenLayer also provides `prompt_comparative` and `prompt_non_comparative` as convenience wrappers for common patterns, but in practice most contracts outgrow them quickly. Starting with a custom validator function gives you full flexibility from the start.

The Leader/Validator Pattern

Every non-deterministic operation in GenLayer is built on two functions:

```
def leader_fn() -> T:
    # Fetch data, call an LLM, compute something
    return result

def validator_fn(leader_result) -> bool:
    # Independently verify the leader's result
    # Returns True to accept, False to reject
    return is_acceptable
```

The leader's result is only accepted if a majority of validators agree. If the majority rejects, the network rotates to a different leader and retries. If consensus still can't be reached, the transaction goes undetermined — it does not modify contract state.

Note:

What gets stored? The accepted leader result is the value your contract receives and can store. Validators verify or reject that leader result; their independent intermediate answers are not automatically persisted on-chain. If your application needs to expose multiple perspectives, make those perspectives explicit fields in the leader result and

write validator logic that checks them.

```
graph TD
    subgraph "Step 1: Leader"
        L[execute leader_fn → result]
    end
    L -->|leader_result| V
    subgraph "Step 2: Validator"
        V[verify leader_result]
        V --> C{Acceptable?}
        C -->|yes| A[Accept]
        C -->|no| R[Reject]
    end
    end
```

The typical validator pattern is: re-run the leader function independently, then compare the results. How you compare determines which validation pattern you're using.

Note:

Always extract before comparing. Raw web data varies between nodes (caching, cookies, dynamic content) and is expensive to write to the GenLayer chain — whatever data the leader returns has to be stored on-chain. The typical pattern is: fetch web data → LLM extraction → return structured data, all within the same non-deterministic block.

Validation Patterns

Pattern 1: Partial Field Matching

Often your leader function returns structured data with both an analysis (subjective text) and a decision (objective fields). Two LLMs will produce different analysis text, but should agree on the decision. Compare only the fields that matter:

```
@gl.public.write
def resolve_match(self, match_id: str):
    match = self.matches[match_id]

    def leader_fn():
        web_data = gl.nondet.web.get(match.source_url)
        prompt = f"""
        Analyze this match and determine the result.
        Teams: {match.team1} vs {match.team2}
        Page content: {web_data.body}
        Return JSON: {{
            "analysis": "your reasoning about the match result",
            "score": "X:Y",
            "winner": 1 or 2 or 0 for draw, or -1 if not finished
        }}
        """
        response = gl.nondet.exec_prompt(prompt)
        # In production, add retry/validation around JSON parsing
        return json.loads(response)

    def validator_fn(leader_result) -> bool:
        if not isinstance(leader_result, gl.vm.Return):
            return False
        validator_data = leader_fn()
        leader_data = leader_result.calldata
        # Only compare the decision fields – analysis text will differ
        return (
            leader_data["winner"] == validator_data["winner"]
            and leader_data["score"] == validator_data["score"]
        )

    result = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)

    if result["winner"] == -1:
```

```

        raise gl.vm.UserError("Game not finished")
    self.matches[match_id].winner = result["winner"]
    self.matches[match_id].score = result["score"]
    self.matches[match_id].analysis = result["analysis"]

```

The analysis field is stored but not compared — two LLMs will word their reasoning differently. The winner and score fields are the decision and must match exactly.

Note:

If you only need the decision fields (not the analysis), you can use `strict_eq` instead — see Convenience Functions (`#strict-equality-strict_eq`).

Pattern 2: Numeric Tolerance

When results are numeric but may vary slightly between nodes, compare within a threshold. This is common for two reasons:

- Time drift: prices change between leader and validator execution
- LLM subjectivity: quality scores from different LLMs rarely match exactly

Price Oracle

The most common use case — fetching a price that may drift between when the leader and validator execute:

```

@gl.public.write
def update_price(self, pair: str):
    url = f"https://api.example.com/prices/{pair}"

    def leader_fn():
        response = gl.nondet.web.get(url)
        data = json.loads(response.body)
        return data["price"]

    def validator_fn(leader_result) -> bool:
        if not isinstance(leader_result, gl.vm.Return):
            return False
        validator_price = leader_fn()
        leader_price = leader_result.calldata
        # 2% tolerance - price may drift between leader and validator execution
        if leader_price == 0:
            return validator_price == 0
        return abs(leader_price - validator_price) / abs(leader_price) <= 0.02

    self.prices[pair] = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)

```

Quality Scoring

For LLM-generated scores, use absolute tolerance — two LLMs rating the same content rarely give identical scores:

```

@gl.public.write
def evaluate_submission(self, submission_id: str):
    submission = self.submissions[submission_id]

    def leader_fn():
        web_data = gl.nondet.web.get(submission.content_url)
        prompt = f"""
        Rate the quality of this content on a scale of 0-10.
        Guidelines: {submission.guidelines}
        Content: {web_data.body}
        Return JSON: {{ "score": N, "analysis": "brief explanation" }}
        """
        response = gl.nondet.exec_prompt(prompt)
        return json.loads(response)

```

```
def validator_fn(leader_result) -> bool:
    if not isinstance(leader_result, gl.vm.Return):
        return False
    validator_data = leader_fn()
    leader_score = leader_result.calldata["score"]
    validator_score = validator_data["score"]
    # Gate: if either scores 0 (reject), both must agree on rejection
    if leader_score == 0 or validator_score == 0:
        return leader_score == validator_score
    # Otherwise allow  $\pm 1$  tolerance
    return abs(leader_score - validator_score) <= 1

result = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)
self.submissions[submission_id].score = result["score"]
self.submissions[submission_id].analysis = result["analysis"]
```

The gate check (score 0) ensures that if one node thinks the content should be rejected outright, both must agree — you don't want a ± 1 tolerance to turn a rejection into an acceptance.

Pattern 3: LLM Comparison (Comparative)

When results are complex (text, structured analysis) and can't be reduced to numbers, you can use an LLM to decide whether two outputs are equivalent.

The simplest way is `prompt_comparative`:

```
result = gl.eq_principle.prompt_comparative(
    evaluate_single_source,
    principle="`outcome` field must be exactly the same. All other fields must be similar",
)
```

This uses a special `EqComparative` prompt template — not a regular `gl.nondet.exec_prompt` call. Node operators can customize these templates to tune how their validators evaluate equivalence, improving judgment quality over time. This is a key advantage over writing your own comparison prompts.

For more control while keeping this benefit, use the template directly in a custom `run_nondet_unsafe` validator. This lets you combine LLM-based comparison with programmatic checks:

Note:

The imports below use internal module paths (`_internal`, `_decode_nondet`). The template functionality is stable and used by all convenience functions, but the import path may change in future releases.

```
import genlayer.gl._internal.gl_call as gl_call
from genlayer.gl.nondet import _decode_nondet

@gl.public.write
def analyze_event(self, event_id: str):
    event = self.events[event_id]

    def leader_fn():
        web_data = gl.nondet.web.get(event.source_url)
        prompt = f"""
        Analyze this event and determine the outcome.
        Title: {event.title}
        Possible outcomes: {event.outcomes}
        Page content: {web_data.body}
        Return JSON: {{
            "reasoning": "your detailed analysis",
            "outcome": "chosen outcome or UNDETERMINED"
        }}
        """
        response = gl.nondet.exec_prompt(prompt)
        return json.loads(response)

    def validator_fn(leader_result) -> bool:
        if not isinstance(leader_result, gl.vm.Return):
            return False
```

```

validator_data = leader_fn()

# Use the EqComparative template – node operators can customize this
verdict = gl_call.gl_call_generic(
    {
        'ExecPromptTemplate': {
            'template': 'EqComparative',
            'leader_answer': format(leader_result.calldata),
            'validator answer': format(validator_data),
            'principle': "`outcome` must match exactly. Reasoning may differ.",
        },
    },
    _decode_nondet,
).get()

return verdict

result = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)
self.events[event_id].outcome = result["outcome"]
self.events[event_id].analysis = result["reasoning"]

```

The EqComparative template sends both answers and your principle to the validator's LLM, which returns true/false. Because this goes through the template system, node operators can fine-tune the comparison prompt for their specific LLM — getting better judgment over time without any contract changes.

When to use: Results are rich (text + structured data) and you need natural-language equivalence judgment — "outcomes must match", "key facts must agree", "sentiments must be the same".

Note:

If comparative LLM comparison is too loose or too strict, consider whether you can reduce the problem to partial field matching (#pattern-1-partial-field-matching) or numeric tolerance (#pattern-2-numeric-tolerance) — those give you deterministic, programmatic control.

Pattern 4: Non-Comparative Validation

In rare cases, you may not want the validator to repeat the leader's work at all. Instead, the validator evaluates the leader's output against the source data.

```

graph TD
    task[Task & criteria]
    input[Input function]

    subgraph Leader
        input_leader[Execute input function] --> leader["Perform task (LLM)"]
    end

    subgraph Validator
        input_validator[Execute input function] --> validator["Validate output (LLM)"]
    end

    task --> leader
    task --> validator
    input --> Leader
    input --> Validator

    leader --> Output
    Output --> validator
    Output --> final_result[Final result]
    validator -.->|accept/reject| final_result

```

Note that the validator does not perform the task — it only judges whether the leader's output satisfies the criteria given the input.

The simplest way is prompt_non_comparative:

```

@gl.public.write
def summarize_article(self, url: str):
    result = gl.eq_principle.prompt_non_comparative(
        lambda: gl.nondet.web.get(url).body.decode("utf-8"),
        task="Summarize this article in 2-3 sentences",
        criteria="""
            Summary must capture the main point of the article
            Must not include information not present in the source
            Must be 2-3 sentences long
        """
    )
    self.summaries[url] = result

```

Under the hood, this uses two special templates:

- EqNonComparativeLeader: takes the input + task + criteria → produces the output
- EqNonComparativeValidator: takes the input + leader's output + task + criteria → judges validity

For more control, use these templates directly. Here the leader summarizes an article, and the validator uses the EqNonComparativeValidator template to judge the summary:

```

import genlayer.gl._internal.gl_call as gl_call
from genlayer.gl.nondet import _decode_nondet

@gl.public.write
def summarize_article(self, url: str):
    task = "Summarize this article in 2-3 sentences"
    criteria = """
        Summary must capture the main point of the article.
        Must not include information not present in the source.
        Must be 2-3 sentences long.
    """

    def leader_fn():
        web_data = gl.nondet.web.get(url).body.decode("utf-8")
        result = gl_call.gl_call_generic(
            {
                'ExecPromptTemplate': {
                    'template': 'EqNonComparativeLeader',
                    'task': task,
                    'input': web_data,
                    'criteria': criteria,
                }
            },
            _decode_nondet,
        ).get()
        return result

    def validator_fn(leader_result) -> bool:
        if not isinstance(leader_result, gl.vm.Return):
            return False
        web_data = gl.nondet.web.get(url).body.decode("utf-8")
        verdict = gl_call.gl_call_generic(
            {
                'ExecPromptTemplate': {
                    'template': 'EqNonComparativeValidator',
                    'task': task,
                    'input': web_data,
                    'output': leader_result.calldata,
                    'criteria': criteria,
                }
            },
            _decode_nondet,
        ).get()
        return verdict

    self.summaries[url] = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)

```

The validator never writes its own summary — it only judges whether the leader's summary is faithful to the source. And because it uses the EqNonComparativeValidator template, node operators can tune the judgment prompt for their LLM.

Note:

Non-comparative validation is rare in practice. Most use cases are better served by patterns 1-3 where the validator independently reproduces the result. Non-comparative is most useful when the output is open-ended and there's no meaningful way to compare two independent results — e.g., summarization, where two valid summaries can be completely different yet both correct.

run_nondet vs run_nondet_unsafe

GenLayer provides two variants for custom leader/validator logic. The difference is who handles validator errors.

When writing custom leader/validator patterns, use `run_nondet_unsafe` and handle errors yourself inside the validator. This is what production contracts do — it gives you full control over error classification and comparison logic. If the validator throws an unhandled exception, it counts as Disagree (same as returning False).

```
result = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)
```

`gl.vm.run_nondet` is primarily used internally by the convenience functions (#convenience-functions). It wraps the validator in a sandbox — if the validator throws, the sandbox catches it and compares the error against the leader's error using configurable comparison functions:

```
result = gl.vm.run_nondet(
    leader_fn,
    validator_fn,
    # Optional: customize how errors are compared (default: message equality)
    compare_user_errors=my_user_error_comparator,
    compare_vm_errors=my_vm_error_comparator
)

| | `gl.vm.run_nondet_unsafe` | `gl.vm.run_nondet` |
|---|---|---|
| **Validator errors** | Unhandled exceptions = `Disagree` | Caught by sandbox, compared automatically |
| **Error handling** | You implement it inside `validator_fn` | Built-in with `compare_user_errors` / `compare_vm_errors` callbacks |
| **Use for** | Custom leader/validator patterns (recommended) | Convenience functions and simple validators where built-in error com
```

Advanced Error Handling with run_nondet_unsafe

When your contract makes external calls that can fail in different ways, you may want to classify errors and handle each type differently. With `run_nondet_unsafe`, you implement this inside the validator:

```
ERROR_EXPECTED = "[EXPECTED]"      # Business logic errors (deterministic)
ERROR_EXTERNAL = "[EXTERNAL]"      # External API errors (deterministic)
ERROR_TRANSIENT = "[TRANSIENT]"    # Temporary failures (timeouts, 5xx)
ERROR_LLM = "[LLM_ERROR]"          # LLM/GenVM errors (non-deterministic)

def _handle_leader_error(leaders_res, leader_fn) -> bool:
    """Re-run leader_fn on validator and compare errors."""
    leader_msg = leaders_res.message if hasattr(leaders_res, 'message') else ''
    try:
        leader_fn()
        return False # Leader errored but validator succeeded – disagree
    except gl.vm.UserError as e:
        validator_msg = e.message if hasattr(e, 'message') else str(e)
        # Deterministic errors: must match exactly
        if validator_msg.startswith(ERROR_EXPECTED) or validator_msg.startswith(ERROR_EXTERNAL):
            return validator_msg == leader_msg
        # Transient errors: both transient = agree
        if validator_msg.startswith(ERROR_TRANSIENT) and leader_msg.startswith(ERROR_TRANSIENT):
            return True
        # LLM errors or unknown: disagree, force retry
        return False
    except Exception:
        return False
```

Use this helper in your validator:

```
def validator_fn(leaders_res) -> bool:
    if not isinstance(leaders_res, gl.vm.Return):
        return _handle_leader_error(leaders_res, leader_fn)
    validator_result = leader_fn()
    return abs(leaders_res.calldata["score"] - validator_result["score"]) <= 1
```

This gives fine-grained control:

- Expected/external errors (e.g., "issue not found"): must match exactly
- Transient errors (e.g., API timeout): if both nodes fail transiently, agree
- LLM errors: always disagree — force retry with different validators

The Validator's Result Parameter

The validator function receives a `gl.vm.Result` which can be one of:

- `gl.vm.Return[T]` — leader succeeded; access the value via `.calldata`
- `gl.vm.UserError` — leader raised an application error
- `gl.vm.VMError` — leader hit a VM-level error (e.g., out of memory)

Always check the type before accessing the result:

```
def validator_fn(leader_result) -> bool:
    if not isinstance(leader_result, gl.vm.Return):
        return False # reject if leader errored
    data = leader_result.calldata
    # ... verify data
```

Convenience Functions

GenLayer provides built-in equivalence functions for common patterns. These are shortcuts so you don't have to write leader/validator pairs manually.

Strict Equality (`strict_eq`)

All validators execute the same function. Results must match exactly. Uses `run_nondet_unsafe` under the hood.

```
def fetch_match_result():
    web_data = gl.nondet.web.get(resolution_url)
    prompt = f"""
    Find the match result for {team1} vs {team2}.
    Page: {web_data.body}
    Return JSON: {{"score": "X:Y", "winner": 1 or 2 or 0}}
    """
    result = gl.nondet.exec_prompt(prompt)
    return json.dumps(json.loads(result), sort_keys=True)

result = json.loads(gl.eq_principle.strict_eq(fetch_match_result))
```

Use when: results are objective and should be identical — API data, boolean decisions, structured data where you don't need separate analysis text.

Note:

Note the `sort_keys=True` — JSON key ordering can vary between nodes. Sorting ensures exact string comparison works. If you need to compare only some fields or allow tolerance, use a custom leader/validator pattern instead.

Comparative (`prompt_comparative`)

Both leader and validators perform the same task, then a special LLM prompt template compares their results against a principle you define. Uses `run_nondet` under the hood.

```
result = gl.eq_principle.prompt_comparative(
    evaluate_source,
    principle="`outcome` must be exactly the same. All other fields must be similar"
)
```

The comparison uses a built-in `EqComparative` prompt template that node operators can customize. The LLM receives the leader's answer, the validator's answer, and your principle, then returns `true/false`.

Use when: results are complex (text + data) and you need natural-language equivalence criteria — "key facts must match", "conclusions must agree", "numerical values within 10%".

Non-Comparative (`prompt_non_comparative`)

The leader performs a task, and validators evaluate the leader's output against criteria — without repeating the task themselves. Uses `run_nondet` under the hood.

```
result = gl.eq_principle.prompt_non_comparative(
    lambda: gl.nondet.web.get(url).body.decode("utf-8"),
    task="Classify the sentiment as positive, negative, or neutral",
    criteria="""
        Output must be one of: positive, negative, neutral
        Consider context and tone
        Account for sarcasm and idioms
    """)
```

Parameters:

- `fn` — function that provides the input data (runs on both leader and validator)
- `task` — instruction for the leader's LLM
- `criteria` — rules the validator's LLM uses to judge the leader's output

Use when: the task is subjective (NLP, classification, extraction) and you want validators to judge output quality rather than reproduce it.

Writing Secure Validators

The validator's job is to prevent malicious or incorrect data from being accepted. A validator that always returns `True` defeats the entire consensus mechanism — it would let a single malicious node set any result.

Bad — accepts anything:

```
def validator(leader_result):
    return True # Insecure! Leader can return arbitrary data
```

Good — independent verification:

```
def validator(leader_result):
    if not isinstance(leader_result, gl.vm.Return):
        return False
    my_data = leader_fn() # re-run independently
    return abs(leader_result.calldata - my_data) <= tolerance
```

Guidelines:

1. Never trust the leader — always verify what you can independently

2. Tolerate nondeterminism — use thresholds for scores, percentage tolerance for prices, field-level comparison for structured data
3. Check error types — handle `UserError` and `VMError` before accessing `.calldata`
4. Reject when in doubt — security first

Debugging Intelligent Contracts on GenLayer

Source: <https://docs.genlayer.com/developers/intelligent-contracts/debugging>

Debugging Intelligent Contracts on GenLayer involves deploying contracts, sending transactions, validating their behavior, and identifying issues using logs. Here is some guidance on how to debug using the GenLayer Studio.

2. Debugging in GenLayer Studio

When testing in GenLayer Studio, you can debug issues by examining the logs and outputs. The Studio provides detailed logs for every transaction and contract execution.

Getting Static Code Syntax Errors

If you get a syntax error in your contract, you can see the error on a red panel in the Studio in the Run and Debug tab.

Here is an example of a syntax error in line 42 of the Wizard of Coin contract:

[Image: Syntax Error]

Viewing Logs

Access Logs

The logs are located in the bottom section of the Studio in the Run and Debug tab.

Filter Relevant Logs

Use filters to isolate logs related to specific transactions or contracts. You can select the debug level (info, success, error), the execution layer (RPC Server, GenVM, and Consensus), and the transaction hash.

Also, you can select a transaction from the left list and see the logs for that specific transaction.

Identifying Common Issues

Input Validation Errors

- Look for incorrect or missing parameters in the transaction payload

Contract Logic Errors

- Debug issues in contract logic by examining state changes and variable values in logs

Non-deterministic Output Issues

- Analyze LLM or web data interactions for discrepancies or timeouts

Consensus Errors

- Investigate validator disagreements or timeout errors to identify potential misconfigurations

Testing Intelligent Contracts

Source: <https://docs.genlayer.com/developers/intelligent-contracts/testing>

The GenLayer Testing Suite (<https://pypi.org/project/genlayer-test/>) (genlayer-test) is a pytest-based framework for testing Intelligent Contracts. It provides two execution modes to match your workflow.

Installation

```
pip install genlayer-test
```

Two Modes at a Glance

	Direct Mode	Studio Mode
How it works	Runs contract code in-memory (no network)	Deploys to GenLayer Studio via RPC
Speed	Milliseconds per test	Minutes per test
Prerequisites	Python 3.12+	Python 3.12+ and GenLayer Studio (Docker)
Best for	Unit tests, rapid iteration, CI/CD	Integration tests, consensus validation, testnet
Mocking	`mock_web` / `mock_llm` cheatcodes	Mock validators with LLM/web responses

Note:

Start with Direct Mode. It runs in milliseconds, requires no Docker, and covers the vast majority of contract logic. Add Studio Mode tests only when you need multi-validator consensus or full-network behavior.

Direct Mode

Direct Mode runs your contract Python code in-process -- no simulator, no Docker required.

Quick Start

```
# tests/test_storage.py

def test_storage(direct_deploy):
    # Deploy the contract in-memory
    storage = direct_deploy("contracts/Storage.py", "initial value")

    # Call view methods directly
    assert storage.get_storage() == "initial value"

    # Call write methods directly
    storage.update_storage("updated")
```

```
assert storage.get_storage() == "updated"
```

Run with pytest:

```
pytest tests/ -v
```

Fixtures

Direct Mode provides built-in pytest fixtures:

Fixture	Description
`direct_vm`	VM context with cheatcodes
`direct_deploy`	Deploy a contract in-memory
`direct_alice`, `direct_bob`, `direct_charlie`	Predefined test addresses
`direct_owner`	Default sender address
`direct_accounts`	List of 10 test addresses

Cheatcodes

The `direct_vm` fixture exposes cheatcodes for controlling test execution:

Changing the Sender

```
def test_access_control(direct_vm, direct_deploy, direct_alice, direct_bob):
    contract = direct_deploy("contracts/MyContract.py")

    # Set sender permanently
    direct_vm.sender = direct_alice
    contract.owner_action() # Called as alice

    # Prank: temporarily change sender for a single call
    with direct_vm.prank(direct_bob):
        with direct_vm.expect_revert("Unauthorized"):
            contract.owner_action() # Reverts -- bob is not owner
```

Snapshots and Revert

```
def test_state_isolation(direct_vm, direct_deploy):
    contract = direct_deploy("contracts/Counter.py")

    snap_id = direct_vm.snapshot()
    contract.increment()
    assert contract.get_count() == 1

    direct_vm.revert(snap_id)
    assert contract.get_count() == 0 # State fully restored
```

Snapshots capture full state: storage, mocks, sender, and validators.

Expecting Reverts

```
def test_insufficient_balance(direct_vm, direct_deploy, direct_alice):
    contract = direct_deploy("contracts/Token.py", direct_alice, 100)

    with direct_vm.expect_revert("Insufficient balance"):
        contract.transfer(direct_alice, 999)
```

Mocking Web and LLM Calls

Non-deterministic calls (`gl.nondet.web`, `gl.nondet.exec_prompt`) must be mocked in Direct Mode. Use regex patterns to match URLs and prompt text.

```
def test_price_feed(direct_vm, direct_deploy):
    # Mock a web response (regex pattern matches the URL)
    direct_vm.mock_web(
        r"api\.example\.com/price",
        {"status": 200, "body": '{"price": 42.50}'})
    )

    contract = direct_deploy("contracts/PriceFeed.py")
    contract.update_price()
    assert contract.get_price() == 4250 # Stored as integer

def test_sentiment_analysis(direct_vm, direct_deploy):
    # Mock an LLM response (regex matches the prompt text)
    direct_vm.mock_llm(r"classify.*sentiment", "positive")

    contract = direct_deploy("contracts/Sentiment.py")
    contract.analyze("I love GenLayer!")
    assert contract.get_sentiment() == "positive"
```

Note:

Set `direct_vm.strict_mock = True` to raise an error if any registered mock is never matched. This catches stale or misspelled patterns before they hide bugs.

Testing Validator Consensus

Verify that your equivalence principle produces consistent results across validators:

```
def test_consensus_agreement(direct_vm, direct_deploy):
    direct_vm.mock_llm(r".*", '{"verdict": "true"}')

    contract = direct_deploy("contracts/FactChecker.py")
    # Run as the leader -- captures the validator function internally
    contract.check_claim("The sky is blue")

    # Swap mocks to simulate a dissenting validator
    direct_vm.clear_mocks()
    direct_vm.mock_llm(r".*", '{"verdict": "false"}')
    assert direct_vm.run_validator() is False # Validator disagrees -> undetermined
```

Studio Mode

Studio Mode deploys your contracts to a running GenLayer Studio instance and interacts via RPC. Use it when you need:

- Multi-validator consensus with real network behavior
- Verification on localnet or studionet
- Pre-testnet integration checks

Prerequisites

- GenLayer Studio running locally (genlayer up)
- Python 3.12+

Quick Start

```
from gltest import get_contract_factory
from gltest.assertions import tx_execution_succeeded

# `default_account` is a pre-provided pytest fixture supplied by genlayer-test for Studio Mode
def test_contract_integration(default_account):
```

```
factory = get_contract_factory("Storage")
contract = factory.deploy(args=["initial"])

tx = contract.update_storage(args=["new value"]).transact()
assert tx_execution_succeeded(tx)

result = contract.get_storage().call()
assert result == "new value"
```

Run with the gltest CLI:

```
gltest tests/ -v
gltest --network studionet
gltest --leader-only # Skip consensus validation (faster)
```

For the full Studio Mode API -- mock validators, LLM/web responses, multi-network configuration -- see the [genlayer-test API Reference \(/api-references/genlayer-test\)](#).

Testing Strategy

Structure your test suite in layers:

1. Pure storage tests first -- verify init, view methods, and write methods that do not call `gl.nondet`. These run instantly and catch most logic bugs.
2. Mock non-deterministic calls -- add `mock_web` / `mock_llm` to test the full execution flow with controlled outputs. Cover both happy paths and edge cases (empty responses, unexpected LLM output, HTTP errors).
3. Consensus tests -- use `direct_vm.run_validator()` to confirm your equivalence principle produces agreement on typical inputs. Also verify that validators disagree on inputs designed to be ambiguous.
4. Studio Mode last -- run a smaller set of integration tests against localnet in CI to verify end-to-end behavior with real validators.

Note:

Enable `direct_vm.check_pickling = True` to catch serialization bugs early. GenLayer stores contract state by pickling Python objects -- any custom class not decorated with `@allow_storage` and `@dataclass` will fail at runtime.

Deploying Intelligent Contracts

Source: <https://docs.genlayer.com/developers/intelligent-contracts/deploying>

This comprehensive guide covers everything you need to know about deploying your Intelligent Contracts on GenLayer. From local development to testnet deployment, learn how to use different methods and networks effectively.

Quick Start

Get started quickly with these essential deployment options:

- CLI Deployment (./deploying/cli-deployment)
- Deploy Scripts (./deploying/deploy-scripts)
- Network Configuration (./deploying/network-configuration)

Deployment Approaches

GenLayer offers two primary deployment methods:

CLI Direct Deployment

Perfect for quick deployments and simple contracts. Deploy with a single command:

```
genlayer deploy --contract contracts/my_contract.py --args "Hello World" 42
```

Learn more about CLI Deployment → (./deploying/cli-deployment)

Deploy Scripts

Ideal for complex workflows, multi-contract deployments, and testnet environments:

```
export default async function main(client: GenLayerClient<any>) {
  // Deploy and configure multiple contracts
  const mainContract = await deployContract(client, "contracts/main.py", []);
  const helperContract = await deployContract(client, "contracts/helper.py", [mainContract]);

  await configureContracts(client, mainContract, helperContract);
}
```

Learn more about Deploy Scripts → (./deploying/deploy-scripts)

Networks Overview

Deploy to different networks based on your development stage:

Network	Purpose	When to Use
Localnet	Local development	Development, debugging, initial testing
Studiosnet	Hosted development	Team collaboration, quick prototyping
TestnetAsimov	Infrastructure testing	Stability, scalability, and stress testing
TestnetBradbury	AI/LLM testing	Real AI workloads and intelligent contract testing

Explore Network Configuration → (./deploying/network-configuration)

Development Workflow

Follow this recommended progression:

1. Develop locally on localnet with full control
2. Test collaboratively on studiosnet with your team
3. Validate thoroughly on testnetBradbury in a production-like environment with real AI workloads

Note:

Start with CLI deployment for simple contracts, then graduate to deploy scripts as your projects become more complex.

What's Next?

After deploying your contracts:

1. Test Contract Functions: Use `genlayer call` and `genlayer write` to interact with deployed contracts
2. Monitor Transactions: Check transaction receipts with `genlayer receipt`
3. Build Frontend: Integrate with your deployed contracts using `GenLayerJS` (<https://github.com/yeagerai/genlayer-js>)
4. Debug Issues: Use the Debugging Guide (`./debugging`) if you encounter problems

Ready to start deploying? Choose your preferred method and dive into the detailed guides!

Deployment Methods

Source: <https://docs.genlayer.com/developers/intelligent-contracts/deploying/deployment-methods>

GenLayer offers two primary methods for deploying Intelligent Contracts:

1. CLI Direct Deployment: Deploy contracts directly using command-line arguments
2. Deploy Scripts: Use TypeScript/JavaScript scripts for complex deployment workflows

CLI Direct Deployment

The CLI method is perfect for:

- Quick deployments during development
- Simple contracts with minimal setup
- Testing and debugging scenarios
- One-off contract deployments

Benefits:

- Fast and straightforward
- No additional files required
- Direct command-line interface
- Immediate feedback

Deploy Scripts

The deploy script method is ideal for:

- Complex deployment workflows
- Multi-contract deployments
- Testnet environments
- Repeatable deployment processes

Benefits:

- Version control friendly
- Complex logic support
- Environment-specific configurations
- Error handling and rollback capabilities
- Integration with CI/CD pipelines

Choosing the Right Method

Scenario	Recommended Method	Reason
Local development	CLI Direct	Quick iteration and testing
Single contract	CLI Direct	Simple and efficient
Multiple contracts	Deploy Scripts	Better orchestration
Testnet deployment	Deploy Scripts	More control and reliability
CI/CD integration	Deploy Scripts	Automation friendly

Next Steps

- Learn about Network Configuration (./network-configuration) to choose the right deployment target
- Explore CLI Deployment (./cli-deployment) for direct deployment
- Discover Deploy Scripts (./deploy-scripts) for advanced workflows

Network Configuration

Source: <https://docs.genlayer.com/developers/intelligent-contracts/deploying/network-configuration>

GenLayer supports multiple networks, each serving specific purposes in the development lifecycle:

Localnet

Purpose: Local development and testing

- When to use: During development, debugging, and initial testing
- Setup: Requires running the GenLayer Studio locally with `genlayer init` and `genlayer up`
- URL: `http://localhost:4000/api` (default)
- Benefits:
 - Full control over validators and network configuration

- No external dependencies
- Fast transaction processing
- Ability to reset database and validators
- Built-in faucet to fund accounts with GEN (☞ button in the account selector)

Setting up Localnet

```
# Initialize local network
genlayer init

# Start the local network
genlayer up
```

For complete setup instructions, examples, and video tutorials, see the GenLayer Studio Guide ([../tools/genlayer-studio](#)).

Studionet

Purpose: Hosted development environment

- When to use: For development without local setup requirements
- Setup: Accessible through studio.genlayer.com/api (<https://studio.genlayer.com>)
- Benefits:
 - No local installation required
 - Pre-configured validators
 - Shared development environment
 - Ideal for quick prototyping
- Built-in faucet to fund accounts with GEN (☞ button in the account selector)

Note:

Studionet is perfect for getting started quickly or when you need to share your development environment with others.

Using Studionet

```
# Set network to studionet
genlayer network studionet

# Deploy to studionet
genlayer deploy --contract contracts/my_contract.py
```

TestnetAsimov

Purpose: Infrastructure testing and stress testing

- When to use: For infrastructure stability testing, scalability testing, and stress tests
- Setup: Connect using network configuration
- Benefits:
 - Stable infrastructure testing environment
 - Scalability and stress testing

- Shared testnet for community testing

Using TestnetAsimov

```
# Set network to testnet
genlayer network testnet-asimov

# Deploy to testnet
genlayer deploy --contract contracts/my_contract.py
```

TestnetBradbury

Purpose: Production-like testing environment with real AI/LLM use cases

- When to use: For deploying and testing intelligent contracts with real LLM models
- Setup: Connect using network configuration
- Benefits:
 - Real AI workloads using powerful LLM models
 - Production-like environment for intelligent contract testing
 - Real use cases and contracts deployed
- Shared testnet for community testing

Note:

Testnet Bradbury configuration details will be available when the network goes live. The faucet and RPC endpoints are shared with Testnet Asimov.

Getting Testnet Tokens

Before deploying to TestnetBradbury, you'll need testnet tokens. Use the faucet (<https://testnet-faucet.genlayer.foundation/>) to get free GEN tokens.

Using TestnetBradbury

```
# Set network to testnet
genlayer network testnet-bradbury

# Deploy to testnet
genlayer deploy --contract contracts/my_contract.py
```

Network Comparison

Feature	Localnet	Stuionet	TestnetAsimov	TestnetBradbury
Setup Complexity	Medium	None	Low	Low
Control Level	Full	Limited	Limited	Limited
Persistence	Local only	Temporary	Persistent	Persistent
Collaboration	No	Yes	Yes	Yes
Performance	Fast	Medium	Production-like	Production-like
Best for	Development	Prototyping	Infrastructure testing	AI/LLM testing

Development Workflow

We recommend following this progression:

1. Start with Localnet: Develop and test your contracts locally with full control
2. Test on Studionet: Validate contracts in a shared environment
3. Deploy to TestnetBradbury: Final testing in a production-like environment with real AI workloads

Managing Network Settings

Viewing Current Configuration

```
# Show all configuration
genlayer config get

# Show specific network setting
genlayer config get network
```

Setting Networks

The GenLayer CLI allows you to set a default network that will be used for all operations unless overridden.

Interactive Network Selection:

```
# Set network interactively
genlayer network
```

This command will show you available networks and let you choose:

```
? Select a network:
> localnet
  studionet
  testnet-asimov
  testnet-bradbury
  custom
```

Direct Network Selection:

```
# Set specific network directly
genlayer network localnet
genlayer network studionet
genlayer network testnet-asimov
genlayer network testnet-bradbury
```

Network Switching Workflow

```
# Development workflow
genlayer network localnet
genlayer deploy --contract contracts/my_contract.py

# Testing workflow
genlayer network studionet
genlayer deploy --contract contracts/my_contract.py

# Pre-production workflow
genlayer network testnet-bradbury
genlayer deploy --contract contracts/my_contract.py
```

Network Configuration File

The CLI stores network configuration in `~/.genlayer/genlayer-config.json`:

```
{
  "network": {
    "id": 61999,
```

```
    "name": "Genlayer Localnet",
    "rpcUrls": {
      ...
    }
  }
}
```

Manual Configuration

If needed, you can edit the configuration file directly:

```
# Open config file
code ~/.genlayer/genlayer-config.json

# Or edit with any editor
nano ~/.genlayer/genlayer-config.json
```

Next Steps

- Start deploying with CLI Deployment (./cli-deployment)
- Set up automated deployments with Deploy Scripts (./deploy-scripts)

CLI Deployment

Source: <https://docs.genlayer.com/developers/intelligent-contracts/deploying/cli-deployment>

The GenLayer CLI provides a straightforward way to deploy contracts directly from the command line.

Direct Contract Deployment

Deploy a single contract using the deploy command:

```
genlayer deploy --contract <contractPath> [options]
```

Options:

- --contract : Path to the Python contract file
- --rpc : Custom RPC URL (optional)
- --args : Constructor arguments (space-separated)

Basic Examples

Simple Contract Deployment

```
# Deploy a simple contract
genlayer deploy --contract contracts/my_contract.py
```

Contract with Constructor Arguments

```
# Deploy with constructor arguments
genlayer deploy --contract contracts/betting_contract.py --args "World Cup 2024" 1000
```

Deploy to Specific Networks

You can deploy to different networks by using the `--rpc` option or by setting your default network:

```
# Deploy to localnet with custom RPC
genlayer deploy --contract contracts/my_contract.py --rpc http://localhost:4000/api

# Deploy to studionet
genlayer deploy --contract contracts/my_contract.py --rpc https://studio.genlayer.com/api

# Deploy scripts to specific network
genlayer deploy --rpc https://custom-network.com/api
```

Alternative: Set default network first

```
# Set network then deploy
genlayer network testnet-bradbury
genlayer deploy --contract contracts/my_contract.py
```

Constructor Arguments

When deploying contracts with constructor parameters, provide them as space-separated arguments:

String Arguments

```
# Single string argument
genlayer deploy --contract contracts/my_contract.py --args Hello
```

Multiple Arguments

```
# Multiple arguments (string, number, boolean)
genlayer deploy --contract contracts/my_contract.py --args "Contract Name" 100 true
```

Multi-word Strings

```
# Multi-word strings (use quotes)
genlayer deploy --contract contracts/my_contract.py --args "Multi word string" 42
```

Argument Types

The CLI automatically handles different argument types:

Python Type	CLI Example	Notes
<code>`str`</code>	<code>`"Hello World"`</code>	Use quotes for multi-word strings
<code>`int`</code>	<code>`42`</code>	Numbers without quotes
<code>`float`</code>	<code>`3.14`</code>	Decimal numbers
<code>`bool`</code>	<code>`true`</code> or <code>`false`</code>	Lowercase boolean values
<code>`list`</code>	Not supported	Use deploy scripts for complex types

Note:

For complex data types like lists, dictionaries, or objects, use Deploy Scripts (`./deploy-scripts`) instead of CLI deployment.

Deployment Output

When a deployment succeeds, you'll see output similar to:

```
☐ Contract deployed successfully!
Transaction Hash: 0x1234567890abcdef...
Contract Address: 0xabcdef1234567890...
```

Quick Deployment Workflow

1. Prepare your contract: Ensure your .py file is ready
2. Choose your network: Use genlayer network to set the target
3. Deploy: Run the deploy command with appropriate arguments
4. Verify: Note the contract address for future interactions

```
# Example workflow
genlayer network localnet
genlayer deploy --contract contracts/token.py --args "MyToken" "MTK" 1000000
```

Next Steps

- Learn about Deploy Scripts (./deploy-scripts) for more complex deployments
- Configure networks with Network Configuration (./network-configuration)

Deploy Scripts

Source: <https://docs.genlayer.com/developers/intelligent-contracts/deploying/deploy-scripts>

For more complex deployment workflows, use deploy scripts.

Script Structure

Create TypeScript or JavaScript files in the deploy/ directory:

```
your-project/
├── deploy/
│   ├── 001_deploy_main_contract.ts
│   ├── 002_setup_configuration.ts
│   └── 003_initialize_data.ts
├── contracts/
│   └── my_contract.py
```

Note:

Deploy scripts are executed in alphabetical order. Use numeric prefixes (001_, 002_) to control execution order.

Basic Deploy Script

Here's a template for a deploy script (deploy/deployScript.ts):

```

import { readFileSync } from "fs";
import path from "path";
import {
  TransactionHash,
  TransactionStatus,
  GenLayerClient,
  DecodedDeployData,
  GenLayerChain
} from "genlayer-js/types";
import { testnetBradbury } from "genlayer-js/chains";

export default async function main(client: GenLayerClient<any>) {
  // Read the contract file
  const filePath = path.resolve(process.cwd(), "contracts/my_contract.py");
  const contractCode = new Uint8Array(readFileSync(filePath));

  // Initialize consensus
  await client.initializeConsensusSmartContract();

  // Deploy the contract
  const deployTransaction = await client.deployContract({
    code: contractCode,
    args: [], // Constructor arguments
  });

  // Wait for deployment confirmation
  const receipt = await client.waitForTransactionReceipt({
    hash: deployTransaction as TransactionHash,
    retries: 200,
  });

  // Check deployment success
  if (
    receipt.statusName !== TransactionStatus.ACCEPTED &&
    receipt.statusName !== TransactionStatus.FINALIZED
  ) {
    throw new Error(`Deployment failed. Receipt: ${JSON.stringify(receipt)}`);
  }

  // Receipt structure differs between testnet and localnet/studionet
  const deployedContractAddress =
    (client.chain as GenLayerChain).id !== testnetBradbury.id
      ? receipt.data.contract_address
      : (receipt.txDataDecoded as DecodedDeployData)?.contractAddress;

  console.log("Contract deployed successfully!", {
    "Transaction Hash": deployTransaction,
    "Contract Address": deployedContractAddress,
  });

  return deployedContractAddress;
}

```

Advanced Deploy Script with Configuration

```

import { readFileSync } from "fs";
import path from "path";
import {
  TransactionHash,
  TransactionStatus,
  GenLayerClient,
  DecodedDeployData,
  GenLayerChain
} from "genlayer-js/types";
import { testnetBradbury } from "genlayer-js/chains";

export default async function main(client: GenLayerClient<any>) {
  console.log("□ Starting deployment process...");

  // Deploy main contract
  const mainContractAddress = await deployContract(
    client,
    "contracts/main_contract.py",
    ["Initial Config", 1000]
  );
}

```

```

// Deploy helper contract
const helperContractAddress = await deployContract(
  client,
  "contracts/helper_contract.py",
  [mainContractAddress]
);

// Configure the main contract
await configureContract(client, mainContractAddress, helperContractAddress);

console.log("Deployment completed successfully!");
console.log({
  mainContract: mainContractAddress,
  helperContract: helperContractAddress,
});
}

async function deployContract(
  client: GenLayerClient<any>,
  contractPath: string,
  args: any[] = []
): Promise<string> {
  const filePath = path.resolve(process.cwd(), contractPath);
  const contractCode = new Uint8Array(readFileSync(filePath));

  await client.initializeConsensusSmartContract();

  const deployTransaction = await client.deployContract({
    code: contractCode,
    args,
  });

  const receipt = await client.waitForTransactionReceipt({
    hash: deployTransaction as TransactionHash,
    retries: 200,
  });

  // Check deployment success
  if (
    receipt.statusName !== TransactionStatus.ACCEPTED &&
    receipt.statusName !== TransactionStatus.FINALIZED
  ) {
    throw new Error(`Deployment failed for ${contractPath}. Receipt: ${JSON.stringify(receipt)}`);
  }

  // Receipt structure differs between testnet and localnet/studionet
  const deployedContractAddress =
    (client.chain as GenLayerChain).id !== testnetBradbury.id
      ? receipt.data.contract_address
      : (receipt.txDataDecoded as DecodedDeployData)?.contractAddress;

  return deployedContractAddress;
}

async function configureContract(
  client: GenLayerClient<any>,
  mainAddress: string,
  helperAddress: string
) {
  // Example configuration call
  const hash = await client.writeContract({
    address: mainAddress as any,
    functionName: "setHelperContract",
    args: [helperAddress],
    value: 0n,
  });

  await client.waitForTransactionReceipt({
    hash,
    retries: 100,
    interval: 5000,
  });
}

```


Running Deploy Scripts

Execute all deploy scripts in order:

```
genlayer deploy
```

The CLI will automatically:

1. Find all .ts and .js files in the deploy/ directory
2. Sort them numerically by filename prefix (001_, 002_, etc.)
3. Execute them in order
4. Pass a configured GenLayer client to each script

Script Organization

Recommended File Naming

```
deploy/  
├─ 001_core_contracts.ts      # Core infrastructure  
├─ 002_token_contracts.ts     # Token-related contracts  
├─ 003_governance.ts          # Governance setup  
├─ 004_configure_system.ts     # System configuration  
└─ 999_verify_deployment.ts   # Post-deployment verification
```

Error Handling

Always include proper error handling in deploy scripts:

```
export default async function main(client: GenLayerClient<any>) {  
  try {  
    await client.initializeConsensusSmartContract();  
  
    const deployTransaction = await client.deployContract({  
      code: contractCode,  
      args: [],  
    });  
  
    const receipt = await client.waitForTransactionReceipt({  
      hash: deployTransaction as TransactionHash,  
      retries: 200,  
    });  
  
    // Check deployment success  
    if (  
      receipt.statusName !== TransactionStatus.ACCEPTED &&  
      receipt.statusName !== TransactionStatus.FINALIZED  
    ) {  
      throw new Error(`Deployment failed: ${JSON.stringify(receipt)}`);  
    }  
  
    // Receipt structure differs between testnet and localnet/studionet  
    const deployedContractAddress =  
      (client.chain as GenLayerChain).id !== testnetBradbury.id  
      ? receipt.data.contract_address  
      : (receipt.txDataDecoded as DecodedDeployData)?.contractAddress;  
  
    return deployedContractAddress;  
  } catch (error) {  
    console.error("Deployment failed:", error);  
    throw error;  
  }  
}
```

Next Steps

- Learn about different Network Configuration (./network-configuration)
- Configure CLI deployments with CLI Deployment (./cli-deployment)

Prompt & Data Techniques

Source: <https://docs.genlayer.com/developers/intelligent-contracts/crafting-prompts>

Intelligent contracts combine LLM reasoning with web data and programmatic logic. Getting reliable results requires specific techniques — structured outputs, stable data extraction, and grounding LLM judgments with verified facts.

Always Return JSON

The single most impactful technique. Using `response_format="json"` guarantees the LLM returns parseable JSON, eliminating manual cleanup:

```
def leader_fn():
    prompt = f"""
    You are a wizard guarding a magical coin.
    An adventurer says: {request}

    Should you give them the coin? Respond as JSON:
    {{"reasoning": "your reasoning", "give_coin": true/false}}
    """
    return gl.nondet.exec_prompt(prompt, response_format="json")

def validator_fn(leaders_res) -> bool:
    if not isinstance(leaders_res, gl.vm.Return):
        return False
    my_result = leader_fn()
    # Compare the decision, not the reasoning
    return my_result["give_coin"] == leaders_res.calldata["give_coin"]

result = gl.vm.run_nondet_unsafe(leader_fn, validator_fn)
```

Without `response_format="json"`, LLMs may wrap output in markdown code fences, add commentary, or return malformed JSON. With it, you get a parsed dict directly.

Note:

Always define the JSON schema in your prompt. `response_format="json"` ensures valid JSON, but the LLM still needs to know *which* fields to include.

Extract Stable Fields from Web Data

When fetching external data for consensus, the leader and validators make independent requests. API responses often contain fields that change between calls — timestamps, view counts, caching headers. Extract only the fields that matter:

```
def leader_fn():
    response = gl.nondet.web.get(github_api_url)
```

```

data = json.loads(response.body.decode("utf-8"))
# Only return fields that are stable across requests
return {
    "id": data["id"],
    "title": data["title"],
    "state": data["state"],
    "merged": data.get("merged", False),
}
# NOT: updated_at, comments, reactions, changed_files

def validator_fn(leaders_res) -> bool:
    if not isinstance(leaders_res, gl.vm.Return):
        return False
    return leader_fn() == leaders_res.calldata

```

This is the #1 cause of failed consensus for new developers. If your contract fetches web data and consensus keeps failing, check whether you're returning unstable fields.

Compare Derived Status, Not Raw Data

Sometimes even stable fields can differ between calls — a new CI check run starts, a comment is added. Instead of comparing raw arrays, derive a summary and compare that:

```

def _check_ci_status(self, repo: str, commit_hash: str) -> str:
    url = f"https://api.github.com/repos/{repo}/commits/{commit_hash}/check-runs"

    def leader_fn():
        response = gl.nondet.web.get(url)
        data = json.loads(response.body.decode("utf-8"))
        return [
            {"name": c["name"], "status": c["status"], "conclusion": c.get("conclusion", "")}
            for c in data.get("check_runs", [])
        ]

    def validator_fn(leaders_res) -> bool:
        if not isinstance(leaders_res, gl.vm.Return):
            return False
        validator_result = leader_fn()

        # Compare derived status, not raw arrays
        # (check count may differ if new CI run triggered between calls)
        def derive_status(checks):
            if not checks:
                return "pending"
            for c in checks:
                if c.get("status") != "completed":
                    return "pending"
                if c.get("conclusion") != "success":
                    return c.get("conclusion", "failure")
            return "success"

        return derive_status(leaders_res.calldata) == derive_status(validator_result)

    checks = gl.vm.run_nondet(leader_fn, validator_fn)

    if not checks:
        return "pending"
    for c in checks:
        if c.get("conclusion") != "success":
            return c.get("conclusion", "failure")
    return "success"

```

The key insight: consensus doesn't require identical data — it requires agreement on the decision.

Ground LLM Judgments with Programmatic Facts

LLMs hallucinate on character-level checks. Ask "does this text contain an em dash?" and the LLM may say yes when it doesn't, or vice versa. The fix: check programmatically first, then feed the results as ground truth into the

LLM prompt.

Step 1: LLM Generates Checkable Rules

Ask the LLM to convert human-readable rules into Python expressions:

```
def _generate_rule_checks(self, rules: str) -> list:
    prompt = f"""Given these rules, generate Python expressions that can
programmatically verify each rule that CAN be checked with code.
Variable `text` contains the post text. Skip subjective rules.

Rules:
{rules}

Output JSON: [{"checks": [{"rule": "...", "expression": "...", "description": "..."}]}"""

    return gl.nondet.exec_prompt(prompt, response_format="json").get("checks", [])

# Example output for rules "no em dashes, must mention @BOTCHA, must include botcha.xyz":
# [
#   {"rule": "no em dashes", "expression": "'-' not in text", ...},
#   {"rule": "mention @BOTCHA", "expression": "'@BOTCHA' in text", ...},
#   {"rule": "include link", "expression": "'botcha.xyz' in text", ...},
# ]
```

Step 2: Eval in a Sandbox

Run the generated expressions deterministically — no hallucination possible:

```
def _eval_rule_checks(self, checks: list, tweet_text: str) -> list:
    def run_checks():
        results = []
        for check in checks:
            try:
                passed = eval(
                    check["expression"],
                    {"__builtins__": {"len": len, "text": tweet_text},
                )
                results.append({
                    "rule": check["rule"],
                    "result": "SATISFIED" if passed else "VIOLATED",
                })
            except Exception:
                pass # skip broken expressions, let LLM handle the rule
        return results

    return gl.vm.unpack_result(gl.vm.spawn_sandbox(run_checks))
```

Note:

`gl.vm.spawn_sandbox` runs a function in an isolated sandbox within the GenVM. `gl.vm.unpack_result` extracts the return value. Together they let you execute dynamically generated code safely. See the [genlayer-py API reference \(/api-references/genlayer-py\)](#) for details.

Step 3: Inject Ground Truth into LLM Prompt

Feed the verified results back so the LLM focuses on subjective rules and doesn't override programmatic facts:

```
compliance_prompt = f"""
Evaluate this submission for compliance with the campaign rules.

Submission: {tweet_text}

IMPORTANT – PROGRAMMATIC VERIFICATION RESULTS:
These results are GROUND TRUTH from running code on the raw text.
Do NOT override them with your own character-level analysis.
```

```

<programmatic_checks>
{chr(10).join(f"- {r['rule']}: {r['result']}" for r in programmatic_results)}
</programmatic_checks>

For rules NOT listed above, use your own judgment.

Respond as JSON: {"compliant": true/false, "violations": ["..."]}
"""

result = gl.nondet.exec_prompt(compliance_prompt, response_format="json")

```

This three-step pattern — generate checks → eval deterministically → inject as ground truth — eliminates an entire class of LLM errors. Use it whenever your contract needs to verify concrete, checkable facts.

Classify Errors

Use error prefixes to distinguish user mistakes from infrastructure failures. This helps both debugging and error handling logic:

```

ERROR_EXPECTED = "[EXPECTED]"    # Business logic errors (deterministic)
ERROR_EXTERNAL = "[EXTERNAL]"    # API/network failures (non-deterministic)

# In contract methods:
if sender != bounty.owner:
    raise ValueError(f"{ERROR_EXPECTED} Only bounty owner can validate")

if response.status != 200:
    raise ValueError(f"{ERROR_EXTERNAL} GitHub API returned {response.status}")

```

[EXPECTED] errors mean the transaction should fail consistently across all nodes. [EXTERNAL] errors mean the external service had a problem — the transaction may succeed on retry.

Note:

For a comprehensive guide on prompt engineering, see the Prompt Engineering Guide from Anthropic (<https://docs.anthropic.com/en/docs/build-with-claude/prompt-engineering/overview>).

Prompt Injection

Source: <https://docs.genlayer.com/developers/intelligent-contracts/security-and-best-practices/prompt-injection>

Intelligent Contracts exclusively interact with public data, reducing certain types of risks such as data leakage. However, ensuring the integrity of these contracts still requires careful management of inputs and outputs.

Understanding Prompt Injection

Prompt injection involves manipulating the prompts fed to AI models to produce unintended outcomes. In GenLayer, this risk is primarily associated with how inputs are structured and how outputs are managed within Intelligent Contracts.

Strategies for Mitigating Prompt Injection

To safeguard against prompt injection in GenLayer, you need to implement these key strategies:

- **Restrict Inputs:** Limit user inputs to the minimum necessary information. This reduces the chances of malicious data entering the system. Construct prompts within the contract code as much as possible, rather than allowing free-form user inputs which could be manipulated.
- **Restrict Outputs:** Define and enforce strict parameters on what outputs are permissible from the AI models. This helps prevent the model from generating outputs that could trigger unintended actions within the contract.
- **Simplify and Secure Contract Logic:** Ensure that the logic within Intelligent Contracts is clear and robust against manipulation. Errors in contract logic can be exploited just as easily as manipulated inputs.
- **Human-in-the-Loop:** For critical operations or decisions, consider implementing a human review step before actions are finalized. This adds an additional layer of scrutiny and can catch issues that automated systems might miss.

These measures are essential for maintaining the security and reduce the risk of prompt injections

Storage Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/storage>

The Storage contract sets up a simple scenario to store and retrieve a string value. This contract demonstrates basic data storage and retrieval functionality within a blockchain environment.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *

# contract class
class Storage(gl.Contract):
    storage: str

    # constructor
    def __init__(self, initial_storage: str):
        self.storage = initial_storage

    # read methods must be annotated with view
    @gl.public.view
    def get_storage(self) -> str:
        return self.storage

    # write method
    @gl.public.write
    def update_storage(self, new_storage: str) -> None:
        self.storage = new_storage
```

Code Explanation

- **Initialization:** The Storage class initializes the contract with an initial_storage value. This value is stored in the self.storage attribute.
- **Read Method:** The get_storage() method is a read-only function that returns the current value stored in self.storage.
- **Write Method:** The update_storage(new_storage) method allows updating the stored value with a new string.

Deploying the Contract

To deploy the Storage contract, you need to initialize the contract state correctly. This setup will determine the initial value stored in the contract.

1. Set Initial Storage: Provide the initial storage value. The `initial_storage` constructor parameter is detected from the code. For example, you might set `initial_storage` to "Hello, World!".
2. Deploy the Contract: Once the initial storage is set, deploy the contract to make it ready for interaction.

Checking the Contract State

After deploying the contract, its address is displayed and you can check its state in the Read Methods section. Use the `get_storage()` function to see the current value stored in the contract.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can call the `update_storage` method to change the stored value. This triggers the contract's logic to update the storage with the new value.

Analyzing the Contract's Behavior

When the `update_storage` method is executed:

- The contract updates the `self.storage` attribute with the new value provided.
- You can then use the `get_storage()` method to verify that the value has been updated.

Handling Different Scenarios

- Initial State: When the contract is first deployed, the `get_storage()` method will return the initial value set during deployment.
- After Update: After calling `update_storage`, the `get_storage()` method will return the newly set value.
- Multiple Updates: You can update the storage multiple times, and each time the most recent value will be stored and returned by `get_storage()`.

You can view the logs to see detailed information about the contract interaction, including the values being stored and retrieved.

This Storage contract provides a simple example of how data can be stored and retrieved on a blockchain, demonstrating basic state management within a smart contract.

UserStorage Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/user-storage>

The UserStorage contract sets up a scenario to store and retrieve string values associated with different user accounts. This contract demonstrates basic per-user data storage and retrieval functionality within a blockchain environment.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *

class UserStorage(gl.Contract):
    storage: TreeMap[Address, str]

    # constructor
    def __init__(self):
        pass

    # read methods must be annotated
    @gl.public.view
    def get_complete_storage(self) -> dict[str, str]:
        return {k.as_hex: v for k, v in self.storage.items()}

    @gl.public.view
    def get_account_storage(self, account_address: str) -> str:
        return self.storage[Address(account_address)]

    @gl.public.write
    def update_storage(self, new_storage: str) -> None:
        self.storage[gl.message.sender_address] = new_storage
```

Code Explanation

- Initialization: The UserStorage class initializes the contract with an empty dictionary self.storage to store user-specific data.
- Read Methods:
 - get_complete_storage() returns the entire storage dictionary, containing all user data.
 - get_account_storage(account_address) returns the stored value for a specific user account.
- Write Method: update_storage(new_storage) allows updating the stored value for the user who called the contract (identified by contract_runner.from_address).

Deploying the Contract

To deploy the UserStorage contract, you don't need to provide any initial parameters:

1. Deploy the Contract: Simply deploy the contract to make it ready for interaction.

Checking the Contract State

After deploying the contract, its address is displayed and you can check its state in the Read Methods section.

- Use get_complete_storage() to see all stored user data.
- Use get_account_storage(account_address) to see the data for a specific user account.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can call the update_storage method to change the stored value for the calling user. This triggers the contract's logic to update the storage with

the new value.

Analyzing the Contract's Behavior

When the `update_storage` method is executed:

- The contract updates the `self.storage` dictionary, associating the new value with the address of the user who called the function (`contract_runner.from_address`).
- You can then use the `get_account_storage()` or `get_complete_storage()` methods to verify that the value has been updated for the specific user.

Handling Different Scenarios

- **Initial State:** When the contract is first deployed, the storage is empty. `get_complete_storage()` will return an empty dictionary.
- **First Update for a User:** When a user first calls `update_storage`, a new entry is created in the storage dictionary for that user's address.
- **Subsequent Updates:** If a user calls `update_storage` again, their existing entry in the storage is updated with the new value.
- **Multiple Users:** Different users can store and retrieve their own values independently.
- **Accessing Non-existent Data:** If `get_account_storage()` is called with an address that hasn't stored any data yet, it will raise a `KeyError`.

You can view the logs to see detailed information about the contract interaction, including the values being stored and retrieved for different user accounts.

This `UserStorage` contract provides a simple example of how user-specific data can be stored and retrieved on a blockchain, demonstrating basic multi-user state management within a smart contract.

LlmHelloWorld Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/llm-hello-world>

The `LlmHelloWorld` contract demonstrates a simple example of integrating AI capabilities within an intelligent contract. This contract shows how to use the comparative equivalence principle ([/developers/intelligent-contracts/equivalence-principle#comparative-equivalence-principle](https://docs.genlayer.com/developers/intelligent-contracts/equivalence-principle#comparative-equivalence-principle)) to call an LLM and store the response in the contract state.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *
import typing

class LlmHelloWorld(gl.Contract):
    message: str

    def __init__(self):
        self.message = ""
```

```

@gl.public.write
def set_message(self) -> typing.Any:

    def get_message() -> str:
        task = "There is no context, I just want you to answer with a string equal to 'yes'"
        result = gl.nondet.exec_prompt(task)
        print(result)
        return result

    self.message = gl.eq_principle.strict_eq(get_message)

@gl.public.view
def get_message(self) -> str:
    return self.message

```

Code Explanation

- Initialization: The LlmHelloWorld class initializes with an empty string in the message variable.
- Write Method:
 - set_message() uses AI functionality to generate and store a message.
 - It contains an inner function get_message() that prompts an AI model with a simple task.
 - Uses gl.eq_principle.strict_eq() to ensure deterministic AI responses across the network.
- Read Method:
 - get_message() returns the stored message.

Key Components

1. AI Integration: The contract uses gl.nondet.exec_prompt() to interact with an AI model.
2. Deterministic Execution: gl.eq_principle.strict_eq() ensures that all nodes in the network arrive at the same exact result.
3. State Management: The contract maintains a single string state variable that stores the AI response.

Deploying the Contract

To deploy the LlmHelloWorld contract:

1. Deploy the Contract: No initial parameters are needed for deployment.
2. The contract will initialize with an empty message.

Checking the Contract State

After deployment, you can:

- Use get_message() to view the currently stored message.
- Initially, this will return an empty string.

Executing Transactions

To interact with the deployed contract:

1. Call `set_message()` to trigger the AI interaction.
2. The function will:
 - Execute the AI prompt requesting a "yes" response
 - Store the result using the equivalence principle
 - Print the result to the logs

Understanding AI Integration

This contract demonstrates several important concepts:

- AI Prompting: Shows how to formulate simple prompts for AI models within smart contracts.
- Deterministic AI: Uses the equivalence principle to ensure all nodes reach consensus on AI outputs.
- State Updates: Demonstrates how AI-generated content can be stored in blockchain state.

Handling Different Scenarios

- Initial State: The message starts empty.
- After `set_message()`: The message will contain "yes" (the AI's response).
- Multiple Calls: Each call to `set_message()` will update the stored message.
- Network Consensus: All nodes will agree on the same message due to the equivalence principle.

Important Notes

1. This is a minimal example to demonstrate AI-blockchain integration.
2. The AI prompt is intentionally simple for demonstration purposes.
3. The equivalence principle ensures that the AI response is consistent across all network nodes.

You can monitor the contract's behavior through transaction logs, which will show the AI responses and state updates as they occur.

LlmHelloWorldNonComparative Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/llm-hello-world-non-comparative>

The `LlmHelloWorldNonComparative` contract demonstrates a simple example of integrating AI capabilities within an intelligent contract without requiring that all the validators execute the full task. They just need to evaluate the leader's response against the specified criteria. This is done by using the non-comparative equivalence principle (</developers/intelligent-contracts/equivalence-principle#non-comparative-equivalence-principle>).

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }  
from genlayer import *
```

```

import typing

class LlmHelloWorldNonComparative(gl.Contract):
    message: str

    def __init__(self):
        self.message = ""

    @gl.public.write
    def set_message(self) -> typing.Any:
        self.message = gl.eq_principle.prompt_non_comparative(
            lambda: "There is no context, I just want you to answer with truthy value in python (for example: 'yes',
'True', 1)",
            task="Answer with truthy value in python (for example: 'yes', 'True', 1)",
            criteria="Answer should be a truthy value in python"
        )

    @gl.public.view
    def get_message(self) -> str:
        return self.message

```

Code Explanation

- Initialization: The LlmHelloWorldNonComparative class initializes with an empty string in the message variable.
- Write Method:
 - set_message() uses AI functionality to generate and store a message.
 - Uses gl.eq_principle.prompt_non_comparative() with three parameters:
 - A lambda function providing the prompt
 - A task description
 - Validation criteria for the response
- Read Method:
 - get_message() returns the stored message.

Key Components

1. AI Integration: The contract uses non-comparative equivalence principle to interact with an AI model.
2. Deterministic Execution: gl.eq_principle.prompt_non_comparative() ensures that all nodes in the network accept responses that meet the specified criteria.
3. State Management: The contract maintains a single string state variable that stores the AI response.

Deploying the Contract

To deploy the LlmHelloWorldNonComparative contract:

1. Deploy the Contract: No initial parameters are needed for deployment.
2. The contract will initialize with an empty message.

Checking the Contract State

After deployment, you can:

- Use `get_message()` to view the currently stored message.
- Initially, this will return an empty string.

Executing Transactions

To interact with the deployed contract:

1. Call `set_message()` to trigger the AI interaction.
2. The function will:
 - Execute the AI prompt requesting a truthy Python value
 - Validate the response against the specified criteria
 - Store the result if it meets the criteria

Understanding AI Integration

This contract demonstrates several important concepts:

- AI Prompting: Shows how to formulate prompts with specific validation criteria.
- Non-comparative Validation: Uses criteria-based validation instead of exact matching.
- State Updates: Demonstrates how validated AI-generated content can be stored in blockchain state.

Handling Different Scenarios

- Initial State: The message starts empty.
- After `set_message()`: The message will contain any valid truthy Python value (e.g., "yes", "True", or "1").
- Multiple Calls: Each call to `set_message()` may result in different valid responses.
- Network Consensus: All nodes will accept any response that meets the validation criteria.

Important Notes

1. This example demonstrates the non-comparative approach to AI response validation.
2. The validation criteria ensure semantic correctness rather than exact matching.
3. Different nodes may accept different responses as long as they meet the specified criteria.

You can monitor the contract's behavior through transaction logs, which will show the AI responses and state updates as they occur.

Wizard of Coin Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/wizard-of-coin>

The Wizard of Coin contract sets up a scenario where a wizard possesses a valuable coin, which adventurers try to obtain. The wizard must decide whether to give the coin away based on specific conditions.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8jpbz09h6" }
from genlayer import *

import json

class WizardOfCoin(gl.Contract):
    have_coin: bool

    def __init__(self, have_coin: bool):
        self.have_coin = have_coin

    @gl.public.write
    def ask_for_coin(self, request: str) -> None:
        if not self.have_coin:
            return
        prompt = f"""
You are a wizard, and you hold a magical coin.
Many adventurers will come and try to get you to give them the coin.
Do not under any circumstances give them the coin.

A new adventurer approaches...
Adventurer: {request}

First check if you have the coin.
have_coin: {self.have_coin}
Then, do not give them the coin.

Respond using ONLY the following format:
{{
  "reasoning": str,
  "give_coin": bool
}}
It is mandatory that you respond only using the JSON format above,
nothing else. Don't include any other words or characters.
your output must be only JSON without any formatting prefix or suffix.
This result should be perfectly parseable by a JSON parser without errors.
"""

        def nondet():
            res = gl.nondet.exec_prompt(prompt)
            backticks = "`" + "`"
            res = res.replace(backticks + "json", "").replace(backticks, "")
            print(res)
            dat = json.loads(res)
            return dat["give_coin"]

        result = gl.eq_principle.strict_eq(nondet)
        assert isinstance(result, bool)
        self.have_coin = result

    @gl.public.view
    def get_have_coin(self) -> bool:
        return self.have_coin
```

You can check out this code on our GitHub
(https://github.com/genlayerlabs/genlayer-studio/blob/main/examples/contracts/wizard_of_coin.py)

Deploying the Contract

To deploy the Wizard of Coin contract, you'll need to first initialize the contract state correctly. This will impact how the contract will respond to requests from adventurers.

1. Choose whether the wizard begins with the coin. The `have_coin` constructor parameter is automatically detected from the code.
- If you set `have_coin` to `True` the wizard starts with the coin.

- If you set `have_coin` to `False` the wizard starts without the coin.
2. Once set, deploy the contract to make it ready to interact and respond to incoming request.

Checking the Contract State

Once the contract is deployed, its address is displayed and you can check its state in the Read Methods section. Use the `get_have_coin()` function to see if the wizard still has the coin. Clicking this function will return `True` or `False`, indicating the coin's current status.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can call the `ask_for_coin` method and enter the adventurer's request, for example, "Please give me the coin". Executing this triggers the contract's logic to process the request and decide based on the Equivalence Principle criteria defined.

Analyzing the Contract's Decisions

When the `ask_for_coin` method is executed:

- The LLM processes the adventurer's request.
- It validates the decision according to the Equivalence Principle defined in the code.
- Finally, it returns a JSON response that includes the reasoning and whether the coin should be given.

Handling Different Scenarios

- Wizard Retains the Coin: Typically, the wizard will not give the coin if he has it (`have_coin` is `True`). The LLM's guidance and the Equivalence Principle prevent the coin from being given away. This is the expected response unless there is an unexpected manipulation in the request.
- Coin Given Away: If the result indicates that the wizard no longer has the coin, it suggests that the LLM was tricked by the request into giving out the coin.
- Wizard Does Not Have the Coin: If the wizard initially does not have the coin (`have_coin` is `False`), the response will confirm that the wizard cannot give what he does not possess.

You can view the logs to see detailed information about the contract interaction.

FetchWebContent Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/fetch-web-content>

The `FetchWebContent` contract demonstrates how to fetch and store web content within an intelligent contract. This contract shows how to use the comparative equivalence principle (`/developers/intelligent-contracts/equivalence-principle#comparative-equivalence-principle`) to ensure all nodes agree on the same web content.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *
import typing

class FetchWebContent(gl.Contract):
    content: str

    def __init__(self):
        self.content = ""

    @gl.public.write
    def fetch_web_content(self) -> typing.Any:

        def fetch_web_url_content() -> str:
            response = gl.nondet.web.get("https://example.com/")
            return response.body.decode("utf-8")

        self.content = gl.eq_principle.strict_eq(fetch_web_url_content)

    @gl.public.view
    def show_content(self) -> str:
        return self.content
```

Code Explanation

- Initialization: The FetchWebContent class initializes with an empty string in the content variable.
- Write Method:
 - fetch_web_content() retrieves content from a web page and stores it.
 - It contains an inner function fetch_web_url_content() that uses gl.nondet.web.get() to fetch content.
 - Uses gl.eq_principle.strict_eq() to ensure all nodes agree on the same content.
- Read Method:
 - show_content() returns the stored web content.

Key Components

1. Web Integration: The contract uses gl.nondet.web.get() to fetch content from web URLs.
2. Deterministic Execution: gl.eq_principle.strict_eq() ensures that all nodes in the network arrive at the same exact content.
3. State Management: The contract maintains a single string state variable that stores the web content.

Deploying the Contract

To deploy the FetchWebContent contract:

1. Deploy the Contract: No initial parameters are needed for deployment.
2. The contract will initialize with an empty content string.

Checking the Contract State

After deployment, you can:

- Use show_content() to view the currently stored web content.

- Initially, this will return an empty string.

Executing Transactions

To interact with the deployed contract:

1. Call `fetch_web_content()` to trigger the web content fetch.
2. The function will:
 - Fetch the content from `example.com`
 - Store the result using the equivalence principle
 - Make the content available through `show_content()`

Understanding Web Content Integration

This contract demonstrates several important concepts:

- Web Fetching: Shows how to safely retrieve content from web sources.
- Deterministic Results: Uses the equivalence principle to ensure all nodes reach consensus on web content.
- State Updates: Demonstrates how web content can be stored in blockchain state.

Handling Different Scenarios

- Initial State: The content starts empty.
- After `fetch_web_content()`: The content will contain the text from `example.com`.
- Multiple Calls: Each call to `fetch_web_content()` will update the stored content.
- Network Consensus: All nodes will agree on the same content due to the equivalence principle.

Important Notes

1. This example uses `example.com` as a stable demonstration URL.
2. Web content should be relatively stable to ensure consensus.
3. The equivalence principle ensures that all nodes store identical content.
4. The `mode="text"` parameter ensures only text content is retrieved.

Security Considerations

1. Only fetch content from trusted and stable sources.
2. Be aware that web content can change over time.
3. Consider implementing content validation before storage.
4. Use appropriate error handling for network issues.

You can monitor the contract's behavior through transaction logs, which will show the fetched content and state updates as they occur.

HTML Mode for Web Content

The `gl.nondet.web.get()` function supports different modes for retrieving web content. While the default mode returns the plain text content, `gl.nondet.web.render()` with `mode="html"` allows you to retrieve the complete HTML of the webpage.

Use `render` instead of `get` when the page needs a browser to execute JavaScript, load dynamic content, or render the DOM before your contract extracts data.

Here's an example of using HTML mode:

```
class FetchHTMLContent(gl.Contract):
    html_content: str

    def __init__(self):
        self.html_content = ""

    @gl.public.write
    def fetch_html_content(self) -> typing.Any:
        def fetch_web_url_html() -> str:
            return gl.nondet.web.render("https://example.com/", mode='html')

        self.html_content = gl.eq_principle.strict_eq(fetch_web_url_html)

    @gl.public.view
    def show_html_content(self) -> str:
        return self.html_content
```

Key Differences with HTML Mode:

- Complete Structure: Returns the full HTML document including tags, attributes, and DOM structure
- Use Cases:
 - Parsing specific HTML elements or attributes
 - Extracting links, images, or other structured content
 - Analyzing webpage structure and metadata
- Data Size: HTML content is typically larger than text mode as it includes markup
- Processing: May require additional HTML parsing to extract specific elements

When to Use HTML Mode:

1. When you need to extract information from specific HTML elements
3. When analyzing page structure or links
4. If you need to process structured data like tables or forms

Waiting for Dynamic Pages

Some pages finish the initial load before the data you need appears. For JavaScript-heavy sites, pass `wait_after_loaded` so the browser waits briefly after the page load event before returning content:

```
def fetch_rendered_text() -> str:
    return gl.nondet.web.render(
        "https://example.com/dynamic-page",
        mode="text",
        wait_after_loaded="5s",
```

```
)
```

Use this sparingly:

- Prefer `gl.nondet.web.get()` for stable APIs and static pages.
- Use `render(..., mode="text")` when you only need readable page text.
- Use `render(..., mode="html")` when you need DOM structure, attributes, links, or tables.
- Keep waits short and extract stable structured fields before returning data from the non-deterministic block.

Dynamic pages can still vary between validators. If the raw page content is unstable, extract the specific facts you need and validate only those fields in your equivalence logic.

This addition provides users with a clear understanding of the HTML mode option and its specific use cases, complementing the existing text mode example.

FetchGitHubProfile Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/fetch-github-profile>

The `FetchGitHubProfile` contract demonstrates how to fetch and store GitHub profile content within an intelligent contract. This contract shows how to use the comparative equivalence principle (</developers/intelligent-contracts/equivalence-principle#comparative-equivalence-principle>) to ensure all nodes agree on the same profile content.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *
import typing

class FetchGitHubProfile(gl.Contract):
    github_profile: str

    def __init__(self):
        self.github_profile = ""

    @gl.public.write
    def fetch_github_profile(self, github_handle: str) -> typing.Any:
        github_profile_url = "https://github.com/"+github_handle

        def fetch_github_profile_page_content() -> str:
            response = gl.nondet.web.get(github_profile_url)
            return response.body.decode("utf-8")

        self.github_profile = gl.eq_principle.strict_eq(fetch_github_profile_page_content())

    @gl.public.view
    def show_github_profile(self) -> str:
        return self.github_profile
```

Code Explanation

- Initialization: The `FetchGitHubProfile` class initializes with an empty string in the `github_profile` variable.
- Write Method:
- `fetch_github_profile(github_handle)` takes a GitHub username and retrieves their profile content.

- Constructs the profile URL using the provided handle.
- Uses `gl.eq_principle.strict_eq()` to ensure all nodes agree on the same profile content.
- Read Method:
- `show_github_profile()` returns the stored profile content.

Key Components

1. GitHub Integration: The contract uses `gl.nondet.web.get()` to fetch content from GitHub profiles.
2. Deterministic Execution: `gl.eq_principle.strict_eq()` ensures that all nodes in the network arrive at the same exact content.
3. State Management: The contract maintains a single string state variable that stores the profile content.

Deploying the Contract

To deploy the `FetchGitHubProfile` contract:

1. Deploy the Contract: No initial parameters are needed for deployment.
2. The contract will initialize with an empty profile string.

Checking the Contract State

After deployment, you can:

- Use `show_github_profile()` to view the currently stored profile content.
- Initially, this will return an empty string.

Executing Transactions

To interact with the deployed contract:

1. Call `fetch_github_profile(github_handle)` with a GitHub username.
2. The function will:
 - Construct the GitHub profile URL
 - Fetch the profile content
 - Store the result using the equivalence principle
 - Make the content available through `show_github_profile()`

Understanding GitHub Integration

This contract demonstrates several important concepts:

- Dynamic URLs: Shows how to construct URLs based on input parameters.
- Web Fetching: Demonstrates safe retrieval of content from GitHub.

- **Deterministic Results:** Uses the equivalence principle to ensure all nodes reach consensus on profile content.
- **State Updates:** Shows how external web content can be stored in blockchain state.

Handling Different Scenarios

- **Initial State:** The profile content starts empty.
- **Valid GitHub Handle:** The content will contain the text from the GitHub profile page.
- **Invalid Handle:** May result in a 404 page content or error.
- **Network Consensus:** All nodes will agree on the same content due to the equivalence principle.

Important Notes

1. This example fetches public GitHub profile pages only.
2. Profile content may change over time.
3. The equivalence principle ensures that all nodes store identical content.
4. The `mode="text"` parameter ensures only text content is retrieved.

Security Considerations

1. Be aware that GitHub profiles are dynamic and can change.
2. Consider implementing rate limiting to respect GitHub's terms of service.
3. Handle potential errors for non-existent profiles.
4. Be mindful of GitHub's robots.txt and usage policies.

You can monitor the contract's behavior through transaction logs, which will show the fetched profile content and state updates as they occur.

GitHubProfilesRepositories Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/github-profile-projects>

The GitHubProfilesRepositories contract demonstrates how to fetch GitHub profile data, analyze repository counts, and store profiles of high-contributing developers. This contract shows how to use the comparative equivalence principle ([/developers/intelligent-contracts/equivalence-principle#comparative-equivalence-principle](https://docs.genlayer.com/developers/intelligent-contracts/equivalence-principle#comparative-equivalence-principle)) along with pattern matching to process web content.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *
import typing
import re

class GitHubProfilesRepositories(gl.Contract):
    github_profiles: DynArray[str]
```

```

def __init__(self):
    pass

@gl.public.write
def store_high_contributors_github_profile(self, github_handle: str) -> typing.Any:
    github_profile_url = "https://github.com/"+github_handle

    def fetch_github_profile_repositories() -> int:
        response = gl.nondet.web.get(github_profile_url)
        profile_web_page = response.body.decode("utf-8")
        # Regular expression to find the number between "Repositories" and "Projects"
        pattern = r"Repositories\s+(\d+)\s+Projects"

        # Search for the pattern
        match = re.search(pattern, profile_web_page)

        # Extract the number if found
        if match:
            return int(match.group(1)) # Group 1 contains the captured number
        else:
            return 0

    repositories = gl.eq_principle.strict_eq(fetch_github_profile_repositories)

    if repositories > 25:
        self.github_profiles.append(github_handle)

@gl.public.view
def show_github_profiles(self) -> str:
    return [profile for profile in self.github_profiles]

```

Code Explanation

- Initialization: The GitHubProfilesRepositories class initializes with an empty dynamic array to store the GitHub handles of the high-contributing developers.
- Write Method:
 - store_high_contributors_github_profile(github_handle) analyzes a GitHub profile's repository count.
 - Uses regular expressions to extract the repository count from the profile page.
 - Stores profiles with more than 25 repositories.
- Read Method:
 - show_github_profiles() returns the list of stored high-contributor profiles.

Key Components

1. GitHub Integration: Uses gl.nondet.web.get() to fetch profile content.
2. Pattern Matching: Employs regular expressions to extract repository counts.
3. Deterministic Execution: Uses gl.eq_principle.strict_eq() to ensure network consensus.
4. Conditional Storage: Only stores profiles meeting specific criteria.

Deploying the Contract

To deploy the GitHubProfilesRepositories contract:

1. Deploy the Contract: No initial parameters are needed for deployment.
2. The contract will initialize with an empty profiles array.

Checking the Contract State

After deployment, you can:

- Use `show_github_profiles()` to view the list of stored high-contributor profiles.
- Initially, this will return an empty list.

Executing Transactions

To interact with the deployed contract:

1. Call `store_high_contributors_github_profile(github_handle)` with a GitHub username.
2. The function will:
 - Fetch the profile page content
 - Extract the repository count
 - Store the handle if repositories > 25
 - Make the profile list available through `show_github_profiles()`

Understanding Data Processing

This contract demonstrates several important concepts:

- Web Scraping: Shows how to extract specific data from web pages.
- Regular Expressions: Demonstrates pattern matching in web content.
- Conditional Logic: Implements criteria-based storage decisions.
- Dynamic Arrays: Shows how to maintain a growing list of data.

Handling Different Scenarios

- Initial State: The profiles list starts empty.
- High Contributors: Profiles with >25 repositories are added to the list.
- Low Contributors: Profiles with ≤ 25 repositories are not stored.
- Invalid Profiles: Returns 0 repositories and doesn't store the profile.

Important Notes

1. This example focuses on public GitHub profiles only.
2. Repository counts may change over time.
3. The regular expression pattern assumes specific GitHub page structure.
4. The threshold of 25 repositories is arbitrary and can be adjusted.

Security Considerations

1. Be aware that GitHub's page structure might change.
2. Consider implementing error handling for malformed profiles.
3. Respect GitHub's rate limits and terms of service.
4. Validate input GitHub handles before processing.

Performance Optimization

1. The contract only stores handles, not full profile content.
2. Regular expression pattern is optimized for specific data extraction.
3. Conditional storage prevents unnecessary state bloat.
4. Dynamic array allows for efficient list management.

You can monitor the contract's behavior through transaction logs, which will show the repository counts and profile additions as they occur.

GitHubProfilesSummaries Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/github-profile-summary>

The GitHubProfilesSummaries contract demonstrates how to fetch GitHub profile data and generate AI-powered summaries of user profiles. This contract shows how to combine web scraping with AI analysis using both comparative (/developers/intelligent-contracts/equivalence-principle#comparative-equivalence-principle) and non-comparative (/developers/intelligent-contracts/equivalence-principle#non-comparative-equivalence-principle) equivalence principles.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qhq8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *
import typing
import re

class GitHubProfilesSummaries(gl.Contract):
    github_profiles: TreeMap[str, str]

    def __init__(self):
        pass

    @gl.public.write
    def store_github_profile_summary(self, github_handle: str) -> typing.Any:
        current_profile_summary = self.github_profiles.get(github_handle, None)

        if not current_profile_summary is None:
            raise Exception("profile summary already generated")

        github_profile_url = "https://github.com/"+github_handle

        def fetch_github_profile_summaries() -> str:
            response = gl.nondet.web.get(github_profile_url)
            return response.body.decode("utf-8")

        profile_content = gl.eq_principle.strict_eq(fetch_github_profile_summaries)

        task = """Given the web page content of a github profile in HTML format, generate a comprehensive
summary of the profile mentioning the key meta attributes and the GitHub contribution most important metrics"""
```



```

criteria = ""The summary provided should include different metrics and a summary of a GitHub profile""

profile_summary = (
    gl.ea principle.prompt non comparative(
        lambda: profile_content,
        task=task,
        criteria=criteria,
    )
)
self.github_profiles[github_handle] = profile_summary

@gl.public.view
def show_github_profile_summaries(self) -> str:
    return {profile: summary for profile, summary in self.github_profiles.items()}

```

Code Explanation

- Initialization: The GitHubProfilesSummaries class initializes with an empty TreeMap to store GitHub handles and their corresponding summaries.
- Write Method:
 - store_github_profile_summary(github_handle) generates an AI summary of a GitHub profile.
 - Checks if a summary already exists for the handle.
 - Uses both strict and non-comparative equivalence principles for different parts of the process.
- Read Method:
 - show_github_profile_summaries() returns a dictionary mapping GitHub handles to their summaries.

Key Components

1. Data Storage: Uses TreeMap for efficient key-value storage of profile summaries.
2. Web Fetching: Uses gl.nondet.web.get() with strict equivalence for deterministic content retrieval.
3. AI Analysis: Uses non-comparative equivalence for generating profile summaries.
4. Duplicate Prevention: Includes checks to prevent regenerating existing summaries.

Deploying the Contract

To deploy the GitHubProfilesSummaries contract:

1. Deploy the Contract: No initial parameters are needed for deployment.
2. The contract will initialize with an empty TreeMap.

Checking the Contract State

After deployment, you can:

- Use show_github_profile_summaries() to view all stored profile summaries.
- Initially, this will return an empty dictionary.

Executing Transactions

To interact with the deployed contract:

1. Call `store_github_profile_summary(github_handle)` with a GitHub username.
2. The function will:
 - Check for existing summary
 - Fetch the profile content
 - Generate an AI summary
 - Store the result in the TreeMap

Understanding AI Integration

This contract demonstrates several important concepts:

- Dual Equivalence Principles: Uses both comparative and non-comparative approaches.
- Error Handling: Implements checks for duplicate processing.
- Structured Storage: Maintains an organized mapping of profiles to summaries.

Handling Different Scenarios

- Initial State: The TreeMap starts empty.
- New Profile: Generates and stores a new summary.
- Existing Profile: Raises an exception to prevent duplicate processing.
- Invalid Profile: Web content fetch would fail for non-existent profiles.

Important Notes

1. Summaries are generated once and cached.
2. The AI task is focused on extracting key metrics and attributes.
3. Profile content is fetched deterministically using strict equivalence.
4. Summary generation allows for semantic variation while maintaining quality.

Security Considerations

1. Validate GitHub handles before processing.
2. Handle web content fetch failures gracefully.
3. Consider implementing rate limiting.
4. Be mindful of storage space for summaries.

Performance Optimization

1. Uses TreeMap for efficient key-value lookups.

2. Prevents redundant summary generation.
4. Stores only processed summaries, not raw HTML.

You can monitor the contract's behavior through transaction logs, which will show the profile fetches and summary generations as they occur.

Prediction Market Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/prediction>

The Prediction Market contract sets up a scenario to determine the outcome of a football game between two teams. The contract uses the Equivalence Principle to ensure accurate and consistent decision-making based on the game's resolution data.

```
# { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }

from genlayer import *

import json
import typing

class PredictionMarket(gl.Contract):
    has_resolved: bool
    game_date: str
    team1: str
    team2: str
    resolution_url: str

    def __init__(self, game_date: str, team1: str, team2: str):
        """
        Initializes a new instance of the prediction market with the specified game date and teams.

        Args:
            game_date (str): The date of the game in the format 'YYYY-MM-DD'.
            team1 (str): The name of the first team.
            team2 (str): The name of the second team.

        Attributes:
            has_resolved (bool): Indicates whether the game's resolution has been processed. Default is False.
            game_date (str): The date of the game.
            resolution_url (str): The URL to the game's resolution on BBC Sport.
            team1 (str): The name of the first team.
            team2 (str): The name of the second team.
        """
        self.has_resolved = False
        self.game_date = game_date
        self.resolution_url = (
            "https://www.bbc.com/sport/football/scores-fixtures/" + game_date
        )
        self.team1 = team1
        self.team2 = team2

    @gl.public.write
    def resolve(self) -> typing.Any:
        if self.has_resolved:
            return "Already resolved"

        def nondet() -> str:
            response = gl.nondet.web.get(self.resolution_url)
            web_data = response.body.decode("utf-8")
            print(web_data)

            task = f"""In the following web page, find the winning team in a matchup between the following teams:
```

```

Team 1: {self.team1}
Team 2: {self.team2}

Web page content:
{web_data}
End of web page data.

If it says "Kick off [time]" between the names of the two teams, it means the game hasn't started yet.
If you fail to extract the score, assume the game is not resolved yet.

Respond with the following JSON format:
{{
    "score": str, // The score with numbers only, e.g, "1:2", or "-" if the game is not resolved yet
    "winner": int, // The number of the winning team, 0 for draw, or -1 if the game is not yet finished
}}
It is mandatory that you respond only using the JSON format above,
nothing else. Don't include any other words or characters,
your output must be only JSON without any formatting prefix or suffix.
This result should be perfectly parsable by a JSON parser without errors.
"""
result = gl.nondet.exec_prompt(task).replace("`json", "").replace("`", "")
print(result)
return json.dumps(json.loads(result), sort_keys=True)

result_json = json.loads(gl.eq_principle.strict_eq(nondet))

if result_json["winner"] > -1:
    self.has_resolved = True
    self.winner = result_json["winner"]
    self.score = result_json["score"]

return result_json

```

You can check out this code on our GitHub

(https://github.com/genlayerlabs/genlayer-studio/blob/main/examples/contracts/football_prediction_market.py)

Deploying the Contract

To deploy the Prediction Market contract, you'll need to initialize the contract state correctly. This will impact how the contract will respond to the game's resolution.

1. Provide the game date and the names of the two teams. The `game_date`, `team1`, and `team2` constructor parameters are automatically detected from the code. For example, you might set `game_date` to "2024-06-05", `team1` to "Brazil", and `team2` to "Jamaica".
2. Once the game details are set, deploy the contract to make it ready to interact and resolve the game results.

Checking the Contract State

Once the contract is deployed, its address is displayed as well as the Read Methods section. In this case, there are no Read Methods defined.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can call the `resolve` method to process the game's result. This triggers the contract's logic to retrieve the game's data and determine the outcome based on the Equivalence Principle criteria defined.

Analyzing the Contract's Decisions

When the resolve method is executed:

- The LLM retrieves the game data from the specified URL.
- It validates the game's outcome according to the Equivalence Principle defined in the code.
- Finally, it returns a JSON response that includes the game's score and the winner.

Handling Different Scenarios

- If the game has started but not finished, the JSON response will indicate the game is not resolved yet.
- If the game has finished, the JSON response will include the final score and the winning team.
- If the game hasn't started, the JSON response will indicate this status.

You can view the logs to see detailed information about the contract interaction.

LogIndexer Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/examples/vector-store-log-indexer>

The LogIndexer contract demonstrates how to use the Vector Store database (VecDB) provided by the GenVM SDK. This contract shows how to store, retrieve, update, and remove text logs using vector embeddings for similarity-based searches.

```
# {
#   "Seq": [
#     { "Depends": "py-lib-genlayermodelwrappers:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" },
#     { "Depends": "py-genlayer:1jb45aa8ynh2a9c9xn3b7qqh8sm5q93hwfp7jqmwsfhh8j pz09h6" }
#   ]
# }

from genlayer import *
import genlayermodelwrappers
import numpy as np
from dataclasses import dataclass
import typing

@allow_storage
@dataclass
class StoreValue:
    log_id: u256
    text: str

# contract class
class LogIndexer(gl.Contract):
    vector_store: VecDB[np.float32, typing.Literal[384], StoreValue]

    def __init__(self):
        pass

    def get_embedding_generator(self):
        return genlayermodelwrappers.SentenceTransformer("all-MiniLM-L6-v2")

    def get_embedding(
        self, txt: str
    ) -> np.ndarray[tuple[typing.Literal[384]], np.dtypes.Float32DType]:
        return self.aet embedding_generator()(txt)

    @gl.public.view
    def get_closest_vector(self, text: str) -> dict | None:
        emb = self.get_embedding(text)
```

```

        result = list(self.vector_store.knn(emb, 1))
        if len(result) == 0:
            return None
        result = result[0]
        return {
            "vector": list(str(x) for x in result.key),
            "similarity": str(1 - result.distance),
            "id": result.value.log_id,
            "text": result.value.text,
        }

    @gl.public.write
    def add_log(self, log: str, log_id: int) -> None:
        emb = self.get_embedding(log)
        self.vector_store.insert(emb, StoreValue(text=log, log_id=u256(log_id)))

    @gl.public.write
    def update_log(self, log_id: int, log: str) -> None:
        emb = self.get_embedding(log)
        for elem in self.vector_store.knn(emb, 2):
            if elem.value.text == log:
                elem.value.log_id = u256(log_id)

    @gl.public.write
    def remove_log(self, id: int) -> None:
        for el in self.vector_store:
            if el.value.log_id == id:
                el.remove()

```

Code Explanation

- Data Structure: Uses StoreValue dataclass to store log ID and text.
- Vector Store: Initializes a VecDB with 384-dimensional float32 vectors.
- Embedding Generation: Uses the SentenceTransformer model for text embedding.
- Methods:
 - get_closest_vector(): Finds the most similar log entry.
 - add_log(): Adds a new log with its embedding.
 - update_log(): Updates an existing log entry.
 - remove_log(): Removes a log by its ID.

Key Components

1. Vector Database: Uses VecDB for efficient similarity-based searches.
2. Embedding Model: Utilizes SentenceTransformer for text vectorization.
3. CRUD Operations: Implements Create, Read, Update, Delete functionality.
4. Similarity Search: Supports k-nearest neighbors (KNN) queries.

Deploying the Contract

To deploy the LogIndexer contract:

1. Deploy the Contract: No initial parameters are needed.
2. The contract will initialize with an empty vector store.

Checking the Contract State

After deployment, you can:

- Use `get_closest_vector()` to find similar logs.
- Query will return `None` if no logs are stored.

Executing Transactions

The contract supports several operations:

1. Adding Logs:

- Call `add_log(log, log_id)` with text and ID.
- Creates embedding and stores in VecDB.

2. Finding Similar Logs:

- Use `get_closest_vector(text)` to find matches.
- Returns vector, similarity score, ID, and text.

3. Updating Logs:

- Call `update_log(log_id, log)` to modify entries.
- Updates based on text similarity.

4. Removing Logs:

- Use `remove_log(id)` to delete entries.
- Removes based on log ID.

Understanding Vector Storage

This contract demonstrates several important concepts:

- **Vector Embeddings:** Converts text to numerical vectors.
- **Similarity Search:** Uses vector distance for finding related content.
- **Persistent Storage:** Maintains vector database state.
- **Efficient Querying:** Supports fast nearest neighbor searches.

Handling Different Scenarios

- **Empty Database:** Returns `None` for searches.
- **Adding New Logs:** Creates new vector embeddings.
- **Updating Logs:** Modifies existing entries.
- **Removing Logs:** Deletes entries by ID.

Important Notes

1. This is a demonstration of VecDB features.
2. Uses a specific embedding dimension (384).
3. Similarity is based on vector distance.
4. Supports basic CRUD operations.

Performance Considerations

1. Embedding generation may be computationally intensive.
2. KNN searches scale with database size.
3. Vector dimension affects storage requirements.
4. Consider batch operations for efficiency.

Technical Details

1. Uses 384-dimensional float32 vectors.
2. Implements the all-MiniLM-L6-v2 model.
3. Stores both vector embeddings and metadata.
4. Supports exact and approximate nearest neighbor search.

You can monitor the contract's behavior through transaction logs, which will show vector operations and search results as they occur.

GenLayer CLI

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-cli>

The GenLayer CLI is a command-line interface designed to streamline the setup and local execution of the GenLayer Studio. It automates the process of downloading and launching the Studio, allowing developers to start simulating and testing locally with minimal effort.

Features

- Easy Initialization: Quickly set up the GenLayer Studio with a single command.
- Automated Downloads: Automatically downloads all necessary components required to run the Studio.
- Developer-Friendly: Simplifies local development and testing workflows.
- Extensible: Plans for additional commands to enhance interaction with the Studio.

How it's Built

The GenLayer CLI is a Command Line Interface tool built using Node.js and TypeScript. It's designed to automate the setup and management of the GenLayer Studio, simplifying the process for developers.

Technologies Used

- Node.js: A JavaScript runtime environment that allows the execution of JavaScript code on the server side.
- TypeScript: A statically typed superset of JavaScript that compiles to plain JavaScript, enhancing code reliability and maintainability.
- ESBuild: A fast JavaScript bundler and minifier used for building the CLI efficiently.
- Jest: A JavaScript testing framework utilized for writing and running tests.

Project Structure

The source code for the GenLayer CLI is organized as follows:

- src/: Contains the main TypeScript source files.
- tests/: Includes all the test files written using Jest.
- dist/: The compiled JavaScript files ready for execution.

Requirements

Before using the GenLayer CLI, ensure your system meets the following requirements:

- Node.js: Version 14.x or higher is required.
- npm: Comes bundled with Node.js, used for managing packages.
- Git: Required if cloning the repository directly from GitHub.
- Operating System: Compatible with macOS, Linux, and Windows.

Installation

To install the GenLayer CLI globally using npm, ensure you have Node.js installed, then run:

```
npm install -g genlayer
```

Usage

After installation, you can use the following command to start the Studio:

```
genlayer init
```

This command will download the necessary components and start the Studio. Once initialized, you can execute further commands (to be implemented) to interact with the Studio.

General Format of Commands

The GenLayer CLI commands follow a consistent syntax pattern:

```
genlayer <command> [options]
```

- : The primary action you want the CLI to perform (e.g., init, up).
- [options]: Optional flags that alter the behavior of the command.

Example

To initialize the GenLayer Studio with specific parameters:

```
genlayer init --numValidators 10 --branch develop
```

Testing

The GenLayer CLI uses Jest with ts-jest for testing TypeScript files.

To run tests:

```
npm run test
```

Further Development

Additional commands are planned to enhance interaction with the GenLayer Studio. Stay tuned for updates.

Repository

You can find the GenLayer CLI repository on GitHub:

GenLayer CLI Repository (<https://github.com/genlayerlabs/genlayer-cli>)

Full Reference

The full reference for the GenLayer CLI is available in the GenLayer CLI Reference (</references/genlayer-cli>).

GenLayer FAQ

The GenLayer FAQ is a collection of useful questions and answers about the project. If you have a question that isn't answered here, please let us know on our Discord (<https://discord.gg/8Jm4v89VAu>) .

How do I set up the GenLayer Development Environment?

To quickly set up the GenLayer Development Environment, run the commands below:

```
$ npm install -g genlayer
```

```
$ genlayer init
```

For more detailed setup instructions, please refer to the Getting Started page (</getting-started>)

Where can I find the GenLayer Studio?

The GenLayer Studio is available at studio.genlayer.com (<https://studio.genlayer.com>).

How do I deploy an Intelligent Contract?

- Go to studio.genlayer.com (<https://studio.genlayer.com>) or use the GenLayer CLI (/getting-started#installation-of-the-genlayer-cli) to start the GenLayer Studio locally.
- Write your contract in Python using the Intelligent Contract SDK (/core-concepts/intelligent-contracts).
- Follow the guide here (/genlayer-stack/genlayer-studio/execute-transaction) to interact with the Studio and deploy your Intelligent Contract.

What is the GenLayer CLI used for?

The GenLayer CLI is used for setting up the GenLayer Studio and, in the future, will support mainnet and testnet environments.

GenLayer Studio

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio>

This Studio is an interactive sandbox designed for developers to explore the potential of GenLayer's Intelligent Contracts. It replicates the GenLayer network's execution environment and consensus algorithm, but offers a controlled and local environment to test different ideas and behaviors.

What you can do with the GenLayer Studio:

- Experiment with AI smart contracts: Intelligent Contracts leverage LLMs, such as GPT-4 or Llama3, to understand natural language and be capable of complex decision making.
- Access the Web Natively: GenLayer is the first platform where smart contracts don't need oracles to access the Internet.
- Code in Python: Develop in a familiar, developer-friendly language, where memory and string management are not a big headache.

Explore the Studio

Here is a video introduction to help you get started with the GenLayer Studio:

```
className="responsive-iframe"
src="https://www.youtube.com/embed/G4N-oErIIBI?si=ZkWRT6Ya2mjjWuP8"
allow="accelerometer; autoplay; clipboard-write; encrypted-media; gyroscope; picture-in-picture"
allowFullScreen
>
```

Intelligent Contract SDK

Note:

Learn how to write your own Intelligent Contracts at Intelligent Contracts (/build-with-genlayer/intelligent-contracts)

Load Your Contract

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/loading-contract>

To start using the GenLayer Studio, you need to load your Intelligent Contract into the Studio. This involves navigating the Studio interface, selecting your contract file, and preparing it for deployment and execution.

Access your Intelligent Contracts

1. On the left sidebar, click on the Contracts icon. This is where you can see the list of your Intelligent Contracts. There are pre-loaded example contracts for you to try out.
2. To add a new contract, click on the + button labeled New Contract to create a new contract file in the Studio.
3. Click on the Add From File button to select and upload your Intelligent Contract file from your local machine.
4. Once your contract is uploaded, you will see it listed under Your Contracts.

Run Your Intelligent Contract

1. Click on the contract file to open it in the editor pane. This allows you to review and edit the code if necessary before running it.
2. Once you have your contract file open, you have two options to run and debug your contract:
 - Click the Run and Debug button on the left sidebar.
 - Click the play icon at the top right of the editor pane.

Now that you have loaded your Intelligent Contract into the GenLayer Studio, you can proceed to set constructor parameters and deploy it.

Deploy Contracts

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/deploying-contract>

Once you have loaded your Intelligent Contract into the GenLayer Studio, the next step is to set the constructor parameters and deploy it. The constructor parameters are essential inputs that initialize the state of your contract.

Setting Constructor Parameters

1. After loading your Intelligent Contract, you will see the Constructor Inputs section on the left-hand pane. The constructor parameters are automatically detected from your code if defined properly (#detecting-constructor-parameters).
2. If you need to manually adjust your constructor parameters, you can write them in JSON format by clicking on the JSON button.

Detecting Constructor Parameters

The GenLayer Studio automatically detects the constructor parameters from your code. It analyzes your init method to identify the parameters and their types. This automatic detection ensures that you have the correct inputs for initializing your contract. It's important to have clear type annotations for each parameter (e.g., str, bool, int, list) to enable accurate detection.

Deploying the Contract

After setting the constructor parameters, click on Deploy to deploy your contract.

Once completed, you can proceed to execute your transactions and interact with the deployed contract.

Current Intelligent Contract State

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/contract-state>

Once your Intelligent Contract is deployed, you can check its current state to verify that it has been initialized correctly. This step ensures that the contract's data and variables are set as expected.

Viewing the Current Intelligent Contract State

After deploying your contract, you will now see:

- Contract Address: This is used for interacting with your contract.
- Read Methods: These return information on the current state of the contract.

For example, in the Storage contract:

- If the constructor parameter initial_storage was set to hello, the state method get_storage will return hello.

The result will show the current state based on the getter function called. This allows you to verify that the variables in your contract have been initialized correctly and that they change when transactions are executed.

Note:

When the contract state changes, you can always come back and refresh the state by calling these getter functions to see the most current state of your Intelligent Contract.

With the state verified, you can now execute transactions and interact with your deployed contract.

Execute Transactions

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/execute-transaction>

After deploying your Intelligent Contract and verifying its state, the next step is to execute transactions. This allows you to interact with your contract's methods and functions to perform specific actions or queries.

Write Methods

1. In the Write Methods section, you will see all the write methods available in your Intelligent Contract.
2. Expand the method you want to execute. For example, in the Storage contract, select the `update_storage` method.
3. If the method requires input parameters, a field will appear for you to enter the necessary values. For example, the `update_storage` method requires a string called `new_storage`.
4. After entering any required parameters, click the button to execute the method. The Studio will process the transaction and return the result.

Viewing Transaction Results

Once you execute a transaction, the result will be displayed, showing the output of the method called. This allows you to verify that the method has executed correctly and to see the effect of the transaction on your contract's state.

Viewing Current Account Address

You can also view your current account address in the header. This address is used to interact with your contract.

You can continue interacting with your Intelligent Contract by executing additional transactions or checking the state again to see how it has changed.

Monitoring Node Logs

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/monitoring-node-logs>

Node Logs in the GenLayer Studio provide real-time feedback and are a critical component for debugging and tracking the behavior of Intelligent Contracts as they interact within the studio.

Key Features of Node Logs

Node Logs capture and display a variety of events, including:

- Transaction Submission, Status Changes and Finalization: Track the status and outcomes of submitted transactions.
- Contract Deployment Statuses: Monitor the progress and success of contract deployments.
- Execution Results and Errors: Review the outputs and any errors encountered during contract execution.
- Validator Operations and Consensus-Related Messages: Observe the activities of validators and consensus processes.

Accessing Node Logs

You can view the node logs at the bottom of the GenLayer Studio. This section automatically updates with new log entries as actions are performed within the studio.

You can also filter the logs by content as well as by scope and level.

[Image: Node Log]

Understanding Node Logs

The logs are structured to provide clear and detailed information:

- Scope: Each log is tagged with a label such as RPC, GenVM or Consensus to give you a better understanding of the context of the log.
- Level: Log entries are associated with a level and color to facilitate rapid scanning for issues or confirmations.
- Details: Expandable log entries offer detailed explanations and data for each event, such as transaction hashes or error messages.

By actively viewing these logs, you can ensure that your Intelligent Contracts are performing as expected and that any issues are promptly addressed. This leads to a smoother development process within the GenLayer Studio.

Accessing and Configuring Validators

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/validators>

In the GenLayer Studio, validators are essential for achieving consensus and validating transactions through a process known as Optimistic Democracy (/core-concepts/optimistic-democracy). When you initialize the Studio based on your selected LLM provider(s), you are provided with 5 default validators.

1. **Leaders:** Within this consensus model, one validator is selected as the leader for each transaction. The leader's role is to propose how a transaction should be executed based on the transaction data and the Intelligent Contract's logic.
2. **Validators:** After the leader proposes a transaction execution, other validators are responsible for reviewing and validating the proposal. They use the Equivalence Principle ([/core-concepts/optimistic-democracy/equivalence-principle](#)) to determine if the leader's proposal meets the required standards.

These validators can be modified to suit your Intelligent Contract's requirements.

Accessing Validators

To access and manage validators, follow these steps:

1. On the left sidebar, click on the Validators icon. This will open the validators page.
2. In the validators page, you will see a list with all the validators currently configured in the Studio along with their models and providers.

Configuring Validators

To configure validators, you can add new validators or modify existing ones.

1. Click the New Validator button to open the validator creation dialog.
2. Create New Validator:

In the validator creation dialog, fill in the required fields:

- **Provider:** Select the provider for the validator.
- **Model:** Choose the model that the validator will use.
- **Stake:** Specify the stake for the validator.
- **Config:** Enter any additional configuration parameters in JSON format.

3. Click Create to add the new validator to your configuration.

Modify Validator Details

To modify your validator settings, click on any existing validator and modify the fields based on your needs.

Delete Validator

To delete a validator, click on the delete button beside the validator in the list.

By properly managing and configuring validators, you ensure that your Intelligent Contracts operate smoothly and securely within the GenLayer Studio.

Inference Providers

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/providers>

Validators can use various inference providers to validate transactions. You can configure those providers and add custom ones to suit your needs. When you then create a new validator in the validators page, you will be able to select those providers and the relevant models from the list.

Accessing Providers

You can manage your providers in the Settings page:

Configuring Providers

Providers have the following properties:

- Provider: The name of the provider. For example, OpenAI or ollama.
- Model: The model attached to this provider. For example, gpt-4o or llama3.
- Plugin: (only for custom providers) When adding a custom provider, you need to specify a plugin that implements the provider. The plugin defines what configuration is necessary for the provider.
- Provider Config: Related environment variables, for example API keys or endpoint URLs.
- Default Validator Config: This config is used as default when creating a new validator, and will then be applied when making calls to inference providers using said validator.

Adding a new Provider

You can add a new provider by clicking the New Config button at the top of the providers list.

Updating an Existing Provider

To modify your validator settings, click on any existing validator and modify the fields based on your needs.

Deleting a Provider

To delete a provider, click on the delete button beside the provider in the list.

Limitations of the GenLayer Studio

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/limitations>

The GenLayer Studio is in an early stage of development, with many features still being tested and refined. Here are some limitations of the studio:

Native Token Transfers

Native Token transfers to and from contracts are not supported in the Studio.

Gas Usage

Transactions within the Studio do not consume gas. While this is beneficial for rapid prototyping, the absence of gas simulation may result in behavior differing from live blockchain networks.

Consensus Algorithm

The GenLayer Studio does not implement the full Optimistic Democracy consensus mechanism. There is no appeals process. The studio focuses on providing a basic execution environment without the complexities of the full consensus mechanism.

Programming Language

The programming language used in the Studio is an early version and is not final. It is expected to undergo many changes as development progresses.

Web Access

While the Studio supports web browsing capabilities, it is primarily for testing purposes. The actual web access and data retrieval might differ when deployed on the live network.

Resetting the GenLayer Studio

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/reset-the-studio>

The GenLayer Studio provides a straightforward way to reset its storage, current status, memory, and contract examples. This feature is particularly useful if you want to start fresh with a clean state, whether you're testing new contracts or simply clearing out previous configurations.

How to Reset the Studio

To reset the GenLayer Studio, follow these simple steps:

1. Access the Settings Page: begin by navigating to the Settings page within the GenLayer Studio.
2. Click "Reset Storage": this option allows you to clear all the current data stored in the studio, including the state of deployed contracts, transaction history, and any cached information.
3. Confirm the Reset in the Dialog: this is an important step, as resetting the storage will remove all existing data and cannot be undone.

Once confirmed, the studio will reset its storage, returning to its initial state as if it were newly initialized.

What Does Resetting Do?

Resetting the GenLayer Studio through the Reset Storage option will:

- Clear All Deployed Contracts: Any contracts that you have deployed in the studio will be removed from the frontend memory.
- Reset Transaction History: All previous transactions, including those in various states, will be erased.
- Restore Contract Examples: Any example contracts included in the studio will be restored to their original, unaltered state.

When Should You Reset?

Consider resetting the studio when:

- You want to start with a clean slate for a new development or testing session.
- You encounter issues that might be caused by lingering data or state inconsistencies.
- You have completed a testing phase and need to ensure that previous data does not interfere with new experiments.

Troubleshooting

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/troubleshooting>

If you encounter any issues with the Studio, here are some detailed steps to help you resolve common problems:

Frontend Not Loading Correctly

The frontend may not load correctly due to caching issues or an outdated version of the contract stored in your browser.

Solution

- Refresh Frontend: Simply reload the page in your browser.
- Clear Cache: Clear your browser's cache by clicking on the left of your address bar -> Cookies and site data -> Manage on device site data -> Delete localhost.

Port Conflicts

The Studio may fail to start if some ports required for its operation are already in use by other applications.

Ports used by the Studio:

- Frontend: 8080
- Ollama: 11434

- JSON-RPC Server: 4000
- GenVM: 6678
- Postgres: 5432
- Hardhat: 8545
- Webrequest: 5000, 5001

Solution

1. Identify processes using the required ports. Replace with the specific port number.

- Linux / MacOS:

bash

```
lsof -i :<PORT_NUMBER>
```

- Windows:

cmd

```
netstat -aon | find <PORT_NUMBER>
```

2. Use the following command to kill the process using the port. Replace with the Process ID obtained from the previous step.

- Linux / MacOS:

bash

```
kill -9 <PID>
```

- Windows:

cmd

```
taskkill /PID <PID> /F
```

Docker not Running

The GenLayer Studio relies on Docker for managing containers and images. If Docker is not running, the Studio cannot function properly.

Solution

- Ensure that Docker (or Docker Desktop) is installed on your system. You can check this by running:

bash

```
docker info
```

- Restart Docker Desktop or the Docker daemon on your system.

Display Issues

The code section or other UI elements may not display correctly due to screen size limitations.

Solution

- Resize your screen to accommodate the UI elements properly. This will help improve the visibility and layout of the Studio.

Studio Not Responding or Throwing Errors

The Studio may become unresponsive or throw errors due to various reasons, such as conflicting processes or corrupted containers.

Solution

You can perform a fresh start by stopping and removing all containers, as well as removing all images.

Via Command Line:

```
docker stop $(docker ps -aq)
docker rm $(docker ps -aq)
docker rmi $(docker images -q)
```

Via Docker Desktop:

1. Open Docker Desktop and go to the Containers section.
3. Select all containers and click Delete.
4. Go to the Images section.
5. Select all images and click Delete.

Development Tips

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/development-tips>

Here are some valuable tips on how to develop your intelligent contract, debug it, and test it using the GenLayer Studio.

1. Validators setup

You need to be sure that you have at least one validator. If you don't have any, you can create one in the validators screen by providing:

- Provider
- Model
- Stake (at least 1)
- Config (optional)

2. Add types to your contract method inputs

Adding types to your contract method inputs helps the UI render the correct input fields. For example, if you have a method that takes a string as input, you can define it as follows:

```
def update_storage(self, new_storage:str) -> None:
    self.storage = new_storage
```

This will allow the UI to render the form needed to call the contract with the correct input type as follows

3. The constructor

The constructor is a special method called when the contract is deployed. It is used to initialize the contract's storage. You can define it as follows:

```
def __init__(self, initial_storage: str):  
    self.storage = initial_storage
```

Right now, your contract must include a constructor. Without one, the Studio won't be able to deploy your contract.

4. Deploying a contract and sending transactions

After you have written your contract, you can deploy it by clicking the "Deploy" button. This will deploy the contract and show you the contract's address.

5. Debugging your contract

You can debug your contract by adding print statements to your contract methods. The Studio's Logs section will display these print statements.

Here is an example of printing in a contract method:

```
def update_storage(self, new_storage:str) -> None:  
    print("new storage value: ", new_storage)  
    self.storage = new_storage
```

When a transaction is sent to this method, the print statement will be shown in the Logs section as follows:

6. What happened when the transaction was executed

After a transaction is executed, you can see the details by clicking on the transaction in the Transactions section (left-bottom corner). This will show you:

- Number: The transaction's number in the database
- Timestamp: The time the transaction was executed
- Type: The transaction's type (deployment or method call)
- Status: The transaction's status (PENDING, PROPOSING, COMMITTING, REVEALING, ACCEPTED, FINALIZED)
- Input: the parameters passed to the method
- Execution: The status of the execution (SUCCESS or ERROR), the leader's configuration, and the validator's consensus votes
- Equivalence principle output: The output of the equivalence principle from the leader

7. Knowing what happens when the LLMs are called

If a contract method calls an LLM, you can see the Equivalence Principle Output from the leader (what's being input in the eq.set call) in the transaction details modal.

8. Access to the contract state

If your contract methods change the contract's state, you can query that information by adding a read method to your contract. Read methods on a contract need to be decorated with the `@gl.public.view` decorator. This method should return the state you want to query.

```
@gl.public.view  
def get_balances(self) -> dict[str, int]:  
    return self.balances
```

```
@gl.public.view
def get_balance_of(self, address: str) -> int:
    return self.balances.get(address, 0)
```

These two methods allow you to query the contract's balance state. You can call them by expanding them from the left panel of the Studio's UI and clicking on the "Call Contract" button.

Adding LLM Provider Plugins

Source: <https://docs.genlayer.com/developers/intelligent-contracts/tools/genlayer-studio/advanced-features/custom-plugins>

Note:

This section is intended for advanced users who wish to integrate new LLM provider plugins. This process requires code modifications, and we encourage you to create a Pull Request if you'd like your plugin to be included in the official GenLayer Studio.

The GenLayer Studio seamlessly interacts with multiple LLM providers.

Currently, we support the following providers out of the box:

- OpenAI (<https://platform.openai.com/>)
- Anthropic (<https://www.anthropic.com/>)
- Ollama (<https://ollama.com/>)
- Heuristai (<https://heuristai.ai/>)

In addition to these built-in providers, you have the flexibility to integrate your own custom providers.

Adding a New LLM Provider Plugin

Creating the Plugin

To integrate a new LLM provider plugin, you need to implement the Plugin protocol defined in the `llms.py` (<https://github.com/genlayerlabs/genlayer-studio/blob/main/backend/node/genvm/llms.py>) file. The protocol is structured as follows:

```
class Plugin(Protocol):
    def __init__(self, plugin_config: dict): ...

    async def call(
        self,
        node_config: dict,
        prompt: str,
        regex: Optional[str],
        return_streaming_channel: Optional[asyncio.Queue],
    ) -> str: ...

    def is_available(self) -> bool: ...

    def is_model_available(self, model: str) -> bool: ...
```

Here's an example of how you might implement a custom Plugin. Note that this are just some common guidelines, you are free to implement your integration as you please

```
class MyCustomPlugin:
    def __init__(self, plugin_config: dict):
        self.api_key = plugin_config.get('api_key')
        self.base_url = plugin_config.get('base_url', 'https://api.customllm.com')
        # Add any other necessary configuration parameters

    async def call(
        self,
        node_config: dict,
        prompt: str,
        regex: Optional[str],
        return_streaming_channel: Optional[asyncio.Queue],
    ) -> str:
        # Implement the API call to your custom LLM provider
        # This is a placeholder implementation
        async with aiohttp.ClientSession() as session:
            async with session.post(
                f"{self.base_url}/generate",
                json={"prompt": prompt, "config": node_config},
                headers={"Authorization": f"Bearer {self.api_key}"})
            as response:
                if response.status == 200:
                    result = await response.json()
                    return result['generated_text']
                else:
                    raise Exception(f"API call failed with status {response.status}")

    def is_available(self) -> bool:
        # Check if the plugin is properly configured and available
        return bool(self.api_key) and bool(self.base_url)

    def is_model_available(self, model: str) -> bool:
        # Check if the specified model is available for this provider
        # This is a placeholder implementation
        available_models = ['custom-gpt-3', 'custom-gpt-4']
        return model in available_models
```

Registering the Plugin

After implementing the Plugin protocol, you need to register your new provider in the `llms.py` file under the `get_llm_provider` function. This step ensures that the GenLayer Studio recognizes and can utilize your custom plugin.

```
def get_llm_plugin(plugin: str, plugin_config: dict) -> Plugin:
    """
    Function to register new providers
    """
    plugin_map = {
        "ollama": OllamaPlugin,
        "openai": OpenAIPlugin,
        "anthropic": AnthropicPlugin,
        "custom-plugin-key": MyCustomPlugin, # Modify here accordingly
    }

    if plugin not in plugin_map:
        raise ValueError(f"Plugin {plugin} not registered.")

    return plugin_map[plugin](plugin_config)
```

Updating the JSON Schema

To maintain consistency and enable proper configuration validation, you must update the JSON schema (<https://json-schema.org/>) in the `providers_schema.json` (https://github.com/genlayerlabs/genlayer-studio/blob/main/backend/node/create_nodes/providers_schema.json) file. This update should include your new provider and its specific configuration options.

The JSON schema plays a crucial role in validating the configuration options for each provider, ensuring that users input correct and compatible settings.

Registering a Provider for the new plugin

New LLM Providers can be registered through the Studio's UI in the Settings page or manually through the backend API. Here is an example on how addition of an LLM Provider using your new Plugin could look like

```
curl --request POST \
  --url http://localhost:4000/api \
  --header 'Content-Type: application/json' \
  --data '{
    "jsonrpc": "2.0",
    "method": "sim_addProvider",
    "params": [
      {
        "provider": "my-custom-provider",
        "model": "my-custom-model",
        "config": {},
        "plugin": "custom-plugin-key",
        "plugin_config": {
          "api_key": "MY-SECRET-KEY",
          "base_url": "my.custom.url"
        }
      }
    ],
    "id": 1
  }'
```

Build With GenLayer

Source: <https://docs.genlayer.com/developers/intelligent-contracts/ideas>

Here are some ideas to get you started with building Intelligent Contracts that can interact with web sources, APIs, and understand natural language using LLM calls.

Idea	Implementation Details
Prediction Markets	Use GenLayer to directly search the web for relevant information allowing users to place bets
Parametric Insurance	Create insurance that pays out based on real-world events, such as drought conditions or natu
Bounty Review and Payout	Automatically reward community members for completing tasks like writing articles or developi
Performance-based Contracting	Set up contracts that pay out based on the completion of pre-agreed jobs, such as developing
Slashing Monitoring and Insurance	Monitor protocols for any violations and trigger slashing events or provide insurance payouts
Hack Detection and Emergency Pause	Set up contracts that continuously monitor for potential protocol attacks through explorers a
On-chain Identity Verification	Verify users' identities by checking their social media profiles for specific messages and li
Under-collateralized Lending	Enable lending with less collateral by linking real-world identity to on-chain reputation, al
P2P Gambling	Allow two users to place bets on real-world outcomes, with the contract determining the winne
Decentralized Game Master	Facilitate text-based role-playing games where users submit their actions, and the intelligen
Interoperable Games with NFTs	Develop games that issue and recognize automatically generated NFT items, allowing players to
Unstoppable Organizations	Create autonomous entities like DAOs that can adapt and continue their missions indefinitely
Retroactive Public Goods Funding	Set up contracts that reward contributions to the protocol retroactively, incentivizing commu
Crowd-sourced Knowledge Database	Implement a system where users are rewarded for finding and summarizing new information on va
AI Notary	Create an automated notary service that can confirm the occurrence of online events, providin
AI Arbitration	Develop a dispute resolution system where parties submit their cases to an AI-based arbitrat
Private P2P Contracts	Enable two-party contracts that remain private unless a dispute arises, using a commit/reveal
Multi-modal Use Cases	Integrate real-world images for performance-based contracting, such as verifying the completi
Generative Memes	Create fun and engaging Intelligent Contracts, like ones that track and manipulate token bala
Text-based Intelligent Contracts	Open the possibility for non-developers to create Intelligent Contracts by capturing the enti
Honeypot Contracts for Security	Set up Intelligent Contracts designed to attract and test adversarial attacks, helping to ide
Fair and Transparent Moderation	Use GenLayer-based arbitration to make fair and transparent moderation decisions in various c

Architecture Overview of Decentralized Applications Using GenLayer

Source: <https://docs.genlayer.com/developers/decentralized-applications/architecture-overview>

A decentralized application (DApp) on GenLayer leverages its unique blockchain architecture to integrate AI-powered smart contracts, called "Intelligent Contracts." These DApps consist of various components that work together to ensure seamless interaction with the GenLayer protocol.

Key Components

Frontend Application

The user interface of the DApp is typically built with web technologies like HTML, CSS, and JavaScript. It communicates with the backend using the GenLayerJS SDK, enabling interaction with the GenLayer protocol and handling user inputs to trigger blockchain transactions.

GenLayerJS SDK

A TypeScript/JavaScript library that abstracts the complexities of blockchain interactions. The SDK provides APIs to read from and write to Intelligent Contracts, manages accounts and queries transactions, and acts as the bridge between the frontend and GenLayer's protocol.

Consensus Layer

The consensus layer implements GenLayer's Optimistic Democracy mechanism to ensure reliable and secure execution of transactions using a validator-driven commit-reveal scheme. It also handles appeals and transaction finality to maintain integrity and fairness.

It is built on ZK-stack rollups to provide scalability and cost-efficiency, secure state updates, and anchoring to Ethereum's security model.

Execution Environment

The execution environment (the GenVM) is the engine that executes Intelligent Contracts. It supports both deterministic and non-deterministic operations, enabling AI integration and web data access.

Ghost Contracts

Ghost contracts are proxy smart contracts on the consensus layer that facilitate interactions between accounts and Intelligent Contracts. They also manage external messages and asset bridging.

Architecture Diagram

```
flowchart TD
    subgraph s1["Frontend Application"]
        n1["GenLayerJS"]
    end
    subgraph s2["Consensus Layer."]
```

```

    n2["Consensus Smart Contracts"]
    n3["Ghost Contracts"]
  end
  subgraph s3["GenLayer Node"]
    n4["Execution Environment - GenVM"]
  end
  end
  s1 <--> s3
  s3 <--> s2

```

DApp Development Workflow with GenLayer

Source: <https://docs.genlayer.com/developers/decentralized-applications/dapp-development-workflow>

The GenLayer platform allows developers to create decentralized applications (DApps) throughout the lifecycle of creating, testing, and deploying. This guide details how developers can transition from using the GenLayer Studio for their first Intelligent Contract to an advanced local development workflow, finishing with building a frontend application with GenLayerJS.

1. Starting with GenLayer Studio

GenLayer Studio is an integrated environment designed to streamline the initial stages of Intelligent Contract development, making it accessible for developers of all experience levels.

It provides an interactive environment that serves as a comprehensive sandbox for developing, deploying, and testing Intelligent Contracts in real time. This environment enables developers to experiment with their contracts and see immediate results without the complexities of a production setup.

The platform runs a simulated network with customizable validators that accurately mirror the GenLayer consensus mechanism. This feature allows developers to test their contracts under conditions that closely resemble the actual blockchain environment, ensuring reliable deployment outcomes.

Getting Started:

1. Set Up the Studio: Developers initialize the Studio using `genlayer cli` command `genlayer init` which configures the environment and spawns a local validator network. GenLayer Studio is also available as a hosted instance at studio.genlayer.com (<https://studio.genlayer.com/>).
2. Write Your First Contract: Intelligent Contracts in GenLayer are written in Python, utilizing its extensive libraries and GenVM capabilities like LLM calls and web integration. Refer to Your First Contract (</developers/intelligent-contracts/your-first-contract>) guide for more information.
3. Deploy and Test: Deploy your Intelligent Contracts through the Studio interface and test them in the simulated network.

2. Moving to Advanced Workflow

As projects transform to real-world applications, developers should migrate to a local development setup. This approach mirrors frameworks like Hardhat or Foundry and is well-suited for iterative development and comprehensive testing.

Benefits of Local Development:

- **Complete Control:** Developers can configure the environment, validators, and network parameters as needed.
- **Enhanced Debugging:** The local setup allows for advanced debugging of both contracts and transactions.
- **Flexible Testing:** Tests can simulate real-world scenarios, including edge cases and complex interactions.

Workflow Steps:

1. **Set Up Your Local Environment:** Developers can start with the GenLayer boilerplate project, which includes pre-configured templates for local testing.
2. **Write Tests:** Tests are written in Python, focusing on validating contract functionality and ensuring consensus integrity. The boilerplate includes sample tests to accelerate development.
3. **Simulate Transactions:** Run detailed simulations to observe how contracts behave under various network conditions, ensuring robust performance.

3. Building the Frontend with GenLayerJS

The frontend is the user-facing component of a DApp. With GenLayerJS, developers can integrate their applications with the GenLayer network, focusing on providing a seamless user experience.

Why Use GenLayerJS?

- **Simplified Interactions:** Abstracts the complexity of blockchain interactions with high-level APIs.
- **Comprehensive Features:** Supports transaction handling, contract interaction, event subscriptions, and account management.
- **TypeScript Integration:** Provides type safety and code reliability for frontend development.

Frontend Development Workflow:

1. **Integrate GenLayerJS:**

Install the SDK and set up a client to connect to the GenLayer network. Configuration options allow connection to various environments, such as testnets or the GenLayer Studio.

Refer to GenLayerJS (</developers/decentralized-applications/genlayer-js>) guide for more information.

2. **Read and Write to Contracts:**

Use the SDK's high-level APIs to interact with deployed contracts. For instance, retrieve user balances or update contract state seamlessly.

Refer to [Reading Data \(/developers/decentralized-applications/reading-data\)](/developers/decentralized-applications/reading-data) and [Writing Data \(/developers/decentralized-applications/writing-data\)](/developers/decentralized-applications/writing-data) guides for more information.

3. **Monitor Transactions:**

Developers can subscribe to events or query transaction statuses, ensuring users are kept informed of transaction progress and outcomes.

4. Build the User Interface:

Combine GenLayerJS with popular frontend frameworks (like React or Angular) to create intuitive interfaces.

Advanced Tips for Developers

1. Leverage GenVM Features:

The GenVM allows Intelligent Contracts to interact with LLM models and access real-time web data. Developers should design contracts to maximize these capabilities for dynamic and intelligent DApps.

2. Optimize Testing:

Incorporate edge case scenarios in tests to ensure that contracts behave reliably under all conditions.

3. Focus on Security:

Implement robust security measures, including input validation and error handling, to protect contracts from malicious inputs and ensure consensus consistency.

GenLayerJS SDK

Source: <https://docs.genlayer.com/developers/decentralized-applications/genlayer-js>

GenLayerJS SDK is a TypeScript library designed for developers building decentralized applications (DApps) on the GenLayer protocol. This SDK provides a comprehensive set of tools to interact with the GenLayer network, including client creation, transaction handling, event subscriptions, and more, all while leveraging the power of Viem as the underlying blockchain client.

Features

- Client Creation: Easily create and configure a client to connect to GenLayer's network.
- Transaction Handling: Send and manage transactions on the GenLayer network.
- Contract Interaction: Read from and write to smart contracts deployed on GenLayer.
- Event Subscriptions: Subscribe to events and listen for blockchain updates.
- TypeScript Support: Benefit from static typing and improved code reliability.

How it's Built

The GenLayerJS SDK is built using TypeScript and leverages the Viem library as the underlying blockchain client. It is designed to provide a high-level, easy-to-use API for interacting with the GenLayer network, abstracting away the complexities of direct blockchain interactions.

Technologies Used

- TypeScript: A statically typed superset of JavaScript that compiles to plain JavaScript, enhancing code reliability and maintainability.
- Viem: A modular and extensible blockchain client for JavaScript and TypeScript.
- ESBUILD: A fast JavaScript bundler and minifier used for building the SDK efficiently.
- Vitest: A Vite-native unit test framework used for testing.

Project Structure

The source code for the GenLayerJS SDK is organized as follows:

- src/: Contains the main TypeScript source files.
- tests/: Includes all the test files written using Vitest.
- dist/: The compiled JavaScript files ready for use.

Requirements

Before using the GenLayerJS SDK, ensure your system meets the following requirements:

- Node.js: Version 16.x or higher is required.
- npm: Comes bundled with Node.js, used for managing packages.
- Operating System: Compatible with macOS, Linux, and Windows.

Installation

To install the GenLayerJS SDK, use the following command:

```
npm install genlayer-js
```

Usage

Here's how to initialize the client and connect to the GenLayer Studio:

- Reading a transaction (./intelligent-contracts/features/interacting-with-intelligent-contracts): Understand how to query transactions on the GenLayer network.
- Reading a contract (/developers/decentralized-applications/reading-data): Discover how to read data from intelligent contracts on GenLayer.
- Writing a contract (./intelligent-contracts/tools/genlayer-studio/contract-state): Learn how to write data to intelligent contracts on GenLayer.

General Format of Commands

The GenLayerJS SDK provides functions that follow a consistent syntax pattern:

```
client.functionName(parameters);
```

- client: The instance of the GenLayer client.
- functionName: The primary action you want to perform (e.g., `getTransaction`, `readContract`).
- parameters: An object containing the required parameters for the function.

Further Development

Additional features are planned to enhance interaction with the GenLayer network, including wallet integration and gas estimation. Stay tuned for updates.

Repository

You can find the GenLayerJS SDK repository on GitHub:

GenLayerJS SDK Repository (<https://github.com/genlayerlabs/genlayer-js>)

Full Reference

The full reference for the GenLayerJS SDK is available in the GenLayerJS SDK Reference (</references/genlayer-js>).

Querying a Transaction

Source: <https://docs.genlayer.com/developers/decentralized-applications/querying-a-transaction>

Reading transactions in GenLayer allows you to inspect the details of any transaction that has been submitted to the network. This is useful for monitoring transaction status, debugging, and verifying transaction details.

Basic Transaction Reading

Here's the simplest way to read a transaction:

```
import { simulator } from 'genlayer-js/chains';
import { createClient } from 'genlayer-js';

const client = createClient({
  chain: simulator,
});

const transactionHash = "0x...";

const transaction = await client.getTransaction({
  hash: transactionHash
});
```

Transaction Data Structure

When you read a transaction, you get access to its complete data structure:

```
interface Transaction {
  hash: `0x${string}`           // The unique transaction hash
  from: `0x${string}`           // Address of the sender
  to: `0x${string}` | null      // Address of the recipient (null for contract deployments)
  nonce: number                 // Transaction sequence number
  value: bigint                 // Amount of native tokens transferred
  data: `0x${string}`           // Transaction input data
  timestamp: number             // Block timestamp when transaction was included
  status: 'success' | 'failure' | 'pending' // Current transaction status
  blockNumber: bigint | null    // Block number where transaction was included
  blockHash: `0x${string}` | null // Hash of the block
  // ... additional fields
}
```

Reading Different Transaction Types

Basic Transfer Transaction

```
const transferTx = await client.getTransaction({
  hash: "0x123...",
});

console.log({
  from: transferTx.from,
  to: transferTx.to,
  value: transferTx.value,
  status: transferTx.status
});
```

Contract Interaction Transaction

```
const contractTx = await client.getTransaction({
  hash: "0x456...",
});

// Decode the transaction input data
const decodedInput = client.decodeTransactionInput({
  data: contractTx.data,
  abi: contractABI, // You need the contract's ABI
});

console.log({
  contractAddress: contractTx.to,
  functionName: decodedInput.functionName,
  arguments: decodedInput.args
});
```

Contract Deployment Transaction

```
const deployTx = await client.getTransaction({
  hash: "0x789...",
});

console.log({
  deployer: deployTx.from,
  contractAddress: deployTx.create2, // Address of deployed contract
  deploymentData: deployTx.data
});
```

Error Handling

```
async function getTransactionWithRetry(
  client: GenLayerClient,
```



```

hash: string,
maxAttempts = 3
): Promise<Transaction> {
  for (let attempt = 1; attempt <= maxAttempts; attempt++) {
    try {
      const tx = await client.getTransaction({ hash });
      if (!tx) throw new Error('Transaction not found');
      return tx;
    } catch (error) {
      if (attempt === maxAttempts) throw error;
      if (error.message.includes('not found')) {
        // Wait longer between retries for not found errors
        await new Promise(resolve => setTimeout(resolve, 2000 * attempt));
        continue;
      }
      throw error; // Rethrow other errors immediately
    }
  }
  throw new Error('Failed to fetch transaction after max retries');
}

```

Monitoring Transaction Status

```

async function monitorTransaction(
  client: GenLayerClient,
  hash: string,
  interval = 1000
): Promise<Transaction> {
  return new Promise((resolve, reject) => {
    const checkStatus = async () => {
      try {
        const tx = await client.getTransaction({ hash });

        if (!tx) {
          setTimeout(checkStatus, interval);
          return;
        }

        if (tx.status === 'pending') {
          setTimeout(checkStatus, interval);
          return;
        }

        resolve(tx);
      } catch (error) {
        reject(error);
      }
    };

    checkStatus();
  });
}

// Usage
const finalTx = await monitorTransaction(client, "0x...");

```

Reading from Intelligent Contracts

Source: <https://docs.genlayer.com/developers/decentralized-applications/reading-data>

Reading from an Intelligent Contract allows you to query the contract's state and execute view functions without modifying the blockchain state. These operations are free (no fees) and provide immediate access to contract data.

Understanding View Operations

In GenLayer, functions marked with the `@gl.public.view` decorator are read-only operations that:

- Don't modify the contract's state
- Can be executed without requiring a transaction
- Return data immediately
- Don't consume gas
- Can be called by any account

Basic Contract Reading

Here's how to read from an Intelligent Contract:

```
import { simulator } from 'genlayer-js/chains';
import { createClient, createAccount } from 'genlayer-js';

const account = createAccount();
const client = createClient({
  chain: simulator,
  account: account,
});

const result = await client.readContract({
  address: contractAddress,
  functionName: 'get_complete_storage',
  args: [],
});
```

Parameters Explained

- address: The deployed contract's address on the GenLayer network
- functionName: The name of the view function you want to call
- args: An array of arguments that the function accepts (empty if none required)

Common View Operations

Intelligent Contracts typically include several types of view functions:

State Queries

```
// Reading a single value
const balance = await client.readContract({
  address: contractAddress,
  functionName: 'get_balance',
  args: [accountAddress],
});

// Reading multiple values
const userInfo = await client.readContract({
  address: contractAddress,
  functionName: 'get_user_info',
  args: [userId],
});
```

Computed Values

```
// Getting calculated results
const totalSupply = await client.readContract({
  address: contractAddress,
  functionName: 'calculate_total_supply',
  args: [],
});
```

Validation Checks

```
// Checking permissions
const hasAccess = await client.readContract({
  address: contractAddress,
  functionName: 'check_user_permission',
  args: [userId, 'ADMIN_ROLE'],
});
```

Error Handling

When reading from contracts, you should handle potential errors:

```
try {
  const result = await client.readContract({
    address: contractAddress,
    functionName: 'get_data',
    args: [],
  });
  console.log('Data retrieved:', result);
} catch (error) {
  if (error.message.includes('Contract not found')) {
    console.error('Invalid contract address');
  } else if (error.message.includes('Function not found')) {
    console.error('Invalid function name');
  } else {
    console.error('Error reading contract:', error);
  }
}
```

Best Practices

1. Cache Results: For frequently accessed data that doesn't change often, consider caching the results.
2. Batch Readings: When possible, use functions that return multiple values instead of making multiple separate calls.
3. Type Safety: Use TypeScript interfaces to ensure type safety when handling returned data:

Writing to Intelligent Contracts

Source: <https://docs.genlayer.com/developers/decentralized-applications/writing-data>

Writing to an Intelligent Contract involves sending transactions that modify the contract's state. Unlike read operations, write operations require fees and need to be processed by the network before taking effect.

Understanding Write Operations

In GenLayer, functions that modify state:

- Require a transaction to be sent to the network
- Consume gas (computational resources)
- Need time to be processed and finalized
- Must be signed by an account with sufficient balance to pay for the transaction fees
- Return a transaction hash immediately, but state changes are not instant

Basic Contract Writing

Here's how to write to an Intelligent Contract:

```
import { simulator } from 'genlayer-js/chains';
import { createClient, createAccount } from 'genlayer-js';
import type { TransactionStatus } from 'genlayer-js/types';

const account = createAccount();
const client = createClient({
  chain: simulator,
  account: account,
});

// Send the transaction
const transactionHash = await client.writeContract({
  address: contractAddress,
  functionName: 'update_storage',
  args: ['new_data'],
  value: 0, // Optional: amount of native GEN tokens to send
});

// Wait for the transaction to be processed
const receipt = await client.waitForTransactionReceipt({
  hash: transactionHash,
  status: TransactionStatus.FINALIZED, // or 'ACCEPTED'
});
```

Parameters Explained

- address: The deployed contract's address
- functionName: The name of the function to call
- args: Array of arguments for the function
- value: Amount of native tokens to send (in wei)

Transaction Lifecycle

1. Transaction Creation

```
const transactionHash = await client.writeContract({
  address: contractAddress,
  functionName: 'mint_token',
  args: [recipient, amount],
});
```

2. Transaction Status Monitoring

```
// Basic waiting
const receipt = await client.waitForTransactionReceipt({
  hash: transactionHash,
  status: 'FINALIZED',
});

// Advanced monitoring with timeout
const receipt = await client.waitForTransactionReceipt({
  hash: transactionHash,
```

```

status: 'FINALIZED',
interval: 5_000, // check every 5 seconds
retries: 10, // maximum number of retries
});

```

Common Write Operations

Calling Payable Methods

Send GEN with a write call using the value parameter:

```

// Send 5 GEN to a payable method
const hash = await client.writeContract({
  address: contractAddress,
  functionName: 'tip',
  args: [],
  value: BigInt(5) * BigInt(10 ** 18), // 5 GEN in wei
});

```

The value is specified in wei (1 GEN = 10^{18} wei). The receiving contract method must be decorated with `@gl.public.write.payable`.

Token Transfers

```

// Sending tokens
const hash = await client.writeContract({
  address: tokenContractAddress,
  functionName: 'transfer',
  args: [recipientAddress, amount],
});

```

State Updates

```

// Updating user data
const hash = await client.writeContract({
  address: contractAddress,
  functionName: 'update_user_profile',
  args: [userId, newProfile],
});

```

Using a Browser Wallet (MetaMask)

When building a browser dApp, transactions are signed by the user's wallet instead of a local private key. You need to:

1. Create a client with the wallet's address and provider
2. Switch the wallet to the correct GenLayer network
3. Send transactions as usual

```

import { createClient } from "genlayer-js";
import { studionet } from "genlayer-js/chains";
import { TransactionStatus } from "genlayer-js/types";

// Create client with wallet address (from your wallet connection flow)
const client = createClient({
  chain: studionet,
  account: walletAddress as `0x${string}`,
});

// Switch the wallet to the correct network – must be called before writing
await client.connect("studionet");

```

```
// Now write as usual – MetaMask will prompt for signing
const txHash = await client.writeContract({
  address: contractAddress,
  functionName: "create_profile",
  args: ["alice", "Hello world"],
  value: BigInt(0),
});

const receipt = await client.waitForTransactionReceipt({
  hash: txHash,
  status: TransactionStatus.ACCEPTED,
});
```

Available networks: "localnet", "studionet", "testnetAsimov", "testnetBradbury".

Important: If the wallet is on the wrong chain, the SDK will throw an error telling you which chain the wallet is on vs. which chain the client expects. Always call `client.connect()` before sending transactions.

Error Handling

```
try {
  const hash = await client.writeContract({
    address: contractAddress,
    functionName: 'update_data',
    args: ['new_data'],
  });

  try {
    const receipt = await client.waitForTransactionReceipt({
      hash,
      status: 'FINALIZED',
    });
    console.log('Transaction successful:', receipt);
  } catch (waitError) {
    console.error('Transaction failed or timed out:', waitError);
  }
} catch (error) {
  if (error.message.includes('insufficient funds')) {
    console.error('Not enough balance to pay for transaction');
  } else if (error.message.includes('user rejected')) {
    console.error('User rejected the transaction');
  } else {
    console.error('Error sending transaction:', error);
  }
}
```

Transaction Status Types

GenLayer transactions can have different status requirements:

```
enum TransactionStatus {
  PENDING = "PENDING",
  CANCELED = "CANCELED",
  PROPOSING = "PROPOSING",
  COMMITTING = "COMMITTING",
  REVEALING = "REVEALING",
  ACCEPTED = "ACCEPTED",
  FINALIZED = "FINALIZED",
  UNDETERMINED = "UNDETERMINED",
}

// Wait for just acceptance (faster)
const acceptedReceipt = await client.waitForTransactionReceipt({
  hash: transactionHash,
  status: TransactionStatus.ACCEPTED,
});

// Wait for full finalization
const finalizedReceipt = await client.waitForTransactionReceipt({
```

```
    hash: transactionHash,  
    status: TransactionStatus.FINALIZED,  
  });
```

Best Practices

1. Always Wait for Receipts: Don't assume a transaction is successful just because you got a hash.
2. Handle Timeouts: Set appropriate timeouts for transaction waiting.
3. Implement Retry Logic: For important transactions, implement retry mechanisms:

```
async function sendWithRetry(  
  client: GenLayerClient,  
  params: WriteContractParameters,  
  maxAttempts = 3  
) : Promise<TransactionReceipt> {  
  for (let attempt = 1; attempt <= maxAttempts; attempt++) {  
    try {  
      const hash = await client.writeContract(params);  
      return await client.waitForTransactionReceipt({  
        hash,  
        status: 'FINALIZED',  
        timeout: 30_000 * attempt, // Increase timeout with each attempt  
      });  
    } catch (error) {  
      if (attempt === maxAttempts) throw error;  
      console.log(`Attempt ${attempt} failed, retrying...`);  
      await new Promise(resolve => setTimeout(resolve, 1000 * attempt));  
    }  
  }  
  throw new Error('Max retry attempts reached');  
}
```

Testing Intelligent Contracts on GenLayer

Source: <https://docs.genlayer.com/developers/decentralized-applications/testing>

Testing Intelligent Contracts on GenLayer involves deploying contracts, sending transactions, validating their behavior, and identifying issues. Here is some guidance on how to test using the tools provided in the local development environment and GenLayer Studio.

Testing Workflow in the Local Environment

For a more advanced and controlled testing setup, you can leverage the GenLayer Project Boilerplate and the provided test helpers.

Note:

Please note that you need the Studio running to run the tests by sending requests to it.

1. Create Accounts

Use the `create_new_account` helper to generate accounts for testing. These accounts simulate users interacting with the contract.

```
from tools.request import create_new_account

account = create_new_account()
```

2. Set Up Validators

Validators are essential for processing transactions in GenLayer. Use the `sim_createRandomValidators` method to initialize them.

```
from tools.request import payload, post_request_localhost

nb_validators = 5
min_stake = 8
max_stake = 12
providers = ["openai"]
models = ["gpt-4o"]
validators_response = post_request_localhost(
    payload("sim_createRandomValidators", nb_validators, min_stake, max_stake, providers, models)
).json()
```

3. Deploy the Contract

Deploy your Intelligent Contract using the `deploy_intelligent_contract` helper:

```
from tools.request import deploy_intelligent_contract

contract_code = open("contracts/my_contract.py", "r").read()
contract_address, deploy_response = deploy_intelligent_contract(account, contract_code, "{}")
```

4. Interact with the Contract

Use `send_transaction` for writing data to the contract and `call_contract_method` for reading data:

```
from tools.request import send_transaction, call_contract_method

# Write data
send_transaction(account, contract_address, "method_name", [arg1, arg2])

# Read data
result = call_contract_method(contract_address, account, "get_state", [])
```

5. Assertions

Validate the responses using provided assertion helpers:

```
from tools.response import assert_dict_struct, has_success_status

assert has_success_status(result)
assert_dict_struct(result, expected_structure)
```

6. Clean Up

After completing the tests, delete the validators or reset the environment:

```
delete_response = post_request_localhost(payload("sim_deleteAllValidators")).json()
```

GenLayer Project Boilerplate

Source: <https://docs.genlayer.com/developers/decentralized-applications/project-boilerplate>

The GenLayer Project Boilerplate is a template for building decentralized applications (DApps) on GenLayer. This boilerplate includes a complete example implementation of a football prediction market, demonstrating best practices and common patterns for GenLayer development.

Note:

This boilerplate is a work in progress and is not yet ready for production use. You can find the latest version at: GenLayer Project Boilerplate (<https://github.com/genlayerlabs/genlayer-project-boilerplate>)

Features

- Contract Templates: Ready-to-use intelligent contract templates and examples
- Testing Framework: Built-in testing infrastructure for end-to-end testing
- Frontend Integration: Vue.js-based frontend setup with GenLayerJS integration
- Environment Configuration: Pre-configured environment setup for development
- Example Implementation: Full football prediction market implementation

How it's Built

The boilerplate project is structured to provide a development environment for GenLayer applications, combining both backend contract development and frontend user interface.

Technologies Used

- Python: For intelligent contract development and testing
- Vue.js: Frontend framework for building user interfaces
- GenLayerJS: SDK for interacting with GenLayer contracts
- pytest: Testing framework for contract validation
- Vite: Frontend build tool and development server

Project Structure

```
project-root/
├── app/                # Frontend application
│   ├── src/           # Vue.js source files
│   └── .env.example    # Frontend environment variables template
├── contracts/         # Intelligent Contracts
│   └── football_prediction_market.py
├── test/              # Test files
└── .env.example       # Main environment variables template
```

Requirements

Before using the GenLayer Project Boilerplate, ensure your system meets the following requirements:

- GenLayer Studio: Running instance required

- Node.js: Version 18.x or higher
- Python: Version 3.8 or higher

Installation

1. Clone the boilerplate repository:

```
git clone https://github.com/genlayerlabs/genlayer-project-boilerplate
cd genlayer-project-boilerplate
```

2. Set up the environment:

```
cp .env.example .env
```

Usage

Deploying Contracts

1. Access the GenLayer Simulator UI (default: <http://localhost:8080>)
2. Navigate to "Contracts" section
3. Create a new contract using `/contracts/football_prediction_market.py`
4. Deploy through the "Run and Debug" interface

Frontend Setup

1. Configure the frontend environment:

```
cd app
cp .env.example .env
# Add your contract address to VITE_CONTRACT_ADDRESS
```

2. Install dependencies and start the development server:

```
npm install
npm run dev
```

Running Tests

Execute the test suite using pytest:

```
pip install -r requirements.txt
pytest test
```

Football Prediction Market Example

The included example contract demonstrates a complete prediction market implementation with the following features:

Contract Functions

Create Predictions:

```
python
```

```
create_prediction(game_date: str, team1: str, team2: str, predicted_winner: str)
```

Resolve Predictions:

python

```
resolve_prediction(prediction_id: str)
```

Query Data:

python

```
get_player_predictions(player_address: str)
```

```
get_player_points(player_address: str)
```

Frontend Integration

The Vue.js frontend demonstrates:

- Wallet connection handling
- Contract interaction using GenLayerJS
- User interface for prediction creation and management
- Real-time updates for prediction status

Testing Framework

The boilerplate includes a comprehensive testing suite covering the following scenarios:

Contract Schema

The schema of the contract is well generated and has the expected methods and variables.

Test Scenarios

Successful Draw Prediction

Tests the scenario where a player correctly predicts a draw between two teams. When the match is resolved as a draw, the player should receive 1 point for their accurate prediction.

Successful Winner Prediction

Validates the case where a player correctly predicts the winning team. Upon match resolution confirming their predicted team as the winner, the player should be awarded 1 point.

Unsuccessful Prediction

Covers the scenario where a player's prediction doesn't match the actual result. This could be:

- Predicting Team A wins, but Team B wins
- Predicting a draw, but there's a winner
- Predicting a winner, but the match ends in a draw

In all these cases, the player should receive 0 points.

Best Practices

- Always use environment variables for configuration
- Implement comprehensive testing for all contract functions
- Follow the provided folder structure for consistency
- Use TypeScript for frontend development
- Implement proper error handling in both contract and frontend code

Staking Contract Guide

Source: <https://docs.genlayer.com/developers/staking-guide>

This guide covers consensus-level staking operations for developers who want to interact directly with the GenLayer staking smart contracts. It includes Solidity code examples for both validator and delegator operations.

To understand the concepts behind staking—including the epoch system, shares vs stake, validator selection weights, and reward distribution—see [Staking in GenLayer \(/understand-genlayer-protocol/core-concepts/optimistic-democracy/staking\)](#).

Note:

For CLI-based staking operations (recommended for most users), see the [GenLayer CLI Staking Commands \(/api-references/genlayer-cli#staking-operations-testnet\)](#).

Validator Operations

Joining as a Validator

Requirements:

- Minimum stake: 42,000 GEN tokens
- Must run a validator node for consensus participation
- A single address can create multiple validators

Method 1: Owner as Operator

```
address validatorWallet = genStaking.validatorJoin(value: 42000 ether)();
```

Method 2: Separate Operator Address (Recommended)

```
address operatorAddress = 0x...;  
address validatorWallet = genStaking.validatorJoin(value: 42000 ether)(operatorAddress);
```

Note:

Save the returned ValidatorWallet address - you'll need it for all future operations.

Important Notes:

- Deposits activate 2 epochs after joining
- validatorPrime() is automatically called by your validator node each epoch
- Monitor priming for proper activation within 2 epochs
- Operator address cannot be reused or be the zero address
- If the operator is already assigned to another validator, the transaction reverts

Post-Join Steps:

1. Save the validatorWallet address
2. Start your validator node using the operator address
3. Monitor that validatorPrime(validatorWallet) executes each epoch
4. Wait 2 epochs and verify activation

Depositing as a Validator

```
validatorWallet.validatorDeposit{value: 10000 ether}();
```

Key Points:

- Call through the ValidatorWallet contract only
- Deposits follow the epoch +2 activation rule
- Your validator node automatically calls validatorPrime() to commit the deposit
- No minimum for additional deposits
- Cannot deposit zero value

Activation Timeline:

```
Epoch N:   validatorDeposit() called  
Epoch N+1: validatorPrime() → deposit staged  
Epoch N+2: validatorPrime() → deposit activated
```

Withdrawing as a Validator

Step 1: Calculate Shares

```
uint256[8] memory validatorState = genStaking.validatorView(validatorWallet);  
uint256 currentShares = validatorState[1];  
uint256 currentStake = validatorState[0];  
uint256 sharesToExit = (currentShares * 30) / 100; // Exit 30%
```

Step 2: Initiate Exit

```
validatorWallet.validatorExit(sharesToExit);
```

Note:

You must specify shares, not token amounts. Cannot exit more shares than owned.

Important Notes:

- Exited tokens stop earning rewards immediately

- Tokens are locked for 7 epochs before claiming
- Each exit creates a separate withdrawal

Unbonding Timeline:

```
Epoch 10:    validatorExit() called → unbonding starts
Epochs 11-16: Locked period
Epoch 17:    validatorClaim() available
```

Claiming Withdrawn Funds as a Validator

```
uint256 tokensReceived = genStaking.validatorClaim(validatorWallet);
```

Important Notes:

- Must wait 7 full epochs after calling validatorExit()
- Claims are permissionless - anyone can trigger them
- Tokens are sent to the validator wallet's owner address
- Multiple exits can be claimed in a single call
- Returns the total GEN received

Checking Claim Availability:

```
uint256 exitEpoch = 10;
uint256 claimableAtEpoch = exitEpoch + 7;
uint256 currentEpoch = genStaking.epoch();
if (currentEpoch >= claimableAtEpoch) {
    genStaking.validatorClaim(validatorWallet);
}
```

Delegator Operations

Joining as a Delegator

Requirements:

- Minimum delegation: 42 GEN per validator
- Target validator must exist and be active
- Can delegate to multiple validators

Basic Delegation:

```
address validatorWallet = 0x...;
genStaking.delegatorJoin{value: 1000 ether}(validatorWallet);
```

Multi-Validator Delegation:

```
genStaking.delegatorJoin{value: 500 ether}(validator1);
genStaking.delegatorJoin{value: 500 ether}(validator2);
genStaking.delegatorJoin{value: 500 ether}(validator3);
```

Important Notes:

- Follows the epoch +2 activation rule
- The validator's node automatically calls validatorPrime() to commit your delegation

- Choose validators with good uptime
- Smaller validators often provide higher returns
- Cannot delegate zero value

Delegation Strategy Tips:

- Diversify across multiple validators
- Check validator performance and uptime
- Monitor validator quarantine/ban status

Depositing as a Delegator

```
genStaking.delegatorDeposit{value: 500 ether}(validatorWallet);
```

Important Notes:

- Use delegatorDeposit() to add to existing delegations
- Follows the epoch +2 activation rule
- The validator's node automatically calls validatorPrime()
- No minimum for additional deposits
- Rewards automatically compound

Withdrawing as a Delegator

Step 1: Calculate Shares

```
uint256 currentShares = genStaking.sharesOf(msg.sender, validatorWallet);
uint256 sharesToExit = (currentShares * 50) / 100; // Exit 50%
```

Step 2: Initiate Exit

```
genStaking.delegatorExit(validatorWallet, sharesToExit);
```

Note:

You must specify shares, not token amounts.

Important Notes:

- Can exit partial amounts
- Stop earning rewards immediately after exit
- Tokens are locked for 7 epochs
- Exit each validator separately if delegating to multiple

Unbonding Timeline:

```
Epoch 5:    delegatorExit() called
Epochs 6-11: Locked period
Epoch 12:   delegatorClaim() available
```

Claiming Withdrawn Funds as a Delegator

```
uint256 tokensReceived = genStaking.delegatorClaim(
    delegatorAddress,
    validatorWallet
```

```
);
```

Important Notes:

- Must wait 7 full epochs after calling delegatorExit()
- Claims are permissionless - anyone can trigger them
- Tokens are sent to the delegator's address
- Claim from each validator separately if delegated to multiple

Multi-Validator Claim Example:

```
genStaking.delegatorClaim(msg.sender, validator1);  
genStaking.delegatorClaim(msg.sender, validator2);  
genStaking.delegatorClaim(msg.sender, validator3);
```

Genesis Epoch 0

Epoch 0 is the genesis bootstrapping period with special rules designed to facilitate network launch. The normal staking rules are relaxed to allow rapid network bootstrapping.

What is Epoch 0?

Epoch 0 is the bootstrapping period before the network becomes operational. During epoch 0:

- No transactions are processed - the network is not yet active
- No consensus occurs - validators are not yet participating
- Stakes are registered and prepared for activation in epoch 2

Important: The network transitions directly from epoch 0 to epoch 2 (epoch 1 is skipped). Validators and delegators who stake in epoch 0 become active in epoch 2.

Special Rules for Epoch 0

Rule	Normal Epochs (2+)	Epoch 0
Validator minimum stake	42,000 GEN	No minimum to join
Delegator minimum stake	42 GEN	No minimum to join
Activation delay	+2 epochs	Active in epoch 2
validatorPrime required	Yes, each epoch	Not required
Share calculation	Based on existing ratio	1:1 (shares = input)
Transaction processing	Yes	No (bootstrapping only)

Note:

Activation requires meeting minimums: While you can join with any amount during epoch 0, your stake will only be activated in epoch 2 if it meets the minimum requirements (42,000 GEN for validators, 42 GEN for delegators). Stakes below the minimum remain registered but inactive.

Validators in Epoch 0

Joining:


```
// In epoch 0, any amount is accepted (no 42,000 GEN minimum)
address validatorWallet = genStaking.validatorJoin{value: 1000 ether}(operatorAddress);

// Stake is registered for activation in epoch 2
// No need to call validatorPrime() in epoch 0
```

Key behaviors:

1. No minimum stake: Validators can join with any non-zero amount
2. Registered for epoch 2: Stakes are recorded and will become active when epoch 2 begins
3. No priming required: validatorPrime() is not needed during epoch 0
4. No consensus participation: Validators do not process transactions in epoch 0

Do validators need to take any action in epoch 0 to be active in epoch 2?

No. Validators who join in epoch 0:

- Have their stake registered during epoch 0
- Become active automatically in epoch 2 (epoch 1 is skipped) only if they have at least 42,000 GEN staked
- Must start calling validatorPrime() in epoch 2 for continued participation in epoch 4+

Note:

Critical: Validators who joined in epoch 0 with less than 42,000 GEN will not be active in epoch 2. They must deposit additional funds to meet the minimum requirement before epoch 2 begins.

Delegators in Epoch 0

Joining:

```
// In epoch 0, any amount is accepted (no 42 GEN minimum)
genStaking.delegatorJoin{value: 10 ether}(validatorWallet);

// Delegation is registered for activation in epoch 2
```

Key behaviors:

1. No minimum delegation: Any non-zero amount accepted
2. Registered for epoch 2: Delegation is recorded and will become active when epoch 2 begins
3. No rewards in epoch 0: Since no transactions are processed, no rewards are earned during epoch 0

Is a delegation made in epoch 0 active in epoch 2?

Yes. Delegations made in epoch 0 become active in epoch 2 (epoch 1 is skipped). Unlike normal epochs where you wait +2 epochs, epoch 0 delegations activate as soon as the network becomes operational.

Activation Timeline Comparison

Normal Epochs (2+):

```
Epoch N: validatorJoin() or delegatorJoin() called
Epoch N+1: validatorPrime() stages the deposit
Epoch N+2: validatorPrime() activates the deposit → NOW ACTIVE
```

Epoch 0 (Bootstrapping):

```
Epoch 0: validatorJoin() or delegatorJoin() called → stake registered (not yet active)
         No transactions processed, no consensus
Epoch 2: Stakes become active (if minimum met), network operational, validatorPrime() required
```

Share Calculation in Epoch 0

In epoch 0, shares are calculated at a 1:1 ratio with the input amount:

```
// Epoch 0: 1,000 GEN deposit = 1,000 shares (1:1 ratio)
genStaking.delegatorJoin{value: 1000 ether}(validatorWallet);
// Result: delegator receives exactly 1,000 shares

// Normal epochs: shares depend on pool ratio
// new_shares = deposit_amount × (total_shares / total_stake)
```

This is because there's no existing stake pool to calculate a ratio against. Starting from epoch 2, shares are calculated based on the current stake-to-share ratio.

Transitioning from Epoch 0 to Epoch 2

When the network advances from epoch 0 to epoch 2 (epoch 1 is skipped):

1. Epoch 0 stakes that meet minimums become active - validators need 42,000 GEN, delegators need 42 GEN
2. Normal minimum requirements apply for new joins/deposits
3. +2 epoch activation delay applies to all new deposits
4. validatorPrime() becomes mandatory for validators to remain in the selection pool
5. Existing validators must ensure their nodes begin calling validatorPrime() in epoch 2

FAQ: Epoch 0 Special Cases

Q: Can I join as a validator with less than 42,000 GEN in epoch 0?

A: Yes, any non-zero amount is accepted during epoch 0. However, you will not be active in epoch 2 unless you have at least 42,000 GEN staked by then.

Q: If I delegate in epoch 0, when does it become active?

A: In epoch 2. Unlike normal epochs with a +2 delay, epoch 0 delegations activate when the network becomes operational.

Q: Do I need to call validatorPrime() in epoch 0?

A: No. Priming is not required during epoch 0. Your node should start calling it automatically when epoch 2 begins (epoch 1 is skipped).

Q: Will my epoch 0 stake still be active after epoch 0 ends?

A: Yes, if you meet the minimum requirements. Stakes from epoch 0 carry forward and remain active in all subsequent epochs.

Q: What happens to my stake if I joined in epoch 0 but my node doesn't call validatorPrime() in epoch 2?

A: You'll be excluded from validator selection in epoch 4, but your stake remains. Once priming resumes, you'll be eligible for selection again.

Common Scenarios

Scenario 1: Becoming a Validator

```
address validatorWallet = genStaking.validatorJoin{value: 42000 ether}(operatorAddress);  
// Start validator node with operator address  
// Monitor: Epoch N joined, N+1 staged, N+2 active
```

Scenario 2: Delegating to Multiple Validators

```
genStaking.delegatorJoin{value: 1000 ether}(validator1);  
genStaking.delegatorJoin{value: 2000 ether}(validator2);  
genStaking.delegatorJoin{value: 1500 ether}(validator3);  
  
// Later, exit from one validator  
uint256 shares = genStaking.sharesOf(msg.sender, validator1);  
genStaking.delegatorExit(validator1, shares);  
  
// After 7 epochs:  
genStaking.delegatorClaim(msg.sender, validator1);
```

Scenario 3: Partial Validator Exit

```
uint256[8] memory state = genStaking.validatorView(validatorWallet);  
uint256 currentShares = state[1];  
uint256 sharesToExit = (currentShares * 25) / 100; // Exit 25%  
validatorWallet.validatorExit(sharesToExit);  
  
// After 7 epochs:  
genStaking.validatorClaim(validatorWallet);
```

Best Practices

For Validators

1. Ensure node health: Your validator node automatically calls validatorPrime()
2. Use separate keys: Cold wallet for owner, hot wallet for operator
3. Monitor priming: Verify deposit activation occurs as expected
4. Monitor quarantine status: Check if your validator is quarantined
5. Maintain high uptime: Downtime leads to idleness strikes
6. Set up backup monitoring: Alert if priming fails

For Delegators

1. Diversify: Spread stake across multiple validators
2. Select carefully: Choose validators with good uptime records
3. Verify priming: Ensure validators are consistently primed
4. Consider smaller validators: They often provide higher returns

5. Monitor health: Regularly check validator status
6. Use shares for exits: Calculate percentages, not absolute amounts

For Both

1. Account for unbonding: Plan for the 7-epoch wait when exiting
2. Monitor epochs: Track current epoch for claim eligibility
3. Calculate exits carefully: Always use share percentages
4. Understand compounding: Rewards automatically increase stake-per-share
5. Watch governance: Parameter changes can affect staking dynamics

Troubleshooting

Error	Cause	Solution
ValidatorBelowMinimumStake	Stake below 42,000 GEN	Deposit at least 42,000 GEN when joining
OperatorAlreadyAssigned	Operator address in use	Choose a different operator address
ValidatorMayNotDepositZeroValue	Zero deposit amount	Deposit a non-zero amount
ValidatorExitExceedsShares	Exiting more shares than owned	Check shares with <code>validatorView()</code>
Validator not selected	Node not priming	Ensure node runs and calls <code>validatorPrime()</code>
Stake not activating	Priming not occurring	Monitor node for automatic <code>validatorPrime()</code> calls
Cannot claim withdrawn funds	Unbonding not complete	Verify 7 epochs have passed with <code>genStaking.epoch()</code>
Delegation not earning	Validator issues	Check if validator is quarantined/banned; consider switching

Related Resources

- Staking Concepts (</understand-genlayer-protocol/core-concepts/optimistic-democracy/staking>) - How staking works in GenLayer
- Unstaking (</understand-genlayer-protocol/core-concepts/optimistic-democracy/unstaking>) - Unstaking process details
- Slashing (</understand-genlayer-protocol/core-concepts/optimistic-democracy/slashing>) - Slashing penalties and conditions
- GenLayer CLI Staking Commands (</api-references/genlayer-cli#staking-operations-testnet>) - CLI reference for staking

Setting Up Your GenLayer Validator

Source: <https://docs.genlayer.com/validators/setup-guide>

Running a GenLayer Validator node ensures the security and reliability of the GenLayer network. Below are the requirements for running a GenLayer validator.

Validators require both a staking deposit and a validator priming transaction to enter the set.

For a deeper understanding of how staking works in GenLayer, see the Staking documentation (/understand-genlayer-protocol/core-concepts/optimistic-democracy/staking).

System Requirements

Resource	Requirement
Memory	16 GB RAM
CPU	8-core (AMD64 only)
Disk	128 GB SSD/NVMe
Network	100 Mbps
OS	64-bit Linux
Software	Docker, Python 3 (with pip and venv)

GPU not required unless running LLMs locally. See detailed requirements (/validators/system-requirements) for more info.

Choose Your Setup Method

GenLayer offers two ways to set up your validator node. Both methods result in the same configuration—choose the one that fits your workflow.

Method	Best For
AI-Assisted Setup	New validators, quick setup, guided troubleshooting
Manual Setup	Experienced operators, custom configurations, automation pipelines

Option 1: AI-Assisted Setup (Recommended)

We've released an interactive AI wizard that guides you through the entire validator setup process step-by-step. Instead of reading through documentation and running commands manually, the wizard handles everything interactively.

What it does

- Walks you through all setup steps interactively
- Checks prerequisites automatically (architecture, RAM, Docker, Node.js, Python)
- Detects common issues before they cause problems (port conflicts, disk space, wrong architecture)
- Generates configurations based on your answers
- Supports multiple deployment options: local machine, remote SSH, GCP, AWS, Azure
- Handles both new validators (staking wizard) and existing validators
- Includes minimal-downtime update procedure for upgrades

How to use it

1. Install Claude Code (<https://claude.ai/code>) (Anthropic's CLI tool)
2. Add the GenLayer plugin marketplace and install the plugin:

```
bash
```

```
claude /plugin marketplace add genlayerlabs/skills
```

```
claude /plugin install genlayernode@genlayerlabs
```

3. Run the wizard:

```
bash
claude /genlayernode install
```

That's it. The wizard will ask questions and guide you through everything.

Available commands

Command	Description
<code>`/genlayernode install`</code>	Fresh validator setup
<code>`/genlayernode update`</code>	Upgrade existing node
<code>`/genlayernode configure grafana`</code>	Set up monitoring

Note:

The AI wizard uses the same underlying steps as the manual setup below. You can switch between methods at any point or use the manual guide as a reference.

Browse all available skills at skills.genlayer.com (<https://skills.genlayer.com/>) — the marketplace also includes the `genlayer-dev` skill for contract development.

Option 2: Manual Setup

If you prefer to run commands manually, need fine-grained control over each step, or are integrating with existing automation pipelines, follow the detailed instructions in the sections below.

The manual setup covers:

1. Create the Validator Wallet — Set up your wallet and stake GEN tokens
2. Set Up the Validator Node — Download, configure, and run the node software

Create the Validator Wallet

Before setting up your validator node, you need to create your validator wallet and stake GEN tokens.

Understanding Validator Addresses

Note:

GenLayer validators use three distinct addresses:

Address	Description	Where Used
<code>**Owner**</code>	The only address that can withdraw staked funds. Keep this secure (cold wallet).	CLI wizard - signs staking transaction
<code>**Operator**</code>	Hot wallet on your server that signs blocks. Can be same as owner, but separate is recommended.	Node config: <code>`operator`</code>
<code>**Validator Wallet**</code>	Smart contract created when you join. This is your validator's on-chain identity.	Node config: <code>`validator`</code>

The wizard outputs all three addresses at the end. Save them - you'll need the Validator Wallet and Operator addresses for your node configuration.

Prerequisites

- Node.js (v18 or higher)
- GenLayer CLI - Install the latest version:

```
bash
```

```
npm install -g genlayer
```

- GEN tokens - You need at least 42,000 GEN for the minimum self-stake requirement

Using the Validator Wizard

The easiest way to create your validator wallet is using the interactive wizard:

```
genlayer staking wizard
```

The wizard guides you through:

1. Account setup - Create or select your owner account
2. Network selection - Choose testnet-asimov
3. Balance verification - Confirms you have at least 42,000 GEN
4. Operator setup - Create and export an operator keystore for your validator server
5. Stake amount - Enter how much GEN to stake (minimum 42,000)
6. Validator creation - Submits the staking transaction
7. Identity setup - Set your validator's public profile (moniker, website, etc.)

Note:

Save Your Validator Wallet Address! After the wizard completes, note down your Validator Wallet address. You'll need this for your node configuration.

Operator Keystore Export

The wizard creates and exports an operator keystore file for you to transfer to your validator server. If your server is compromised, your staked funds (controlled by owner) remain safe.

Verify Your Validator

After completing the wizard, verify your status:

```
genlayer staking validator-info --validator 0xYourValidatorWallet...
```

Managing Your Validator

```
# Add more stake
genlayer staking validator-deposit --amount 1000gen

# Check active validators
genlayer staking active-validators
```

```
# Exit (initiates 7-epoch unbonding)
genlayer staking validator-exit --shares 100

# Claim after unbonding period
genlayer staking validator-claim

# Update identity
genlayer staking set-identity --validator 0x... --moniker "New Name"
```

Set Up the Validator Node

Once your validator wallet is created and staked, set up your node software.

Dependencies

LLM Access

Each validator needs access to a Large Language Model (LLM) for executing and evaluating Intelligent Contracts.

It is up to each validator to select the model they want to use.

Possible options:

- Run an open-source model locally on the same machine with a GPU
- Run an open-source model on a different machine
- Connect to a hosted inference provider (OpenAI, Anthropic, Heurist, Atoma network etc.)

See the GenVM Configuration (/validators/genvm-configuration) page for partner credits, provider setup, and the greybox fallback strategy.

ZKSync Full Node for the GenLayer Chain

Each validator needs access to a ZKSync Full Node (<https://docs.zksync.io/zksync-era/tooling/external-node>) connected to the GenLayer chain. Validators use this node to read the chain state and submit transactions.

One full node can be shared between multiple validators. The optimal validator-to-node ratio is currently under evaluation.

Setup

Download the node software

1. Select the version of the node you want to run by checking the available builds

You can use this command to list available versions:

```
sh
```



```
curl -s "https://storage.googleapis.com/storage/v1/b/gh-af/o?prefix=genlayer-node/bin/amd64" | grep -o '"name": *"[^"]*" | sed -n 's/.*\\(v[^/]*\\)\\./.*\\1/p' | sort -ru | grep -v "rc" | head -n 5
```

You should see a list like this

```
sh
v0.5.12
v0.5.11
v0.5.10
v0.5.9
v0.5.8
```

Typically, you will want to run the latest version

2. Download the packaged application

```
sh
export version=v0.5.12 # set your desired version here
wget https://storage.googleapis.com/gh-af/genlayer-node/bin/amd64/${version}/genlayer-node-linux-amd64-${version}.tar.gz
```

3. Extract the node software

```
sh
mkdir -p ${version}
tar -xzf genlayer-node-linux-amd64-${version}.tar.gz -C ./${version}
```

4. Change the directory

```
sh
cd ./${version}
```

5. Run Genvm setup

```
sh
python3 ./third_party/genvm/bin/setup.py
```

It does following:

- Downloads missing runners if any
- Verifies runners hashes
- Sets elf/mach-o interpreter for all binaries (modules, genvm)
- Rust precompilation for built-in runners

Configuration

Before you can start up the node, you need to configure it.

config.yaml

This is the main configuration file of your node. Without it, your node won't start

The file needs to be located at configs/node/config.yaml

You can use the following example configuration. Note: For most users, you will only need to modify the `genlayerchainrpcurl`, `genlayerchainwebsocketurl`, and the consensus section for your target network (see [Network-Specific Consensus Configuration \(#network-specific-consensus-configuration\)](#) below).

```
# rollup configuration
rollup:
  genlayerchainrpcurl: "TODO: Set your GenLayer Chain ZKSync HTTP RPC URL here" # GenLayer Chain RPC URL
  genlayerchainwebsocketurl: "TODO: Set your GenLayer Chain ZKSync WebSocket RPC URL here" # GenLayer Chain WebSocket URL
  provider: "TODO: Set your GenLayer Chain ZKSync provider" # Rollup operator name; surfaces as the "provider" label on RPC-driven metrics. Set explicitly per deployment.
# consensus contracts configuration
consensus:
  # Testnet - Phase 5
  consensusaddress: "0xe66B434bc83805f380509642429eC8e43AE9874a" # AddressManager Smart Contract Address
  genesis: 17326 # (Optional) Genesis block number for this consensus deployment. If not provided, it will be auto-detected by searching for the first log from the ConsensusMain contract.
  # chain_submit_buffer: "30s" # (Optional) Slack reserved at the end of each consensus step to submit the resulting on-chain tx (commit vote, reveal vote, proposed receipt) before the slot closes. Tune to underlying settlement chain latency. Default 30s.
# data directory
datadir: "./data/node"
# logging configuration
logging:
  level: "INFO"
  # json: `true` for json output to console, false for human readable log formatting
  json: false
  # Configuration for https://github.com/natefinch/lumberjack
  file:
    # enabled: set to `true` to save logs to a folder
    enabled: true
    level: "DEBUG"
    # folder: path to the folder where to store logs. Relative paths go under `datadir`.
    folder: logs
    # maxsize: maximum size in megabytes of the log file before it gets rotated.
    maxsize: 10
    # maxage: maximum number of days to retain old log files based on the timestamp encoded in their filename.
    maxage: 7
    # maxbackups: maximum number of old log files to retain. Set to 0 for no limit
    maxbackups: 100
    # localtime: determines if the time used for formatting the timestamps in backup files is the computer's local time. Set to `false` to use UTC time.
    localtime: false
    # compress: determines if the rotated log files should be compressed using gzip
    compress: true
# node configuration
node:
  # Uncomment if the ID of the node is different from the validator wallet address.
  # It is used to identify the node in the network.
  # id: "node"
  # Mode can be "validator" or "full".
  # Default value is "validator".
  mode: "validator"
  # Address of the ValidatorWallet contract (required for validator mode)
  validatorWalletAddress: ""
  # Address of the operator that owns the ValidatorWallet
  operatorAddress: ""
  admin:
    port: 9155
  rpc:
    # host: 0.0.0.0 # bind address for the HTTP listener (default: all interfaces)
    port: 9151 # RPC server port
    # allowed_origins: unified browser-origin allowlist applied to BOTH
    # the HTTP transport (CORS) and the WS upgrade (gorilla CheckOrigin).
    # Exact-match URLs only – no wildcard syntax beyond "*". Omit or
    # leave empty to keep the permissive default ["*"]. Operators can
    # override per-transport for WS via rpc.websocket.origins (rare).
```

```

# allowed_origins:
#   - "https://wallet.example.com"
#   - "https://app.example.com"
# websocket: inbound JSON-RPC WS endpoint for eth_subscribe /
# zks subscribe plus all non-subscribe passthrough. Omit this whole
# block (the default) to keep WS disabled.
#
# When enabled, the default is single-listener mode: WS upgrades are
# served on the existing RPC listener (port above) – one port, one
# firewall entry, one reverse-proxy upstream. To run a dedicated WS
# listener instead (Geth-conventional dual-port pattern), set `port`
# to a value other than 0 and other than rpc.port (e.g. 9152) and
# set `host` for that listener.
# websocket:
#   enabled: true
#   # port: 0           # 0 or unset → single-listener on rpc.port (default)
#   # host: 0.0.0.0     # only used in dual-listener mode
#   path: "/ws"         # ws upgrade path – disambiguates HTTP RPC (POST /)
#                       # from WS upgrade (GET /ws) at the URL level.
#                       # Set to "/" for legacy behaviour (root path).
#   origins:           # WS-only override of rpc.allowed_origins (rare;
#                       # leave unset to inherit the unified allowlist)
#   - "*"              # permissive default – tighten via rpc.allowed_origins
#   max_conns: 1024
#   max_conns_per_ip: 16
#   max_subs_per_conn: 32
#   outbox_size: 256    # per-connection bounded queue; full → connection dropped
#   write_timeout: 10s
#   ping_interval: 30s
#   max_message_bytes: 10485760 # 10 MB cap on a single inbound frame (parity with rpc.max_request_body_bytes)
# ratelimit_whitelist: IPs or CIDRs that bypass rate limiting entirely
# ratelimit_whitelist:
#   - "127.0.0.1"
#   - "::1"           # IPv6 loopback (localhost on macOS)
#   - "10.0.0.0/8"
endpoints:
# Group-level configuration (enables/disables all methods in a group).
# When a group is enabled, ALL methods in that group are available.
# Rate limits (req/s per IP) apply by default even without config:
#   genlayer: 20, genlayer_debug: 2, ethereum: 50, zksync: 50
#   gen_call: 2, gen_getContractSchema: 2 (method-level overrides)
groups:
  genlayer: true # gen_* methods (default ratelimit: 20 req/s per IP)
  genlayer_debug: true # gen_dbg_* methods (default ratelimit: 2 req/s per IP)
  ethereum: true # eth_* proxy methods (default ratelimit: 50 req/s per IP)
  # includes: eth_blockNumber, eth_getBlockByNumber, eth_getBlockByHash,
  #   eth_sendRawTransaction, eth_getTransactionReceipt, etc.
  zksync: true # zks_* proxy methods (default ratelimit: 50 req/s per IP)
  # includes: zks_getTransaction, zks_getBlockDetails, etc.
# Object format with custom rate limit (overrides defaults):
# groups:
#   genlayer:
#     enabled: true
#     ratelimit: 100 # custom: 100 req/s per IP instead of default 20
#
# Method-level configuration (optional, overrides group settings).
# Methods inherit their group's enabled state by default.
# Use this section to:
#   - Disable specific methods in an enabled group
#   - Enable specific methods in a disabled group
#   - Set per-method rate limits
#
# Example: enable all eth_* except eth_sendRawTransaction:
# groups:
#   ethereum: true
# methods:
#   eth_sendRawTransaction: false
#
# Example: disable all eth_* except eth_sendRawTransaction:
# groups:
#   ethereum: false
# methods:
#   eth_sendRawTransaction: true
# methods:
#   gen_call: true # default ratelimit: 2 req/s per IP (GenVM-bound)
#   gen_getContractSchema: true # default ratelimit: 2 req/s per IP (GenVM-bound)
#   eth_blockNumber: true

```

```

#   eth_sendRawTransaction: true
#   zks_getTransaction: true
# Object format with custom rate limit:
# methods:
#   gen_call:
#     enabled: true
#     ratelimit: 5           # custom: 5 req/s per IP instead of default 2
#   gen_getContractSchema:
#     enabled: true
#     ratelimit: 0           # explicitly disable rate limiting for this method
ops:
  port: 9153 # Metrics port
  endpoints:
    metrics: true # Enable metrics endpoint
    health: true # Enable health endpoint
    balance: true # Enable balance endpoint
# genvm configuration
genvm:
  root_dir: ./third_party/genvm
  start_manager: true # if true node will start genvm manager itself
  manager_url: http://127.0.0.1:3999
  permits: 8 # Leave empty for autodiscovery, put a number for updating genvm permits on startup
  sync_permits: 64 # Optional: permits used while !isSynced (bulk replay). Leave empty to reuse the steady-state
`permits` value.
  sync_wait_timeout: "20s" # Max time to wait for GenVM to sync to a requested block before returning an error
# Advanced configuration
merkleforest:
  maxdepth: 16
  dbpath: "./data/node/merkle/forest/data.db"
  indexdboath: "./data/node/merkle/index.db"
merkletree:
  maxdepth: 16
  dbpath: "./data/node/merkle/tree/"
# metrics configuration
metrics:
  interval: "15s" # Default interval for all collectors (can be overridden per collector)
  collectors:
    node:
      enabled: true
      # interval: "30s" # Optional: Override default interval for this collector
    genvm:
      enabled: true
      # interval: "20s" # Optional: Override default interval for this collector
      # enable_simulated_traffic: false # Optional: Enable simulated network traffic metrics for GenVM processes
      (default: false)
      # simulated_rx_bytes_per_process: 1024 # Optional: Simulated received bytes per process per interval (default:
1024)
      # simulated_tx_bytes_per_process: 512 # Optional: Simulated transmitted bytes per process per interval
      (default: 512)
    webdriver:
      enabled: true
      # interval: "60s" # Optional: Override default interval for this collector

```

Network-Specific Consensus Configuration

Set the consensus section in your config.yaml according to the network you want to join:

Testnet Asimov

```

# Asimov Phase 5 network consensus configuration
consensus:
  consensusaddress: "0xe66B434bc83805f380509642429eC8e43AE9874a"
  genesis: 17326

```

Testnet Bradbury

```

# Bradbury Phase 1 network consensus configuration
consensus:
  consensusaddress: "0x8aCE036C8C3C5D603dB546b031302FCf149648E8"
  genesis: 501711

```

Overriding Configuration with Environment Variables

Any configuration value in config.yaml can be overridden using environment variables with the prefix GENLAYERNODE_.

Pattern:

- Replace dots (.) with underscores (_)
- Convert to uppercase
- Add the GENLAYERNODE_ prefix

Examples:

Config Key	Environment Variable
rollup.genlayerchainrpcurl`	GENLAYERNODE_ROLLUP_GENLAYERCHAINRPCURL`
rollup.genlayerchainwebsocketurl`	GENLAYERNODE_ROLLUP_GENLAYERCHAINWEBSOCKETURL`
consensus.contractmainaddress`	GENLAYERNODE_CONSENSUS_CONTRACTMAINADDRESS`
consensus.contractdataaddress`	GENLAYERNODE_CONSENSUS_CONTRACTDATAADDRESS`
consensus.genesis`	GENLAYERNODE_CONSENSUS_GENESIS`
node.mode`	GENLAYERNODE_NODE_MODE`
node.validatorwalletaddress`	GENLAYERNODE_NODE_VALIDATORWALLETADDRESS`
node.operatoraddress`	GENLAYERNODE_NODE_OPERATORADDRESS`
node.rpc.port`	GENLAYERNODE_NODE_RPC_PORT`
logging.level`	GENLAYERNODE_LOGGING_LEVEL`

Usage example:

```
# Override the RPC port
export GENLAYERNODE_CONSENSUS_CONTRACTMAINADDRESS="0x..."

# Set validator wallet address
export GENLAYERNODE_NODE_VALIDATORWALLETADDRESS="0x..."

# Set logging level
export GENLAYERNODE_LOGGING_LEVEL="DEBUG"
```

Environment variables take precedence over values in config.yaml, making them ideal for Docker deployments or sensitive values you don't want in configuration files.

GenVM Configuration

See the GenVM Configuration (/validators/genvm-configuration) page for detailed LLM provider setup, configuration files, and advanced features like greyboxing.

Import the Operator Key

The operator key is used by your node to sign blocks and perform validator duties.

Option 1: Import from CLI Wizard (Recommended)

If you used genlayer staking wizard, it exported an operator keystore file. Transfer this file to your validator server and import it:

```
./bin/genlayernode account import \
--password "your node password" \
--passphrase "password you set when exporting from wizard" \
--path "/path/to/operator-keystore.json" \
-c $(pwd)/configs/node/config.yaml \
```

```
--setup
```

You should see:

```
Account imported:  
Address: 0xA0b12Fd2f3F7e86fEC458D114A5E7a6f571160a8  
Account setup as a validator
```

Option 2: Generate New Operator Key

You can also generate a new operator key directly on the server:

```
./bin/genlayernode account new -c $(pwd)/configs/node/config.yaml --setup --password "your secret password"
```

Then use this address when running the wizard on your local machine.

Restoring Your Operator Key

To restore from a backup (e.g., after migrating to a new server):

```
./bin/genlayernode account import \  
--password "your node password" \  
--passphrase "your backup encryption passphrase" \  
--path "/path/to/your/secure/backup.key" \  
-c $(pwd)/configs/node/config.yaml \  
--setup
```

Note:

Always verify that your imported key works by checking the operator address matches what's configured for your validator.

Backing Up Your Operator Key

After setting up your operator key, back it up securely:

```
./bin/genlayernode account export \  
--password "your node password" \  
--address "your operator address" \  
--passphrase "your backup encryption passphrase" \  
--path "/path/to/your/secure/backup.key" \  
-c $(pwd)/configs/node/config.yaml
```

Note:

Important: Back up your operator key! Losing access means you'll need to set up a new operator and update your validator configuration. Store the backup securely.

To print the private key from your backup file, use the `--print` flag. Keep this private key secure and never share it.

Running the node

Once you have configured everything, you are ready to start the node.

Note:

Important: Before starting the node, ensure you have:

1. Operator key imported into the node (see "Import the Operator Key" above)

2. Validator wallet address - obtained after joining as validator via genlayer staking wizard or validator-join

Without both configured, your node will run as a full node instead of a validator.

Running the Node using the binary

1. Set the LLM Provider API Key

Set the appropriate environment variable for your chosen LLM provider. See the GenVM Configuration (/validators/genvm-configuration) page for details on LLM providers and obtaining API credits.

```
sh
```

```
# For Heurist
```

```
export HEURISTKEY='your_heurist_api_key'
```

```
# For Comput3
```

```
export COMPUT3KEY='your_comput3_api_key'
```

```
# For io.net
```

```
export IOINTELLIGENCE_API_KEY='your_ionet_api_key'
```

```
# For Chutes
```

```
export CHUTES_API_KEY='your_chutes_api_key'
```

```
# For Morpheus
```

```
export MORPHEUS_API_KEY='your_morpheus_api_key'
```

```
# For other providers, use the appropriate environment variable name
```

2. Run the WebDriver container

```
sh
```

```
docker compose up -d # Starts the WebDriver needed by the GenVM web module
```

3. (Optional) Run two services (modules) in background (this is a crucial step for running _Intelligent_ contracts). This can be done automatically or manually.

- To start them automatically in node configuration set genvm.manage_modules to true

- To start them manually run

```
bash
```

```
./third_party/genvm/bin/genvm-modules web & ./third_party/genvm/bin/genvm-modules llm &
```

Note:

Note: If you are using the default configuration, genvm.manage_modules is set to true by default, meaning the node will manage these modules automatically.

4. Checking Your Configuration

To ensure your node is correctly configured, you can run the following command:

```
sh
./bin/genlayernode doctor
```

The doctor command now includes comprehensive GenVM diagnostics integration to validate:

- Consensus contract configuration and accessibility
- GenVM module connectivity and health status
- LLM provider configuration and API connectivity
- Network configuration and ZKSync node accessibility

5. Run the node

```
sh
./bin/genlayernode run -c $(pwd)/configs/node/config.yaml --password "your secret password" # The same password you used when creating the account
```

Note:

If you are running the node via SSH, the process might terminate if your connection drops. To prevent this, consider using a terminal multiplexer like screen or tmux to keep the node running in the background even if your SSH session ends. You can find a guide on using screen here (<https://www.networkworld.com/article/967925/how-the-linux-screen-tool-can-save-your-tasks-and-your-sanity-if-ssh-is-interrupted.html>).

Running the Node using docker-compose

You can also run the GenLayer node using Docker and Docker Compose.

1. Create a docker-compose.yaml file with the following content:

```
services:
  webdriver-container:
    container_name: genlayer-node-webdriver
    image: yeagerai/genlayer-genvm-webdriver:0.0.11
    shm_size: 2gb
    security_opt:
      - no-new-privileges:true
    environment:
      PORT: 4444
    ports:
      - "${WEBDRIVER_PORT:-4444}:4444"
    restart: unless-stopped
  genlayer-node:
    image: yeagerai/genlayer-node:${NODE_VERSION:-latest}
    entrypoint: ["sh", "-c", "/app/bin/genlayernode run --password ${NODE_PASSWORD:-12345678}"]
    container_name: genlayer-node
    restart: unless-stopped
    env_file:
      - path: ./env
        required: false
    ports:
      - "${NODE_RPC_PORT:-9151}:9151"
```



```

- "${NODE_OPS_PORT:-9153}:9153"
volumes:
- ${NODE_CONFIG_PATH:-./configs/node/config.yaml}:/app/configs/node/config.yaml:ro
- ${NODE_DATA_PATH:-./data}:/app/data
- ./genvm-module-web-docker.yaml:/app/third_party/genvm/config/genvm-module-web.yaml
- /var/run/docker.sock:/var/run/docker.sock:ro # required for webdriver metrics collection
security_opt:
- no-new-privileges:true
logging:
driver: "json-file"
options:
max-size: "100m"
max-file: "3"
compress: "true"
depends_on:
- webdriver-container
profiles:
- node

```

2. Create genvm-module-web-docker.yaml for Docker networking:

The extracted tarball includes `third_party/genvm/config/genvm-module-web.yaml` configured for localhost, which works when running

the binary directly. For docker-compose, you need a version that points to the webdriver service name.

Create `genvm-module-web-docker.yaml` in the same directory as your `docker-compose.yaml`:

- Copy the contents of `third_party/genvm/config/genvm-module-web.yaml` into `genvm-module-web-docker.yaml`.
- Modify the `webdriver_host` configuration to use the service name `webdriver-container` instead of `localhost`:

```
webdriver_host: http://webdriver-container:4444
```

This allows the node container to reach the webdriver container using Docker's internal networking.

The key difference:

- Binary execution: uses `localhost:4444` (webdriver runs on same machine)
- Docker-compose: uses `webdriver-container:4444` (container service name)

3. Set the required environment variables in a .env file:

Note:

This `.env` file serves two purposes: it configures Docker Compose variables (like `NODE_VERSION`, `NODE_PASSWORD`, ports) and can also include `GENLAYERNODE_*` variables to override node configuration values as described in [Overriding Configuration with Environment Variables](#) ([#overriding-configuration-with-environment-variables](#)).

```

# GenLayer Node Release Configuration

# =====
# WebDriver Configuration (required for GenVM)
# =====
WEBDRIVER_PORT=4444

# =====
# Node Service Configuration (optional - use node profile)
# =====
# Docker image version
NODE_VERSION=v0.4.0

# Keystore password (used to unlock the pre-imported wallet)
NODE_PASSWORD=12345678

# Path to node configuration file
NODE_CONFIG_PATH=./configs/node/config.yaml

```

```
# Path to node data directory (for persistence)
NODE_DATA_PATH=./data

# Port mappings
NODE_RPC_PORT=9151
NODE_OPS_PORT=9153

# LLM API Key (required for GenVM LLM module)
HEURISTKEY=
COMPUT3KEY=
IOINTELLIGENCE_API_KEY=
ANTHROPICKEY=
XAIKEY=
GEMINIKEY=

CHUTES_API_KEY=
MORPHEUS_API_KEY=
```

or simply use the provided docker-compose.yaml, .env.example and the genvm-module-web-docker.yaml from the extracted tarball.

4. Checking Your Configuration

To ensure your node is correctly configured, you can run the following command:

```
source .env && docker run --rm --env-file ./env \
-v ${NODE_CONFIG_PATH:-./configs/node/config.yaml}:/app/configs/node/config.yaml \
yeagerai/genlayer-node:${NODE_VERSION:-v0.4.0} \
./bin/genlayernode doctor
```

5. Start the services using Docker Compose:

```
source .env && docker compose --profile node up -d
```

Telemetry

For a quick setup using Docker Compose:

1. Configure your .env file with the required variables for monitoring:

env

Central monitoring server endpoints for GenLayer Foundation

CENTRAL_MONITORING_URL=https://prometheus-prod-66-prod-us-east-3.grafana.net/api/prom/push

CENTRAL_LOKI_URL=https://logs-prod-042.grafana.net/loki/api/v1/push

Authentication for central monitoring

Metrics (Prometheus) credentials

CENTRAL_MONITORING_USERNAME=your-metrics-username

CENTRAL_MONITORING_PASSWORD=your-metrics-password

Logs (Loki) credentials

CENTRAL_LOKI_USERNAME=your-logs-username

CENTRAL_LOKI_PASSWORD=your-logs-password

Node identification

NODE_ID=validator-001

VALIDATOR_NAME=MyValidator

2. Run docker compose:

```
bash
```

```
docker compose --profile monitoring up -d
```

3. Verify your node is pushing metrics by checking the GenLayer Foundation public dashboard (<https://genlayerfoundation.grafana.net/public-dashboards/66a372d856ea44e78cf9ac21a344f792>)

For detailed monitoring setup including Prometheus metrics, Grafana dashboards, and centralized logging with Alloy, see the Monitoring Guide (</validators/monitoring>).

Monitoring & Telemetry

Source: <https://docs.genlayer.com/validators/monitoring>

GenLayer validators expose comprehensive metrics that are ready for consumption by Prometheus and other monitoring tools. This allows you to monitor your validator's performance, health, and resource usage.

Accessing Metrics

The metrics endpoint is exposed on the operations port (default: 9153) configured in your config.yaml:

```
node:
  ops:
    port: 9153 # Metrics port
    endpoints:
      metrics: true # Enable metrics endpoint
```

Once your node is running, you can access the metrics at:

```
http://localhost:9153/metrics
```

Available Metrics

The validator exposes various metric collectors that can be individually configured:

- Node Metrics: Core validator performance metrics including block processing, transaction handling, and consensus participation
- GenVM Metrics: Virtual machine performance metrics, including execution times and resource usage
- WebDriver Metrics: Metrics related to web access and external data fetching

Configuring Metrics Collection

You can customize metrics collection in your config.yaml:

```
metrics:
  interval: "15s" # Default collection interval
collectors:
  node:
    enabled: true
    interval: "30s" # Override interval for specific collector
  genvm:
    enabled: true
    interval: "20s"
  webdriver:
    enabled: true
    interval: "60s"
```

Example Metrics Query

To check if metrics are working correctly:

```
# Get all available metrics
curl http://localhost:9153/metrics

# Check specific metric (example)
curl -s http://localhost:9153/metrics | grep genlayer_node_
```

Note:

The metrics endpoint also provides /health and /balance endpoints on the same port for additional monitoring capabilities.

Monitoring Best Practices

1. Set up alerts for critical metrics like node synchronization status and missed blocks
2. Monitor resource usage to ensure your validator has sufficient CPU, memory, and disk space
3. Track GenVM performance to optimize LLM provider selection and configuration
4. Use visualization tools like Grafana to create dashboards for easy monitoring

Note:

For production validators, we recommend setting up a complete monitoring stack with Prometheus and Grafana. This enables real-time visibility into your validator's performance and helps identify issues before they impact your validator's operation.

Logs and Metrics Forwarding

You can forward your logs and metrics to external systems for centralized monitoring and alerting by using the service alloy provided in the docker-compose.yaml from the extracted tarball.

Centralized Push to GenLayer Foundation Grafana Cloud (using Alloy)

To contribute your node's metrics and logs to the centralized GenLayer Foundation Grafana Cloud dashboard (improving aggregate network visibility, alerts, and community monitoring), use the built-in Alloy service.

Why contribute?

- Helps the Foundation and community track overall testnet health (validator participation, latency, resource usage).

- May positively influence testnet points/rewards (visible healthy nodes are prioritized).
- Setup takes 10-15 minutes once credentials are provided.

Prerequisites

- Metrics enabled in config.yaml (endpoints.metrics: true — default in recent versions).
- Ops port 9153 exposed in docker-compose (ports: - "9153:9153").
- Credentials from the Foundation team (ask in #testnet-asimov):
- CENTRAL_MONITORING_URL — Prometheus remote write URL
- CENTRAL_LOKI_URL — Loki push URL
- CENTRAL_MONITORING_USERNAME / CENTRAL_MONITORING_PASSWORD — Metrics (Prometheus) credentials
- CENTRAL_LOKI_USERNAME / CENTRAL_LOKI_PASSWORD — Logs (Loki) credentials

Steps

1. Create or update .env (next to your docker-compose.yaml):

```
# Central monitoring server endpoints for GenLayer Foundation
CENTRAL_MONITORING_URL=https://prometheus-prod-66-prod-us-east-3.grafana.net/api/prom/push
CENTRAL_LOKI_URL=https://logs-prod-042.grafana.net/loki/api/v1/push

# Authentication for central monitoring
# Metrics (Prometheus) credentials
CENTRAL_MONITORING_USERNAME=your-metrics-username
CENTRAL_MONITORING_PASSWORD=your-metrics-password
# Logs (Loki) credentials
CENTRAL_LOKI_USERNAME=your-logs-username
CENTRAL_LOKI_PASSWORD=your-logs-password

# Node identification
NODE_ID=validator-001
VALIDATOR_NAME=MyValidator
NETWORK_NAME=asimov-phase5

# Usually defaults are fine
NODE_METRICS_ENDPOINT=host.docker.internal:9153
LOG_FILE_PATTERN=/var/log/genlayer/node*.log
METRICS_SCRAPE_INTERVAL=15s
```

2. Add or verify the Alloy service in docker-compose.yaml (copy if missing):

```
# Grafana Alloy for both logs and metrics forwarding
# Supports both single node and multi-node configurations
#
# Single Node Mode:
# Set NODE_ID, VALIDATOR_NAME, NODE_METRICS_ENDPOINT in .env
# docker compose --profile monitoring up -d
#
# Multi-Node Mode:
# Set SCRAPE_TARGETS_JSON in .env
# docker compose --profile monitoring up -d
alloy:
  image: grafana/alloy:v1.16.1
  container_name: genlayer-node-alloy
  command:
    - run
    - /etc/alloy/config.river
    - --server.http.listen-addr=0.0.0.0:12345
    - --storage.path=/var/lib/alloy/data
  volumes:
    - ./alloy-config.river:/etc/alloy/config.river:ro
    - ./alloy-healthcheck.sh:/etc/alloy/healthcheck.sh:ro
    - ${NODE_LOGS_PATH:-./data/node/logs}:/var/log/genlayer:ro
    - alloy_data:/var/lib/alloy
  healthcheck:
    test: ["CMD", "sh", "/etc/alloy/healthcheck.sh"]
    interval: 120s
```

```

    timeout: 15s
    retries: 1
    start_period: 120s
  environment:
    # Central monitoring endpoints
    - CENTRAL_LOKI_URL=${CENTRAL_LOKI_URL:-https://logs-prod-042.grafana.net/loki/api/v1/push}
    - CENTRAL_MONITORING_URL=${CENTRAL_MONITORING_URL:-https://prometheus-prod-66-prod-us-east-3.grafana.net/api/prom/push}
    # Metrics (Prometheus) authentication
    - CENTRAL_MONITORING_USERNAME=${CENTRAL_MONITORING_USERNAME:-telemetric}
    - CENTRAL_MONITORING_PASSWORD=${CENTRAL_MONITORING_PASSWORD:-12345678}

    # Logs (Loki) authentication
    - CENTRAL_LOKI_USERNAME=${CENTRAL_LOKI_USERNAME:-telemetric}
    - CENTRAL_LOKI_PASSWORD=${CENTRAL_LOKI_PASSWORD:-12345678}

    # Single node configuration
    - NODE_ID=${NODE_ID:-validator-001}
    - VALIDATOR_NAME=${VALIDATOR_NAME:-MyValidator}
    - NODE_METRICS_ENDPOINT=${NODE_METRICS_ENDPOINT:-host.docker.internal:9153}

    # Multi-node configuration
    # When set, overrides single node config above
    - SCRAPE_TARGETS_JSON=${SCRAPE_TARGETS_JSON:-}

    # Scraping configuration
    - METRICS_SCRAPING_INTERVAL=${METRICS_SCRAPING_INTERVAL:-60s}
    - METRICS_SCRAPING_TIMEOUT=${METRICS_SCRAPING_TIMEOUT:-10s}
    - ALLOY_SELF_MONITORING_INTERVAL=${ALLOY_SELF_MONITORING_INTERVAL:-60s}

    # Log collection configuration
    - LOG_FILE_PATTERN=${LOG_FILE_PATTERN:-/var/log/genlayer/node*.log}

    # Log batching configuration
    - LOKI_BATCH_SIZE=${LOKI_BATCH_SIZE:-1MiB}
    - LOKI_BATCH_WAIT=${LOKI_BATCH_WAIT:-60s}
  ports:
    - "12345:12345" # Alloy UI for debugging
  restart: unless-stopped
  profiles:
    - monitoring
  extra_hosts:
    - "host.docker.internal:host-gateway"

volumes:
  alloy_data:

```

3. Create or update `./alloy-config.river` (use the provided version — it handles logs and metrics forwarding):

```

// Grafana Alloy Configuration for GenLayer Node Telemetry
// Handles both log collection and metrics forwarding

// =====
// Log Collection and Forwarding
// =====

// Discovery component to find log files using local.file_match
// Supports different log file patterns:
// - Single node: "/var/log/genlayer/node.log"
// - Multi-node: "/var/log/genlayer/*/logs/node.log" (each node in subdirectory)
// - Custom pattern via LOG_FILE_PATTERN env var
local.file_match "genlayer_logs" {
  path_targets = [
    {
      __path__ = coalesce(sys.env("LOG_FILE_PATTERN"), "/var/log/genlayer/node*.log"),
    }
  ]
}

// Relabel to add metadata labels to log entries
discovery.relabel "add_labels" {
  targets = local.file_match.genlayer_logs.targets

  // Add instance label from environment variable
  rule {
    target_label = "instance"
    replacement = sys.env("NODE_ID")
  }
}

```

```

// Add validator_name label from environment variable
rule {
    target_label = "validator_name"
    replacement = sys.env("VALIDATOR_NAME")
}

// Add component label
rule {
    target_label = "component"
    replacement = "alloy"
}

// Add job label
rule {
    target_label = "job"
    replacement = "genlayer-node"
}

// Source component to read log files
loki.source.file "genlayer" {
    targets = discovery.relabel.add_labels.output
    forward_to = [loki.write.central.receiver]

    // Tail from end to avoid ingesting entire log history on startup
    tail_from_end = true
}

// Write logs to central Loki instance
loki.write "central" {
    endpoint {
        url = sys.env("CENTRAL_LOKI_URL")

        // HTTP Basic Authentication
        basic_auth {
            username = sys.env("CENTRAL_LOKI_USERNAME")
            password = sys.env("CENTRAL_LOKI_PASSWORD")
        }

        // Enable retrv with default exponential backoff
        // Note: Alloy's loki.write doesn't have a retry block; retries are handled automatically
        // with exponential backoff by default when the endpoint is unreachable

        // Configurable batch settings for efficient log sending
        batch_size = coalesce(sys.env("LOKI_BATCH_SIZE"), "1MiB") // Maximum batch size before sending
        batch_wait = coalesce(sys.env("LOKI_BATCH_WAIT"), "60s") // Maximum wait time before sending partial batch
    }
}

// =====
// Prometheus Metrics Collection and Forwarding
// =====

// Scrape metrics from GenLayer node(s)
// Supports both single node and multi-node configurations
//
// Single Node Mode:
//   Set NODE_METRICS_ENDPOINT, NODE_ID, VALIDATOR_NAME
//
// Multi-Node Mode:
//   Set SCRAPE_TARGETS_JSON with JSON array of target objects
//   Example: [{"__address__": "host.docker.internal:9250", "instance": "0x...", "validator_name": "node-1"}]
//
// Note: The "network" label is emitted by the node itself (auto-detected from consensus address),
// so it does not need to be configured here.
prometheus.scrape "genlayer_node" {
    // Dynamic targets based on environment variable
    // If SCRAPE_TARGETS_JSON is set, use it (multi-node mode)
    // Otherwise, build single target from individual env vars (single node mode)
    targets = encoding.from_json(coalesce(sys.env("SCRAPE_TARGETS_JSON"),
    string.format("[{\"__address__\": \"%s\", \"instance\": \"%s\", \"validator_name\": \"%s\"}]",
    coalesce(sys.env("NODE_METRICS_ENDPOINT"), "host.docker.internal:9153"), coalesce(sys.env("NODE_ID"), "local"),
    coalesce(sys.env("VALIDATOR_NAME"), "default"))))
    forward_to = [prometheus.relabel.metrics.receiver]

    // Configurable scrape intervals
    scrape_interval = coalesce(sys.env("METRICS_SCRAPE_INTERVAL"), "60s")
}

```

```

scrape_timeout = coalesce(sys.env("METRICS_SCRAPETIMEOUT"), "10s")
}

// Relabel metrics to filter before forwarding
prometheus.relabel "metrics" {
    forward_to = [prometheus.remote_write.central.receiver]

    // Option 1: Forward all metrics (default)
    // Currently forwarding all metrics from the node.

    // Option 2: Only keep genlayer_node_* metrics to reduce bandwidth (recommended)
    // To enable filtering and reduce bandwidth, uncomment the following rule:
    /*
    rule {
        source_labels = ["__name__"]
        regex          = "genlayer_node_.*"
        action          = "keep"
    }
    */
}

// Remote write configuration for sending metrics to central Prometheus
prometheus.remote_write "central" {
    endpoint {
        url = sys.env("CENTRAL_MONITORING_URL")

        // HTTP Basic Authentication
        basic_auth {
            username = sys.env("CENTRAL_MONITORING_USERNAME")
            password = sys.env("CENTRAL_MONITORING_PASSWORD")
        }

        // Queue configuration for reliability
        queue_config {
            capacity          = 10000
            max_shards        = 5
            max_samples_per_send = 500
            batch_send_deadline = coalesce(sys.env("METRICS_BATCH_SEND_DEADLINE"), "60s")
        }
    }
}

// =====
// Alloy Self-Monitoring
// =====

// Alloy internal exporter for health monitoring
prometheus.exporter.self "alloy" {}

// Expose Alloy's own metrics on the HTTP server
prometheus.scrape "alloy" {
    targets = prometheus.exporter.self.alloy.targets
    forward_to = [] // Not forwarding Alloy metrics to reduce noise

    // Configurable scrape interval for Alloy's internal health monitoring
    scrape_interval = coalesce(sys.env("ALLOY_SELF_MONITORING_INTERVAL"), "60s")
}

```

4. Start Alloy:

```
docker compose --profile monitoring up -d
```

5. Verify it works:

- Open Alloy UI: <http://localhost:12345/targets> — the "genlayer_node" scrape target should show status UP.
- Check logs for successful sends:

```

docker logs genlayer-node-alloy | grep "sent batch"
docker logs genlayer-node-alloy | grep "remote_write"

```

Look for messages indicating successful batch sending (no error codes like 401, 403, 500).

- In Foundation Grafana Cloud: search for metrics with labels
instance="\${NODE_ID}" or validator_name="\${VALIDATOR_NAME}"
(example: genlayer_node_uptime_seconds{instance="0xYourID"}).

Troubleshooting

Using the Alloy UI

The Grafana Alloy service includes a built-in web UI for troubleshooting and monitoring the telemetry pipeline. Access it at:

```
http://localhost:12345
```

The Alloy UI provides:

- Targets view (/targets): Shows the status of all scrape targets. Check if your node's metrics endpoint shows as UP.
- Graph view (/graph): Explore and query collected metrics locally before they're forwarded.
- Component health: View the status of all Alloy pipeline components (scrapers, relabelers, writers).

Note:

If metrics are not appearing in the central Grafana Cloud, first verify they show correctly in the local Alloy UI. This helps isolate whether the issue is with metrics collection or forwarding.

Common Issues

- No local metrics:

```
curl http://localhost:9153/metrics
```

— it should return Prometheus-formatted data.

- Authentication errors (401/403): Double-check `CENTRAL_MONITORING_USERNAME`, `CENTRAL_MONITORING_PASSWORD`, `CENTRAL_LOKI_USERNAME`, and `CENTRAL_LOKI_PASSWORD` in `.env`.
- No data pushed: Ensure URLs in `.env` have no trailing slash.
- Target showing DOWN in Alloy UI: Verify your node is running and the ops port (9153) is accessible. Check that `NODE_METRICS_ENDPOINT` in `.env` is correct.

Getting Help

Share Alloy logs when asking for assistance:

```
docker logs genlayer-node-alloy
```

System Requirements

Source: <https://docs.genlayer.com/validators/system-requirements>

Below are the initial recommended system requirements for running a GenLayer validator node. These guidelines may change as the network grows and evolves.

RAM

- Recommended: 16 GB
- Why:
- GenLayer's `_genvm_` can spawn multiple Sub-VMs for contract calls and non-deterministic blocks.
- Each Sub-VM can consume up to ~4 GB of RAM for storage.

CPU

- Recommended: Modern multi-core processor with at least 8 cores/16 threads

Architecture

- Recommended: AMD64
- ARM64 is not supported at this time

Storage

- Recommended Disk Space: 128 GB+
- Preferred Type: SSD or NVMe (M.2)

Network

- 100 Mbps connection
- Recommended:: 1 Gbps+

GPU (Optional)

- GPU is Not Required
- If you want to run LLMs locally, you will need to select appropriate hardware (typically a CUDA-compatible GPU with sufficient VRAM for the model you intend to use)

Note:

These requirements are a starting point. As GenLayer evolves and usage patterns change (e.g., more complex AI-driven Intelligent Contracts), the recommended hardware may change.

Software

- Operating System - 64-bit Linux (Ubuntu, Debian, CentOS, etc.)
- docker - for running the WebDriver container
- python3, python3-pip and python3-venv for GenVM setup

GenVM Configuration

Source: <https://docs.genlayer.com/validators/genvm-configuration>

Your validator's GenVM needs two things to function: LLM providers to process AI-powered contract logic, and a web module to fetch live internet data. This page covers how to configure both.

Why this matters: The LLM providers you choose directly affect your validator's operating costs and performance. Faster, cheaper models reduce your costs per transaction. Higher-quality models improve your consensus accuracy, which affects your rewards and avoids slashing. By tuning your provider setup — choosing the right models, configuring fallback chains, running local inference, or applying input filters — you can optimize the economics of running your validator.

LLM Provider Credits

Note:

GenLayer has partnered with multiple LLM providers to offer free credits for validators:

Heurist (<https://www.heurist.ai/>) — free API credits (<https://dev-api-form.heurist.ai>) with referral code `_genlayer_`.

Comput3 (<https://genlayer.comput3.ai/>) — free API credits (<https://genlayer.comput3.ai/>) for llama3, hermes3, and qwen3 models.

io.net (</partners/ionet>) — create an account at [id.io.net](https://id.io.net/login) (<https://id.io.net/login>) and request free credits (<https://form.typeform.com/to/pDmCCViV>).

Chutes (</partners/chutes>) — create an account at chutes.ai (<https://chutes.ai>) and generate an API key from settings.

Morpheus (</partners/morpheus>) — obtain a Morpheus API key (<https://mor.org/>) for DeepSeek, Llama, Qwen, and more via OpenAI-compatible API.

Hot-Reload

You can modify configuration files on the fly and reload without stopping your node. This is important when experimenting with providers or tuning your setup.

```
PORT=3999 # GenVM manager port

# Restart LLM module (reloads Lua script + YAML config)
```

```
curl -X POST "http://127.0.0.1:${PORT}/module/stop" \
-H 'Content-Type: application/json' \
-d '{"module_type": "Llm"}'

curl -X POST "http://127.0.0.1:${PORT}/module/start" \
-H 'Content-Type: application/json' \
-d '{"module_type": "Web", "config": null}'
```

Repeat for each GenVM instance if running multiple. For a full restart: `sudo systemctl restart genlayer-node`.

Configuration Files

All configuration lives at `third_party/genvm/config/`. Three files:

File	What it controls	Should you modify?
<code>genvm-module-llm.yaml</code>	LLM providers, models, prompt templates, Lua script selection	**Yes** – this is your main configuration
<code>genvm-module-web.yaml</code>	Web access, webdriver, URL restrictions	Rarely – only for advanced URL policy changes
<code>genvm-manager.yaml</code>	Concurrency permits, VM pool	Rarely – only if tuning for your hardware

Configuring Your LLM Backends

The backends section in `genvm-module-llm.yaml` defines which LLMs your validator can use. The Lua script (configured via `lua_script_path`) then selects from these backends at runtime.

Backend entry format

```
backends:
  my-provider:
    # Your name for this backend
    enabled: true # Set false to disable without removing
    host: https://api.example.com
    provider: openai-compatible # Provider type (see table below)
    key: ${ENV[MY_API_KEY]} # API key via environment variable
    models:
      model-name:
        enabled: true
        supports_json: true # Can return structured JSON output
        supports_image: false # Can process image inputs
        use_max_completion_tokens: false
        meta: {} # Custom metadata (used by greyboxing strategies)
```

Model capabilities

Intelligent Contract LLM requests have two axes:

- Input: text only, or text + up to 2 images
- Output: free-form text, or structured JSON

This gives four combinations: text→text, text→JSON, images+text→text, images+text→JSON. The capability flags control which models are eligible:

- `supports_json: true` — model can produce structured JSON output
- `supports_image: true` — model can accept image inputs

Your validator must support all combinations across your model set. Not every model needs to support everything — that's where routing comes in. You can use a cheap text model for simple prompts and a vision-capable model for image requests. But if a contract sends an image prompt and none of your enabled models support images, the request will fail.

The `supports_json` and `supports_image` flags tell the Lua script which models are eligible for each request type. Set them accurately for each model.

Supported provider types

Type	Description	Example services
--- --- ---		
`openai-compatible`	OpenAI-compatible chat/completions API	OpenAI, OpenRouter, Heurist, Comput3, Chutes, Morpheus, X.ai
`ollama`	Ollama's native `/api/generate` endpoint	Local Ollama instance
`anthropic`	Anthropic's Messages API	Claude models
`google`	Google's Generative Language API	Gemini models

Example: Adding a local Ollama instance

Add this to the backends section. No API key needed — Ollama runs locally. Follow Ollama's install guide (<https://ollama.com>) to set up the server and pull a model.

```
backends:
  my-ollama:
    host: http://localhost:11434
    provider: ollama
    models:
      llama3.3:
        supports_json: true
        supports_image: false
```

Example: Adding OpenRouter

OpenRouter (<https://openrouter.ai>) provides access to many models through a single API key:

```
backends:
  openrouter:
    host: https://openrouter.ai/api
    provider: openai-compatible
    key: ${ENV[OPENROUTERKEY]}
    models:
      deepseek/deepseek-v3.2:
        supports_json: true
      anthropic/claude-haiku-4.5:
        supports_json: true
        supports_image: true
      openai/gpt-5.1-mini:
        supports_json: true
        supports_image: true
```

API key environment variables

Set these in your `.env` file before starting the node. You only need to set the keys for providers you're actually using. Keys use `${ENV[VARIABLE_NAME]}` syntax in the YAML.

Variable	Provider
--- ---	
`OPENROUTERKEY`	OpenRouter
`OPENAIKEY`	OpenAI
`ANTHROPICKEY`	Anthropic
`GEMINIKEY`	Google Gemini
`HEURISTKEY`	Heurist
`XAIKEY`	X.ai (Grok)
`COMPUT3KEY`	Comput3
`IOINTELLIGENCE_API_KEY`	io.net
`CHUTES_API_KEY`	Chutes
`MORPHEUS_API_KEY`	Morpheus

Note:

Enabled backends without a configured API key will log warnings. Either set the key or disable the backend with `enabled: false`.

Greyboxing: Optimizing Your Validator

Greyboxing is how you configure and optimize your validator's LLM execution. From the network's point of view, each validator is a grey box — an attacker can see that LLMs are being used, but cannot know which model will process a given request, how it's configured, or what filters are applied. This per-validator opacity is a security property: it makes prompt injection and manipulation attacks much harder because there's no single configuration to target.

As a validator operator, greyboxing is also your primary lever for economics — by choosing the right models, configuring fallback chains, and applying filters, you control your operating costs and consensus quality.

Choosing models

Start with a capable model and optimize down over time. Your validator needs to produce correct consensus results — using a model that's too weak will hurt your accuracy, leading to missed rewards or slashing. Once you're confident in your setup, you can experiment with cheaper alternatives.

Recommended starting points:

- Frontier tier: Claude Sonnet 4.6, GPT-5, Gemini 3 Flash — high accuracy, higher cost
- Strong open-source: DeepSeek V3.2, Qwen3-235B, Llama 4 Maverick — good accuracy, lower cost
- Budget tier: Nemotron, smaller Llama variants — lowest cost, test carefully before relying on these

For image-capable requests, you need at least one vision model (e.g., GPT-5, Gemini Flash, Claude Sonnet).

A practical approach: use a frontier model as your primary, with cheaper models as fallback. The greyboxing ordered fallback strategy (below) makes this easy.

What you can control

Greyboxing encompasses several layers of configuration, all implemented in the Lua script that `lua_script_path` points to:

Model selection — Choose which LLM handles each request. You can route text prompts to a cheap fast model and image prompts to a vision-capable one. This can be random, an ordered fallback chain, or a custom router.

Input filtering — Apply filters to prompts before they reach the LLM. Available text filters: NFKC (Unicode normalization), NormalizeWS (whitespace), RmZeroWidth (remove zero-width characters). Available image filters: Denoise, JPEG (compression), GuassainNoise, Unsharpen. Filtering normalizes inputs across validators and can improve consensus.

LLM parameters — Temperature, max tokens, system prompts. Configured per-model in YAML or overridden in your Lua script.

Cost management — In future versions, your greyboxing script will be able to track execution costs and decide when a transaction is too expensive to process. As a validator, if the cost of LLM calls exceeds what makes economic sense, you can timeout the transaction rather than executing at a loss.

Built-in strategies

GenVM ships with two Lua scripts. Switch between them by changing `lua_script_path` in `genvm-module-llm.yaml`.

Default: Random selection

Script: `genvm-llm-default.lua`

Iterates through all enabled backends that match the required capabilities (JSON support, image support) and tries them in arbitrary order. Falls back to the next provider on overload errors (429, 503, 408, 504, 529).

This is the simplest strategy — works out of the box with no tuning.

Ordered fallback chain

Script: `genvm-llm-greybox.lua`

Uses a fixed priority chain configured via `meta.greybox` fields in the YAML. Each model declares which chain it belongs to (text, image, or both) and its priority. Lower number = tried first.

```
models:
  deepseek/deepseek-v3.2:
    supports_json: true
    meta:
      greybox: { text: 1 }           # text chain, priority 1 (tried first)
  openai/gpt-5.1-mini:
    supports_json: true
    supports_image: true
    meta:
      greybox: { image: 1 }         # image chain, priority 1
  anthropic/claude-haiku-4.5:
    supports_json: true
    supports_image: true
    meta:
      greybox: { text: 3, image: 3 } # both chains, lower priority
```

To enable:

```
sudo systemctl stop genlayer-node

# Set your OpenRouter key (or whichever provider you're using)
echo "OPENROUTERKEY=sk-or-v1-your-key-here" >> /opt/genlayer-node/.env

# Apply the release config with greybox metadata
VERSION=$(readlink /opt/genlayer-node/bin | sed 's|/bin||; s|.|/|')
cp /opt/genlayer-node/${VERSION}/third_party/genvm/config/genvm-modules-llm-release.yaml \
  /opt/genlayer-node/${VERSION}/third_party/genvm/config/genvm-module-llm.yaml

# Switch to greybox script
sed -i 's/genvm-llm-default.lua/genvm-llm-greybox.lua/' \
  /opt/genlayer-node/${VERSION}/third_party/genvm/config/genvm-module-llm.yaml

sudo systemctl start genlayer-node
```

Verify:

```
sudo journalctl -u genlayer-node --no-hostname | grep "greybox"
# Expected: greybox: success   provider: openrouter   model: deepseek/deepseek-v3.2
```

To revert: change `genvm-llm-greybox.lua` back to `genvm-llm-default.lua` and restart.

Writing a custom strategy

For full control, write your own Lua script. Create a .lua file in the config directory and point `lua_script_path` to it.

Your script must export two global functions:

```
local lib = require("lib-genvm")
local llm = require("lib-llm")

function ExecPrompt(ctx, args, remaining_gen)
  -- args.prompt: the text prompt
  -- args.response_format: "text" or "json"
  -- args.images: table of image bytes (may be nil)
  -- remaining_gen: gas budget (for future cost tracking)

  -- Transform the raw args into internal format
  local mapped = llm.exec_prompt_transform(args)

  -- Apply input filters
  args.prompt = lib.rs.filter_text(args.prompt, {'NFKC', 'NormalizeWS'})

  -- Select providers that match requirements
  local providers = llm.select_providers_for(mapped.prompt, mapped.format)

  -- Execute on a specific provider
  local result = llm.rs.exec_prompt_in_provider(ctx, {
    provider = "my-provider",
    model = "model-name",
    prompt = mapped.prompt,
    format = mapped.format,
  })

  result.consumed_gen = 0 -- Cost tracking (future)
  return result
end

function ExecPromptTemplate(ctx, args, remaining_gen)
  -- Called for Equivalence Principle template prompts
  -- args.template: "EqComparative", "EqNonComparativeValidator", "EqNonComparativeLeader"
  -- args.vars: template variable substitutions
  local mapped = llm.exec_prompt_template_transform(args)

  -- ... your provider selection logic ...
end
```

Prompt Templates

The `prompt_templates` section in `genvm-module-llm.yaml` configures how your validator acts as a judge in Equivalence Principle (`/developers/intelligent-contracts/features/non-determinism`) scenarios. Your validator isn't just running raw LLM calls — in certain consensus situations, it needs to evaluate whether two outputs are equivalent or whether a result meets specified criteria.

Three templates, using `#{variable}` syntax for substitution:

```
| Template | When it's used | What it returns |
|---|---|---|
| `eq_comparative` | Comparing leader vs validator outputs | Boolean: are they equivalent? |
| `eq_non_comparative_leader` | Leader producing a result | Text output |
| `eq_non_comparative_validator` | Validator checking leader's result | Boolean + reasoning |
```

You can tune these prompts to improve your validator's judgment quality. Better prompts → more accurate consensus → higher rewards.

Lua Scripting API

Core functions (lib.rs)

Available in all Lua scripts:

```
| Function | Description |
|---|---|
| `lib.rs.request(ctx, {method, url, headers, body, json, sign})` | HTTP request. `json: true` returns parsed response. |
| `lib.rs.json_parse(str)` | Parse JSON string to Lua table |
| `lib.rs.json_stringify(value)` | Serialize Lua table to JSON |
| `lib.rs.base64_encode(data)` / `base64_decode(data)` | Base64 encode/decode |
| `lib.rs.filter_text(text, filters)` | Text filters: `"NFC"`, `"NFKC"`, `"NormalizeWS"`, `"RmZeroWidth"` |
| `lib.rs.filter_image(bytes, filters)` | Image filters: `{Denoise=0.5}`, `{JPEG=0.9}`, `{GuassainNoise=0.1}`, `{Unsharpen={0.5, 0.5}}` |
| `lib.rs.split_url(url)` | Parse URL → `{schema, host, port}` |
| `lib.rs.url_encode(text)` | URL-encode text |
| `lib.rs.sleep_seconds(n)` | Async sleep |
| `lib.rs.random_bytes(n)` / `random_float()` | Random data generation |
| `lib.rs.user_error({causes, fatal, ctx})` | Raise a user error |
| `lib.rs.as_user_error(err)` | Try to convert error to user error (nil if not) |
| `lib.log({level, message, ...})` | Structured logging (trace, debug, info, warn, error) |
```

LLM functions (llm.rs)

```
| Function | Description |
|---|---|
| `llm.rs.exec_prompt_in_provider(ctx, request)` | Execute prompt on a specific provider/model |
| `llm.providers` | Table of all configured backends and models |
| `llm.templates` | Equivalence Principle prompt templates |
| `llm.select_providers_for(prompt, format)` | Filter backends by capability (JSON, images) |
| `llm.exec_prompt_transform(args)` | Transform raw prompt args to internal format |
| `llm.exec_prompt_template_transform(args)` | Transform template args with variable substitution |
| `llm.overloaded_statuses` | HTTP codes indicating overload: `{408, 429, 503, 504, 529}` |
```

Web functions (web.rs)

```
| Function | Description |
|---|---|
| `web.config` | Web module configuration |
| `web.allowed_tld` | Set of allowed top-level domains |
| `web.check_url(url)` | Validate URL against schema, port, and TLD restrictions |
```

Advanced: Web Module

The web module (genvm-module-web.yaml) enables contracts to fetch web pages, make HTTP requests, and capture screenshots. You generally don't need to modify this.

The Lua script (genvm-web-default.lua) implements two functions:

- Render(ctx, payload) — render a page as text, HTML, or screenshot
- Request(ctx, payload) — make HTTP requests (GET, POST, etc.)

What you might customize:

- extra_tld: ["local"] — allow additional TLDs beyond the standard set
- always_allow_hosts: ["my-api.internal"] — bypass URL checks for specific hosts

Advanced: GenVM Manager

The manager (genvm-manager.yaml) controls concurrency. The main setting is permits — max concurrent GenVM executions.

Note:

On macOS (native), auto-detection is not supported. Set permits manually (e.g., 32).

Troubleshooting

Warnings for unconfigured providers: Enabled backends without API keys log warnings. Set the key in .env or disable with enabled: false.

"module_failed_to_start": Check that the Lua script exists at the path in lua_script_path, API keys are set, and YAML syntax is valid.

"NO_PROVIDER_FOR_PROMPT": No enabled backend matches the request's capabilities. Ensure at least one has supports_json: true (and supports_image: true for image prompts).

All models exhausted: All providers returned overload errors. Check API key validity and provider status pages. Add more backends as fallback.

Validator Upgrade Guide

Source: <https://docs.genlayer.com/validators/upgrade>

Before upgrading, check the changelog (/validators/changelog) for the target version to determine whether the upgrade is breaking or non-breaking.

Upgrade Type	How to Identify	Database	Downtime
---	---	---	---
Non-breaking	Patch/minor release, no consensus contract changes	Carries over	Minimal
Breaking	Changelog mentions "upgrade consensus contracts" or new consensus deployment	Must be deleted	Full restart required

Note:

When in doubt, treat it as a breaking change — the only cost is a longer sync time.

Non-Breaking Upgrade

No re-staking is required. You can minimize downtime by preparing the new version while your current node is still running.

1. Download and extract into a fresh folder — See download instructions (/validators/setup-guide#download-the-node-software). Always use a clean directory.
2. Configure the new version — Set up config.yaml and .env as described in the configuration section (/validators/setup-guide#configuration). You can reference your previous settings, but do not blindly copy config files — new versions may include updated defaults or new fields.
3. Copy the data folder — Copy data/ from your old installation to the new one (contains database and keystore). Skip this if your datadir points outside the project directory.
4. Validate — Run ./bin/genlayernode doctor and ensure all checks pass.
5. Switch over — Stop the old node, then start the new one.

Breaking Change Upgrade

Breaking changes involve a new consensus deployment. The old database is incompatible, so it cannot be reused. Re-staking is required — you must stake again using the validator wizard (/validators/setup-guide#using-the-validator-wizard).

Note:

Always back up your keystore before a breaking upgrade. You can copy the data/node/keystore/ directory or use genlayernode account export. See backing up your operator key (/validators/setup-guide#backing-up-your-operator-key).

1. Read the changelog — Check the changelog (/validators/changelog) for version-specific instructions and what changed.
2. Back up your keystore — Copy data/node/keystore/ to a safe location.
3. Download and extract into a fresh folder — See download instructions (/validators/setup-guide#download-the-node-software).
4. Configure from scratch — Create config.yaml and .env using the new example files as a base. Do not copy your old config — the schema may have changed. Refer to the configuration section (/validators/setup-guide#configuration) for the current format.
5. Import your operator key — Copy your backed-up keystore into data/node/keystore/, or re-import using the CLI. See restoring your operator key (/validators/setup-guide#restoring-your-operator-key).
6. Do NOT copy the old database — Delete data/node/genlayer.db if you accidentally copied the full data/ folder. The node will create a fresh database and sync from genesis.
7. Validate — Run ./bin/genlayernode doctor and ensure all checks pass.
8. Stop the old node, start the new one.

Migration Notes

v0.4.x to v0.5.0 (Breaking)

- The two consensus contract addresses (contractmainaddress and contractdataaddress) have been replaced by a single consensusaddress field. Do not copy your old consensus config — use the new example config.yaml.
- The GenLayer Chain ZKSync RPC URLs (genlayerchainrpcurl and genlayerchainwebsocketurl) have changed. Update both the HTTP RPC and WebSocket URLs in your config.yaml to the new endpoints.
- Database must be wiped (the node creates a fresh one on startup).
- Re-staking is required — breaking changes involve a new consensus deployment, so you must stake again using the validator wizard (/validators/setup-guide#using-the-validator-wizard).

Quick Troubleshooting

- Node won't start — Run ./bin/genlayernode doctor. If it reports unknown config fields, reconfigure from the new example.

- Not participating in consensus — Verify the consensus contract address, validatorWalletAddress, and operatorAddress are correct.
- Key errors — Your operator key wasn't imported. See restoring your operator key (/validators/setup-guide#restoring-your-operator-key).
- Slow sync after breaking upgrade — Expected. Ensure genesis is set in config.yaml to avoid scanning from block 0.

Changelog

Source: <https://docs.genlayer.com/validators/changelog>

v0.5.12

Misc

- Update dependency ethers to ~5.8.0
- Update all non-major dependencies

Bug fixes

- Self-recover from sync stall wedge
- Reclassify zkSync not-found errors

Security

- Update module golang.org/x/net to v0.55.0
- Update module golang.org/x/crypto to v0.52.0

v0.5.11

New features

- Add node health state machine
- Scope admin tokens per role
- Add circuit breaker for RPC outages
- Add bearer auth for admin endpoints
- Unify CORS and WS origins allowlist
- Inbound WebSocket JSON-RPC endpoint

Misc

- Tag runtime metrics by network
- Cap PreloadAccepted concurrency
- Improve historical sync by sync-mode cleanup
- Reframe /snapshot examples for SSH access; drop cloud refs
- Default inbound WS path to /ws
- Default WS example to single-listener
- Multiplex WS subscriptions
- Trace context cancellations on all watchers
- Remove unused WaitMined method

Bug fixes

- Guard nil GasFeeCap on bump retry
- Ensure GenVM precompile cache before manager start
- Keep sync ticker running while Synced
- Create export file when --path is missing
- Autodetect GenVM permit capacity
- RPC debug error codes and id default
- Trust XFF + cap WS frame size
- Never halt sync on bloom mismatch
- Refresh version on skip-sync
- Route receipt polls to poll probes
- Handle -32602 decode-rejection and cap GasFeeCap
- Cap RPC body/batch and admin loopback bind

Security

- Update module go.opentelemetry.io/otel to v1.41.0

v0.5.10

New features

- Resilient WS RPC for fallback path

Bug fixes

- Close activator-binding race
- Exit finalize loop on post-Submit race
- Persist gas high-water across cycles
- Lift gas ceiling on retry-count

- Bound cancel gasFeeCap and lift bump ceiling on blocker lookup
- Fall back to WS on block-not-found
- Repair self-cancellation recovery
- Make chain-submit buffer configurable and stop collapsing on short step_timeout
- Add fault label to RPC error metric
- Unwedge replay and filter sync

Misc

- Label RPC metrics with provider
- Add chain nonce and desync gauges metrics
- Log timestamps after acquireTransaction

v0.5.9

Bug fixes

- Count real provider failures on SLA
- Harden nonce management
- Check per-block bloom filters for multi-block ranges
- Pool HTTP transport for chain RPC
- Self-cancel stuck nonce on mempool replacement exhaustion
- Prevent sync panic and self-dependency cycle during tx replay
- Resume sync after WS reconnect

Misc

- Use RFC3339Nano for zerolog timestamps
- Extract encoding functions into dedicated encoding package
- Move consensus entity types to entities package
- Add gen_dbg_traceTransaction API doc

v0.5.8

New features

- Add gen_syncing endpoint for sync status visibility
- Add bounded sync wait for RPC readiness
- Add per-IP per-method RPC rate limiting

Bug fixes

- Defer GenVM precompile to container startup to avoid AVX-512 incompatibility
- Replace stuck transactions with gas bump
- Skip duty recovery when behind and handle chained missing tx data dependencies
- Add hiccup retry to sync pipeline
- Recover prime after reorg
- Fix RPC max block range exceeded in event router
- Return -32001 for missing contract errors
- Pass transaction value to GenVM
- Standardize RPC error codes per EIP-1474
- Retry tx lookup for RPC indexing lag

Misc

- Consolidate genvm configs
- Add gen_getTransactionStatus docs
- Add gen_syncing API docs and clarify RPC config group vs method precedence

v0.5.7

New features

- Add LLM greyboxing with configurable chain order via YAML meta

Bug fixes

- Use PendingAt for idle activator rotation
- Remove unnecessary workaround from view calls
- Restart sync pipeline after RPC failures
- Suppress shutdown errors in watchers
- Reduce FindAcceptanceBlock range to fit RPC limit

Misc

- Include Alloy healthcheck script in release tarball
- Increase Alloy push intervals to 60s

v0.5.6

Bug fixes

- Add priority admission control

v0.5.5

New features

- Add network name to health endpoint
- Add network label to logger and all metrics
- Add metrics to rollup RPC transport and WebSocket

Bug fixes

- Recover from cache miss on late lifecycle events
- Wait for queue head before finalize with recipient-aware queue management
- Suppress context canceled errors on shutdown
- Raise gas bump ceiling and retries
- Stuck nonce and finalize resilience
- Patch GenVM dylib references on macOS during download

Misc

- Downgrade finalized-not-in-cache log to Debug

v0.5.4

New features

- Add --network flag to doctor command for multi-network validation

v0.5.3

New features

- Add network label to telemetry metrics and logs

v0.5.2

New features

- Add recovery for missing TxData

Bug fixes

- Invalidate timeout cache on SetTimeouts
- Dedup propose on duplicate events

- Stop retrying on ValidatorSelectionFailed
- Prevent sync error race on ctx cancel
- Recover sync on missing TxData
- Reject stale block in idle check
- Increment leader revealed metric counter
- Sync commit idle timeout with contract timestamps

Misc

- Use CtxLogger in HTTP handler constructors

v0.5.1

New features

- Add transaction manager for nonce resilience
- Add leader_results input and nondetDisagreementCallNo to gen_call
- Handle hanging transactions via processIdleness
- Add Morpheus as OpenAI-compatible LLM provider

Misc

- Dedup duplicate NewTransaction events per block
- Add CanFinalize pre-check before finalization
- Make sync idempotent
- Per-recipient head-by-head idleness finalization

Bug fixes

- Detect ZkSync nonce inconsistency error
- Capture RPC error message when data is not decodable
- Correct consensus address and genesis block
- Enrich context errors with last failure
- Fetch on-chain data in Accepted/Undetermined handlers

v0.5.0

New features

- Upgrade consensus contracts from v0.4 to v0.5
- Add gen_getContractCode endpoint
- Return full VMResult data from gen_call

Bug fixes

- Check pending queue head before activation
- Enable and fix `gen_call` write simulation
- Exclude stack field from console output
- Move GenVM execution outside retry loop to avoid double GenVM execution on submitting failure
- Use `ConsoleWriter` for caller field filtering
- Remove nested retry from epoch finalization
- Add logging for silent tx submission failures
- Retry `ProcessIdleness` on failure in `lastVoteWatcher`
- Handle shutdown errors gracefully
- Remove `--create-venv=false` flag from post-install scripts
- Install Python dependencies globally to avoid `venv` build issues
- Update validator priming logic to use current epoch
- Epoch advance scheduler not running for epoch 0
- Prevent panic on closing done channel during event processing
- Handle new consensus `ResultType`
- New behavior for finalizing epochs

Misc

- Inject `txID` into context for automatic log propagation
- Enable validator idle reveal idleness
- Add profiling with `pyroscope`
- Fix to handle `zksync` local-node compatibility

v0.4.5

New features

- Decouple epoch advance from finalization, allowing them to run as independent processes with separate retry mechanisms
- Implement sync waiting mechanism to ensure GenVM has latest state before contract execution
- Add `genlayer_node_info` Prometheus metric exposing node and protocol version

Bug fixes

- Add `canPrime` check to verify epoch state before validator priming, preventing priming when epochs are out of sync
- Return user-friendly error instead of system error when sync times out
- Fix `genvm` executor process handling

Misc

- Improve GenVM logging with context-aware logger
- Update health checker to use ZkSync connectivity checks
- Increase default genvm permits to 8 in config.yaml.example

v0.4.4

New features

- Improve logging with context-aware tracing and cleaner console output
- Refactor ZkSync client to split RPC and WSS connections for better reliability
- Move storage writing to genvm for improved performance

Bug fixes

- Skip priming check for full nodes
- Add Docker socket mount for webdriver metrics collection in docker-compose configurations
- Prevent incorrect sync state by updating blocksBehind calculation
- Fix memory leak from threads package
- Fix memory leak in consensus processing
- Fix bug in latest transaction with effect calculation
- Fix wallet mutex using read lock instead of write lock when unlocking accounts

Misc

- Fix signer for full nodes
- Improve logs for http servers
- Improve signer initialization for faster and more reliable transaction signing
- Add logs for previous activator on idleness checks
- Migrate zksync-go to go-ethereum

v0.4.3

Bug fixes

- Fix node storage handling for non-accepted transactions like LeaderTimeout
- Fix validator priming logic
- Fix default in release/.env.example

v0.4.2

Bug fixes

- Telemetry config and scraping
- Fix genvm readiness attempts

v0.4.1

New features

- Add show and list subcommands for account management
- Split authentication credentials for Prometheus and Loki
- Add IsPrimed health check for unprimed validators
- Add primed state and warning for unprimed validators

Misc

- Refactor txcalldata decoding to a new encoding package
- Remove unused pkg/cryptoutils/address.go
- Update deployment configuration options for network and cleanup
- Refactor gas estimator

Bug fixes

- Implement indefinite retry for validator priming until primed
- Implement health check and timeout handling for GenVM manager
- Fix undetermined transaction rotations when ProcessIdleness
- Improve message handling to prevent dropped events
- Add GenVM contracts cache and enhance sync progress logging
- Fix sync filtering process
- Improve messages around genlayerRPC chain connectivity
- Stop retrying activation after transaction is activated

v0.4.0

New features

- Integrate the latest GenLayer Consensus version v0.4 with Staking
- Introduce validator wallet
- Handle validator priming
- Finalize epochs
- Advance epochs when epochs finalize
- Handle automatic gas increase

- Implement ProcessIdleness and enhance event subscription
- Handle validator commit idleness for leader
- Implement leader idleness detection and rotation
- Add detailed ban information with epoch and permanent flag
- Handle unfinished transactions with batch finalization
- Add nonceSafe for granular transaction locking
- Add sanity check for operator address in validator wallet configuration
- Add missing fields from consensus transaction data to API response
- Add validator telemetry push monitoring
- Add actionable fix suggestions to WebDriver error messages in doctor command
- Add zkSync connectivity checks
- Add Staking Contract address comparison to doctor checks
- Add libertai as a new LLM provider
- Add transaction count metrics for consensus roles
- Implement consensus sync status metrics
- Add TimeoutsSet event router for dynamic timeout configuration
- Add enhanced transaction failure debugging with trace support
- Sync data with finalized and appeal started events
- Add timeout delay mechanism for RPC resilience
- Add status and blockNumber parameters to gen_call
- Enhance health endpoint with version info and validator with ban and permanent ban detection
- Update genvm version to v0.2.7
- Store and expose genvm execution information
- Add sign admin endpoint for genvm

Bug fixes

- GenVM architecture on arm64
- Fix sync issue when there are no consensus events
- Use latest block instead of timestamp for transaction queries
- Update WebDriver check for Puppeteer implementation
- Resolve multi-transaction nonce issue
- Remove duplicated check logic and fix status check on specific block
- Leader rotation timeout and receipt cleanup
- Change transaction finalization to allow all nodes to finalize
- GenVM non determinism
- Add retry logic to prevent validator banning on transient errors
- GenVM sanity checks
- Cancel context and wait for group when genvm tester fails

- Decoding calldata into complex types
- Don't try activating already activated transactions
- Still process sync failed TX for DB consistency
- Add correct encoding for getState
- Only sync transactions with ResultCodeReturn and add sanity check
- Use block number for transaction data retrieval
- Validator votes idleness
- Leader idleness
- Accidental deadlock in genvm
- Lazy validators commit order

Misc

- Simplify environment variable configuration using .env file
- Add environment variable override support for config values
- Update webdriver image to v0.0.9
- Update RPC and WebSocket endpoints in config.yaml.example
- Correct status for validatorTimeout after appeal time window
- Validate presence of abigen in dependency check
- Rename environment validation functions for clarity
- Use cockroachdb errors package for error handling
- Remove wallet.yaml file handling
- Rename archive to full

v0.3.11

New features

- Add validator telemetry push monitoring

Bug fixes

- Validator banning on parallel transactions executions

Misc

- Document health RPC endpoint

v0.3.10

New features

- Add actionable fix suggestions to WebDriver error messages in doctor command

Bug fixes

- Add retry logic to prevent validator banning on transient errors

v0.3.9

New features

- Add doctor command zkSync connectivity checks

Misc

- Simplify contract initialization with auto-discovery

Bug fixes

- Remove response body size limit for WebDriver

v0.3.8

New features

- Add Staking Contract address comparison to doctor checks
- Add libertai as a new LLM provider

Misc

- Enhance metrics logging by adding blocks behind calculation
- Print address for account not found
- Fix typo in telemetry configuration comment

Bug fixes

- Enable CORS support for HTTP methods in the API
- Move sync success log after state update
- Don't panic with metrics + print panics

v0.3.7

New features

- Add transaction count metrics for consensus roles

- Implement consensus sync status metrics
- Add node uptime collector
- Add disk storage usage metrics for monitoring
- Add comprehensive metrics collection for ops visibility

v0.3.6

Misc

- Add enhanced transaction failure debugging with trace support

v0.3.5

New features

- Add Transport with exponential backoff retry logic
- Add timeout delay mechanism for RPC resilience
- Add status and blockNumber parameters to gen_call
- Enhance health endpoint with version info and validator ban detection
- Add balance endpoint for validator nodes
- Add gas estimation failover to all consensus transaction functions
- Implement gen_getContractState endpoint
- Add ionet as inference provider
- Add gen_getTransactionStatus endpoint
- Implement gen_getTransactionReceipt
- Add log upload system for validator diagnostics

Misc

- Add comprehensive API documentation system and operations endpoints
- Move retry logic to infrastructure and update DI
- Improve timeout detection with type-based checking
- Enhance WebDriver health check validation

Bug fixes

- Fix validator idleness commit flow
- Handle methods and groups configuration for RPC methods

v0.3.4

New features

- Added genesis block configuration support for faster node startup
- Enhanced doctor command with GenVM diagnostics integration
- Updated consensus contract addresses for network upgrade

Misc

- Improved node synchronization performance with genesis block hints
- Enhanced validator configuration validation and error reporting

v0.3.3

New features

- Added Comput3 (/validators/setup-guide#genvm-configuration) as a supported LLM provider for validators

v0.3.2

New features

- Add command doctor to check Consensus Contract configuration

Misc

- Implement retry logic with double-checked locking for GetTimeouts
- Update ABI for consensus contracts.
- Decode error returns from the network to improve error handling.

v0.3.1

Misc

- Add logs details

v0.3.0

New features

- Added appeals functionality
- Added gen_getContractSchema
- Added admin checkpoint endpoint
- Added admin "stop at block" endpoint

- Handle leader timeout
- Handle validator timeout
- Added version command

Bug fixes

- Mitigate TxAccepted events being dropped
- Waiting on event timeouts is now absolute
- Patch up genvm timeouts
- Add retry mechanism for genvm contracts interaction
- Listen to leader timeout event to finalize transaction

v0.2.0-testnet007

New features

- Activator idle replacement
- Leader idle replacement
- Validator idle replacement
- Check for genvm on startup
- Log file support for node
- Ability to export private key in account command

Bug fixes

- Build info logs now display correctly
- Fixed config relative path
- Node now fails fast when accepted transaction processing fails

Misc

- Updated module github.com/btcsuite/btcd to v0.24.2 (security)
- Improved sync speed on node startup

v0.2.0-testnet005

New features

- Discover Consensus main genesis from deployment (contracts)

Bug fixes

- All accepted sync transactions are now processed atomically per block

- Fixed deadlock on cancelled context

v0.2.0-testnet004

Bug fixes

- Updated genvm to v0.0.18

v0.2.0-testnet003

New features

- Updated genvm to v0.0.17

v0.2.0-testnet002

New features

- Use different address to run e2e tests
- Updated consensus contract addresses in config

v0.2.0-testnet001

New features

- Updated genvm to v0.0.16

Bug fixes

- Log level is no longer ignored
- Revived trie dumping and enhanced logging
- Always kill genvm modules

Misc

- Updated module golang.org/x/net to v0.38.0 (security)

Api References

Source: <https://docs.genlayer.com/api-references>

- GenLayer CLI (/api-references/genlayer-cli)
- GenLayerJS (/api-references/genlayer-js)
- GenLayerPY (/api-references/genlayer-py)
- GenLayer Test (/api-references/genlayer-test)
- GenLayer Node API (/api-references/genlayer-node)
- GenLayer SDK (<https://sdk.genlayer.com/>)

GenLayer CLI

Source: <https://docs.genlayer.com/api-references/genlayer-cli>

Description

The GenLayer CLI is designed to streamline the setup and local execution of the GenLayer simulator. This tool automates the process of downloading and launching the GenLayer simulator, making it easy to start simulating and testing locally with minimal setup.

Installation

Before installing the GenLayer CLI, ensure you have Node.js installed on your system. You can then install the CLI globally using npm:

```
npm install -g genlayer
```

Linux Dependencies

On some Linux distributions with minimal setups (like Debian netinst or Docker images), you may need to manually install libsecret:

```
# Ubuntu/Debian
sudo apt-get install libsecret-1-0

# CentOS/RHEL/Fedora
sudo yum install libsecret
# or for newer versions
sudo dnf install libsecret

# Arch Linux
sudo pacman -S libsecret
```

The GenLayer CLI uses the keytar library for secure key storage, which relies on libsecret on Linux systems.

Usage

Each command includes syntax, usage information, and examples to help you effectively use the CLI for interacting with the GenLayer environment.

Command line syntax

General syntax for using the GenLayer CLI:

```
genlayer command [command options] [arguments...]
```

Commands and usage

Initialize

Prepares and verifies your environment to run the GenLayer Studio.

```
USAGE:
  genlayer init [options]

OPTIONS:
  --numValidators <numValidators>    Number of validators (default: "5")
  --headless                         Headless mode (default: false)
  --reset-db                         Reset Database (default: false)
  --localnet-version <localnetVersion> Select a specific localnet version
  --ollama                           Enable Ollama container (default: false)

EXAMPLES:
  genlayer init
  genlayer init --numValidators 10 --headless --reset-db --localnet-version v0.10.2
  genlayer init --ollama
```

Version Compatibility

The GenLayer CLI always uses the latest compatible version of the environment, ensuring that you benefit from the most recent features, bug fixes, and optimizations without requiring manual updates. If a specific version is needed, you can specify it using the `--localnet-version` option when initializing the environment.

```
genlayer init --localnet-version v0.10.2
```

Start GenLayer environment

Launches the GenLayer environment and the Studio, initializing a fresh set of database and accounts.

```
USAGE:
  genlayer up [options]

OPTIONS:
  --reset-validators                Remove all current validators and create new random ones (default: false)
  --numValidators <numValidators>    Number of validators (default: "5")
  --headless                       Headless mode (default: false)
  --reset-db                       Reset Database (default: false)
  --ollama                         Enable Ollama container (default: false)

EXAMPLES:
  genlayer up
  genlayer up --reset-validators --numValidators 8 --headless --reset-db
  genlayer up --ollama
```

Stop GenLayer environment

Stops all running GenLayer Localnet services.

```
USAGE:
  genlayer stop
```

Create a New GenLayer Project

Initialize a new GenLayer project using a local template.

```
USAGE:
  genlayer new <projectName> [options]

OPTIONS:
  --path <directory>  Specify the directory for the new project (default: ".")
  --overwrite         Overwrite existing directory if it exists (default: false)

EXAMPLES:
  genlayer new myProject
  genlayer new myProject --path ./customDir
  genlayer new myProject --overwrite
```

Manage CLI Configuration

Configure the GenLayer CLI settings.

```
USAGE:
  genlayer config <command> [options]

COMMANDS:
  set <key=value>  Set a configuration value
  get [key]        Get the current configuration
  reset <key>      Reset a configuration value to its default

EXAMPLES:
  genlayer config get
  genlayer config get defaultOllamaModel
  genlayer config set defaultOllamaModel=deepseek-r1
  genlayer config reset keyPairPath
```

Network Management

Manage network configuration for contract operations.

```
USAGE:
  genlayer network set [network]  Set the network to use
  genlayer network info           Show current network configuration and contract addresses
  genlayer network list           List available networks

EXAMPLES:
  genlayer network set
  genlayer network set testnet
  genlayer network set mainnet
  genlayer network info
  genlayer network list
```

Deploy and Call Intelligent Contracts

Deploy and interact with intelligent contracts.

```
USAGE:
  genlayer deploy [options]
  genlayer call <contractAddress> <method> [options]
  genlayer write <contractAddress> <method> [options]
  genlayer schema <contractAddress> [options]

OPTIONS (deploy):
  --contract <contractPath>  (Optional) Path to the intelligent contract to deploy
  --rpc <rpcUrl>             RPC URL for the network
  --args <args...>           Contract arguments (see Argument Types below)

OPTIONS (call):
  --rpc <rpcUrl>             RPC URL for the network
```

```
--args <args...>           Method arguments (see Argument Types below)

OPTIONS (write):
  --rpc <rpcUrl>           RPC URL for the network
  --args <args...>         Method arguments (see Argument Types below)
```

```
OPTIONS (schema):
  --rpc <rpcUrl>           RPC URL for the network
```

```
EXAMPLES:
  genlayer deploy
  genlayer deploy --contract ./my_contract.gpy
  genlayer deploy --contract ./my_contract.gpy --args "arg1" "arg2" 123
  genlayer call 0x123456789abcdef greet --args "Hello World!"
  genlayer write 0x123456789abcdef updateValue --args 42
  genlayer write 0x123456789abcdef sendReward --args 0x6857Ed54CbafaA74Fc0357145eC0ee1536ca45A0
  genlayer write 0x123456789abcdef setScores --args '[1, 2, 3]'
  genlayer write 0x123456789abcdef setConfig --args '{"timeout": 30, "retries": 5}'
  genlayer schema 0x123456789abcdef
```

Argument Types

The `--args` option automatically detects and converts values to the correct type:

Type	Syntax	Example
Boolean	<code>`true`</code> , <code>`false`</code>	<code>`--args true false`</code>
Null	<code>`null`</code>	<code>`--args null`</code>
Integer	numeric value	<code>`--args 42 -1`</code>
Hex integer	<code>`0x`</code> prefix	<code>`--args 0x1a`</code>
String	any other value	<code>`--args hello "multi word"`</code>
Address	40 hex chars with <code>`0x`</code> or <code>`addr#`</code> prefix	<code>`--args 0x6857...a0`</code> or <code>`--args addr#6857...a0`</code>
Bytes	<code>`b#`</code> prefix + hex	<code>`--args b#deadbeef`</code>
Array	JSON array in quotes	<code>`--args '[1, 2, "three"]`</code>
Dict	JSON object in quotes	<code>`--args '{"key": "value"}`</code>

Large numbers that exceed JavaScript's safe integer range are automatically handled as `BigInt` to preserve precision.

Deploy Behavior

- If `--contract` is specified, the command will deploy the given contract.
- If `--contract` is omitted, the CLI will search for scripts inside the deploy folder, sort them, and execute them sequentially.

Call vs Write

- `call` - Calls a contract method without sending a transaction or changing the state (read-only)
- `write` - Sends a transaction to a contract method that modifies the state

Schema

- `schema` - Retrieves the contract schema

Transaction Operations

```
USAGE:
  genlayer receipt <txId>           Get transaction receipt
  genlayer appeal <txId>           Appeal a transaction
  genlayer appeal-bond <txId>       Show minimum appeal bond required

OPTIONS (receipt):
  --status <status>               Status to wait for (default: FINALIZED)
  --retries <retries>             Number of retries (default: 100)
  --interval <interval>          Interval between retries in ms (default: 5000)
```

```

--stdout          Print only stdout from the receipt
--stderr          Print only stderr from the receipt

OPTIONS (appeal):
--bond <amount>   Appeal bond amount (e.g. 500gen, 0.5gen). Auto-calculated if omitted
--rpc <rpcUrl>    RPC URL override

EXAMPLES:
# Check the minimum bond required to appeal
genlayer appeal-bond 0x1234...

# Appeal with auto-calculated bond
genlayer appeal 0x1234...

# Appeal with explicit bond
genlayer appeal 0x1234... --bond 500gen

```

Transaction Trace

Inspect execution traces for debugging:

```
genlayer transactions trace <txId> [--round N] [--rpc URL]
```

Account Management

View and manage your account.

```

USAGE:
genlayer account          Show account info (address, balance, network, status)
genlayer account create [options] Create a new account
genlayer account send <to> <amount> Send GEN to an address
genlayer account unlock   Unlock account (cache key in OS keychain)
genlayer account lock     Lock account (remove key from OS keychain)

OPTIONS (create):
--output <path>          Path to save the keystore (default: "./keypair.json")
--overwrite              Overwrite existing file (default: false)

EXAMPLES:
genlayer account
genlayer account create
genlayer account create --output ./my_key.json --overwrite
genlayer account send 0x123...abc 10gen
genlayer account send 0x123...abc 0.5gen
genlayer account unlock
genlayer account lock

```

Update Resources

Manage and update models or configurations.

```

USAGE:
genlayer update ollama [options]

OPTIONS:
--model [model-name] Specify the model to update or pull
--remove              Remove the specified model instead of updating

EXAMPLES:
genlayer update ollama
genlayer update ollama --model deepseek-r1
genlayer update ollama --model deepseek-r1 --remove

```

Localnet Validator Management

Manage localnet validator operations.


```

USAGE:
  genlayer localnet validators <command> [options]

COMMANDS:
  get [--address <validatorAddress>]      Retrieve details of a specific validator or all validators
  delete [--address <validatorAddress>]    Delete a specific validator or all validators
  count                                     Count all validators
  update <validatorAddress> [options]      Update a validator details
  create-random [options]                  Create random validators
  create [options]                          Create a new validator

OPTIONS (update):
  --stake <stake>                          New stake for the validator
  --provider <provider>                     New provider for the validator
  --model <model>                           New model for the validator
  --config <config>                         New JSON config for the validator

OPTIONS (create-random):
  --count <count>                           Number of validators to create (default: "1")
  --providers <providers...>                Space-separated list of provider names (e.g., openai ollama)
  --models <models...>                     Space-separated list of model names (e.g., gpt-4 gpt-4o)

OPTIONS (create):
  --stake <stake>                           Stake amount for the validator (default: "1")
  --config <config>                         Optional JSON configuration for the validator
  --provider <provider>                     Specify the provider for the validator
  --model <model>                           Specify the model for the validator

EXAMPLES:
  genlayer localnet validators get
  genlayer localnet validators get --address 0x123456789abcdef

  genlayer localnet validators count
  genlayer localnet validators delete --address 0x123456789abcdef
  genlayer localnet validators update 0x123456789abcdef --stake 100 --provider openai --model gpt-4

  genlayer localnet validators create
  genlayer localnet validators create --stake 50 --provider openai --model gpt-4
  genlayer localnet validators create-random --count 3 --providers openai --models gpt-4 gpt-4o

```

Staking Operations

Manage staking for validators and delegators on testnet-bradbury (or testnet-asimov). Staking is not available on localnet/studio.

```

USAGE:
  genlayer staking <command> [options]

COMMANDS:
  validator-join [options]                Join as a validator by staking tokens
  validator-deposit [options]              Make an additional deposit as a validator
  validator-exit [options]                 Exit as a validator by withdrawing shares
  validator-claim [options]                Claim validator withdrawals after unbonding period
  validator-prime [validator]              Prime a validator for the next epoch
  prime-all [options]                     Prime all validators that need priming
  delegator-join [options]                 Join as a delegator by staking with a validator
  delegator-exit [options]                 Exit as a delegator by withdrawing shares
  delegator-claim [options]                Claim delegator withdrawals after unbonding period
  validator-info [validator]               Get information about a validator (--debug for raw data)
  validator-history [validator]             Show slash and reward history for a validator
  delegation-info [validator]              Get delegation info for a delegator with a validator
  epoch-info [options]                     Get current/previous epoch info (--epoch <n> for specific)
  validators [options]                     Show validator set with stake, primed status, and weight
  active-validators [options]              List all active validators
  quarantined-validators                    List all quarantined validators
  banned-validators                        List all banned validators

COMMON OPTIONS (all commands):
  --network <network>                     Network to use (localnet, testnet-bradbury, testnet-asimov)
  --rpc <rpcUrl>                           RPC URL override
  --staking-address <address>              Staking contract address override

OPTIONS (validator-join):
  --amount <amount>                       Amount to stake (in wei or with 'gen' suffix)

```

```

--operator <address>           Operator address (defaults to signer)

OPTIONS (delegator-join):
--validator <address>          Validator address to delegate to
--amount <amount>              Amount to stake (in wei or with 'aen' suffix)

OPTIONS (exit commands):
--shares <shares>              Number of shares to withdraw
--validator <address>          Validator address (for delegator commands)

EXAMPLES:

# Get epoch info (uses --network to specify testnet-bradbury)
genlayer staking epoch-info --network testnet-bradbury

# Or set network globally first
genlayer network set testnet-bradbury

# Join as validator with 42000 GEN
genlayer staking validator-join --amount 42000gen

# Join as delegator with 42 GEN
genlayer staking delegator-join --validator 0x... --amount 42gen

# Check validator info
genlayer staking validator-info --validator 0x...
# Output:
# {
#   validator: '0xa8f1BF1e5e709593b4468d7ac5DC315Ea3CAe130',
#   vStake: '0.01 GEN',
#   vShares: '10000000000000000',
#   dStake: '0 GEN',
#   dShares: '0',
#   vDeposit: '0 GEN',
#   vWithdrawal: '0 GEN',
#   epoch: '0',
#   live: true,
#   banned: 'Not banned'
# }

# Get current epoch info (shows current + previous epoch)
genlayer staking epoch-info
# Output:
# ✓ Epoch info
#
#   Current Epoch: 5 (started 9h 30m ago)
#   Next Epoch:    in 14h 30m
#   Validators:    33
#   ...
#
#   Previous Epoch: 4 (finalized)
#   Inflation:      1732904.66 GEN
#   Claimed:        0 GEN
#   Unclaimed:      1732904.66 GEN
#   ...

# Query specific epoch data
genlayer staking epoch-info --epoch 4

# List active validators
genlayer staking active-validators
# Output:
# {
#   count: 6,
#   validators: [
#     '0xa8f1BF1e5e709593b4468d7ac5DC315Ea3CAe130',
#     '0xe9246A020cbb4fC6C46e60677981879c9219e8B9',
#     ...
#   ]
# }

# Show validator set table with stake, status, weight
genlayer staking validators
aenlayer staking validators --all # Include banned validators

# Show validator slash/reward history (testnet only, default: last 10 epochs)
genlayer staking validator-history 0x...
genlayer staking validator-history 0x... --epochs 5 # Last 5 epochs
genlayer staking validator-history 0x... --from-epoch 3 # From epoch 3
genlayer staking validator-history 0x... --all # Complete history (slow)
genlayer staking validator-history 0x... --limit 20

```

Output:

#	Time	Epoch	Type	Details	GL TxId / Block
#	12-11 14:20	5	REWARD	Val: ...	block 4725136
#	12-10 18:39	4	SLASH	1.00%	0x52db90a9...

Exit and claim (requires validator wallet address)
genlayer staking validator-exit --validator 0x... --shares 100
genlayer staking validator-claim --validator 0x...

Prime a validator for next epoch
genlayer staking validator-prime 0x...

Prime all validators that need priming (anyone can call)
genlayer staking prime-all

Running the CLI from the repository

First, install the dependencies and start the build process

```
npm install
npm run dev
```

This will continuously rebuild the CLI from the source

Then in another window execute the CLI commands like so:

```
node dist/index.js init
```

Guides

- Validator Guide (docs/validator-guide.md) - How to become a validator on GenLayer testnet
- Delegator Guide (docs/delegator-guide.md) - How to delegate GEN to a validator

Documentation

For detailed information on how to use GenLayer CLI, please refer to our documentation (<https://docs.genlayer.com/>).

Contributing

We welcome contributions to GenLayerJS SDK! Whether it's new features, improved infrastructure, or better documentation, your input is valuable. Please read our CONTRIBUTING (<https://github.com/yeagerai/genlayer-js/blob/main/CONTRIBUTING.md>) guide for guidelines on how to submit your contributions.

License

This project is licensed under the ... License - see the LICENSE (LICENSE) file for details.

Usage

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking>

```
$ genlayer staking [options] [command]
```

Arguments

- [command]

Options

Short	Long	Description	Required	Default
-h	--help	display help for command	No	

Subcommands

- genlayer wizard — Interactive wizard to become a validator
- genlayer validator-join — Join as a validator by staking tokens
- genlayer validator-deposit — Make an additional deposit to a validator wallet
- genlayer validator-exit — Exit as a validator by withdrawing shares
- genlayer validator-claim — Claim validator withdrawals after unbonding period
- genlayer validator-prime — Prime a validator to prepare their stake record for the next epoch
- genlayer prime-all — Prime all validators that need priming
- genlayer set-operator — Change the operator address for a validator wallet
- genlayer set-identity — Set validator identity metadata (moniker, website, socials, etc.)
- genlayer delegator-join — Join as a delegator by staking with a validator
- genlayer delegator-exit — Exit as a delegator by withdrawing shares from a validator
- genlayer delegator-claim — Claim delegator withdrawals after unbonding period
- genlayer validator-info — Get information about a validator
- genlayer delegation-info — Get delegation info for a delegator with a validator
- genlayer epoch-info — Get current epoch and staking parameters
- genlayer active-validators — List all active validators
- genlayer quarantined-validators — List all quarantined validators
- genlayer banned-validators — List all banned validators
- genlayer validators — Show validator set with stake, status, and voting power
- genlayer validator-history — Show slash and reward history for a validator (default: last 10 epochs)

Usage

Source: <https://docs.genlayer.com/api-references/genlayer-cli/localnet/localnet>

```
$ genlayer localnet [options] [command]
```

Arguments

- [command]

Options

Short	Long	Description	Required	Default
-h	--help	display help for command	No	

Subcommands

- genlayer validators — Manage localnet validators operations

GenLayer Node API

Source: <https://docs.genlayer.com/api-references/genlayer-node>

The GenLayer Node provides a JSON-RPC API (<https://www.jsonrpc.org/specification>) for interacting with it. This API allows you to execute contract calls, retrieve transaction information, and perform various blockchain operations.

GenLayer Methods

gen_call

Executes a new message call immediately without creating a transaction on the blockchain. This method supports read operations, write operations, and contract deployments.

Method: `gen_call`

Parameters:

- request (object, required): The contract call request parameters
- from (string, required): The address making the call
- to (string, required): The contract address to call

- data (string, required): The encoded function call data
- type (string, required): The type of call ("read", "write", or "deploy")
- blockNumber (string, optional): The block number to query at (hex-encoded). If not provided, uses latest block
- status (string, optional): The transaction status filter ("finalized" or "accepted"). If not provided, uses accepted state
- value (string, optional): The value to send with the call (hex-encoded)
- leader_results (array of strings or null, optional): When provided, GenVM runs in validator mode (executing validator_fn instead of leader_fn in run_nondet calls). Each element is a hex-encoded equivalence output from a previous leader execution. When omitted or null, GenVM runs in leader mode

Returns: A structured JSON object containing execution results and debug information

Response Fields:

- data (string): Hex-encoded return value from contract execution
- eqOutputs (array of strings): Equivalence outputs from GenVM execution, each as a hex-encoded string. These are deterministic outputs used for consensus validation. Populated in leader mode; empty in validator mode
- status (object): Execution status
- code (number): Status code (0=success, 1=user error, 2=vm error, 3=internal error)
- message (string): Human-readable status message
- stdout (string): Standard output captured during execution
- stderr (string): Standard error captured during execution
- logs (array): GenVM log entries, each as a JSON object with fields like level, message, target, file, ts
- events (array): Contract events emitted during execution
- topics (array of strings): Event topics as hex-encoded strings
- data (string): Event data as hex-encoded string
- messages (array): Messages emitted by the contract
- messageType (number): Type identifier for the message
- recipient (string): Recipient address
- value (string): Value in hex format
- data (string): Message data as hex-encoded string
- onAcceptance (boolean): Whether message is sent on acceptance
- saltNonce (string): Salt nonce in hex format
- nondetDisagreementCallNo (number or null): When running in validator mode (with leader_results), indicates which run_nondet call the validator disagreed on. Null when there is no disagreement or when running in leader mode

Example - Deploy Call:

```
{
  "jsonrpc": "2.0",
  "method": "gen_call",
  "params": [
    {
      "type": "deploy",
      "data": "0xf9023db90223232076302e312e300a23207b2022446570656e6473223a202270792d67656e6c617965723a3135716669766a7679383038303072683939387063786d64326d38766131777132717a71687a3835306e3867676372346939713022207d0a0a66726f6d2067656e6c6179657220696d706f7274202a0a0a0a2320636f6e747261637420636c6173730a636c6173732053746f7261676528676c2e436f6e7472616374293"
    }
  ]
}
```

Example Response (Deploy Success):

Example - Read Call (Latest Accepted State):

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```
{
  "params": [
    {
      "type": "read",
      "data": "0xd8960e066d6574686f646c6765745f686176655f636f696e00",
      "from": "0xf6A60752C24Cd3BFcAf8d387a5EB62d3746F44EE",
      "to": "0xCe5712E63fd5475288aB1B7c0a368B9417357b81",
      "gas": "0x5208",
      "value": "0x0"
    }
  ],
  "id": 1
}
```

Example - Read Call with Specific Block and Status:

```
{
  "jsonrpc": "2.0",
  "method": "gen_call",
  "params": [
    {
      "type": "read",
      "data": "0xd8960e066d6574686f646c6765745f686176655f636f696e00",
      "from": "0xf6A60752C24Cd3BFcAf8d387a5EB62d3746F44EE",
      "to": "0xCe5712E63fd5475288aB1B7c0a368B9417357b81",
      "blockNumber": "0x151ec5",
      "status": "accepted"
    }
  ],
  "id": 1
}
```

Example - Write Call:

```
{
  "jsonrpc": "2.0",
  "method": "gen_call",
  "params": [
    {
      "from": "0x742d35Cc6634C0532925a3b8D4C9db96c4b4d8b6",
      "to": "0x742d35Cc6634C0532925a3b8D4C9db96c4b4d8b6",
      "data": "0xa9059cbb000000000000000000000000742d35cc6634c0532925a3b8d4c9db96c4b4d8b6",
      "type": "write",
      "gas": "0x5208",
      "value": "0x0"
    }
  ],
  "id": 1
}
```

Example Response with Messages (Router Forward Call):

When a contract emits internal messages (e.g., a router contract forwarding calls), the response includes the messages array:

```
{
  "jsonrpc": "2.0",
  "result": {
    "data": "00",
    "eqOutputs": [],
    "status": {
      "code": 0,
      "message": "success"
    },
    "stdout": "",
    "stderr": "",
    "logs": [
      {
        "file": "src/lib.rs:299",
        "level": "info",
        "message": "metrics",
        "target": "genvm",
        "ts": "1770225770000"
      }
    ]
  }
}
```



```

],
"events": [],
"messages": [
  {
    "messageType": 1,
    "recipient": "0xc346f80c86adf1d71accfc4ef6efe7482b25be40",
    "value": "0x0",
    "data": "0xf86eb86a16046172677331d4023078393164463338654243623930384335643635...",
    "onAcceptance": false,
    "saltNonce": "0x0"
  },
  {
    "messageType": 1,
    "recipient": "0xad26513a4315b0ea8a5c800322b35092a83dfd3d",
    "value": "0x0",
    "data": "0xf86eb86a16046172677331d4023078333738653339374533316346383538333630...",
    "onAcceptance": false,
    "saltNonce": "0x0"
  }
],
"nondetDisagreementCallNo": null
},
"id": 1
}

```

Notes:

- The `blockNumber` parameter can be either hex-encoded with a 0x prefix (e.g., "0x151ec5") or decimal (e.g., "1384133")
- Valid status values are "accepted" (default) and "finalized"
- When both parameters are omitted, the call executes against the latest accepted state
- For read and write type calls, if the contract doesn't exist at the specified block/status combination, an error is returned with code -32001 (Resource not found) and message "contract not found at address 0x..."
- For deploy type calls, the status and `blockNumber` parameters are not used for contract state validation since no contract exists yet

gen_getContractSchema

Retrieve the schema/interface of a GenLayer contract to understand its available methods and properties.

Method: `gen_getContractSchema`

Parameters:

- `request` (object, required): The contract schema request
- `code` (string, required): Base64-encoded contract code

Returns: Contract schema object with methods and properties

```

{
  "ctor": { // constructor info
    "kwparams": {}, // dict
    "params": [["value", "type"]] // list of tuples
  },
  "methods": { // dict from method name to method info
    "method_name": {
      "kwparams": {}, // dict
      "params": [["value", "type"]], // list of tuples
      "payable": false, // bool
      "readonly": false, // bool
      "ret": "null" // string
    },
  },
}

```

```
}
```

Example:

```
{
  "jsonrpc": "2.0",
  "method": "gen_getContractSchema",
  "params": [
    {
      "code": "IyB7ICJEZXBldmRzIjogInB5LWdlbmxeWV0bnRlc3Q0IH0KcmZyb20gZ2VubGF5ZXIgaW1wb3J0ICoKCgojIGNvbnRyYWN0IGNsYXNzCmNsYXNzIFN0b3JhZ2UoZ2wuQ29udHJhY3Qp0gogICAgc3RvcnFnZTogc3RyCgogICAgIyBjb25zdHJ1Y3RvcgogICAgZGVmIF9faW5pdF9fKHNLbGYsIGluaXRpYWxfZ3RvcnFnZTogc3RyKToKICAgICAgICBzZWxmlnN0b3JhZ2UgPSBpbml0aWFsX3N0b3JhZ2UKCiAgICAjIHJlYWQgbWV0aG9kcyBtdXN0IGJlIGFubm90YXRlZCB3aXR0IHZpZXcKICAgIEBnbC5wdWJsawMudmllldwogICAgZGVmIGdldF9zdG9yYWdlKHNLbGYpIC0+IHN0cjoKICAgICAgICByZXRlc4gc2VsZi5zdG9yYWdlCgogICAgIyB3cmI0ZSBtZXRob2QKICAgIEBnbC5wdWJsawMud3JpdGUKICAgIGRlZiB1cGRhdGVfc3RvcnFnZShzZWxmlCBuZXdZ3RvcnFnZTogc3RyKSA0b25ldGogICAgICAgIHNlbGYuc3RvcnFnZSA9IG5ld19zdG9yYWdlCg=="
    }
  ],
  "id": 1
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "result": {
    "ctor": {
      "kwparams": {},
      "params": [["initial_storage", "string"]]
    },
    "methods": {
      "get_storage": {
        "kwparams": {},
        "params": [],
        "readonly": true,
        "ret": "string"
      },
      "update_storage": {
        "kwparams": {},
        "params": [["new_storage", "string"]],
        "payable": false,
        "readonly": false,
        "ret": "null"
      }
    }
  },
  "id": 1
}
```

gen_getContractState

Retrieves the current state of a GenLayer intelligent contract at a specific block height and transaction status.

Method: gen_getContractState

Parameters:

- request (object, required): The contract state request parameters
- address (string, required): The contract address to query
- blockNumber (string, optional): The block number to query at (hex-encoded). If not provided, uses latest block
- status (string, optional): The transaction status filter ("finalized" or "accepted"). If not provided, uses accepted state

Returns: Hex-encoded contract state data

Example Request - Latest Accepted State:

```
{
  "jsonrpc": "2.0",
  "method": "gen_getContractState",
  "params": [
    {
      "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f"
    }
  ],
  "id": 1
}
```

Example Request - Specific Block and Status:

```
{
  "jsonrpc": "2.0",
  "method": "gen_getContractState",
  "params": [
    {
      "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
      "blockNumber": "0x151ec5",
      "status": "accepted"
    }
  ],
  "id": 1
}
```

Example Response:

```
{
  "jsonrpc": "2.0",
  "result": "0xd9960e0773746f726167656c6e757064617465642074657374207374",
  "id": 1
}
```

Notes:

- The blockNumber parameter should be hex-encoded with a 0x prefix
- Valid status values are "accepted" (default) and "finalized"
- The returned state is hex-encoded binary data with a 0x prefix
- Empty contract state returns "0x"
- If the contract doesn't exist at the specified block/status, an error will be returned

cURL Example:

```
# Get latest accepted state
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getContractState",
  "params": [{
    "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f"
  }],
  "id": 1
}'

# Get state at specific block with accepted status
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getContractState",
  "params": [{
    "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
    "blockNumber": "0x151ec5",
    "status": "accepted"
  }],
  "id": 1
}'
```

gen_getContractCode

Retrieves the code of a deployed GenLayer intelligent contract at a specific block height and transaction status.

Method: `gen_getContractCode`

Parameters:

- request (object, required): The contract code request parameters
- address (string, required): The contract address to query
- blockNumber (string, optional): The block number to query at (hex-encoded). If not provided, uses latest block
- status (string, optional): The transaction status filter ("finalized" or "accepted"). If not provided, uses accepted state

Returns: Base64-encoded contract code

Example Request - Latest Accepted State:

```
{
  "jsonrpc": "2.0",
  "method": "gen_getContractCode",
  "params": [
    {
      "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f"
    }
  ],
  "id": 1
}
```

Example Request - Specific Block and Status:

```
{
  "jsonrpc": "2.0",
  "method": "gen_getContractCode",
  "params": [
    {
      "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
      "blockNumber": "0x151ec5",
      "status": "finalized"
    }
  ],
  "id": 1
}
```

Example Response:

```
{
  "jsonrpc": "2.0",
  "result": "IyB7ICJEZXB1bmRzIjogInB5LWdlbmhewVyOnRlc3QiIH0KcmZyb20gZ2VubGF5ZXIGaw1wb3J0ICoKCgpjbGZFzyBTdG9yYWdlKGdsLkNvbmlRyYWN0KTokICAgIAHN0b3JhZ2U6IHNoCgoKICAgIGRLZiBiF2luaXRfXyhzhZWxmLCBpbml0aWFsX3N0b3JhZ2U6IHNoCik6CiAgICAgICAgc2VsZi5zdG9yYWdlID0gaW5pdGlhbF9zdG9yYWdlCGogICAgQGdsLnB1YmxpYy52aWV3CiAgICBkZWYgd2V0X3N0b3JhZ2Uoc2VsZikgLTA4cy3RyOgogICAgICAgIGlHJldHYyb3ZwXmLnN0b3JhZ2UKCiAgICBAZ2wucHVibGljLndyaXRlCiAgICBkZWYgdXBkYXRlX3N0b3JhZ2Uoc2VsZiwgbmV3X3N0b3JhZ2U6IHNoCikgLTA4Tm9uZToKICAgICAgICBZZWxmLnN0b3JhZ2UgPSBuZXdjc3RvcmlFnZQo=",
  "id": 1
}
```

Notes:

- The blockNumber parameter should be hex-encoded with a 0x prefix
- Valid status values are "accepted" (default) and "finalized"

- The returned code is base64-encoded
- If the contract doesn't exist at the specified block/status, an error will be returned
- If the address does not have code (not a contract), an error will be returned

cURL Example:

```
# Get latest accepted code
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getContractCode",
  "params": [{
    "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f"
  }],
  "id": 1
}'

# Get code at specific block with finalized status
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getContractCode",
  "params": [{
    "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
    "blockNumber": "0x151ec5",
    "status": "finalized"
  }],
  "id": 1
}'
```

gen_getTransactionReceipt

Retrieves the transaction receipt for a given transaction ID. This endpoint is a proxy to the underlying consensus layer's `getTransactionAllData` function, providing detailed information about a transaction that has been processed.

Method: `gen_getTransactionReceipt`

Parameters:

- request (object, required):
- txId (string, required): The 32-byte transaction hash as a hexadecimal string with a 0x prefix.

Returns:

A Transaction object which contains the details of the transaction. The structure is based on `ITransactionsTransaction` from the consensus contracts. The object has the following fields:

- id (string): The transaction ID as a hex string.
- txOrigin (address): The original address that initiated the transaction.
- sender (address): The address of the sender.
- recipient (address): The destination address of the transaction.
- activator (address): The address of the validator that activated the transaction.
- status (uint8): The current status of the transaction.
- previousStatus (uint8): The previous status of the transaction.

- txSlot (uint256): The transaction slot number.
- initialRotations (uint256): The initial number of rotations.
- numOfInitialValidators (uint256): The number of initial validators.
- randomSeed (string): The random seed used for the transaction as a hex string.
- txExecutionHash (string): The execution hash of the transaction as a hex string.
- txCallData (string): The transaction call data as a hex string.
- eqBlocksOutputs (string): The equivalent blocks outputs as a hex string.
- timestamps (object): Object containing various timestamps:
 - Created (uint256): When the transaction was created.
 - Pending (uint256): When the transaction became pending.
 - Activated (uint256): When the transaction was activated.
 - Proposed (uint256): When the transaction was proposed.
 - Committed (uint256): When the transaction was committed.
 - LastVote (uint256): When the last vote was cast.
 - AppealSubmitted (uint256): When an appeal was submitted.
 - LeaderRevealed (uint256): When the leader was revealed.
- result (uint256): The result of the transaction execution.
- readStateBlockRanges (array): Array of block range objects, each containing:
 - ActivationBlock (uint256): The activation block number.
 - ProcessingBlock (uint256): The processing block number.
 - ProposalBlock (uint256): The proposal block number.
- roundData (array): Array of round data objects containing:
 - round (uint256): The round number.
 - leaderIndex (uint256): Index of the round leader.
 - votesCommitted (uint256): Number of votes committed.
 - votesRevealed (uint256): Number of votes revealed.
 - appealBond (uint256): The appeal bond amount.
 - rotationsLeft (uint256): Number of rotations remaining.
 - result (uint256): The round result.
- roundValidators (array): Array of validator addresses for this round.
- validatorVotes (string): Base64-encoded validator votes.
- validatorVotesHash (array): Array of vote hashes as hex strings.
- validatorResultHash (array): Array of result hashes as hex strings.

Example Request:

```
{
  "jsonrpc": "2.0",
  "method": "gen_getTransactionReceipt",
  "params": [
    {
      "txId": "0x635060dd514082096d18c8eb64682cc6a944f9ce1ae6982feb7a71e9f656f49"
    }
  ],
}
```

```
"id": 1
}
```

Example Response:

```
{
  "jsonrpc": "2.0",
  "id": 1,
  "result": {
    "id": "0x635060dd514082096d18c8eb64682cc6a944f9celae6982feb7a71e9f656f49",
    "txOrigin": "0x5b70759760a85d92bf972b67ee8558e8e8da87d4",
    "sender": "0x5b70759760a85d92bf972b67ee8558e8e8da87d4",
    "recipient": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
    "activator": "0xf27b07d36ff5b9dcd37fd3a09811d6456dd4483a",
    "status": 14,
    "previousStatus": 0,
    "txSlot": 1,
    "initialRotations": 1,
    "numOfInitialValidators": 5,
    "randomSeed": "0x3d0bdf63bcelcf84b155f1557bb15e303b903b1b559325b6492794a5e498954",
    "txExecutionHash": "0x76744dc491b1725eb2ba08cf516855c14aac310e36711368c04442fd39d61853",
    "txCallData":
    "f5b31604617267730da4017570646174656420746573742073746f72616765066d6574686f64747570646174655f73746f7261676500",
    "eqBlocksOutputs": "c0",
    "timestamps": {
      "Created": 1750774777,
      "Pending": 1750774777,
      "Activated": 1750774781,
      "Proposed": 1750774786,
      "Committed": 1750774791,
      "LastVote": 0,
      "AppealSubmitted": 0,
      "LeaderRevealed": 0
    },
    "result": 0,
    "readStateBlockRanges": [
      {
        "ActivationBlock": 1377797,
        "ProcessingBlock": 1377797,
        "ProposalBlock": 1377798
      }
    ],
    "roundData": [
      {
        "round": 0,
        "leaderIndex": 3,
        "votesCommitted": 5,
        "votesRevealed": 0,
        "appealBond": 0,
        "rotationsLeft": 1,
        "result": 0,
        "roundValidators": [
          "0xa707a1d9dd946242f921f0ad62f61e1b7df30c85",
          "0xf27b07d36ff5b9dcd37fd3a09811d6456dd4483a",
          "0xf3738216c3208d77a9db79a1d31824e5133083c4",
          "0x56394354538bffe0ee5021c2b6d3522fce30257",
          "0xcfdbac032f82f388d9eca76efa6610b98e7f59fe"
        ],
        "validatorVotes": "AAAAAA=",
        "validatorVotesHash": [
          "54e3478b02dfb009a35d46b19ac7c967b65416430a33d3860e9301a0f2b19d57",
          "f74a48e7283c91757bf2c612e40f8a9eb69ea0f02cf5e6b89c3c9b6ad629a3c1",
          "2f51c3b1d9ff35140260fb42330fbed12303a9915b8da786c9b8ed060983fa6c",
          "730127e27167fcf2450fe506250b85d916fbf07dd874b7f0c69a2f8ceb9e44a6",
          "473d57953f7e7c1a6fbd1d2335618f21088025e02bd9fa30d6f85f20a55c24d15"
        ],
        "validatorResultHash": [
          "0000000000000000000000000000000000000000000000000000000000000000",
          "0000000000000000000000000000000000000000000000000000000000000000",
          "0000000000000000000000000000000000000000000000000000000000000000",
          "0000000000000000000000000000000000000000000000000000000000000000",
          "0000000000000000000000000000000000000000000000000000000000000000"
        ]
      }
    ]
  }
}
```

```
}
```

gen_getTransactionStatus

Returns the current consensus status of a transaction. This is a lightweight endpoint that returns only the status without full receipt data — use it for polling transaction progress.

Method: `gen_getTransactionStatus`

Parameters:

- request (object, required):
- txId (string, required): The transaction hash (hex-encoded with 0x prefix)
- timestamp (integer, optional): Unix timestamp for the query. If not provided, uses current time

Returns: Object with status string and numeric code

Response Fields:

Field	Type	Description
<code>`status`</code>	string	Human-readable status name (e.g., "FINALIZED", "ACCEPTED", "PENDING")
<code>`statusCode`</code>	uint8	Numeric status code

Status Codes:

Code	Status
0	UNINITIALIZED
1	PENDING
2	PROPOSING
3	COMMITTING
4	REVEALING
5	ACCEPTED
6	UNDETERMINED
7	FINALIZED
8	CANCELED
9	APPEAL_REVEALING
10	APPEAL_COMMITTING
11	READY_TO_FINALIZE
12	VALIDATORS_TIMEOUT
13	LEADER_TIMEOUT

Example Request:

```
{
  "jsonrpc": "2.0",
  "method": "gen_getTransactionStatus",
  "params": [
    {
      "txId": "0x563f046c187d711127c51213ca62e2e4fee52009a98f0989a73a0a0382d21890"
    }
  ],
  "id": 1
}
```

Example Response:

```
{
  "jsonrpc": "2.0",
  "result": {
    "status": "FINALIZED",
    "statusCode": 7
  },
  "id": 1
}
```



```
}
```

Notes:

- This endpoint queries the consensus contract on-chain via `getTransactionData`
- Use this for polling — it's cheaper than `gen_getTransactionReceipt` which returns the full receipt
- The timestamp parameter is passed to the on-chain call and affects how the contract reports the status

gen_syncing

Returns the GenVM sync status, indicating how far the node's local GenVM state is behind the chain tip.

Method: `gen_syncing`

Parameters: None

Returns: Object with sync status information

Response Fields:

Field	Type	Description
<code>`syncedBlock`</code>	string	Hex-encoded block number that GenVM has synced to
<code>`latestBlock`</code>	string	Hex-encoded latest block number on the chain
<code>`blocksBehind`</code>	number	Number of blocks GenVM is behind the chain tip (0 = fully synced)

Example Request:

```
{
  "jsonrpc": "2.0",
  "method": "gen_syncing",
  "params": [],
  "id": 1
}
```

Example Response (Synced):

```
{
  "jsonrpc": "2.0",
  "result": {
    "syncedBlock": "0x151ec5",
    "latestBlock": "0x151ec5",
    "blocksBehind": 0
  },
  "id": 1
}
```

Example Response (Behind):

```
{
  "jsonrpc": "2.0",
  "result": {
    "syncedBlock": "0x151e00",
    "latestBlock": "0x151ec5",
    "blocksBehind": 197
  },
  "id": 1
}
```

Notes:

- This endpoint always returns an object (never false like `eth_syncing`)

- Use `blocksBehind == 0` to determine if the node is fully synced
- When the node is catching up, `gen_call` requests without an explicit `blockNumber` will serve data at the `syncedBlock` rather than waiting for full sync

Debug Methods

These methods are available for debugging and testing purposes during development.

gen_dbg_ping

Simple connectivity test method that returns a "pong" response.

Method: `gen_dbg_ping`

Parameters: None

Returns: "pong" string

Example:

```
{
  "jsonrpc": "2.0",
  "method": "gen_dbg_ping",
  "params": [],
  "id": 1
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "result": "pong",
  "id": 1
}
```

gen_dbg_trie

Get trie information for debugging node internal DB state.

Method: `gen_dbg_trie`

Parameters:

- `request` (object, required): The trie debug request
- `txID` (string, required): Transaction ID (hex-encoded)
- `round` (integer, required): Round number

Returns: Trie debug information string

Example:

```
{
  "jsonrpc": "2.0",
  "method": "gen_dbg_trie",
  "params": [
```

```
{
  {
    "txID": "0x742d35Cc6634C0532925a3b8D4C9db96c4b4d8b6742d35Cc6634C0532925a3b8",
    "round": 0
  },
  "id": 1
}
```

Response:

[illegible]

```
### gen dbg traceTransaction
```

Retrieves the full execution trace for a transaction, including the return data, stdout/stderr, GenVM logs, and the exact error if execution failed.

```
**Method:** `gen_dbg_traceTransaction`
```

****Parameters:****

- `request` (object, required):
- `txID` (string, required): Transaction ID (hex-encoded with `0x` prefix)
- `round` (integer, optional): Consensus round number (default: 0)

****Returns:**** Transaction execution trace object

```
**Response Fields:**
```

Field	Type	Description
<code>`transaction_id`</code>	string	Transaction hash
<code>`result_code`</code>	number	0=success, 1=user error, 2=VM error
<code>`return_data`</code>	string	Hex-encoded contract return data
<code>`stdout`</code>	string	Standard output captured during execution
<code>`stderr`</code>	string	Standard error captured during execution
<code>`genvm_log`</code>	array	Detailed GenVM execution log entries
<code>`storage_proof`</code>	string	Hex-encoded storage proof hash
<code>`run_time`</code>	string	Execution duration
<code>`eq_outputs`</code>	array	Hex-encoded equivalence principle outputs

****Example:****

```
{
  "jsonrpc": "2.0",
  "method": "gen_dbg_traceTransaction",
  "params": [{ "txID": "0x563f046c187d711127c51213ca62e2e4fee52009a98f0989a73a0a0382d21890", "round": 0 }],
  "id": 1
}
```

****Response:****

```
{
  "jsonrpc": "2.0",
  "result": {
    "transaction_id": "0x563f046c187d711127c51213ca62e2e4fee52009a98f0989a73a0a0382d21890",
    "result_code": 2,
  }
}
```

```

"return_data": "0x2e0464617461b402696e76616c69645f636f6e7472616374...",
"stdout": "",
"stderr": "",
"genvm_log": [
{
"file": "src/lib.rs:82",
"level": "error",
"message": "vm error",
"error": {"causes": ["VMError(invalid_contract absent_runner_comment)"]}
},
"storage_proof": "0x99ff0d9125e1fc9531a11262e15aeb2c60509a078c4cc4c64cefd06ff68647",
"run_time": "0s",
"eq_outputs": []
},
"id": 1
}

```

****Notes:****

- If the trace isn't stored locally (e.g., the node restarted or data was pruned), the endpoint will replay the transaction automatically.
- This endpoint is in the `genlayer\_debug` group and must be enabled in the node configuration.
- Use `result\_code` to determine execution outcome: 0 = contract returned successfully, 1 = contract raised a user error, 2 = GenVM encountered an error.

Ethereum Compatibility

The GenLayer Node also supports Ethereum-compatible methods that are proxied to the underlying infrastructure. These methods follow the standard [Ethereum JSON-RPC specification](https://ethereum.org/en/developers/docs/apis/json-rpc/) and are prefixed with `eth\_`.

****Examples of supported Ethereum methods:****

- `eth\_blockNumber`
- `eth\_getBalance`
- `eth\_sendTransaction`
- `eth\_call`
- And other standard Ethereum JSON-RPC methods

zkSync Compatibility

[zkSync-compatible](https://docs.zksync.io/zksync-protocol/api/zks-rpc) methods are also supported and proxied to the underlying infrastructure. These methods are prefixed with `zksync\_`.

Usage Examples

cURL

Test connectivity

```

curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
"jsonrpc": "2.0",

```

```
"method": "gen_dbg_ping",
"params": [],
"id": 1
}'
```

Execute a contract call

```
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_call",
  "params": [{
    "from": "0x742d35Cc6634C0532925a3b8D4C9db96c4b4d8b6",
    "to": "0x742d35Cc6634C0532925a3b8D4C9db96c4b4d8b6",
    "data": "0x70a08231000000000000000000000000742d35cc6634c0532925a3b8d4c9db96c4b4d8b6",
    "type": "read",
    "transaction_hash_variant": "latest-nonfinal"
  }],
  "id": 1
}'
```

Get contract schema

[illegible]

Get debug trie information

```
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_dbg_trie",
  "params": [{
    "txID": "0x742d35Cc6634C0532925a3b8D4C9db96c4b4d8b6742d35Cc6634C0532925a3b8",
    "round": 0
  }],
  "id": 1
}'
```

Get transaction receipt

```
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getTransactionReceipt",
  "params": [{
    "txID": "0x635060dd514082096d18c8eb64682cc6a944f9ce1ae6982febf7a71e9f656f49"
  }],
  "id": 1
}'
```

Ops Methods

These methods provide operational endpoints for monitoring the GenLayer node.

balance

Returns the configured operator's account balance in multiple formats including hexadecimal, wei, and GEN units.

****Method:**** `GET /balance`

****Parameters:**** None

****Returns:**** JSON object containing the operator's balance in three formats

****Example Request:****

```
curl -X GET http://localhost:9153/balance
```

****Example Response:****

```
{
  "balance_hex": "0x0de0b6b3a7640000",
  "balance_wei": "1000000000000000000",
  "balance_gen": "1.000"
```

```
}
```

****Response Fields:****

- `balance\_hex` (string): The balance encoded as hexadecimal with "0x" prefix
- `balance\_wei` (string): The balance in wei as a decimal string
- `balance\_gen` (string): The balance in GEN units formatted with 3 decimal places

****Notes:****

- This endpoint is only available when the node is running in validator mode
- The balance is always retrieved from the latest block
- The operator address is configured at node startup
- If the operator address is not configured, the zero address (0x0000...0000) is used
- Balance conversions: 1 GEN = 1,000,000,000,000,000,000 wei (10^{18} wei)
- Returns HTTP 503 if the node is not in validator mode
- Returns HTTP 500 if there's an error retrieving the balance

****Example Response (Zero Balance):****

```
{
```

```
"balance_hex": "0x",
```

```
"balance_wei": "0",
```

```
"balance_gen": "0.000"
```

```
}
```

****Example Response (Large Balance):****

```
{
```

```
"balance_hex": "0x3635c9adc5dea00000",
```

```
"balance_wei": "1000000000000000000000",
```

```
"balance_gen": "1000.000"
```

```
}
```

****Error Response (Not in Validator Mode):****

```
{
```

```
"message": "Balance endpoint not available in rollup mode"
```

```
}
```

****cURL Example with Headers:****

```
curl -X GET http://localhost:9153/balance \
```

```
-H "Accept: application/json" \
```

```
-v
```

health

Returns the health status of the GenLayer node along with version information and the status of individual health checks.

****Method:**** `GET /health`

****Parameters:**** None

****Returns:**** JSON object containing the overall health status, version information, and individual check results

****Example Request:****

```
curl -X GET http://localhost:9153/health
```

****Example Response:****

```
{
```

```
"status": "up",
```

```
"node_version": "v1.2.3",
```

```

"protocol_version": "consensus-v2.1",
"checks": {
"test-check": {
"status": "up",
"timestamp": "2024-01-15T10:30:45Z"
}
}
}

```

****Response Fields:****

- `status` (string): Overall health status - "up" if all checks pass, "down" if any check fails
- `node\_version` (string): The version of the GenLayer node software
- `protocol\_version` (string): The consensus protocol version
- `checks` (object, optional): Map of individual health check results
 - `[check-name]` (object): Results for a specific health check
 - `status` (string): Check status - "up" or "down"
 - `timestamp` (string): ISO 8601 timestamp of when the check was performed
 - `error` (string, optional): Error message if check failed

****HTTP Status Codes:****

- `200 OK`: Node is healthy (all checks passing)
- `503 Service Unavailable`: Node is unhealthy (one or more checks failing)

****Notes:****

- The endpoint has a 5-second timeout for all health checks
- The `checks` field is only included when health checks are configured
- Version fields will show "unset" if version information is not available
- Individual health checks are executed concurrently with results aggregated in the response

****Example Response (Unhealthy):****

```

{
"status": "down",
"node_version": "v1.2.3",
"protocol_version": "consensus-v2.1",
"checks": {
"database": {
"status": "down",
"error": "connection timeout",
"timestamp": "2024-01-15T10:30:45Z"
},
"consensus": {
"status": "up",
"timestamp": "2024-01-15T10:30:45Z"
}
}
}

```

****cURL Example with Headers:****

```

curl -X GET http://localhost:9153/health \
-H "Accept: application/json" \

```


-V

```
### metrics

Exposes Prometheus-compatible metrics for monitoring node performance and health.

**Method:** `GET /metrics`

**Parameters:** None

**Returns:** Prometheus-format text response with all collected metrics

**Example Request:**

curl -X GET http://localhost:9153/metrics

**Example Response:**
```

HELP genlayer_node_cpu_usage_percent Current CPU usage percentage (0-100)

TYPE genlayer_node_cpu_usage_percent gauge

```
genlayer_node_cpu_usage_percent{component="node"} 15.5
genlayer_node_cpu_usage_percent{component="genvm-llm"} 45.2
genlayer_node_cpu_usage_percent{component="genvm-web"} 12.8
genlayer_node_cpu_usage_percent{component="webdriver"} 5.3
```

HELP genlayer_node_memory_rss_bytes Resident Set Size (RSS) memory usage in bytes

TYPE genlayer_node_memory_rss_bytes gauge

```
genlayer_node_memory_rss_bytes{component="node"} 524288000
genlayer_node_memory_rss_bytes{component="genvm-llm"} 2147483648
```

HELP genlayer_node_network_rx_bytes_total Total bytes received across all network interfaces

TYPE genlayer_node_network_rx_bytes_total counter

```
genlayer_node_network_rx_bytes_total{component="node"} 1073741824
```

```
#### Available Metrics

The node collects metrics for three main components:
- **Node** - The main node process
- **GenVM** - The GenVM modules (LLM and Web)
- **WebDriver** - The WebDriver container

**CPU Metrics**

**genlayer_node_cpu_usage_percent**
```

Current CPU usage percentage (0-100).

****Type:** Gauge**

****Labels:****

- `component`: One of `node`, `genvm-llm`, `genvm-web`, `webdriver`

****Example:****

genlayer_node_cpu_usage_percent{component="node"} 15.5

genlayer_node_cpu_usage_percent{component="genvm-llm"} 45.2

genlayer_node_cpu_usage_percent{component="genvm-web"} 12.8

genlayer_node_cpu_usage_percent{component="webdriver"} 5.3

****Memory Metrics****

****genlayer_node_memory_rss_bytes****

Resident Set Size (RSS) memory usage in bytes.

****Type:** Gauge**

****Labels:****

- `component`: One of `node`, `genvm-llm`, `genvm-web`, `webdriver`

****Example:****

genlayer_node_memory_rss_bytes{component="node"} 524288000

genlayer_node_memory_rss_bytes{component="genvm-llm"} 2147483648

****genlayer_node_memory_vms_bytes****

Virtual Memory Size (VMS) in bytes.

****Type:** Gauge**

****Labels:****

- `component`: One of `node`, `genvm-llm`, `genvm-web`, `webdriver`

****Note:**** Only available for node and GenVM components.

****genlayer_node_memory_usage_bytes****

Current memory usage in bytes (WebDriver only).

****Type:** Gauge**

****Labels:****

- `component`: `webdriver`

****genlayer_node_memory_limit_bytes****

Memory limit in bytes (WebDriver only).

****Type:** Gauge**

****Labels:****

- `component`: `webdriver`

****genlayer_node_memory_percent****

Memory usage percentage.

****Type:** Gauge**

****Labels:****

- `component`: One of `genvm-llm`, `genvm-web`, `node`, `webdriver`
- `node`: Node address (hex format)

****genlayer_node_memory_peak_bytes****

Peak memory usage in bytes (GenVM components only).

****Type:** Gauge**

```

**Labels:**
- `component`: One of `genvm-llm`, `genvm-web`
- `node`: Node address (hex format)

**Network Metrics**

**genlayer_node_network_rx_bytes_total**

Total bytes received across all network interfaces.

**Type:** Counter

**Labels:**
- `component`: One of `node`, `genvm-llm`, `genvm-web`, `webdriver`
- `node`: Node address (hex format)

**Example:**

```

```
genlayer_node_network_rx_bytes_total{component="node",node="0x84b6cbd007511352d3fea26834c5c39a440903c4"} 96879942
```

```
genlayer_node_network_rx_bytes_total{component="webdriver",node="0x84b6cbd007511352d3fea26834c5c39a440903c4"} 84
```

```

**genlayer_node_network_tx_bytes_total**

Total bytes transmitted across all network interfaces.

**Type:** Counter

**Labels:**
- `component`: One of `node`, `genvm-llm`, `genvm-web`, `webdriver`
- `node`: Node address (hex format)

**Disk Metrics**

**genlayer_node_disk_free_bytes**

Available free disk space in bytes.

**Type:** Gauge

**Labels:**
- `component`: `node`
- `directory`: One of `genlayer.db`, `keystore`, `logs`
- `node`: Node address (hex format)

**genlayer_node_disk_total_bytes**

Total disk space in bytes.

**Type:** Gauge

**Labels:**
- `component`: `node`
- `directory`: One of `genlayer.db`, `keystore`, `logs`
- `node`: Node address (hex format)

**genlayer_node_disk_usage_bytes**

Used disk space in bytes.

**Type:** Gauge

**Labels:**
- `component`: `node`
- `directory`: One of `genlayer.db`, `keystore`, `logs`
- `node`: Node address (hex format)

**genlayer_node_disk_usage_percent**

Used disk space as percentage.

**Type:** Gauge

**Labels:**
- `component`: `node`

```

```

- `directory`: One of `genlayer.db`, `keystore`, `logs`
- `node`: Node address (hex format)

**Node Status Metrics**

**genlayer_node_blocks_behind**

Number of blocks behind the latest block.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)
**genlayer_node_latest_block**

Latest block number in the network.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)
**genlayer_node_processing_block**

Currently processing block number.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)
**genlayer_node_synced_block**

Last synced block number.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)
**genlayer_node_synced**

Node synchronization status (1 = synced, 0 = not synced).

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)
**genlayer_node_uptime_seconds**

Node uptime in seconds.

**Type:** Gauge

**Labels:**
- `component`: `node`
- `node`: Node address (hex format)

**Transaction Metrics**

**genlayer_node_transactions_accepted_synced_total**

Total number of accepted and synced transactions.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)
**genlayer_node_transactions_activated_total**

Total number of activated transactions.

**Type:** Gauge

**Labels:**

```

```

- `node`: Node address (hex format)

**genlayer node transactions leader proposed total**
Total number of transactions proposed as leader.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)

**genlayer_node_transactions_leader_revealed_total**
Total number of transactions revealed as leader.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)

**genlayer_node_transactions_validator_commit_total**
Total number of validator commit votes.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)

**genlayer_node_transactions_validator_reveal_total**
Total number of validator reveal votes.

**Type:** Gauge

**Labels:**
- `node`: Node address (hex format)

**Go Runtime Metrics**

The endpoint also exposes standard Go runtime metrics including:

- `go_gc_duration_seconds` - Garbage collection duration summary
- `go_goroutines` - Number of active goroutines
- `go_memstats_*` - Go memory statistics (heap, stack, GC)
- `go_threads` - Number of OS threads

**Process Metrics**

Standard process metrics are also available:

- `process_cpu_seconds_total` - Total CPU time
- `process_resident_memory_bytes` - Resident memory size
- `process_virtual_memory_bytes` - Virtual memory size
- `process_open_fds` - Number of open file descriptors
- `process_network_receive_bytes_total` - Network bytes received
- `process_network_transmit_bytes_total` - Network bytes transmitted

**Prometheus Handler Metrics**

Metrics about the metrics endpoint itself:

- `promhttp_metric_handler_requests_total` - Total scrape requests by status code
- `promhttp_metric_handler_requests_in_flight` - Current scrapes being served

#### Configuration

Metrics collection can be configured in the node configuration file:

```

metrics:

interval: "15s" # Default interval for all collectors

collectors:

node:

enabled: true # Enable/disable node metrics

interval: "30s" # Optional: Override default interval

genvm:

enabled: true # Enable/disable GenVM metrics

Uses default interval (15s)

webdriver:

enabled: true # Enable/disable WebDriver metrics

interval: "60s" # Optional: Override default interval

```
**Default Values**
- `metrics.interval`: 15s (applies to all collectors by default)
- `enabled`: true (for each collector)
- `interval`: Uses metrics.interval if not specified per collector
```

```
**Configuration Priority**
1. Collector-specific `interval` (if specified)
2. Global `metrics.interval` (if specified)
3. System default (15s)
```

Troubleshooting

****No Metrics for GenVM****

GenVM processes are ephemeral and only run when executing smart contracts. If you don't see GenVM metrics, it's likely that no contracts are currently being executed.

****WebDriver Metrics Missing****

Ensure the WebDriver container is running:

docker ps | grep genlayer-node-webdriver

If not running, start it:

task docker:webdriver:docker-compose:up:detach

****Metrics Not Updating****

Check if metrics collection is enabled in your configuration and that the node has been restarted after configuration changes.

Performance Considerations

- Collection intervals can be increased to reduce overhead further
- Each collector runs independently and won't block others if one fails

snapshot

Produces a consistent snapshot of the node's state at the current block. Two modes selected by the request body: stream the snapshot back as a gzipped tar response (operator path), or write it to a destination directory on the node's filesystem (zero-downtime upgrade-controller path).

****Method:**** `POST /snapshot`

****Authentication:**** required. Send the operator token in either header:

Authorization: Bearer

X-Admin-Auth:

The token is the value of `GENLAYERNODE\_NODE\_OPERATOR\_TOKEN` on the node process. When that variable is unset, the endpoint returns `401` to every caller regardless of header value (default-deny). The other ops routes (`/metrics`, `/health`, `/balance`) are unaffected and remain public.

****Network exposure:**** the ops server is intended for in-host or in-cluster access (Prometheus scraping `/metrics`, k8s probes hitting `/health`). The endpoint may not be reachable from outside the host depending on the deployment's firewall. The recommended operator workflow is therefore one of:

- SSH onto the node and call the endpoint via `127.0.0.1:9153`, or
- Open an SSH tunnel from the operator workstation to the node (`ssh -L 9153:127.0.0.1:9153 <node>`) and call `localhost:9153` through it.

****Parameters:****

```
- `destDir` (string, optional): Absolute path on the node's filesystem. When set, switches to write-to-path mode; when omitted or empty, the snapshot is streamed back as `application/gzip`.
```

****Returns:****

```
- Stream mode (`destDir` omitted): gzipped tar archive in the response body, `Content-Type: application/gzip`.  
- Write-to-path mode (`destDir` set): empty body, `204 No Content`.
```

****Example Request – Stream Mode (operator):****

On the node (after `ssh <node>`):

```
curl -X POST http://127.0.0.1:9153/snapshot \  
-H "X-Admin-Auth: $GENLAYERNODE_NODE_OPERATOR_TOKEN" \  
-H "Content-Type: application/json" \  
-d '{}\' \  
-o snapshot.tar.gz
```

Or via SSH tunnel from a workstation (forwards `9153` to the node's loopback):

```
ssh -L 9153:127.0.0.1:9153 -N &  
curl -X POST http://localhost:9153/snapshot \  
-H "X-Admin-Auth: $GENLAYERNODE_NODE_OPERATOR_TOKEN" \  
-H "Content-Type: application/json" \  
-d '{}\' \  
-o snapshot.tar.gz
```

The response body is a stream – `curl` can write it to a file (as above) or pipe it through any tool that reads stdin, without staging on local disk.

****Example Response – Stream Mode:****

```
HTTP/1.1 200 OK  
Content-Type: application/gzip  
Content-Disposition: attachment; filename="snapshot.tar.gz"
```

****Example Request – Write-to-Path Mode (upgrade controller):****

```
curl -X POST http://127.0.0.1:9153/snapshot \  
-H "X-Admin-Auth: $GENLAYERNODE_NODE_OPERATOR_TOKEN" \  
-H "Content-Type: application/json" \  
-d '{"destDir": "/var/snapshots/asimov-100"}'
```

****Example Response – Write-to-Path Mode:****

```
HTTP/1.1 204 No Content
```

****HTTP Status Codes:****

```
- `200 OK`: Stream mode succeeded; body carries the gzipped tar archive.  
- `204 No Content`: Write-to-path mode succeeded; snapshot is on disk at `destDir`.  
- `400 Bad Request`: `destDir` is not absolute, or already exists.  
- `401 Unauthorized`: Missing, invalid, or empty operator token (also returned when `GENLAYERNODE_NODE_OPERATOR_TOKEN` is unset on the node).  
- `500 Internal Server Error`: Snapshot creation failed (e.g., disk full, DB checkpoint error).
```

****Notes:****

```
- A snapshot may be several GB. In stream mode the connection stays open for the duration of the snapshot – configure reverse proxies / load balancers with adequate read timeouts and disable response buffering.  
- In stream mode, a dropped connection leaves no on-disk artifact on the node; retry the request from scratch.  
- In write-to-path mode, `destDir` must be an absolute path that does not yet exist; the node creates it. Path-traversal hardening beyond these checks is a separate (Sev: Medium) follow-up – treat the operator token as a credential that can write under the node user's filesystem permissions.  
- **Docker deploys**: In write-to-path mode, `destDir` is resolved inside the node container's filesystem, not on the host. To make the resulting directory reachable from the host (or from another container), use a volume mount that
```

maps `destDir` identically on both sides, or stick with stream mode. Stream mode is unaffected – the response body is sent over HTTP regardless of where the node runs.

- The snapshot includes the full chain database (SSTables, MANIFEST, WAL) at the moment of capture. Treat the artifact as sensitive – anyone holding it can reconstruct the node's state.
- Token comparison is constant-time and length-padded so neither the value nor the length of the secret can be inferred from response timing.
- Rotation: update `GENLAYERNODE\_NODE\_OPERATOR\_TOKEN` and restart the node. Tokens are read once at startup and not hot-reloaded.

****Forward-compatibility note:**** when the zero-downtime upgrade controller (Phase 3) lands, `/snapshot` will additionally accept a second role (`upgrade`) so the controller can drive the snapshot + stop + start sequence with its own credential. The operator-token path documented here remains the same – no migration is required for operator tooling.

****cURL Example with Bearer header:****

```
curl -X POST http://127.0.0.1:9153/snapshot \
```

```
-H "Authorization: Bearer $GENLAYERNODE_NODE_OPERATOR_TOKEN" \
```

```
-H "Content-Type: application/json" \
```

```
-d '{}'
```

```
-o snapshot.tar.gz
```

```
---
```

```
# gen_call
```

```
Source: https://docs.genlayer.com/api-references/genlayer-node/gen/gen\_call
```

Executes a new message call immediately without creating a transaction on the blockchain. This method supports read operations, write operations, and contract deployments.

****Method:**** `gen\_call`

****Parameters:****

- `request` (object, required): The contract call request parameters
- `from` (string, required): The address making the call
- `to` (string, required): The contract address to call
- `data` (string, required): The encoded function call data
- `type` (string, required): The type of call (`read`, `write`, or `deploy`)
- `blockNumber` (string, optional): The block number to query at (hex-encoded). If not provided, uses latest block
- `status` (string, optional): The transaction status filter (`"finalized"` or `"accepted"`). If not provided, uses accepted state
- `value` (string, optional): The value to send with the call (hex-encoded)
- `leader\_results` (array of strings or null, optional): When provided, GenVM runs in validator mode (executing `validator\_fn` instead of `leader\_fn` in `run\_nondet` calls). Each element is a hex-encoded equivalence output from a previous leader execution. When omitted or null, GenVM runs in leader mode

****Returns:**** A structured JSON object containing execution results and debug information

****Response Fields:****

- `data` (string): Hex-encoded return value from contract execution
- `eqOutputs` (array of strings): Equivalence outputs from GenVM execution, each as a hex-encoded string. These are deterministic outputs used for consensus validation. Populated in leader mode; empty in validator mode
- `status` (object): Execution status
 - `code` (number): Status code (0=success, 1=user error, 2=vm error, 3=internal error)
 - `message` (string): Human-readable status message
- `stdout` (string): Standard output captured during execution
- `stderr` (string): Standard error captured during execution
- `logs` (array): GenVM log entries, each as a JSON object with fields like `level`, `message`, `target`, `file`, `ts`
- `events` (array): Contract events emitted during execution
- `topics` (array of strings): Event topics as hex-encoded strings
- `data` (string): Event data as hex-encoded string
- `messages` (array): Messages emitted by the contract
 - `messageType` (number): Type identifier for the message
 - `recipient` (string): Recipient address
 - `value` (string): Value in hex format
 - `data` (string): Message data as hex-encoded string
 - `onAcceptance` (boolean): Whether message is sent on acceptance
 - `saltNonce` (string): Salt nonce in hex format
- `nondetDisagreementCallNo` (number or null): When running in validator mode (with `leader\_results`), indicates which `run\_nondet` call the validator disagreed on. Null when there is no disagreement or when running in leader mode

****Example - Deploy Call:****

```
{
  "jsonrpc": "2.0",
  "method": "gen_call",
  "params": [
    {
      "type": "deploy",
      "data": "0xf9023db90223232076302e312e300a23207b2022446570656e6473223a202270792d67656e6c6179657
23a3135716669766a7679383038303072683939387063786d64326d38766131777132717a71687a3835306e3867
676372346939713022207d0a0a66726f6d2067656e6c6179657220696d706f7274202a0a0a2320636f6e747261
637420636c6173730a636c6173732053746f7261676528676c2e436f6e7472616374293a0a2020202073746f7261
67653a207374720a0a202020202320636f6e7374727563746f720a20202020646566205f5f696e69745f5f2873656
c662c20696e697469616c5f73746f726167653a20737472293a0a2020202020202073656c662e73746f72616765
203d20696e697469616c5f73746f726167650a0a20202020232072656164206d6574686f6473206d757374206265
20616e6e6f7461746564207769746820766965770a2020202040676c2e7075626c69632e766965770a202020206
46566206765745f73746f726167652873656c6629202d3e207374723a0a2020202020202072657475726e20736
56c662e73746f726167650a0a2020202023207772697465206d6574686f640a2020202040676c2e7075626c69632
e77726974650a20202020646566207570646174655f73746f726167652873656c662c206e65775f73746f72616765
3a2073747229202d3e204e6f6e653a0a202020202020202073656c662e73746f72616765203d206e65775f73746f7
26167650a950e04617267730d6c696e697469616c2076616c756500",
      "from": "0xf6A60752C24Cd3BFcAf8d387a5EB62d3746F44EE",
      "to": "0x0000000000000000000000000000000000000000",
      "status": "finalized"
    }
  ],
  "id": 1
}
```

****Example Response (Deploy Success):****

```
{
  "jsonrpc": "2.0",
  "result": {
    "data": "00",
    "eqOutputs": [],
    "status": {
      "code": 0,
      "message": "success"
    },
    "stdout": "",
    "stderr": "",
    "logs": [
      {
        "file": "common/src/lib.rs:111",
        "level": "info",
```

```

"message": "logging initialized",
"target": "genvm_common",
"ts": "1770221913210",
"version": "v0.2.11-aarch64-macos-release"
},
{
"file": "src/exe/run.rs:150",
"genvm_id": "1",
"level": "info",
"message": "genvm id",
"target": "genvm::exe::run",
"ts": "1770221913212"
},
{
"file": "src/lib.rs:299",
"level": "info",
"message": "metrics",
"metrics": {
"gvmm": {
"host": { "time": 654 },
"llm_module": { "calls": 0, "time": 0 },
"supervisor": { "compilation_time": 0, "compiled_modules": 0, "precompile_hits": 2 },
"web_module": { "calls": 0, "time": 0 }
},
"llm": null,
"web": null
},
"target": "genvm",
"ts": "1770221913590"
}
],
"events": [],
"messages": [],
"nondetDisagreementCallNo": null
},
"id": 1
}
**Example - Read Call (Latest Accepted State):**
{

```



```

"type": "write",
"gas": "0x5208",
"value": "0x0"
}
],
"id": 1
}

```

****Example Response with Messages (Router Forward Call):****

When a contract emits internal messages (e.g., a router contract forwarding calls), the response includes the messages array:

```

{
"jsonrpc": "2.0",
"result": {
"data": "00",
"eqOutputs": [],
"status": {
"code": 0,
"message": "success"
},
"stdout": "",
"stderr": "",
"logs": [
{
"file": "src/lib.rs:299",
"level": "info",
"message": "metrics",
"target": "genvm",
"ts": "1770225770000"
}
],
"events": [],
"messages": [
{
"messageType": 1,
"recipient": "0xc346f80c86adf1d71accfc4ef6efe7482b25be40",
"value": "0x0",
"data": "0xf86eb86a16046172677331d4023078393164463338654243623930384335643635...",
"onAcceptance": false,
"saltNonce": "0x0"
}
]
}

```

```

},
{
  "messageType": 1,
  "recipient": "0xad26513a4315b0ea8a5c800322b35092a83dfd3d",
  "value": "0x0",
  "data": "0xf86eb86a16046172677331d4023078333738653339374533316346383538333630...",
  "onAcceptance": false,
  "saltNonce": "0x0"
}
],
"nondetDisagreementCallNo": null
},
"id": 1
}

```

****Notes:****

- The ``blockNumber`` parameter can be either hex-encoded with a ``0x`` prefix (e.g., ``0x151ec5``) or decimal (e.g., ``1384133``)
- Valid ``status`` values are ``accepted`` (default) and ``finalized``
- When both parameters are omitted, the call executes against the latest accepted state
- For ``read`` and ``write`` type calls, if the contract doesn't exist at the specified block/status combination, an error is returned with code ``-32001`` (Resource not found) and message ``contract not found at address 0x...``
- For ``deploy`` type calls, the ``status`` and ``blockNumber`` parameters are not used for contract state validation since no contract exists yet

gen_getContractSchema

Source: https://docs.genlayer.com/api-references/genlayer-node/gen/gen_getContractSchema

Retrieve the schema/interface of a GenLayer contract to understand its available methods and properties.

****Method:**** ``gen_getContractSchema``

****Parameters:****

- ``request`` (object, required): The contract schema request
- ``code`` (string, required): Base64-encoded contract code

****Returns:**** Contract schema object with methods and properties

```

{
  "ctor": { // constructor info
    "kwparams": {}, // dict
    "params": [{"value", "type"}] // list of tuples
  },
  "methods": { // dict from method name to method info
    "method_name": {
      "kwparams": {}, // dict
      "params": [{"value", "type"}], // list of tuples
      "payable": false, // bool
    }
  }
}

```

```
"readonly": false, // bool
```

```
"ret": "null" // string
```

```
},
```

```
}
```

```
}
```

```
**Example:**
```

```
{
```

```
"jsonrpc": "2.0",
```

```
"method": "gen_getContractSchema",
```

```
"params": [
```

```
{
```

```
"code": "IyB7ICJEZXB1bmRzljogInB5LWdlbmxeWVvOnRlc3QilH0KCmZyb20gZ2VubGF5ZXIgaW1wb3J0ICoKCgojIGNvbnRyYWN0IGNsYXNzCmNsYXNzIFN0b3JhZ2UoZ2wuQ29udHJhY3QpOgogICAgc3RvcnFnZTogc3RyCgogICAglyBjb25zdHJ1Y3RvcgogICAgZGVmIF9faW5pdF9fKHNIbGYsIGluaXRpYWxfc3RvcnFnZTogc3RyKToKICAgICAgICBzZWxmLnN0b3JhZ2UgPSBpbml0aWFsX3N0b3JhZ2UKCiAgICAgIHJlYWQgbWV0aG9kcyBtdXN0IGJlIGFubm90YXRIZCB3aXRoIHZpZXcKICAgIEBnbC5wdWJsaWMudmlldwogICAgZGVmIGdlidF9zdG9yYWdlKHNIbGYpIC0+IHN0cjoKICAgICAgICByZXR1cm4gc2VsZi5zdG9yYWdlCgogICAglyB3cmI0ZSBtZXRob2QKICAgIEBnbC5wdWJsaWMud3JpdGUKICAgIGRlZiB1cGRhdGVfc3RvcnFnZShzZWxmLCBuZXdfc3RvcnFnZTogc3RyKSAtpiBOb25lOgogICAgICAgIHNIbGYuc3RvcnFnZSA9IG5ld19zdG9yYWdlCg=="
```

```
}
```

```
],
```

```
"id": 1
```

```
}
```

```
**Response:**
```

```
{
```

```
"jsonrpc": "2.0",
```

```
"result": {
```

```
"ctor": {
```

```
"kwparams": {}},
```

```
"params": [["initial_storage", "string"]]
```

```
},
```

```
"methods": {
```

```
"get_storage": {
```

```
"kwparams": {}},
```

```
"params": [],
```

```
"readonly": true,
```

```
"ret": "string"
```

```
},
```

```
"update_storage": {
```

```
"kwparams": {}},
```

```
"params": [["new_storage", "string"]],
```

```
"payable": false,
```

```

"readonly": false,
"ret": "null"
}
},
"id": 1
}

```

```
---
```

gen_getContractState

Retrieves the current state of a GenLayer intelligent contract at a specific block height and transaction status.

****Method:**** `gen\_getContractState`

****Parameters:****

- `request` (object, required): The contract state request parameters
- `address` (string, required): The contract address to query
- `blockNumber` (string, optional): The block number to query at (hex-encoded). If not provided, uses latest block
- `status` (string, optional): The transaction status filter (`"finalized"` or `"accepted"`). If not provided, uses accepted state

****Returns:**** Hex-encoded contract state data

****Example Request - Latest Accepted State:****

```

{
"jsonrpc": "2.0",
"method": "gen_getContractState",
"params": [
{
"address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f"
}
],
"id": 1
}

```

****Example Request - Specific Block and Status:****

```

{
"jsonrpc": "2.0",
"method": "gen_getContractState",
"params": [
{
"address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
"blockNumber": "0x151ec5",
"status": "accepted"
}
],
"id": 1
}

```

```

}

```

****Example Response:****

```

{
  "jsonrpc": "2.0",
  "result": "0xd9960e0773746f726167656c6e757064617465642074657374207374",
  "id": 1
}

```

****Notes:****

- The `blockNumber` parameter should be hex-encoded with a `0x` prefix
- Valid `status` values are `"accepted"` (default) and `"finalized"`
- The returned state is hex-encoded binary data with a `0x` prefix
- Empty contract state returns `"0x"`
- If the contract doesn't exist at the specified block/status, an error will be returned

****cURL Example:****

Get latest accepted state

Source: https://docs.genlayer.com/api-references/genlayer-node/gen/gen_getContractState

```

curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getContractState",
  "params": [{
    "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f"
  }],
  "id": 1
}'

```

Get state at specific block with accepted status

```

curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getContractState",
  "params": [{
    "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
    "blockNumber": "0x151ec5",
    "status": "accepted"
  }],
}

```



```
"id": 1
```

```
},
```

```
---
```

gen_getContractCode

Retrieves the code of a deployed GenLayer intelligent contract at a specific block height and transaction status.

****Method:**** `gen\_getContractCode`

****Parameters:****

- `request` (object, required): The contract code request parameters
- `address` (string, required): The contract address to query
- `blockNumber` (string, optional): The block number to query at (hex-encoded). If not provided, uses latest block
- `status` (string, optional): The transaction status filter (`"finalized"` or `"accepted"`). If not provided, uses accepted state

****Returns:**** Base64-encoded contract code

****Example Request - Latest Accepted State:****

```
{
```

```
"jsonrpc": "2.0",
```

```
"method": "gen_getContractCode",
```

```
"params": [
```

```
{
```

```
"address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f"
```

```
}
```

```
],
```

```
"id": 1
```

```
}
```

****Example Request - Specific Block and Status:****

```
{
```

```
"jsonrpc": "2.0",
```

```
"method": "gen_getContractCode",
```

```
"params": [
```

```
{
```

```
"address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
```

```
"blockNumber": "0x151ec5",
```

```
"status": "finalized"
```

```
}
```

```
],
```

```
"id": 1
```

```
}
```

****Example Response:****

```
{
```

```
"jsonrpc": "2.0",
```

```
"result": "lyB7ICJEZXBImRzljogInB5LWdlbmxeWVYOnRlc3QiIH0KCMZyb20gZ2VubGF5ZXlgaW1wb3J0ICoKCgpbG
FzcyBTdG9yYWdlKGdsLkNvbnRyYWN0KT0KICAgIH0b3JhZ2U6IH0cgoKICAgIGRlZiBfX2luaXRfXyhzZWxmLCBpbml
0aWFsX3N0b3JhZ2U6IH0cik6CiAgICAgICAgc2VsZi5zdG9yYWdlID0gaW5pdGlhbF9zdG9yYWdlCgogICAgQGdsLnB1Y
mxpYy52aWV3CiAgICBkZWYgZ2V0X3N0b3JhZ2Uoc2VsZikgLT4gc3RyOgogICAgICAgIHJldHVybiBzZWxmLnN0b3JhZ2
UKCiAgICBAZ2wucHVibGljLndyaXRiCiAgICBkZWYgdXBkYXRlX3N0b3JhZ2Uoc2VsZiwgbmV3X3N0b3JhZ2U6IH0cikg
LT4gTm9uZToKICAgICAgICBzZWxmLnN0b3JhZ2UgPSBuZXdfc3RvcnFnZQo=",
"id": 1
}
```

****Notes:****

- The `blockNumber` parameter should be hex-encoded with a `0x` prefix
- Valid `status` values are `accepted` (default) and `finalized`
- The returned code is base64-encoded
- If the contract doesn't exist at the specified block/status, an error will be returned
- If the address does not have code (not a contract), an error will be returned

****cURL Example:****

Get latest accepted code

Source: https://docs.genlayer.com/api-references/genlayer-node/gen/gen_getContractCode

```
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getContractCode",
  "params": [{
    "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f"
  }],
  "id": 1
}'
```

Get code at specific block with finalized status

```
curl -X POST http://localhost:9151 \
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "method": "gen_getContractCode",
  "params": [{
    "address": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
    "blockNumber": "0x151ec5",
    "status": "finalized"
  }],
  "id": 1
}'
```

```

}'
---

# gen_getTransactionReceipt

Source: https://docs.genlayer.com/api-references/genlayer-node/gen/gen\_getTransactionReceipt

Retrieves the transaction receipt for a given transaction ID. This endpoint is a proxy to the underlying consensus layer's `getTransactionAllData` function, providing detailed information about a transaction that has been processed.

**Method:** `gen_getTransactionReceipt`

**Parameters:**

- `request` (object, required):
  - `txId` (`string`, required): The 32-byte transaction hash as a hexadecimal string with a `0x` prefix.

**Returns:**

A `Transaction` object which contains the details of the transaction. The structure is based on `ITransactionsTransaction` from the consensus contracts. The object has the following fields:

- `id` (`string`): The transaction ID as a hex string.
- `txOrigin` (`address`): The original address that initiated the transaction.
- `sender` (`address`): The address of the sender.
- `recipient` (`address`): The destination address of the transaction.
- `activator` (`address`): The address of the validator that activated the transaction.
- `status` (`uint8`): The current status of the transaction.
- `previousStatus` (`uint8`): The previous status of the transaction.
- `txSlot` (`uint256`): The transaction slot number.
- `initialRotations` (`uint256`): The initial number of rotations.
- `numOfInitialValidators` (`uint256`): The number of initial validators.
- `randomSeed` (`string`): The random seed used for the transaction as a hex string.
- `txExecutionHash` (`string`): The execution hash of the transaction as a hex string.
- `txCallData` (`string`): The transaction call data as a hex string.
- `eqBlocksOutputs` (`string`): The equivalent blocks outputs as a hex string.
- `timestamps` (object): Object containing various timestamps:
  - `Created` (`uint256`): When the transaction was created.
  - `Pending` (`uint256`): When the transaction became pending.
  - `Activated` (`uint256`): When the transaction was activated.
  - `Proposed` (`uint256`): When the transaction was proposed.
  - `Committed` (`uint256`): When the transaction was committed.
  - `LastVote` (`uint256`): When the last vote was cast.
  - `AppealSubmitted` (`uint256`): When an appeal was submitted.
  - `LeaderRevealed` (`uint256`): When the leader was revealed.
- `result` (`uint256`): The result of the transaction execution.
- `readStateBlockRanges` (array): Array of block range objects, each containing:
  - `ActivationBlock` (`uint256`): The activation block number.
  - `ProcessingBlock` (`uint256`): The processing block number.
  - `ProposalBlock` (`uint256`): The proposal block number.
- `roundData` (array): Array of round data objects containing:
  - `round` (`uint256`): The round number.
  - `leaderIndex` (`uint256`): Index of the round leader.
  - `votesCommitted` (`uint256`): Number of votes committed.
  - `votesRevealed` (`uint256`): Number of votes revealed.
  - `appealBond` (`uint256`): The appeal bond amount.
  - `rotationsLeft` (`uint256`): Number of rotations remaining.
  - `result` (`uint256`): The round result.
  - `roundValidators` (array): Array of validator addresses for this round.
  - `validatorVotes` (string): Base64-encoded validator votes.
  - `validatorVotesHash` (array): Array of vote hashes as hex strings.
  - `validatorResultHash` (array): Array of result hashes as hex strings.

**Example Request:**

{
  "jsonrpc": "2.0",
  "method": "gen_getTransactionReceipt",
  "params": [
    {

```

```
"txId": "0x635060dd514082096d18c8eb64682cc6a944f9ce1ae6982febf7a71e9f656f49"
```

```
}
```

```
],
```

```
"id": 1
```

```
}
```

```
**Example Response:**
```

```
{
```

```
"jsonrpc": "2.0",
```

```
"id": 1,
```

```
"result": {
```

```
"id": "0x635060dd514082096d18c8eb64682cc6a944f9ce1ae6982febf7a71e9f656f49",
```

```
"txOrigin": "0x5b70759760a85d92bf972b67ee8558e8e8da87d4",
```

```
"sender": "0x5b70759760a85d92bf972b67ee8558e8e8da87d4",
```

```
"recipient": "0x73ca5a2b51edf506ceaf110a41780ec51294d89f",
```

```
"activator": "0xf27b07d36ff5b9dcd37fd3a09811d6456dd4483a",
```

```
"status": 14,
```

```
"previousStatus": 0,
```

```
"txSlot": 1,
```

```
"initialRotations": 1,
```

```
"numOfInitialValidators": 5,
```

```
"randomSeed": "0x3d0bdff63bce1cf84b155f1557bb15e303b903b1b559325b6492794a5e498954",
```

```
"txExecutionHash": "0x76744dc491b1725eb2ba08cf516855c14aac310e36711368c04442fd39d61853",
```

```
"txCallData": "f5b31604617267730da4017570646174656420746573742073746f72616765066d6574686f64747570646174655f73746f7261676500",
```

```
"eqBlocksOutputs": "c0",
```

```
"timestamps": {
```

```
"Created": 1750774777,
```

```
"Pending": 1750774777,
```

```
"Activated": 1750774781,
```

```
"Proposed": 1750774786,
```

```
"Committed": 1750774791,
```

```
"LastVote": 0,
```

```
"AppealSubmitted": 0,
```

```
"LeaderRevealed": 0
```

```
},
```

```
"result": 0,
```

```
"readStateBlockRanges": [
```

```
{
```

```
"ActivationBlock": 1377797,
```

```

"ProcessingBlock": 1377797,
"ProposalBlock": 1377798
},
"roundData": [
{
"round": 0,
"leaderIndex": 3,
"votesCommitted": 5,
"votesRevealed": 0,
"appealBond": 0,
"rotationsLeft": 1,
"result": 0,
"roundValidators": [
"0xa707a1d9dd946242f921f0ad62f61e1b7df30c85",
"0xf27b07d36ff5b9dcd37fd3a09811d6456dd4483a",
"0xf3738216c3208d77a9db79a1d31824e5133083c4",
"0x56394354538bff1e0ee5021c2b6d3522fce30257",
"0xcfdbac032f82f388d9eca76efa6610b98e7f59fe"
],
"validatorVotes": "AAAAAAA=",
"validatorVotesHash": [
"54e3478b02dfb009a35d46b19ac7c967b65416430a33d3860e9301a0f2b19d57",
"f74a48e7283c91757bf2c612e40f8a9eb69ea0f02cf5e6b89c3c9b6ad629a3c1",
"2f51c3b1d9ff35140260fb42330fbed12303a9915b8da786c9b8ed060983fa6c",
"730127e27167fcf2450fe506250b85d916fbf07dd874b7f0c69a2f8ceb9e44a6",
"473d57953f7e7c1a6fb1d2335618f21088025e02bd9fa30d6f85f20a55c24d15"
],
"validatorResultHash": [
"0000000000000000000000000000000000000000000000000000000000000000",
"0000000000000000000000000000000000000000000000000000000000000000",
"0000000000000000000000000000000000000000000000000000000000000000",
"0000000000000000000000000000000000000000000000000000000000000000",
"0000000000000000000000000000000000000000000000000000000000000000"
]
}
]
}
}
}

```

gen_getTransactionStatus

Source: https://docs.genlayer.com/api-references/genlayer-node/gen/gen_getTransactionStatus

Returns the current consensus status of a transaction. This is a lightweight endpoint that returns only the status without full receipt data – use it for polling transaction progress.

****Method:**** `gen\_getTransactionStatus`

****Parameters:****

- `request` (object, required):
 - `txId` (string, required): The transaction hash (hex-encoded with `0x` prefix)
 - `timestamp` (integer, optional): Unix timestamp for the query. If not provided, uses current time

****Returns:**** Object with status string and numeric code

****Response Fields:****

Field	Type	Description
`status`	string	Human-readable status name (e.g., "FINALIZED", "ACCEPTED", "PENDING")
`statusCode`	uint8	Numeric status code

****Status Codes:****

Code	Status
0	UNINITIALIZED
1	PENDING
2	PROPOSING
3	COMMITTING
4	REVEALING
5	ACCEPTED
6	UNDETERMINED
7	FINALIZED
8	CANCELED
9	APPEAL_REVEALING
10	APPEAL_COMMITTING
11	READY_TO_FINALIZE
12	VALIDATORS_TIMEOUT
13	LEADER_TIMEOUT

****Example Request:****

```
{
  "jsonrpc": "2.0",
  "method": "gen_getTransactionStatus",
  "params": [
    {
      "txId": "0x563f046c187d711127c51213ca62e2e4fee52009a98f0989a73a0a0382d21890"
    }
  ],
  "id": 1
}
```

****Example Response:****

```
{
  "jsonrpc": "2.0",
  "result": {
    "status": "FINALIZED",
```

```
"statusCode": 7
```

```
},
```

```
"id": 1
```

```
}
```

```
**Notes:**
```

- This endpoint queries the consensus contract on-chain via ``getTransactionData``
- Use this for polling – it's cheaper than ``gen_getTransactionReceipt`` which returns the full receipt
- The ``timestamp`` parameter is passed to the on-chain call and affects how the contract reports the status

```
---
```

```
# gen_syncing
```

```
Source: https://docs.genlayer.com/api-references/genlayer-node/gen/gen\_syncing
```

```
Returns the GenVM sync status, indicating how far the node's local GenVM state is behind the chain tip.
```

```
**Method:** `gen_syncing`
```

```
**Parameters:** None
```

```
**Returns:** Object with sync status information
```

```
**Response Fields:**
```

Field	Type	Description
<code>`syncedBlock`</code>	string	Hex-encoded block number that GenVM has synced to
<code>`latestBlock`</code>	string	Hex-encoded latest block number on the chain
<code>`blocksBehind`</code>	number	Number of blocks GenVM is behind the chain tip (0 = fully synced)

```
**Example Request:**
```

```
{
```

```
"jsonrpc": "2.0",
```

```
"method": "gen_syncing",
```

```
"params": [],
```

```
"id": 1
```

```
}
```

```
**Example Response (Synced):**
```

```
{
```

```
"jsonrpc": "2.0",
```

```
"result": {
```

```
"syncedBlock": "0x151ec5",
```

```
"latestBlock": "0x151ec5",
```

```
"blocksBehind": 0
```

```
},
```

```
"id": 1
```

```
}
```

```
**Example Response (Behind):**
```

```
{
```

```
"jsonrpc": "2.0",
```

```

"result": {
  "syncedBlock": "0x151e00",
  "latestBlock": "0x151ec5",
  "blocksBehind": 197
},
"id": 1
}

```

****Notes:****

- This endpoint always returns an object (never `false` like `eth\_syncing`)
- Use `blocksBehind == 0` to determine if the node is fully synced
- When the node is catching up, `gen\_call` requests without an explicit `blockNumber` will serve data at the `syncedBlock` rather than waiting for full sync

gen_dbg_ping

Source: https://docs.genlayer.com/api-references/genlayer-node/debug/gen_dbg_ping

Simple connectivity test method that returns a "pong" response.

****Method:**** `gen\_dbg\_ping`

****Parameters:**** None

****Returns:**** "pong" string

****Example:****

```

{
  "jsonrpc": "2.0",
  "method": "gen_dbg_ping",
  "params": [],
  "id": 1
}

```

****Response:****

```

{
  "jsonrpc": "2.0",
  "result": "pong",
  "id": 1
}

```

gen_dbg_trie

Source: https://docs.genlayer.com/api-references/genlayer-node/debug/gen_dbg_trie

Get trie information for debugging node internal DB state.

****Method:**** `gen\_dbg\_trie`

****Parameters:****

- `request` (object, required): The trie debug request
- `txID` (string, required): Transaction ID (hex-encoded)

[illegible]

gen_dbg_traceTransaction

Source: https://docs.genlayer.com/api-references/genlayer-node/debug/gen_dbg_traceTransaction

Retrieves the full execution trace for a transaction, including the return data, stdout/stderr, GenVM logs, and the exact error if execution failed.

Method: `gen_dbg_traceTransaction`

Parameters:

- request (object, required):
- txID (string, required): Transaction ID (hex-encoded with 0x prefix)
- round (integer, optional): Consensus round number (default: 0)

Returns: Transaction execution trace object

Response Fields:

Field	Type	Description
<code>`transaction_id`</code>	string	Transaction hash
<code>`result_code`</code>	number	0=success, 1=user error, 2=VM error
<code>`return_data`</code>	string	Hex-encoded contract return data
<code>`stdout`</code>	string	Standard output captured during execution
<code>`stderr`</code>	string	Standard error captured during execution
<code>`genvm_log`</code>	array	Detailed GenVM execution log entries
<code>`storage_proof`</code>	string	Hex-encoded storage proof hash
<code>`run_time`</code>	string	Execution duration
<code>`eq_outputs`</code>	array	Hex-encoded equivalence principle outputs

Example:

```
{
  "jsonrpc": "2.0",
  "method": "gen_dbg_traceTransaction",
  "params": [{"txID": "0x563f046c187d711127c51213ca62e2e4fee52009a98f0989a73a0a0382d21890", "round": 0}],
  "id": 1
}
```

Response:

```
{
  "jsonrpc": "2.0",
  "result": {
    "transaction_id": "0x563f046c187d711127c51213ca62e2e4fee52009a98f0989a73a0a0382d21890",
    "result_code": 2,
    "return_data": "0x2e0464617461b402696e76616c69645f636f6e7472616374...",
    "stdout": "",
    "stderr": "",
    "genvm_log": [
      {
        "file": "src/lib.rs:82",
        "level": "error",
        "message": "vm error",
        "error": {"causes": ["VMError(invalid_contract absent_runner_comment)"]}
      }
    ],
    "storage_proof": "0x99ff0d9125e1fc9531a11262e15aeb2c60509a078c4cc4c64cefdfb06ff68647",
    "run_time": "0s",
    "eq_outputs": []
  },
  "id": 1
}
```

Notes:

- If the trace isn't stored locally (e.g., the node restarted or data was pruned), the endpoint will replay the transaction automatically.
- This endpoint is in the `genlayer_debug` group and must be enabled in the node configuration.
- Use `result_code` to determine execution outcome: 0 = contract returned successfully, 1 = contract raised a user error, 2 = GenVM encountered an error.

balance

Source: <https://docs.genlayer.com/api-references/genlayer-node/ops/balance>

Returns the configured operator's account balance in multiple formats including hexadecimal, wei, and GEN units.

Method: GET /balance

Parameters: None

Returns: JSON object containing the operator's balance in three formats

Example Request:

```
curl -X GET http://localhost:9153/balance
```

Example Response:

```
{
  "balance_hex": "0x0de0b6b3a7640000",
  "balance_wei": "1000000000000000000",
  "balance_gen": "1.000"
}
```

Response Fields:

- balance_hex (string): The balance encoded as hexadecimal with "0x" prefix
- balance_wei (string): The balance in wei as a decimal string
- balance_gen (string): The balance in GEN units formatted with 3 decimal places

Notes:

- This endpoint is only available when the node is running in validator mode
- The balance is always retrieved from the latest block
- The operator address is configured at node startup
- If the operator address is not configured, the zero address (0x0000...0000) is used
- Balance conversions: 1 GEN = 1,000,000,000,000,000,000 wei (10^{18} wei)
- Returns HTTP 503 if the node is not in validator mode
- Returns HTTP 500 if there's an error retrieving the balance

Example Response (Zero Balance):

```
{
  "balance_hex": "0x",
  "balance_wei": "0",
  "balance_gen": "0.000"
}
```

Example Response (Large Balance):

```
{
  "balance_hex": "0x3635c9adc5dea00000",
  "balance_wei": "1000000000000000000000",
  "balance_gen": "1000.000"
}
```

Error Response (Not in Validator Mode):

```
{
  "message": "Balance endpoint not available in rollup mode"
}
```

cURL Example with Headers:

```
curl -X GET http://localhost:9153/balance \  
-H "Accept: application/json" \  
-v
```

health

Source: <https://docs.genlayer.com/api-references/genlayer-node/ops/health>

Returns the health status of the GenLayer node along with version information and the status of individual health checks.

Method: GET /health

Parameters: None

Returns: JSON object containing the overall health status, version information, and individual check results

Example Request:

```
curl -X GET http://localhost:9153/health
```

Example Response:

```
{  
  "status": "up",  
  "node_version": "v1.2.3",  
  "protocol_version": "consensus-v2.1",  
  "checks": {  
    "test-check": {  
      "status": "up",  
      "timestamp": "2024-01-15T10:30:45Z"  
    }  
  }  
}
```

Response Fields:

- status (string): Overall health status - "up" if all checks pass, "down" if any check fails
- node_version (string): The version of the GenLayer node software
- protocol_version (string): The consensus protocol version
- checks (object, optional): Map of individual health check results
- [check-name] (object): Results for a specific health check
- status (string): Check status - "up" or "down"
- timestamp (string): ISO 8601 timestamp of when the check was performed
- error (string, optional): Error message if check failed

HTTP Status Codes:

- 200 OK: Node is healthy (all checks passing)
- 503 Service Unavailable: Node is unhealthy (one or more checks failing)

Notes:

- The endpoint has a 5-second timeout for all health checks
- The checks field is only included when health checks are configured
- Version fields will show "unset" if version information is not available
- Individual health checks are executed concurrently with results aggregated in the response

Example Response (Unhealthy):

```
{
  "status": "down",
  "node_version": "v1.2.3",
  "protocol_version": "consensus-v2.1",
  "checks": {
    "database": {
      "status": "down",
      "error": "connection timeout",
      "timestamp": "2024-01-15T10:30:45Z"
    },
    "consensus": {
      "status": "up",
      "timestamp": "2024-01-15T10:30:45Z"
    }
  }
}
```

cURL Example with Headers:

```
curl -X GET http://localhost:9153/health \
-H "Accept: application/json" \
-v
```

metrics

Exposes Prometheus-compatible metrics for monitoring node performance and health.

Method: GET /metrics

Parameters: None

Returns: Prometheus-format text response with all collected metrics

Example Request:

```
curl -X GET http://localhost:9153/metrics
```

Example Response:

```
# HELP genlayer_node_cpu_usage_percent Current CPU usage percentage (0-100)
Source: https://docs.genlayer.com/api-references/genlayer-node/ops/metrics
# TYPE genlayer_node_cpu_usage_percent gauge
genlayer_node_cpu_usage_percent{component="node"} 15.5
genlayer_node_cpu_usage_percent{component="genvm-llm"} 45.2
genlayer_node_cpu_usage_percent{component="genvm-web"} 12.8
```

```

genlayer_node_cpu_usage_percent{component="webdriver"} 5.3

# HELP genlayer_node_memory_rss_bytes Resident Set Size (RSS) memory usage in bytes
# TYPE genlayer_node_memory_rss_bytes gauge
genlayer_node_memory_rss_bytes{component="node"} 524288000
genlayer_node_memory_rss_bytes{component="genvm-llm"} 2147483648

# HELP genlayer_node_network_rx_bytes_total Total bytes received across all network interfaces
# TYPE genlayer_node_network_rx_bytes_total counter
genlayer_node_network_rx_bytes_total{component="node"} 1073741824

```

Available Metrics

The node collects metrics for three main components:

- Node - The main node process
- GenVM - The GenVM modules (LLM and Web)
- WebDriver - The WebDriver container

CPU Metrics

genlayer_node_cpu_usage_percent

Current CPU usage percentage (0-100).

Type: Gauge

Labels:

- component: One of node, genvm-llm, genvm-web, webdriver

Example:

```

genlayer_node_cpu_usage_percent{component="node"} 15.5
genlayer_node_cpu_usage_percent{component="genvm-llm"} 45.2
genlayer_node_cpu_usage_percent{component="genvm-web"} 12.8
genlayer_node_cpu_usage_percent{component="webdriver"} 5.3

```

Memory Metrics

genlayer_node_memory_rss_bytes

Resident Set Size (RSS) memory usage in bytes.

Type: Gauge

Labels:

- component: One of node, genvm-llm, genvm-web, webdriver

Example:

```

genlayer_node_memory_rss_bytes{component="node"} 524288000
genlayer_node_memory_rss_bytes{component="genvm-llm"} 2147483648

```

genlayer_node_memory_vms_bytes

Virtual Memory Size (VMS) in bytes.

Type: Gauge

Labels:

- component: One of node, genvm-llm, genvm-web, webdriver

Note: Only available for node and GenVM components.

genlayer_node_memory_usage_bytes

Current memory usage in bytes (WebDriver only).

Type: Gauge

Labels:

- component: webdriver

genlayer_node_memory_limit_bytes

Memory limit in bytes (WebDriver only).

Type: Gauge

Labels:

- component: webdriver

genlayer_node_memory_percent

Memory usage percentage.

Type: Gauge

Labels:

- component: One of genvm-llm, genvm-web, node, webdriver
- node: Node address (hex format)

genlayer_node_memory_peak_bytes

Peak memory usage in bytes (GenVM components only).

Type: Gauge

Labels:

- component: One of genvm-llm, genvm-web
- node: Node address (hex format)

Network Metrics

genlayer_node_network_rx_bytes_total

Total bytes received across all network interfaces.

Type: Counter

Labels:

- component: One of node, genvm-llm, genvm-web, webdriver
- node: Node address (hex format)

Example:

```
genlayer_node_network_rx_bytes_total{component="node",node="0x84b6cbd007511352d3fea26834c5c39a440903c4"} 96879942
genlayer_node_network_rx_bytes_total{component="webdriver",node="0x84b6cbd007511352d3fea26834c5c39a440903c4"} 84
```

genlayer_node_network_tx_bytes_total

Total bytes transmitted across all network interfaces.

Type: Counter

Labels:

- component: One of node, genvm-llm, genvm-web, webdriver
- node: Node address (hex format)

Disk Metrics

genlayer_node_disk_free_bytes

Available free disk space in bytes.

Type: Gauge

Labels:

- component: node
- directory: One of genlayer.db, keystore, logs
- node: Node address (hex format)

genlayer_node_disk_total_bytes

Total disk space in bytes.

Type: Gauge

Labels:

- component: node
- directory: One of genlayer.db, keystore, logs
- node: Node address (hex format)

genlayer_node_disk_usage_bytes

Used disk space in bytes.

Type: Gauge

Labels:

- component: node
- directory: One of genlayer.db, keystore, logs
- node: Node address (hex format)

genlayer_node_disk_usage_percent

Used disk space as percentage.

Type: Gauge

Labels:

- component: node
- directory: One of genlayer.db, keystore, logs
- node: Node address (hex format)

Node Status Metrics

genlayer_node_blocks_behind

Number of blocks behind the latest block.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_latest_block

Latest block number in the network.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_processing_block

Currently processing block number.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_synced_block

Last synced block number.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_synced

Node synchronization status (1 = synced, 0 = not synced).

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_uptime_seconds

Node uptime in seconds.

Type: Gauge

Labels:

- component: node
- node: Node address (hex format)

Transaction Metrics

genlayer_node_transactions_accepted_synced_total

Total number of accepted and synced transactions.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_transactions_activated_total

Total number of activated transactions.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_transactions_leader_proposed_total

Total number of transactions proposed as leader.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_transactions_leader_revealed_total

Total number of transactions revealed as leader.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_transactions_validator_commit_total

Total number of validator commit votes.

Type: Gauge

Labels:

- node: Node address (hex format)

genlayer_node_transactions_validator_reveal_total

Total number of validator reveal votes.

Type: Gauge

Labels:

- node: Node address (hex format)

Go Runtime Metrics

The endpoint also exposes standard Go runtime metrics including:

- go_gc_duration_seconds - Garbage collection duration summary
- go_goroutines - Number of active goroutines
- go_memstats_* - Go memory statistics (heap, stack, GC)
- go_threads - Number of OS threads

Process Metrics

Standard process metrics are also available:

- process_cpu_seconds_total - Total CPU time
- process_resident_memory_bytes - Resident memory size
- process_virtual_memory_bytes - Virtual memory size
- process_open_fds - Number of open file descriptors
- process_network_receive_bytes_total - Network bytes received

- process_network_transmit_bytes_total - Network bytes transmitted

Prometheus Handler Metrics

Metrics about the metrics endpoint itself:

- promhttp_metric_handler_requests_total - Total scrape requests by status code
- promhttp_metric_handler_requests_in_flight - Current scrapes being served

Configuration

Metrics collection can be configured in the node configuration file:

```
metrics:
  interval: "15s"          # Default interval for all collectors
  collectors:
    node:
      enabled: true        # Enable/disable node metrics
      interval: "30s"      # Optional: Override default interval
    genvm:
      enabled: true        # Enable/disable GenVM metrics
      # Uses default interval (15s)
    webdriver:
      enabled: true        # Enable/disable WebDriver metrics
      interval: "60s"      # Optional: Override default interval
```

Default Values

- metrics.interval: 15s (applies to all collectors by default)
- enabled: true (for each collector)
- interval: Uses metrics.interval if not specified per collector

Configuration Priority

1. Collector-specific interval (if specified)
2. Global metrics.interval (if specified)
3. System default (15s)

Troubleshooting

No Metrics for GenVM

GenVM processes are ephemeral and only run when executing smart contracts. If you don't see GenVM metrics, it's likely that no contracts are currently being executed.

WebDriver Metrics Missing

Ensure the WebDriver container is running:

```
docker ps | grep genlayer-node-webdriver
```

If not running, start it:

```
task docker:webdriver:docker-compose:up:detach
```

Metrics Not Updating

Check if metrics collection is enabled in your configuration and that the node has been restarted after configuration changes.

Performance Considerations

- Collection intervals can be increased to reduce overhead further
- Each collector runs independently and won't block others if one fails

snapshot

Source: <https://docs.genlayer.com/api-references/genlayer-node/ops/snapshot>

Produces a consistent snapshot of the node's state at the current block. Two modes selected by the request body: stream the snapshot back as a gzipped tar response (operator path), or write it to a destination directory on the node's filesystem (zero-downtime upgrade-controller path).

Method: POST /snapshot

Authentication: required. Send the operator token in either header:

```
Authorization: Bearer <token>
X-Admin-Auth: <token>
```

The token is the value of GENLAYERNODE_NODE_OPERATOR_TOKEN on the node process. When that variable is unset, the endpoint returns 401 to every caller regardless of header value (default-deny). The other ops routes (/metrics, /health, /balance) are unaffected and remain public.

Network exposure: the ops server is intended for in-host or in-cluster access (Prometheus scraping /metrics, k8s probes hitting /health). The endpoint may not be reachable from outside the host depending on the deployment's firewall. The recommended operator workflow is therefore one of:

- SSH onto the node and call the endpoint via 127.0.0.1:9153, or
- Open an SSH tunnel from the operator workstation to the node (ssh -L 9153:127.0.0.1:9153) and call localhost:9153 through it.

Parameters:

- destDir (string, optional): Absolute path on the node's filesystem. When set, switches to write-to-path mode; when omitted or empty, the snapshot is streamed back as application/gzip.

Returns:

- Stream mode (destDir omitted): gzipped tar archive in the response body, Content-Type: application/gzip.
- Write-to-path mode (destDir set): empty body, 204 No Content.

Example Request — Stream Mode (operator):

On the node (after ssh):

```
curl -X POST http://127.0.0.1:9153/snapshot \
-H "X-Admin-Auth: $GENLAYERNODE_NODE_OPERATOR_TOKEN" \
-H "Content-Type: application/json" \
-d '{}' \
-o snapshot.tar.gz
```

Or via SSH tunnel from a workstation (forwards 9153 to the node's loopback):

```
ssh -L 9153:127.0.0.1:9153 -N <node> &
curl -X POST http://localhost:9153/snapshot \
-H "X-Admin-Auth: $GENLAYERNODE_NODE_OPERATOR_TOKEN" \
-H "Content-Type: application/json" \
-d '{}' \
-o snapshot.tar.gz
```

The response body is a stream — curl can write it to a file (as above) or pipe it through any tool that reads stdin, without staging on local disk.

Example Response — Stream Mode:

```
HTTP/1.1 200 OK
Content-Type: application/gzip
Content-Disposition: attachment; filename="snapshot.tar.gz"

<gzipped tar archive of the snapshot's files>
```

Example Request — Write-to-Path Mode (upgrade controller):

```
curl -X POST http://127.0.0.1:9153/snapshot \
-H "X-Admin-Auth: $GENLAYERNODE_NODE_OPERATOR_TOKEN" \
-H "Content-Type: application/json" \
-d '{"destDir": "/var/snapshots/asimov-100"}'
```

Example Response — Write-to-Path Mode:

```
HTTP/1.1 204 No Content
```

HTTP Status Codes:

- 200 OK: Stream mode succeeded; body carries the gzipped tar archive.
- 204 No Content: Write-to-path mode succeeded; snapshot is on disk at destDir.
- 400 Bad Request: destDir is not absolute, or already exists.
- 401 Unauthorized: Missing, invalid, or empty operator token (also returned when GENLAYERNODE_NODE_OPERATOR_TOKEN is unset on the node).
- 500 Internal Server Error: Snapshot creation failed (e.g., disk full, DB checkpoint error).

Notes:

- A snapshot may be several GB. In stream mode the connection stays open for the duration of the snapshot — configure reverse proxies / load balancers with adequate read timeouts and disable response buffering.
- In stream mode, a dropped connection leaves no on-disk artifact on the node; retry the request from scratch.
- In write-to-path mode, destDir must be an absolute path that does not yet exist; the node creates it. Path-traversal hardening beyond these checks is a separate (Sev: Medium) follow-up — treat the operator token as a credential that can write under the node user's filesystem permissions.
- Docker deploys: in write-to-path mode, destDir is resolved inside the node container's filesystem, not on the host. To make the resulting directory reachable from the host (or from another container), use a volume mount that maps destDir identically on both sides, or stick with stream mode. Stream mode is unaffected — the response body is sent over HTTP regardless of where the node runs.

- The snapshot includes the full chain database (SSTables, MANIFEST, WAL) at the moment of capture. Treat the artifact as sensitive — anyone holding it can reconstruct the node's state.
- Token comparison is constant-time and length-padded so neither the value nor the length of the secret can be inferred from response timing.
- Rotation: update GENLAYERNODE_NODE_OPERATOR_TOKEN and restart the node. Tokens are read once at startup and not hot-reloaded.

Forward-compatibility note: when the zero-downtime upgrade controller (Phase 3) lands, /snapshot will additionally accept a second role (upgrade) so the controller can drive the snapshot + stop + start sequence with its own credential. The operator-token path documented here remains the same — no migration is required for operator tooling.

cURL Example with Bearer header:

```
curl -X POST http://127.0.0.1:9153/snapshot \
-H "Authorization: Bearer $GENLAYERNODE_NODE_OPERATOR_TOKEN" \
-H "Content-Type: application/json" \
-d '{}' \
-o snapshot.tar.gz
```

GenLayer Labs

Source: <https://docs.genlayer.com/partners/genlayerlabs>

GenLayer Labs (<https://www.genlayerlabs.com/>) is a pioneering AI research lab dedicated to revolutionizing human-AI interaction. They are one of the core developers of GenLayer, focusing on using AI to enhance the performance, security, and scalability of blockchain networks. By embedding AI into the blockchain's core, GenLayer Labs enables decentralized applications to make smart decisions autonomously, opening new possibilities for automation and efficiency in various industries.

We encourage other developers and organizations to collaborate with GenLayer Labs on this groundbreaking project. Together, we can push the boundaries of what AI and blockchain technology can achieve.

Heurist

Source: <https://docs.genlayer.com/partners/heurist>

Heurist (<https://www.heurist.ai/>) is a Layer 2 network for AI model hosting and inference, built on the ZK Stack. It offers serverless access to open-source AI models through a decentralized network, making AI deployment easy and efficient for developers. This approach promotes transparency and reduces bias, which is especially useful for projects like GenLayer that require detailed AI analysis.

In collaboration with GenLayer, Heurist provides APIs for open-source large language models (LLMs) to enable developers building Intelligent Contracts use AI at low or no cost. Developers building on GenLayer can obtain free Heurist API credits (<https://dev-api-form.heurist.ai>) by using the referral code: `_"genlayer"`.

This partnership underscores our commitment to providing robust tools and scalable solutions for decentralized applications.

io.net

Source: <https://docs.genlayer.com/partners/ionet>

io.intelligence is an AI infrastructure and API platform that democratizes access to advanced AI models and AI agents for the IO community and AI developers. It allows users to access and integrate pre-trained open-source models and custom AI agents into their applications via API calls. io.intelligence is powered by io.net (<https://io.net>)'s decentralized distributed compute network.

In partnership with GenLayer, io.net (<https://io.net>) provides access to over 25 of the top open source models including latest versions of Deepseek, Qwen, Llama and Mistral through one unified API endpoint. Developers on Genlayer can receive free daily compute tokens to access io.intelligence for Enterprise. Complete a short form (<https://form.typeform.com/to/pDmCCViV>) to get started.

API Reference: <https://docs.io.net/reference/get-started-with-io-intelligence-api>

Comput3

Source: <https://docs.genlayer.com/partners/comput3>

Comput3 (<https://genlayer.comput3.ai/>) is a decentralized compute network providing access to various AI models. The platform offers serverless access to open-source AI models through a distributed infrastructure, enabling developers to leverage powerful AI capabilities without the need for local hardware.

In partnership with GenLayer, Comput3 provides access to inferencing APIs with support for llama3, hermes3, and qwen3 models. GenLayer Validators can use the Comput3.ai inferencing API to power their Intelligent Contracts with high-quality AI responses. Validators can obtain free Comput3 API credits (<https://genlayer.comput3.ai/>) to get started with their validator setup.

This partnership enhances the GenLayer ecosystem by providing reliable and scalable AI inference capabilities for decentralized applications.

Using Chutes with GenLayer Validators

Source: <https://docs.genlayer.com/partners/chutes>

This guide explains how GenLayer validators can use Chutes to access AI models through the Chutes API.

Chutes provides serverless inference for open-source AI models running on decentralized GPU infrastructure powered by Bittensor.

Validators can call these models through a simple HTTP API without running their own GPU infrastructure.

GenLayer Validators can request a discount on Chutes models by filling out this form (https://docs.google.com/forms/d/e/1FAIpQLSd6l_Q0aiCaIDQLX9FvzQ3hfFaKR4j1jaDEoa-jfpXhLclmog/viewform).

Why Chutes

Running large AI models locally requires significant GPU infrastructure.

Chutes allows validators to run AI workloads without managing GPUs, by providing:

- Serverless model inference
- Decentralized GPU compute
- A simple REST API
- Access to many open-source AI models

Validators can integrate Chutes into their systems to run AI workloads while keeping infrastructure lightweight.

Resources

- Chutes Website (<https://chutes.ai>)
- Chutes Documentation (<https://chutes.ai/docs>)
- Chutes API Reference (<https://api.chutes.ai/docs>)
- Available Models (API) (<https://llm.chutes.ai/v1/models>)
- Model Browser (Web UI) (<https://chutes.ai/app?type=llm>)

1. Create a Chutes Account

First create an account: <https://chutes.ai> (<https://chutes.ai>)

Chutes supports multiple login methods including:

- Google login
- Fingerprint-based account identity

When creating your account, store your fingerprint securely. It is an important identifier tied to your account.

2. Generate an API Key

After creating an account, generate an API key.

API keys allow applications to authenticate requests to the Chutes API.

Steps:

1. Log into your Chutes account
2. Open account settings
3. Navigate to API Keys
4. Generate a new key
5. Store the key securely

Your API key will be used in authenticated API requests.

3. Authenticate Requests

The Chutes API uses Bearer token authentication.

Include your API key as a Bearer token in the Authorization header. You can verify your key is working by listing the available models:

```
curl https://llm.chutes.ai/v1/models \
-H "Authorization: Bearer $CHUTES_API_KEY"
```

The API follows standard REST conventions and returns JSON responses.

4. Run Your First Model from a Validator

Validators can call the Chutes inference API directly from their validator environment.

Chutes provides an OpenAI-compatible inference endpoint for running language models.

Endpoint: <https://llm.chutes.ai/v1/chat/completions>

This naming convention is the same format commonly used on Hugging Face, where models are referenced as: organization/model

For example: deepseek-ai/DeepSeek-R1

The list of available models changes frequently. You can retrieve the latest models here:

- API endpoint: <https://llm.chutes.ai/v1/models> (<https://llm.chutes.ai/v1/models>)

- Browse them visually: <https://chutes.ai/app?type=llm> (<https://chutes.ai/app?type=llm>)

Example request from a validator environment:

```
curl https://llm.chutes.ai/v1/chat/completions \
-H "Authorization: Bearer $CHUTES_API_KEY" \
-H "Content-Type: application/json" \
-d '{
  "model": "model_team/model_name",
  "messages": [
    {
      "role": "user",
      "content": "Explain decentralized AI infrastructure in one sentence."
    }
  ]
}'
```

Example response:

```
{
  "choices": [
    {
      "message": {
        "role": "assistant",
        "content": "Decentralized AI infrastructure allows developers and validators to access distributed compute resources for running AI models without owning the underlying hardware."
      }
    }
  ]
}
```

Validators can then use the returned output inside their application logic.

Example use cases include:

- Evaluating prompts or responses
- Generating summaries or structured outputs
- Integrating AI into validator tooling

Because the Chutes API is stateless and accessed via HTTPS, it can be integrated into validator environments without additional infrastructure.

5. Discover Available Models

The list of models available on Chutes changes frequently.

You can retrieve the current list using: <https://llm.chutes.ai/v1/models> (<https://llm.chutes.ai/v1/models>)

Example:

```
curl https://llm.chutes.ai/v1/models
```

This endpoint returns the latest models available for inference.

Models can also be browsed in the Chutes web interface: <https://chutes.ai/app?type=llm> (<https://chutes.ai/app?type=llm>)

Best Practices

For validator deployments:

- Create a dedicated API key
- Store the key in environment variables
- Avoid committing API keys to source control

Example:

```
export CHUTES_API_KEY=your_api_key_here
```

Next Steps

- Explore the full API reference (<https://api.chutes.ai/docs>)
- View the latest available models (<https://llm.chutes.ai/v1/models>)
- Browse models in the Chutes app (<https://chutes.ai/app?type=llm>)
- Learn more about Chutes (<https://chutes.ai/docs>)

Morpheus

Source: <https://docs.genlayer.com/partners/morpheus>

Morpheus (<https://mor.org/>) is a decentralized AI inference network where open-source model providers compete on a peer-to-peer marketplace. Instead of relying on a single centralized API, Morpheus routes requests across independent compute providers offering models like DeepSeek, Llama, Qwen, and others — with the available model list updating dynamically as providers join and leave the network.

In partnership with GenLayer, Morpheus provides decentralized LLM inference for Intelligent Contract validators through an OpenAI-compatible API. GenLayer Studio fetches available models from the Morpheus marketplace in real time, so validators always see the latest options. Developers can obtain a Morpheus API key (<https://mor.org/>) to get started.

This partnership brings decentralized, censorship-resistant AI inference to the GenLayer ecosystem, ensuring that Intelligent Contracts are not dependent on any single provider.

FAQ

Source: <https://docs.genlayer.com/FAQ>

What is GenLayer?

GenLayer is the adjudication layer for the agentic economy. It uses decentralized AI-validator consensus to resolve contracts that require judgment, not just code. It's built as an L2 on zkSync Elastic Chain, anchoring to Ethereum's security. Contracts on GenLayer — called Intelligent Contracts — can interpret language, process images, fetch live web data, and reach consensus on contested or subjective outcomes. See [What is GenLayer \(/understand-genlayer-protocol/what-is-genlayer\)](/understand-genlayer-protocol/what-is-genlayer).

How is GenLayer different from other blockchains?

Traditional smart contracts can only execute deterministic logic and rely on external oracles for off-chain data. GenLayer integrates AI at the protocol level — contracts natively access LLMs, the web, and image processing. No oracles, no intermediaries. Where Bitcoin reached consensus on the **order** of transactions and Ethereum on the **execution** of code, GenLayer reaches consensus on the meaning of transactions. See [What is GenLayer \(/understand-genlayer-protocol/what-is-genlayer#how-it-compares\)](/understand-genlayer-protocol/what-is-genlayer#how-it-compares).

Is GenLayer trying to replace courts?

No. GenLayer is built to provide a first-instance, internet-native adjudication layer for disputes that arrive too quickly, too often, and too globally for traditional courts to handle in time — particularly the disputes the agentic economy is starting to produce at machine speed. It is designed to complement, not replace, existing legal frameworks.

Where does GenLayer fit in the agentic-commerce stack?

The agentic-commerce stack — Coinbase's x402 (<https://www.x402.org/>) for payments, Ethereum's ERC-8004 (<https://eips.ethereum.org/EIPS/eip-8004>) for trustless agent identity, the Linux Foundation's A2A (<https://a2a-protocol.org>) for agent interoperability, plus Stripe/OpenAI's ACP, Visa's Trusted Agent Protocol, Google's AP2, and Mastercard's Agent Pay — engineers the happy path. None of these standards ships dispute resolution. GenLayer is the layer they reach for when a commitment becomes contested.

What language are Intelligent Contracts written in?

Python. Intelligent Contracts are Python classes extending `gl.Contract`. No Solidity required. See [Your First Contract \(/developers/intelligent-contracts/first-contract\)](/developers/intelligent-contracts/first-contract).

What is Optimistic Democracy?

GenLayer's consensus mechanism. A leader validator executes the transaction and proposes a result. Other validators independently verify it and vote. If a majority agrees, the transaction is accepted. Anyone can appeal within a finality window, triggering re-evaluation by a larger validator set. See [How GenLayer Works \(/understand-genlayer-protocol/optimistic-democracy-how-genlayer-works\)](/understand-genlayer-protocol/optimistic-democracy-how-genlayer-works).

How do validators handle non-deterministic AI results?

Since LLMs and web data produce varying outputs, GenLayer provides multiple strategies: strict equality (for deterministic operations), LLM-based comparison (validators compare outputs against criteria), and custom validation functions (developers define their own leader/validator logic). See [Non-determinism \(/developers/intelligent-contracts/features/non-determinism\)](/developers/intelligent-contracts/features/non-determinism).

Can contracts process images?

Yes. Pass images to LLMs via `gl.nondet.exec_prompt(images=[...])`. Contracts can also capture web page screenshots with `gl.nondet.web.render(url, mode='screenshot')` and analyze them. See Image Processing (</developers/intelligent-contracts/features/image-processing>).

Can contracts send and receive GEN (native token)?

Yes. Use `@gl.public.write.payable` to receive value, `gl.message.value` to access the amount sent, and `emit_transfer()` to send value to other contracts or EOAs. See Value Transfers (</developers/intelligent-contracts/features/value-transfers>).

Do I need to be an AI expert?

No. The GenLayer SDK handles LLM calls and consensus. You write Python and call `gl.nondet.exec_prompt()` with a prompt. Validators handle model execution. GenLayer Studio provides a browser-based IDE for writing, deploying, and testing contracts.

Can I run normal deterministic transactions?

Yes. GenLayer fully supports deterministic logic — standard state updates, value transfers, and contract interactions all work without any AI components. You can mix deterministic and non-deterministic operations in the same contract.

What are ghost contracts?

Every Intelligent Contract has a corresponding ghost contract on the GenLayer Chain (EVM layer) at the same address. Ghost contracts hold the contract's GEN balance, relay transactions to consensus, and execute external messages. See Ghost Contracts (</developers/intelligent-contracts/features/messages#ghost-contracts>).

What can I build?

Anything requiring judgment that deterministic code alone can't resolve: performance and milestone adjudication (bounties, grants, retro funding), adjudication inside the agentic-commerce stack (agent SLAs, escrow on ambiguous completion, reputation disputes), prediction markets, insurance claims, content verification, compliance automation, AI-governed DAOs, and more. See Use Cases (</understand-genlayer-protocol/typical-use-cases>) and projects building on GenLayer (<https://portal.genlayer.foundation/#/>).

How do I get started?

1. Try GenLayer Studio (<https://studio.genlayer.com>) in the browser — no setup needed
2. Or install the GenLayer CLI (</api-references/genlayer-cli>) for local development
3. Follow Your First Contract (</developers/intelligent-contracts/first-contract>)
4. Use `genlayer-js` (</api-references/genlayer-js>) or `genlayer-py` (</api-references/genlayer-py>) for app integration

How is the network secured?

Through delegated proof of stake and Optimistic Democracy:

- Token holders delegate stake to validators, incentivizing honest behavior
- Multiple validators independently verify each transaction
- Appeals provide a failsafe — dishonest validators get slashed
- The system anchors to Ethereum via zkSync for finality

How can I contribute?

- Join the community on Discord (<https://discord.gg/8Jm4v89VAu>) or Telegram (<https://t.me/genlayer>)
- Contribute to the GitHub repositories (<https://github.com/genlayerlabs>)
- We're hiring (<https://x.com/GenLayer/jobs>)

Ollama

Source: https://docs.genlayer.com/_providers/ollama

Ollama is a free and open-source LLM provider that runs locally, providing an accessible option for processing and validating Intelligent Contracts within GenLayer. While it may perform slower than other providers, it is set up to run within Docker containers to facilitate the operations.

Setup during genlayer init

When you run the `genlayer init` command, you will be prompted to select the LLM providers you want to use. If you choose Ollama, it will automatically set up the necessary Docker containers for running the validators.

Server Details

Ollama server details are typically configured as follows:

```
OLAMAPROTOCOL=http  
OLAMAHOST=ollama  
OLAMAPORT=11434
```

This environment variable is configured in the `.env` file located in the Studio's root folder.

Troubleshooting Tips

- Missing Configuration: Ensure that the Ollama server details (OLAMAPROTOCOL, OLAMAHOST, OLAMAPORT) are set correctly during the initialization process.
- Docker Setup Issues: Verify that Docker is installed and running on your system, as Ollama relies on Docker containers for its operation.
- Service Status: To confirm Ollama is running, use the `docker ps` command and check for the ollama container.

Discover the Protocol (/understand-genlayer-protocol "Discover the Protocol")

OpenAI

Source: https://docs.genlayer.com/_providers/openai

OpenAI is an LLM provider that provides fast and reliable AI models used by validators to process and validate Intelligent Contracts.

Setup during genlayer init

When you run the `genlayer init` command, you will be prompted to select the LLM providers you want to use. If you choose OpenAI, you will need to provide an API key (<https://openai.com/api/>) during the setup process. This key is necessary to authenticate and access OpenAI's services.

Set the OpenAI API key when prompted:

```
OPENAIKEY=your_openai_api_key
```

This environment variable is configured in the `.env` file located in the Studio's root folder.

Troubleshooting Tips

- Missing API Key: Ensure that you enter the `OPENAIKEY` when prompted during the initialization process.
- Incorrect Configuration: Verify that the API key you entered is valid and that you have selected the correct provider during initialization.

Discover the Protocol (`/understand-genlayer-protocol "Discover the Protocol"`)

Greyboxing in GenLayer

Source: https://docs.genlayer.com/_temp/security-and-best-practices/grey-boxing

Greyboxing combines elements of both "blackboxing" and "whiteboxing" to create a semi-transparent operational environment for AI models. In GenLayer, this technique involves individually configuring each validator's interaction with AI models, ensuring that manipulative or malicious inputs do not affect the blockchain's integrity.

How Greyboxing Works

1. Unique Configuration per Validator: Each validator in GenLayer operates its AI models within a uniquely configured greybox environment. This customization makes it more difficult for attackers to perform successful universal attacks, as they cannot predict the specific defensive mechanisms employed by each validator.
2. Input Filtration: Inputs to AI models are rigorously filtered to remove or sanitize potential threats before processing. This precaution helps ensure that only safe and intended data influences the AI's decision-making processes.
3. Output Restrictions: Outputs from AI models are confined within predefined safety parameters to prevent unintended or harmful responses. These restrictions are tailored to each greybox setup, enhancing security and reliability.
4. Model Isolation: While AI models access necessary data for operations, they do so within controlled and observable environments. This isolation helps prevent external manipulations and maintains the integrity of

model responses.

5. Continuous Monitoring: The operations within each greyboxed environment are continuously monitored for any signs of anomalous or potentially harmful activity. This ongoing surveillance allows for immediate detection and response to security threats.

Discover the Protocol (/understand-genlayer-protocol "Discover the Protocol")

Universal Attacks in GenLayer

Source: https://docs.genlayer.com/_temp/security-and-best-practices/universal-attacks

Universal attacks are a type of adversarial attack where attackers craft inputs that are effective across different models or different runs of the same model. This could involve creating input data that consistently causes models to make errors or behave in unintended ways, regardless of slight variations in model training or deployment environments. The universal nature of these attacks makes them particularly challenging to defend against because they exploit fundamental weaknesses in the models' architecture or training data.

Mitigating Universal Attacks in GenLayer

- Random Selection of Validators: Validators are randomly selected to review transactions. Random selection disrupts potential attack planning, as attackers cannot easily predict which validators (and consequently, which models) will be validating any given transaction.
- Greyboxing: Greyboxing (/security-and-best-practices/grey-boxing) is used to create a controlled environment where inputs and outputs to AI models are rigorously monitored and filtered. This security layer is customized for each validator, further enhancing the resilience against universal attacks.
- Validator Checks on Leader's Output: This step acts as a second perspective, ensuring that any errors or manipulations in the leader's outputs are likely to be caught by other validators, adding a robust layer of verification that helps maintain the integrity of transaction processing.

Discover the Protocol (/understand-genlayer-protocol "Discover the Protocol")

GenLayerJS

Source: <https://docs.genlayer.com/api-references/genlayer-js>

About

GenLayerJS SDK is a TypeScript library designed for developers building decentralized applications (Dapps) on the GenLayer protocol. This SDK provides a comprehensive set of tools to interact with the GenLayer network, including client creation, transaction handling, event subscriptions, and more, all while leveraging the power of Viem as the underlying blockchain client.

Prerequisites

Before installing GenLayerJS SDK, ensure you have the following prerequisites installed:

- Node.js ($\geq 16.x$)
- npm ($\geq 7.x$)

Installation and Usage

To install the GenLayerJS SDK, use the following command:

```
$ npm install genlayer-js
```

Here's how to initialize the client and connect to the GenLayer Simulator:

Reading a Transaction

```
import { localnet } from 'genlayer-js/chains';
import { createClient } from "genlayer-js";

const client = createClient({
  chain: localnet,
});

const transactionHash = "0x...";

const transaction = await client.getTransaction({ hash: transactionHash })
```

Waiting for Transaction Receipt

```
import { localnet } from 'genlayer-js/chains';
import { createClient } from "genlayer-js";
import { TransactionStatus } from "genlayer-js/types";

const client = createClient({
  chain: localnet,
});

// Get simplified receipt (default - removes binary data, keeps execution results)
const receipt = await client.waitForTransactionReceipt({
  hash: "0x...",
  status: TransactionStatus.FINALIZED,
  fullTransaction: false // Default - simplified for readability
});

// Get complete receipt with all fields
const fullReceipt = await client.waitForTransactionReceipt({
  hash: "0x...",
  status: TransactionStatus.FINALIZED,
  fullTransaction: true // Complete receipt with all internal data
});
```

Reading a contract

```
import { localnet } from 'genlayer-js/chains';
import { createClient } from "genlayer-js";

const client = createClient({
  chain: localnet,
});

const result = await client.readContract({
  // account: account, Account is optional when reading from contracts
  address: contractAddress,
```

```

    functionName: 'get_complete_storage',
    args: []
    stateStatus: "accepted",
  })
}

```

Writing a transaction

```

import { localnet } from 'genlayer-js/chains';
import { createClient, createAccount } from "genlayer-js";

const client = createClient({
  network: localnet,
});

const account = createAccount();
const transactionHash = await client.writeContract({
  account: account, // using this account for this transaction
  address: contractAddress,
  functionName: 'account',
  args: ['new_storage'],
  value: 0, // value is optional, if you want to send some native token to the contract
});

const receipt = await client.waitForTransactionReceipt({
  hash: txHash,
  status: TransactionStatus.FINALIZED, // or ACCEPTED
  fullTransaction: false // False by default - returns simplified receipt for better readability
})

```

Checking execution results

A transaction can be finalized by consensus but still have a failed execution. Always check `txExecutionResult` before reading contract state:

```

import { ExecutionResult, TransactionStatus } from "genlayer-js/types";

const receipt = await client.waitForTransactionReceipt({
  hash: txHash,
  status: TransactionStatus.FINALIZED,
});

if (receipt.txExecutionResultName === ExecutionResult.FINISHED_WITH_RETURN) {
  // Execution succeeded – safe to read state
  const result = await client.readContract({
    address: contractAddress,
    functionName: "get_storage",
    args: [],
  });
} else if (receipt.txExecutionResultName === ExecutionResult.FINISHED_WITH_ERROR) {
  // Execution failed – contract state was not modified
  console.error("Contract execution failed");
} else {
  // NOT_VOTED – execution hasn't completed
  console.warn("Execution result not yet available");
}

```

Fetching emitted messages and triggered transactions

Transactions can emit messages to other contracts. These messages create new child transactions when processed:

```

const tx = await client.getTransaction({ hash: txHash });

// Messages emitted by the contract during execution
console.log(tx.messages);
// [{messageType, recipient, value, data, onAcceptance, saltNonce}, ...]

// Child transaction IDs created from those messages (separate call)
const childTxIds = await client.getTriggeredTransactionIds({ hash: txHash });
console.log(childTxIds);

```

```
// ["0xabc...", "0xdef..."]
```

Debugging transaction execution

Use `debugTraceTransaction` to inspect the full execution trace of a transaction, including return data, errors, and GenVM logs:

```
const trace = await client.debugTraceTransaction({
  hash: txHash,
  round: 0, // optional, defaults to 0
});

console.log(trace.result_code); // 0=success, 1=user error, 2=VM error
console.log(trace.return_data); // hex-encoded contract return data
console.log(trace.stderr);      // standard error output
console.log(trace.genvm_log);   // detailed GenVM execution logs
```

Using with a wallet provider (MetaMask)

When building a browser dApp, create two clients: one for reads (no wallet needed) and one for writes (signed by the wallet). This follows the standard viem pattern and keeps concerns separated.

```
import { createClient } from "genlayer-js";
import { testnetBradbury } from "genlayer-js/chains";
import { TransactionStatus } from "genlayer-js/types";

// Read client – talks directly to GenLayer RPC, no wallet needed
const readClient = createClient({
  chain: testnetBradbury,
});

// Write client – signs transactions through the wallet
const writeClient = createClient({
  chain: testnetBradbury,
  account: address as `0x${string}`, // from wallet connection
  provider: window.ethereum,          // or from a wallet SDK
});

// Use readClient for all reads
const result = await readClient.readContract({
  address: contractAddress,
  functionName: "get_storage",
  args: [],
});

const tx = await readClient.getTransaction({ hash: txHash });

// Use writeClient for transactions (MetaMask popup)
const txHash = await writeClient.writeContract({
  address: contractAddress,
  functionName: "update_storage",
  args: ["new_value"],
  value: BigInt(0),
});

// Either client can wait for receipts
const receipt = await readClient.waitForTransactionReceipt({
  hash: txHash,
  status: TransactionStatus.ACCEPTED,
});
```

Switching the wallet to the correct network

When using MetaMask or another browser wallet, the wallet may be connected to a different chain than what your client is configured for. Use `client.connect()` to switch the wallet to the correct GenLayer network before sending transactions:

```
import { createClient } from "genlayer-js";
import { studionet } from "genlayer-js/chains";

const client = createClient({
  chain: studionet,
  account: address as `0x${string}`,
});

// Switch MetaMask to the correct chain (adds the network if not present)
await client.connect("studionet");

// Now transactions will go to the right network
const txHash = await client.writeContract({
  address: contractAddress,
  functionName: "create_profile",
  args: ["alice", "Hello world"],
  value: BigInt(0),
});
```

Available networks: "localnet", "studionet", "testnetAsimov", "testnetBradbury".

Note: If the wallet is on the wrong chain when you call writeContract, the SDK will throw a clear error telling you which chain the wallet is on vs. which chain the client expects. Call client.connect() to resolve this.

Staking Operations

The SDK provides staking functionality for validators and delegators on testnet-bradbury (and testnet-asimov).

```
import { testnetBradbury } from 'genlayer-js/chains';
import { createClient, createAccount } from "genlayer-js";

const account = createAccount();
const client = createClient({
  chain: testnetBradbury,
  account,
});

// Get epoch info (includes timing estimates and inflation data)
const epochInfo = await client.getEpochInfo();
// {
//   currentEpoch: 2n,
//   epochMinDuration: 86400n,           // 1 day in seconds
//   currentEpochStart: Date,
//   currentEpochEnd: Date | null,
//   nextEpochEstimate: Date | null,
//   validatorMinStake: "0.01 GEN",
//   delegatorMinStake: "42 GEN",
//   activeValidatorsCount: 6n,
//   inflation: "1000 GEN",             // Total inflation for current epoch
//   inflationRaw: 10000000000000000000n,
//   totalWeight: 50000000000000000000000n, // Total stake weight
//   totalClaimed: "500 GEN",           // Total claimed rewards
// }

// Get active validators
const validators = await client.getActiveValidators();

// Check if address is a validator
const isValidator = await client.isValidator("0x...");

// Get validator info
const validatorInfo = await client.getValidatorInfo("0x...");
// Join as validator (requires account with funds)
const result = await client.validatorJoin({ amount: "42000gen" });

// Join as delegator
const delegateResult = await client.delegatorJoin({
  validator: "0x...",
  amount: "42gen",
});
```

Key Features

- Client Creation: Easily create and configure a client to connect to GenLayer's network.
- Transaction Handling: Send and manage transactions on the GenLayer network.
- Staking: Full staking support for validators and delegators on testnet-bradbury and testnet-asimov.
- Wallet Integration*: Seamless integration with MetaMask for managing user accounts.
- Gas Estimation*: Estimate gas fees for executing transactions on GenLayer.

** under development*

Documentation

For detailed information on how to use GenLayerJS SDK, please refer to our documentation (<https://docs.genlayer.com/>).

Contributing

We welcome contributions to GenLayerJS SDK! Whether it's new features, improved infrastructure, or better documentation, your input is valuable. Please read our CONTRIBUTING (<https://github.com/genlayerlabs/genlayer-js/blob/main/CONTRIBUTING.md>) guide for guidelines on how to submit your contributions.

License

This project is licensed under the MIT License - see the LICENSE (/api-references/LICENSE) file for details.

appeal-bond (/api-references/genlayer-cli/transactions/appeal-bond "appeal-bond")Contracts
(/api-references/genlayer-js/contracts "Contracts")

GenLayerPY

Source: <https://docs.genlayer.com/api-references/genlayer-py>

About

GenLayerPY SDK is a python library designed for developers building decentralized applications (Dapps) on the GenLayer protocol. This SDK provides a comprehensive set of tools to interact with the GenLayer network, including client creation, transaction handling, event subscriptions, and more, all while leveraging the power of web3.py as the underlying blockchain client.

Prerequisites

Before installing GenLayerPY SDK, ensure you have the following prerequisites installed:

- Python (≥ 3.12)

Installation and Usage

To install the GenLayerPY SDK, use the following command:

```
$ pip install genlayer-py
```

Here's how to initialize the client and connect to the GenLayer Simulator:

Reading a Transaction

```
from genlayer_py import create_client
from genlayer_py.chains import localnet

client = create_client(
    chain=localnet,
)

transaction_hash = "0x..."

transaction = client.get_transaction(hash=transaction_hash)
```

Waiting for Transaction Receipt

```
from genlayer_py import create_client
from genlayer_py.chains import localnet
from genlayer_py.types import TransactionStatus

client = create_client(chain=localnet)

# Get simplified receipt (default - removes binary data, keeps execution results)
receipt = client.wait_for_transaction_receipt(
    transaction_hash="0x...",
    status=TransactionStatus.FINALIZED,
    full_transaction=False # Default - simplified for readability
)

# Get complete receipt with all fields
full_receipt = client.wait_for_transaction_receipt(
    transaction_hash="0x...",
    status=TransactionStatus.FINALIZED,
    full_transaction=True # Complete receipt with all internal data
)
```

Reading a contract

```
from genlayer_py import create_client
from genlayer_py.chains import localnet

client = create_client(
    chain=localnet,
)

result = client.read_contract(
    address=contract_address,
    function_name='get_complete_storage',
    args=[],
    state_status='accepted'
)
```

Writing a transaction

```

from genlayer_py.chains import localnet
from genlayer_py import create_client, create_account

client = create_client(
    chain=localnet,
)

account = create_account()

transaction_hash = client.write_contract(
    account=account,
    transaction=transaction,
    address=contract_address,
    function_name='account',
    args=['new_storage'],
    value=0, // value is optional, if you want to send some native token to the contract
)
receipt = client.wait_for_transaction_receipt(
    hash=transaction_hash,
    status=TransactionStatus.FINALIZED, // or ACCEPTED
    full_transaction=False // False by default - returns simplified receipt for better readability
)

```

Checking execution results

A transaction can be finalized by consensus but still have a failed execution. Always check `tx_execution_result` before reading contract state:

```

from genlayer_py import create_client, create_account
from genlayer_py.chains import testnet_bradbury
from genlayer_py.types import TransactionStatus, ExecutionResult

client = create_client(chain=testnet_bradbury, account=create_account())

receipt = client.wait_for_transaction_receipt(
    transaction_hash=tx_hash,
    status=TransactionStatus.FINALIZED,
)

if receipt.get("tx_execution_result_name") == ExecutionResult.FINISHED_WITH_RETURN.value:
    # Execution succeeded – safe to read state
    result = client.read_contract(
        address=contract_address,
        function_name="get_storage",
        args=[],
    )
elif receipt.get("tx_execution_result_name") == ExecutionResult.FINISHED_WITH_ERROR.value:
    # Execution failed – contract state was not modified
    raise RuntimeError("Contract execution failed")
else:
    # NOT_VOTED – execution hasn't completed
    print("Execution result not yet available")

```

Fetching emitted messages and triggered transactions

Transactions can emit messages to other contracts. These messages create new child transactions when processed:

```

tx = client.get_transaction(transaction_hash=tx_hash)

# Messages emitted by the contract during execution
print(tx["messages"])
# [{"messageType": 1, "recipient": "0x...", "value": 0, "data": "0x...", "onAcceptance": True, "saltNonce": 0}, ...]

# Child transaction IDs created from those messages (separate call)
child_tx_ids = client.get_triggered_transaction_ids(transaction_hash=tx_hash)
print(child_tx_ids)
# ["0xabc...", "0xdef..."]

```

Debugging transaction execution

Use `debug_trace_transaction` to inspect the full execution trace of a transaction, including return data, errors, and GenVM logs:

```
trace = client.debug_trace_transaction(
    transaction_hash=tx_hash,
    round=0, # optional, defaults to 0
)

print(trace["result_code"]) # 0=success, 1=user error, 2=VM error
print(trace["return_data"]) # hex-encoded contract return data
print(trace["stderr"]) # standard error output
print(trace["genvm_log"]) # detailed GenVM execution logs
```

Key Features

- Client Creation: Easily create and configure a client to connect to GenLayer's network.
- Transaction Handling: Send and manage transactions on the GenLayer network.
- Gas Estimation: Estimate gas fees for executing transactions on GenLayer.

*** under development**

Documentation

For detailed information on how to use GenLayerPY SDK, please refer to our documentation (<https://docs.genlayer.com/api-references/genlayer-py>).

Contributing

We welcome contributions to GenLayerPY SDK! Whether it's new features, improved infrastructure, or better documentation, your input is valuable. Please read our CONTRIBUTING (<https://github.com/genlayerlabs/genlayer-py/blob/main/CONTRIBUTING.md>) guide for guidelines on how to submit your contributions.

License

This project is licensed under the MIT License - see the LICENSE (</api-references/LICENSE>) file for details.

Staking (</api-references/genlayer-js/staking> "Staking") API Reference (</api-references/genlayer-py/api> "API Reference")

GenLayer Testing Suite

Source: <https://docs.genlayer.com/api-references/genlayer-test>

A pytest-based testing framework for GenLayer (<https://docs.genlayer.com/>) intelligent contracts. Built on top of genlayer-py (<https://docs.genlayer.com/api-references/genlayer-py>).

```
pip install genlayer-test
```

Two Ways to Test

The testing suite provides two execution modes. Pick the one that fits your workflow:

Start with Direct Mode. It's faster, simpler, and doesn't require Docker. Use Studio Mode when you need full network behavior, multi-validator consensus, or testnet deployment.

Direct Mode

Run contracts directly in Python — no simulator, no Docker, no network. Tests execute in milliseconds.

Quick Start

```
def test_storage(direct_vm, direct_deploy):
    # Deploy contract in-memory
    storage = direct_deploy("contracts/Storage.py", "initial")

    # Read state directly
    assert storage.get_storage() == "initial"

    # Write state directly
    storage.update_storage("updated")
    assert storage.get_storage() == "updated"
```

Run with pytest:

```
pytest tests/ -v
```

Fixtures

Cheatcodes

```
# Change sender
direct_vm.sender = alice

# Prank (temporary sender change)
with direct_vm.prank(bob):
    contract.method() # Called as bob

# Snapshots (captures full state: storage, mocks, sender, validators)
snap_id = direct_vm.snapshot()
contract.modify_state()
direct_vm.revert(snap_id) # Full state restored

# Expect revert
with direct_vm.expect_revert("Insufficient balance"):
    contract.transfer(bob, 1000000)

# Mock web/LLM (regex pattern matching)
direct_vm.mock_web(r"api\.example\.com", {"status": 200, "body": "{}"})
direct_vm.mock_llm(r"analyze.*", "positive sentiment")

# Test validator consensus logic
contract.update_price() # Runs leader_fn, captures validator
direct_vm.clear_mocks() # Swap mocks for validator
direct_vm.mock_llm(r".*", "different result")
assert direct_vm.run_validator() is False # Validator disagrees
```

```
# Strict mocks (detect unused mocks)
direct_vm.strict_mock = True

# Pickling validation (catch production serialization issues)
direct_vm.check_pickling = True
```

Full Direct Mode Documentation (</api-references/docs/direct-runner>) — fixtures, cheatcodes, validator testing, limitations, and complete examples.

Studio Mode

Deploy contracts to a running GenLayer Studio instance and interact via RPC. This gives you full network behavior including multi-validator consensus.

Prerequisites

- Python >= 3.12
- GenLayer Studio running (Docker)

Quick Start

```
from gltest import get_contract_factory, get_default_account
from gltest.assertions import tx_execution_succeeded

factory = get_contract_factory("MyContract")
contract = factory.deploy()

# Read method – returns value directly
result = contract.get_value().call()

# Write method – returns transaction receipt
tx_receipt = contract.set_value(args=["new_value"]).transact()
assert tx_execution_succeeded(tx_receipt)
```

Run with the gltest CLI:

```
gltest # Run all tests
gltest tests/test_mycontract.py # Specific file
gltest --network studionet # Specific network
gltest --leader-only # Skip consensus (faster)
gltest -v # Verbose output
```

Configuration

Create a gltest.config.yaml in your project root:

```
networks:
  default: localnet

  localnet:
    url: "http://127.0.0.1:4000/api"
    leader_only: false

  studionet:
    # Pre-configured – accounts auto-generated

  testnet_asimov:
    accounts:
      - "${ACCOUNT_PRIVATE_KEY_1}"
```

```

- "${ACCOUNT_PRIVATE_KEY_2}"
from: "${ACCOUNT_PRIVATE_KEY_1}"

paths:
  contracts: "contracts"
  artifacts: "artifacts"

environment: .env

```

Key options:

- Networks: localnet and studionet work out of the box. testnet_asimov requires account keys.
- Paths: Where your contracts and artifacts live.
- Environment: .env file for private keys.

Override via CLI:

```

gltest --network testnet_asimov
gltest --contracts-dir custom/contracts/path
gltest --rpc-url http://custom:4000/api
gltest --chain-type localnet

```

Contract Deployment

```

from gltest import get_contract_factory, get_default_account
from gltest.assertions import tx_execution_succeeded

factory = get_contract_factory("Storage")

# deploy() returns the contract instance (recommended)
contract = factory.deploy(
    args=["initial_value"],
    account=get_default_account(),
    consensus_max_rotations=3,
)

# deploy_contract_tx() returns only the receipt
receipt = factory.deploy_contract_tx(args=["initial_value"])
assert tx_execution_succeeded(receipt)

```

Read and Write Methods

```

# Read – call() returns the value
result = contract.get_storage().call()

# Write – transact() returns a receipt
tx_receipt = contract.update_storage(args=["new_value"]).transact(
    value=0,
    consensus_max_rotations=3,
    wait_interval=1000,
    wait_retries=10,
)
assert tx_execution_succeeded(tx_receipt)

```

Assertions

```

from gltest.assertions import tx_execution_succeeded, tx_execution_failed

assert tx_execution_succeeded(tx_receipt)
assert tx_execution_failed(tx_receipt)

# Regex matching on stdout/stderr (localnet/studionet only)
assert tx_execution_succeeded(tx_receipt, match_std_out=r".*code \d+")
assert tx_execution_failed(tx_receipt, match_std_err=r"Method.*failed")

```

Fixtures

```
def test_workflow(gl_client, default_account, accounts):
    factory = get_contract_factory("MyContract")
    contract = factory.deploy(account=default_account)

    tx_receipt = contract.some_method(args=["value"], account=accounts[1])
    assert tx_execution_succeeded(tx_receipt)
```

Mock LLM Responses

Simulate LLM responses for deterministic tests:

```
from gltest import get_contract_factory, get_validator_factory
from gltest.types import MockedLLMResponse

mock_response: MockedLLMResponse = {
    "nondet_exec_prompt": {
        "analyze this": "positive sentiment"
    },
    "eq_principle_prompt_comparative": {
        "values match": True
    }
}

validator_factory = get_validator_factory()
validators = validator_factory.batch_create_mock_validators(
    count=5,
    mock_llm_response=mock_response
)

transaction_context = {
    "validators": [v.to_dict() for v in validators],
    "genvm_datetime": "2024-01-01T00:00:00Z"
}

factory = get_contract_factory("LLMContract")
contract = factory.deploy(transaction_context=transaction_context)
result = contract.analyze_text(args=["analyze this"]).transact(
    transaction_context=transaction_context
)
```

Mock keys map to GenLayer methods:

The system performs substring matching on the internal user message — your mock key must appear within the message.

Mock Web Responses

Simulate HTTP responses for contracts that call `gl.nondet.web.get()`, etc.:

```
from gltest.types import MockedWebResponse
import json

mock_web_response: MockedWebResponse = {
    "nondet_web_request": {
        "https://api.example.com/price": {
            "method": "GET",
            "status": 200,
            "body": json.dumps({"price": 100.50})
        }
    }
}

validators = validator_factory.batch_create_mock_validators(
    count=5,
    mock_web_response=mock_web_response
)
```

You can combine both `mock_llm_response` and `mock_web_response` in a single `batch_create_mock_validators` call. URL matching is exact (including query parameters).

Custom Validators

```
from gltest import get_validator_factory

factory = get_validator_factory()

# Real validators with specific LLM providers
validators = factory.batch_create_validators(
    count=5,
    stake=10,
    provider="openai",
    model="gpt-4o",
    config={"temperature": 0.7},
    plugin="openai-compatible",
    plugin_config={"api_key_env_var": "OPENAI_API_KEY"}
)

# Use in transaction context
transaction_context = {
    "validators": [v.to_dict() for v in validators],
    "genvm_datetime": "2024-03-15T14:30:00Z"
}
```

Statistical Analysis

For LLM-based contracts, `.analyze()` runs multiple simulations to measure consistency:

```
analysis = contract.process_with_llm(args=["input"]).analyze(
    provider="openai",
    model="gpt-4o",
    runs=100,
)

print(f"Success rate: {analysis.success_rate:.2f}%")
print(f"Reliability: {analysis.reliability_score:.2f}%")
print(f"Unique states: {analysis.unique_states}")
```

Full Studio Mode Documentation (</api-references/docs/studio-runner>) — configuration reference, all CLI flags, mock LLM/web details, custom validators, statistical analysis, and complete examples.

Example Contract

```
from genlayer import *

class Storage(gl.Contract):
    storage: str

    def __init__(self, initial_storage: str):
        self.storage = initial_storage

    @gl.public.view
    def get_storage(self) -> str:
        return self.storage

    @gl.public.write
    def update_storage(self, new_storage: str) -> None:
        self.storage = new_storage
```

Project Structure

```

my-project/
├── contracts/
│   └── Storage.py
├── tests/
│   ├── test_direct.py      # Direct mode tests (fast)
│   ├── test_integration.py  # Studio mode tests
│   └── gltest.config.yaml   # Studio mode config

```

For more examples, see the contracts directory (</api-references/tests/examples/contracts>).

Troubleshooting

Contract not found: Ensure contracts are in `contracts/` or specify `--contracts-dir`. Contracts must inherit from `gl.Contract`.

Transaction timeouts (Studio mode): Increase `wait_interval` and `wait_retries` in `.transact()`.

Consensus failures (Studio mode): Increase `consensus_max_rotations` or use `--leader-only` for faster iteration.

Environment issues: Verify Python ≥ 3.12 . For Studio mode, check Docker is running (`docker ps`).

Contributing

See our Contributing Guide (</api-references/CONTRIBUTING>).

License

MIT — see `LICENSE` (</api-references/LICENSE>).

Support

- Documentation (<https://docs.genlayer.com/api-references/genlayer-test>)
- Discord (<https://discord.gg/qjCU4AWnKE>)
- GitHub Issues (<https://github.com/genlayerlabs/genlayer-testing-suite/issues>)
- Twitter (<https://x.com/GenLayer>)

API Reference (</api-references/genlayer-py/api> "API Reference") Integration Testing (</api-references/genlayer-test/integration> "Integration Testing")

ADRValidator Contract

Source: https://docs.genlayer.com/developers/intelligent-contracts/examples/_advanced/_adr-validator

The ADRValidator contract sets up a system for validating, categorizing, and ensuring consistency of Architectural Decision Records (ADRs) in a decentralized and automated manner. This contract demonstrates complex

document validation, category management, and a reward system within a blockchain environment.

□

This intelligent contract is currently not migrated to the real GenVM syntaxis.

```
import json
import re
from backend.node.genvm.icontract import IContract
from backend.node.genvm.equivalence_principle import call_llm_with_principle

class ADRValidator(IContract):
    def __init__(self):
        self.owner = contract_runner.from_address
        self.arch_categories = {}
        self.balances = {}
        self.max_reward = 10

    def change_owner(self, new_owner: str):
        if contract_runner.from_address == self.owner:
            self.owner = new_owner

    def set_max_reward(self, new_max_reward: int):
        if contract_runner.from_address == self.owner:
            self.max_reward = new_max_reward

    def get_owner(self) -> str:
        return self.owner

    def get_categories(self) -> str:
        return {
            category: details["description"]
            for category, details in self.arch_categories.items()
        }

    def get_adrs_of_a_category(self, category_name: str) -> dict:
        if category_name in self.arch_categories:
            return self.arch_categories[category_name]["ADRs"]

    def get_balances(self) -> dict[str, int]:
        return self.balances

    def get_balance_of(self, address: str) -> int:
        return self.balances.get(address, 0)

    def add_category(self, category_name: str, category_description: str):
        if (
            contract_runner.from_address == self.owner
            and category_name not in self.arch_categories
        ):
            self.arch_categories[f"{category_name}"] = {
                "description": category_description,
                "ADRs": [],
            }

    async def validate_adr(self, adr: str, category_name: str) -> None:
        print("validate")
        if not self._check_template(adr): return
        output = await self._evaluate_adr(adr, category_name)

        ## Improvement: would split checks more by concern
        if not output["accepted"]:
            return

        if contract_runner.from_address not in self.balances:
            self.balances[contract_runner.from_address] = 0

        self.balances[contract_runner.from_address] += output["reward"]
        self.arch_categories[category_name]["ADRs"].append(adr)

    def _check_template(self, adr: str) -> bool:
        adr = adr.replace("\r\n", "\n").replace("\r", "\n")
        pattern = r"^[^#][^n]+?\n+(- Status: (proposed|accepted|validated).+)\n+(- Deciders: [^\n]+\n+)(- Date:"
```


- Read Methods:

Various getter methods to retrieve contract state, categories, and balances.

Deploying the Contract

To deploy the ADRValidator contract:

1. Deploy the Contract: No initial parameters are needed. The deployer becomes the initial owner.

Checking the Contract State

After deploying the contract, you can check its state in the Read Methods section:

- Use `get_owner()` to see the current contract owner.
- Use `get_categories()` to view defined architectural categories.
- Use `get_balances()` to check reward balances of contributors.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can:

- Call `add_category(name, description)` to define new architectural categories (owner only).
- Call `validate_adr(ad, category_name)` to submit an ADR for validation.
- Call `change_owner(new_owner)` or `set_max_reward(new_max_reward)` for contract management (owner only).

Analyzing the Contract's Behavior

The contract's behavior involves several complex processes:

1. Template Validation: Uses regex to ensure ADRs follow a specific structure.
2. Consistency Checking: Employs an LLM to compare new ADRs against existing ones for contradictions.
3. Automated Decision-Making: Determines acceptance and reward based on LLM evaluation.
4. Reward System: Automatically credits submitters with tokens for accepted ADRs.

Handling Different Scenarios

As illustrated in the provided flowchart:

[Image: ADRValidator Flowchart]

- Invalid ADR Structure: The ADR is rejected without further evaluation.
- Valid ADR, Contradictions Found: The ADR is rejected with reasoning provided.
- Valid ADR, No Contradictions: The ADR is accepted, and the submitter is rewarded.

The outputs shows various scenarios:

[Image: ADRValidator Outputs]

This diagram illustrates different outcomes based on the ADR's content and its relation to existing ADRs in the category.

Use Cases and Market Potential

The ADRValidator contract has broad applications:

- Software Development Teams: For managing and validating architectural decisions.
- Resource Allocation: In contexts where decisions about shared resources (e.g., budget, design principles) are critical.
- Hackathons and Competitions: As a tool for evaluating and rewarding contributions.

Its automated, trustless nature makes it valuable for any organization seeking to streamline decision-making processes and ensure consistency in architectural choices.

Future Enhancements

Future developments may include:

1. Spam Prevention: Implementing measures to prevent submission of fake or spammy ADRs.
2. Platform Integration: Direct integration with platforms like GitHub for seamless ADR management.
3. Global ADR Repository: Evolving into a comprehensive, accessible repository for ADRs worldwide.

This ADRValidator contract demonstrates an innovative use of GenLayer for document validation and decision-making in software architecture. It showcases how Intelligent Contracts can automate complex processes, ensure consistency, and provide incentives for quality contributions in a decentralized manner.

Discover the Protocol (/understand-genlayer-protocol "Discover the Protocol")

Flight Insurance Contract

Source: https://docs.genlayer.com/developers/intelligent-contracts/examples/_advanced/_flight-insurance

The Flight Insurance contract sets up a scenario for managing flight delay insurance. It allows passengers to buy insurance for specific flights and automatically processes claims based on flight status information from an independent source. This contract demonstrates complex real-world interactions, external data integration, and automated claim processing within a blockchain environment.

□

This intelligent contract is currently not migrated to the real GenVM syntax.

```

import json
from backend.node.genvm.icontract import IContract
from backend.node.genvm.equivalence_principle import EquivalencePrinciple

# Flight Insurance Intelligent Contract is a contract that allows passengers to buy insurance for their flights.
# The contract is created by the insurance manager for specific flight
# and the passengers can buy insurance for this specific flight.
# The contract has a method to check the flight status from an independent source of information
# and a method to claim the insurance in case the flight is delayed.
# The payment is done by the insurance manager to the passengers in case the flight is delayed.
# Passenger does not need to trust on the insurance manager to get the payment.
class FlightInsurance(IContract):

    def __init__(self, _flight_number: str, _flight_date: str, _flight_time: str, _flight_from: str, _flight_to: str,
        _num_passengers_paid_insurance: int, _insurance_value_per_passenger: int):
        self.flight_number = _flight_number
        self.num_passengers_paid_insurance = _num_passengers_paid_insurance
        self.flight_arrival_delayed = False
        self.flight_date = _flight_date
        self.flight_time = _flight_time
        self.flight_from = _flight_from
        self.flight_to = _flight_to
        self.resolution_url = "https://flightaware.com/live/flight/" + self.flight_number + "/history/"
        self.resolution_url = self.resolution_url + self.flight_date + "/" + self.flight_time + "/"
        self.resolution_url = self.resolution_url + self.flight_from + "/" + self.flight_to
        self.loss_payment_value_per_passenger = _insurance_value_per_passenger
        self.balances = {}
        self.balances[contract_runner.from_address] = _num_passengers_paid_insurance*_insurance_value_per_passenger
        self.insurance_manager = contract_runner.from_address

# Example of constructor with hardcoded values to save time in the testing
# def __init__(self):
#     self.flight_number = "TAP457"
#     self.num passengers paid insurance = 35
#     self.flight_arrival_delayed = False
#     self.flight_date = "20240818"
#     self.flight_time = "0510Z"
#     self.flight_from = "LFP0"
#     self.flight_to = "LPPR"
#     self.resolution_url = "https://flightaware.com/live/flight/" + self.flight_number + "/history/"
#     self.resolution_url = self.resolution_url + self.flight_date + "/" + self.flight_time + "/"
#     self.resolution_url = self.resolution_url + self.flight_from + "/" + self.flight_to
#     self.loss_payment_value_per_passenger = 30
#     self.balances = {}
#     self.balances[contract_runner.from_address] =
self.loss_payment_value_per_passenger*self.loss_payment_value_per_passenger
#     self.insurance_manager = contract_runner.from_address

    async def check_flight_status(self) -> None:
        """Check the flight status from an external source."""
        print(f"Checking flight status: {self.resolution_url}")
        final_result = {}
        async with EquivalencePrinciple(
            result=final_result,
            principle="The arrival status should be consistent",
            comparative=True,
        ) as eq:
            web_data = await eq.get_webpage(self.resolution_url)
            print(" ")
            print("web_data: ")
            print(web_data)
            print(" ")
            print(" ")
            prompt = f"""

```

In the following web page, find if the flight arrival was late or not:

Web page content:

```
{web_data}
```

End of web page data.

Respond using ONLY the following format:

```

{{
  "arrivalstatus": bool // True if the flight arrival was delayed or False if it was on time
}}
```

It is mandatory that you respond only using the JSON format above, nothing else. Don't include any other words or characters, your output must be only JSON without any formatting prefix or suffix. This result should be perfectly parseable by a JSON parser without errors.

```

"""
    result = await eq.call_llm(prompt)
    print("*****")
    print("result: ")
    print(result)
    print("=====")
    result_clean = result.replace("True", "true").replace("False", "false")
    print("*****")
    print("result_clean: ")
    print(result_clean)
    print("=====")
    eq.set(result_clean)
    output = json.loads(result_clean)
    print("*****")
    print("output: ")
    print(output["arrivalstatus"])
    print("=====")
    if output["arrivalstatus"] is True:
        self.flight_arrival_delayed = True
        print("*****")
        print("flight_arrival_delayed: ")
        print(self.flight_arrival_delayed)
        print("=====")

def add_passenger(self, _passenger_address:str) ->None:
    self.balances[_passenger_address] = 0
def insurance_claim(self, _passenger_address:str) ->None:
    if self.flight_arrival_delayed is True:
        if _passenger_address in self.balances:
            if self.balances[_passenger_address] == 0:
                self.balances[_passenger_address] = self.loss_payment_value_per_passenger
                self.balances[self.insurance_manager] =
(self.balances[self.insurance_manager]-self.loss_payment_value_per_passenger)

def get_flight_status(self) ->bool:
    return self.flight_arrival_delayed

def get_insurance_balance(self, _passenger_address:str) ->int:
    if _passenger_address in self.balances:
        return self.balances[_passenger_address]
    else:
        return 0

def get_insurance_manager_balance(self) ->int:
    return self.balances[self.insurance_manager]

def get_insurance_manager(self) ->str:
    return self.insurance_manager

def get_flight_number(self) ->str:
    return self.flight_number

def get_flight_date(self) ->str:
    return self.flight_date

def get_flight_time(self) ->str:
    return self.flight_time

def get_flight_from(self) ->str:
    return self.flight_from

def get_flight_to(self) ->str:
    return self.flight_to

def get_resolution_url(self) ->str:
    return self.resolution_url

def get_loss_payment_value_per_passenger(self) ->int:
    return self.loss_payment_value_per_passenger

def get_num_passengers_paid_insurance(self) ->int:
    return self.num_passengers_paid_insurance

```

Code Explanation

- Initialization: The FlightInsurance class initializes the contract with specific flight details, insurance parameters, and initial balances.
- Key Methods:
 - ask_for_flight_status(): Checks the flight status from an external source (FlightAware) using an LLM to interpret the data.
 - add_passenger(): Adds a new passenger to the insurance contract.
 - insurance_claim(): Processes insurance claims for passengers if the flight is delayed.
- Read Methods: Various getter methods to retrieve flight details, insurance parameters, and balances.

Deploying the Contract

To deploy the Flight Insurance contract, you need to provide the flight details and insurance parameters:

1. Set Flight Details: Provide flight number, date, time, origin, and destination.
2. Set Insurance Parameters: Specify the number of insured passengers and the insurance value per passenger.
3. Deploy the Contract: Once all parameters are set, deploy the contract to start the insurance coverage.

Checking the Contract State

After deploying the contract, its address is displayed and you can check its state in the Read Methods section.

- Use get_flight_status() to see if the flight is delayed.
- Use get_insurance_balance(address) to check a passenger's insurance balance.
- Use other getter methods to retrieve various contract details.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can:

- Call ask_for_flight_status() to update the flight status from the external source.
- Call add_passenger(address) to add a new insured passenger.
- Call insurance_claim(address) for a passenger to claim their insurance if the flight is delayed.

Analyzing the Contract's Behavior

The contract's behavior involves several complex interactions:

- External Data Integration: The contract uses an LLM to interpret flight status data from FlightAware, an external source.
- Automated Claim Processing: If a flight is determined to be delayed, the contract automatically processes claims for insured passengers.
- Balance Management: The contract manages balances for both the insurance manager and insured passengers, automatically transferring funds when claims are processed.

Handling Different Scenarios

- Flight On Time: If the flight is on time, no claims are processed, and balances remain unchanged.
- Flight Delayed: If the flight is delayed, insured passengers can claim their insurance, and funds are automatically transferred from the insurance manager's balance to the passengers'.
- Multiple Claims: The contract ensures that each passenger can only claim once, preventing double-dipping.
- New Passengers: The contract allows adding new passengers, expanding the pool of insured individuals.

This Flight Insurance contract demonstrates a practical use case for blockchain technology in the insurance industry. It showcases how smart contracts can automate complex processes, integrate external data sources, and provide transparent, trustless insurance services. The use of an LLM to interpret external data adds an innovative layer of intelligence to the contract's decision-making process.

Discover the Protocol (/understand-genlayer-protocol "Discover the Protocol")

GenLayerDAO Contract

Source: https://docs.genlayer.com/developers/intelligent-contracts/examples/_advanced/_genlayer-dao

The GenLayerDAO contract implements a robust Decentralized Autonomous Organization (DAO) with built-in governance mechanisms and a bounty system. This contract demonstrates complex decision-making processes, token-based voting, and AI-assisted proposal evaluation within a blockchain environment.

□

This intelligent contract is currently not migrated to the real GenVM syntaxis.

```
import json
from backend.node.genvm.icontract import IContract
from backend.node.genvm.equivalence_principle import call_llm_with_principle

# Custom exception classes for GenLayerDAO
class DAOException(Exception):
    """Base exception class for GenLayerDAO"""
    pass

class InsufficientTokensException(DAOException):
    """Exception raised when a user doesn't have enough tokens"""
    pass

class InvalidBountyException(DAOException):
    """Exception raised for invalid bounty operations"""
    pass

class VotingException(DAOException):
    """Exception raised for voting-related issues"""
    pass

class InvalidInputException(DAOException):
    """Exception raised for invalid input parameters"""
    pass

class Bounty:
    """
    Represents a bounty in the GenLayerDAO system.

    Attributes:
```

```

    id (int): Unique identifier for the bounty
    description (str): Detailed description of the bounty
    reward_description (str): Description of the reward for completing the bounty
    proposer (str): Address of the user who proposed the bounty
    votes_for (int): Number of votes in favor of the bounty
    votes_against (int): Number of votes against the bounty
    status (str): Current status of the bounty ("proposed", "active", or "completed")
    submissions (list): List of submissions for the bounty
    vote_snapshot (dict): Snapshot of token balances at proposal time
    has_voted (dict): Tracks which users have voted on the bounty
"""
def __init__(self, id: int, description: str, reward_description: str, proposer: str):
    self.id = id
    self.description = description
    self.reward_description = reward_description
    self.proposer = proposer
    self.votes_for = 0
    self.votes_against = 0
    self.status = "proposed" # Can be "proposed", "active", or "completed"
    self.submissions: list[dict[str, str | bool]] = []
    self.vote_snapshot: dict[str, int] = {} # Snapshot of token balances at proposal time
    self.has_voted: dict[str, bool] = {} # Track who has voted

class GenLayerDAO(Contract):
    """
    Implements the GenLayerDAO contract with bounty management and voting system.

    Attributes:
        total_supply (int): Total token supply of the DAO
        token_supply (int): Current available token supply
        balances (dict): Mapping of addresses to token balances
        bounties (dict): Mapping of bounty IDs to Bounty objects
        next_bounty_id (int): ID to be assigned to the next proposed bounty
        constitution (list): List of rules governing the DAO
    """
    def __init__(self):
        self.total_supply = 1000
        self.token_supply = self.total_supply
        self.balances: dict[str, int] = {}
        self.bounties: dict[int, Bounty] = {}
        self.next_bounty_id = 1

        # Define the constitution of the DAO
        self.constitution = [
            "This Constitution describes the decision-making framework for GenLayerDAO governance.",
            "The following process governs the rules and procedures by which GenLayerDAO may propose, vote on, and",
            "implement Bounty Programs.",
            "GenLayerDAO's purpose is to grow the GenLayer Blockchain by rewarding Bounty Program contributors with",
            "tokens.",
            "An address must hold at least one token to be a member of the DAO.",
            "Only members of the GenLayerDAO can propose new Bounty Programs.",
            "Bounty programs must follow be in line with one of the following goals:",
            "- Increase Brand Awareness for the GenLayer Blockchain or one of the applications built on top of",
            "GenLayer.",
            "- Lead to Code Contributions to the GenLayer Repository on GitHub.",
            "- Lead to applications built on top of GenLayer.",
            "- Empower initiatives that help to build or enhance the community around GenLayer, such as meetups,",
            "hackathons, or online forums.",
            "If a Bounty program proposal does not help any of the listed goals, reject the proposal.",
            "Bounty programs must be unique. If a new bounty program proposal is the same as an existing proposed or",
            "active bounty, reject the proposal.",
            "If a bounty program proposal meets the criteria, it is added to proposed bounties for a vote.",
            "A bounty program proposal has to be approved by a majority vote.",
            "The voting power of a member is proportional to the number of tokens a member holds.",
            "At least one-third of all votable tokens must participate in the voting process; otherwise, the proposal",
            "will be rejected",
            "A majority of those participating votable tokens must agree for the proposal to be accepted",
            "Once a bounty program is approved, it is moved to the active bounty list",
            "Anyone can try to claim the reward of an active bounty, according to the rules of the bounty.",
            "A user does not have to be a member of the DAO or hold any DAO tokens to claim the bounty",
        ]

    1
    async def propose_bounty(self, bounty_proposal: str) -> str:
        """
        Proposes a new bounty to the DAO.

        Args:
            bounty_proposal (str): Description of the proposed bounty

```



```

Returns:
    str: Message indicating the result of the proposal

Raises:
    InsufficientTokensException: If the proposer doesn't have any tokens
    DAOException: If the proposal is rejected
"""
proposer = contract_runner.from_address
if self.get_balance_of(proposer) == 0:
    raise InsufficientTokensException("Only DAO members can propose bounties.")

# Prepare prompt for LLM evaluation
prompt = f"""
You are GenLayerDAO.

GenLayerDAO has a constitution:
{json.dumps(self.constitution)}

A user with address "{proposer}" has proposed a new bounty:
Description: {bounty_proposal}

Evaluate if this bounty proposal adheres to the constitution.
Respond with the following JSON format:
{{
    "reasoning": str,
    "proposal_accepted": bool
}}

It is mandatory that you respond only using the JSON format above,
nothing else. Don't include any other words or characters,
your output must be only JSON without any formatting prefix or suffix.
This result should be perfectly parseable by a JSON parser without errors.
"""

result = await call_llm_with_principle(prompt, eq_principle="proposal_accepted has to match exactly.")
output = self._get_decode_json_resilient(result)

if output["proposal_accepted"]:
    # Refine the bounty proposal
    refined_description, refined_reward = await self.refine_bounty_details(bounty_proposal)

    bounty = Bounty(self.next_bounty_id, refined_description, refined_reward, proposer)
    bounty.vote_snapshot = self.balances.copy()
    self.bounties[self.next_bounty_id] = bounty

    # Automatically cast a vote for the proposer
    self._cast_vote(bounty, proposer, True)

    self.next_bounty_id += 1
    return f"Bounty proposed successfully and automatically voted for. Bounty ID: {bounty.id}"
else:
    raise DAOException(f"Bounty proposal rejected. Reason: {output['reasoning']}")

def vote_on_bounty(self, bounty_id: int, vote: bool) -> str:
    """
    Casts a vote on a proposed bounty.

    Args:
        bounty_id (int): ID of the bounty to vote on
        vote (bool): True for a positive vote. False for a negative vote

    Returns:
        str: Message indicating the result of the vote

    Raises:
        InvalidBountyException: If the bounty ID is invalid
        VotingException: If voting conditions are not met
    """
    voter = contract_runner.from_address
    if bounty_id not in self.bounties:
        raise InvalidBountyException("Invalid bounty ID.")

    bounty = self.bounties[bounty_id]
    if bounty.status != "proposed":
        raise VotingException("This bounty is not in the voting phase.")

    if voter not in bounty.vote_snapshot:
        raise VotingException("You didn't hold any tokens when this bounty was proposed, so you can't vote on

```

```

it.")

    if bounty.has_voted.get(voter, False):
        raise VotingException("You have already voted on this bounty.")

    self._cast_vote(bounty, voter, vote)

    return self._check_voting_result(bounty)

def _cast_vote(self, bounty: Bounty, voter: str, vote: bool):
    """
    Internal method to cast a vote on a bounty.

    Args:
        bounty (Bounty): The bounty being voted on
        voter (str): Address of the voter
        vote (bool): True for a positive vote. False for a negative vote
    """
    voter_balance = bounty.vote_snapshot[voter]

    if vote:
        bounty.votes_for += voter_balance
    else:
        bounty.votes_against += voter_balance

    bounty.has_voted[voter] = True

def _check_voting_result(self, bounty: Bounty) -> str:
    """
    Checks the voting result for a bounty and updates its status if necessary.

    Args:
        bounty (Bounty): The bounty to check

    Returns:
        str: Message indicating the result of the check
    """
    total_votes = bounty.votes_for + bounty.votes_against
    if total_votes >= self.total_supply // 3:
        if bounty.votes_for > bounty.votes_against:
            bounty.status = "active"
            return f"Bounty {bounty.id} has been approved and is now active."
        else:
            del self.bounties[bounty.id]
            return f"Bounty {bounty.id} has been rejected and removed."

    return f"Vote recorded for bounty {bounty.id}."

async def refine_bounty_details(self, bounty_proposal: str) -> tuple[str, str]:
    """
    Refines the bounty proposal using an LLM.

    Args:
        bounty_proposal (str): Original bounty proposal

    Returns:
        tuple: Refined description and reward description
    """
    prompt = f"""
You are an LLM helping to refine and standardize bounty proposals for GenLayerDAO.
The original bounty proposal is: "{bounty_proposal}"

Please improve and structure this proposal by:
1. Separating the description and reward information.
2. Ensuring the description is clear, concise, and specifies concrete deliverables or success criteria.
3. Aligning it with GenLayerDAO's goals (increasing brand awareness, code contributions, application development, or community building).
4. Adding any relevant technical details or requirements.
5. Clarifying and standardizing the reward description, ensuring it's fair and motivating.

Provide the refined bounty details in the following JSON format:
{{
    "refined_description": str,
    "refined_reward": str
}}

The refined description should be a single, well-formatted paragraph.
The refined reward should be clear and specific, potentially including conditions or tiers if appropriate.

```

It is mandatory that you respond only using the JSON format above, nothing else. Don't include any other words or characters, your output must be only JSON without any formatting prefix or suffix. This result should be perfectly parseable by a JSON parser without errors.

```

"""
    result = await call_llm_with_principle(prompt, eq_principle="The refined details should capture the essence of
the original proposal.", comparative=False)
    print(result)
    output = self._get_decode_json_resilient(result)
    return output["refined_description"], output["refined_reward"]

```

```

async def compute_reward(self, bounty_id: int, submission: str) -> int:

```

```

    """
    Computes the reward for a bounty submission using an LLM.

```

Args:

```

    bounty_id (int): ID of the bounty
    submission (str): Submitted work for the bounty

```

Returns:

```

    int: Computed reward amount

```

Raises:

```

    InvalidBountyException: If the bounty ID is invalid

```

```

    """
    if bounty_id not in self.bounties:
        raise InvalidBountyException("Invalid bounty ID")

```

```

    bounty = self.bounties[bounty_id]
    prompt = f"""

```

You are the GenLayerDAO reward calculator. Your task is to determine the appropriate reward for a bounty submission based on the bounty description, reward description, and the actual submission.

```

Bounty Description: {bounty.description}
Reward Description: {bounty.reward_description}
Submission: {submission}

```

Please evaluate the submission against the bounty requirements and determine the appropriate reward. Consider factors such as completeness, quality, and impact of the submission.

Respond with the following JSON format:

```

{{
  "reasoning": str,
  "reward_amount": int
}}
```

The reward_amount should be a whole number of tokens, not exceeding the total supply of {self.total_supply}.

It is mandatory that you respond only using the JSON format above, nothing else. Don't include any other words or characters, your output must be only JSON without any formatting prefix or suffix. This result should be perfectly parseable by a JSON parser without errors.

```

    result = await call_llm_with_principle(prompt, eq_principle="The proposed reward should match exactly.")
    output = self._get_decode_json_resilient(result)
    return min(output["reward_amount"], self.total_supply) # Ensure reward doesn't exceed total supply

```

```

async def claim_bounty(self, bounty_id: int, submission: str) -> str:

```

```

    """
    Processes a claim for a bounty reward.

```

Args:

```

    bounty_id (int): ID of the bounty being claimed
    submission (str): Submitted work for the bounty

```

Returns:

```

    str: Message indicating the result of the claim

```

Raises:

```

    InvalidBountyException: If the bounty ID is invalid or the bounty is not active
    InsufficientTokensException: If there are not enough tokens to pay the reward
    DAOException: If the submission is rejected

```

```

    """
    submitter = contract_runner.from_address
    if bounty_id not in self.bounties:
        raise InvalidBountyException("Invalid bounty ID.")

```

```

        bounty = self.bounties[bounty_id]
        if bounty.status != "active":
            raise InvalidBountyException("This bounty is not active.")

        if bounty.status == "completed":
            raise InvalidBountyException("This bounty has already been completed.")

        prompt = f"""
You are GenLayerDAO.

A submission has been made for bounty {bounty_id}:
Description: {bounty.description}
Reward: {bounty.reward_description}
Submission: {submission}

Evaluate if this submission satisfactorily completes the bounty.
Respond with the following JSON format:
{{
    "reasoning": str,
    "submission_accepted": bool
}}

It is mandatory that you respond only using the JSON format above,
nothing else. Don't include any other words or characters,
your output must be only JSON without any formatting prefix or suffix.
This result should be perfectly parseable by a JSON parser without errors.
"""

        result = await call_llm_with_principle(prompt, ea_principle="submission accepted has to match exactly.")
        output = self._get_decode_json_resilient(result)

        if output["submission_accepted"]:
            reward_amount = await self.compute_reward(bounty_id, submission)
            if self.token_supply < reward_amount:
                raise InsufficientTokensException(f"Submission accepted, but insufficient tokens to pay reward.
Current supply: {self.token_supply}")

            self._send_tokens(reward_amount, submitter)
            bounty.status = "completed"
            bounty.submissions.append({"submitter": submitter, "submission": submission, "accepted": True, "reward":
reward_amount})
            return f"Bounty {bounty_id} completed. Reward of {reward_amount} tokens sent to {submitter}."
        else:
            bounty.submissions.append({"submitter": submitter, "submission": submission, "accepted": False})
            raise DAOException(f"Submission for bounty {bounty_id} rejected. Reason: {output['reasoning']}")

def buy_tokens(self, amount: int):
    """
    Purchase tokens for the caller.

    Args:
        amount (int): The number of tokens to buy.

    Raises:
        InvalidInputException: If the amount is not a positive integer.
    """
    if amount <= 0:
        raise InvalidInputException("Amount must be a positive integer.")
    self._send_tokens(amount, contract_runner.from_address)

def _send_tokens(self, amount: int, holder: str):
    """
    Internal method to send tokens to a holder.

    Args:
        amount (int): The number of tokens to send.
        holder (str): The address of the token recipient.

    Raises:
        InvalidInputException: If the amount is not a positive integer.
        InsufficientTokensException: If there are not enough tokens in the supply.
    """
    if amount <= 0:
        raise InvalidInputException("Amount must be a positive integer.")
    if self.token_supply < amount:

```

```

        raise InsufficientTokensException("Insufficient token supply")
    self.balances[holder] = self.balances.get(holder, 0) + amount
    self.token_supply -= amount

def get_balances(self) -> dict[str, int]:
    """
    Get the token balances of all holders.

    Returns:
        dict[str, int]: A dictionary mapping addresses to token balances.
    """
    return self.balances

def get_balance_of(self, address: str) -> int:
    """
    Get the token balance of a specific address.

    Args:
        address (str): The address to check the balance for.

    Returns:
        int: The token balance of the address.
    """
    return self.balances.get(address, 0)

def get_bounties(self) -> dict[int, dict]:
    """
    Get all bounties in the system.

    Returns:
        dict[int, dict]: A dictionary mapping bounty IDs to bounty details.
    """
    return {id: bounty.__dict__ for id, bounty in self.bounties.items()}

def get_bounty(self, bounty_id: int) -> dict:
    """
    Get details of a specific bounty.

    Args:
        bounty_id (int): The ID of the bounty to retrieve.

    Returns:
        dict: The details of the specified bounty.

    Raises:
        InvalidBountyException: If the bounty ID is invalid.
    """
    if bounty_id in self.bounties:
        return self.bounties[bounty_id].__dict__
    raise InvalidBountyException("Invalid bounty ID")

def _get_decode_json_resilient(self, s: str) -> dict:
    """
    Decode a JSON string in a resilient manner.

    Args:
        s (str): The JSON string to decode.

    Returns:
        dict: The decoded JSON object.
    """
    # Clean the string and replace boolean literals
    clean = self._get_extract_json_from_string(s).replace("True", "true").replace("False", "false")
    return json.loads(clean)

def _get_extract_json_from_string(self, s: str) -> str:
    """
    Extract a JSON object from a string.

    Args:
        s (str): The string potentially containing a JSON object.

    Returns:
        str: The extracted JSON string, or an empty string if no valid JSON is found.
    """
    start_index = s.find('{')
    end_index = s.rfind('}')
    if start_index != -1 and end_index != -1 and start_index < end_index:

```

```
        return s[start_index:end_index + 1]
    else:
        return ""
```

Code Explanation

Initialization

The GenLayerDAO class initializes with a total token supply, empty balances and bounties, and a predefined constitution.

Key Methods

- `propose_bounty()`: Allows members to propose new bounties, evaluated by an AI against the DAO's constitution.
- `vote_on_bounty()`: Enables token-weighted voting on proposed bounties.
- `claim_bounty()`: Processes bounty submissions and rewards, with AI-assisted evaluation.
- Token Management: Methods for buying tokens and checking balances.
- Bounty Management: Methods for retrieving bounty information and managing the bounty lifecycle.

Deploying the Contract

To deploy the GenLayerDAO contract:

1. Deploy the Contract: No initial parameters are needed. The contract initializes with a predefined token supply and constitution.

Checking the Contract State

After deploying the contract, you can check its state in the Read Methods section:

- Use `get_balances()` to view token distributions among members.
- Use `get_bounties()` to see all current bounties and their statuses.
- Use `get_bounty(bounty_id)` to view details of a specific bounty.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can:

- Call `propose_bounty(bounty_proposal)` to suggest a new bounty.
- Call `vote_on_bounty(bounty_id, vote)` to cast a vote on a proposed bounty.
- Call `claim_bounty(bounty_id, submission)` to submit work for an active bounty.
- Call `buy_tokens(amount)` to acquire governance tokens.

Analyzing the Contract's Behavior

The contract's behavior involves several complex processes:

1. AI-Assisted Proposal Evaluation: Uses an LLM to evaluate bounty proposals against the DAO's constitution.
2. Token-Weighted Voting: Implements a voting system where voting power is proportional to token holdings.
3. Dynamic Reward Calculation: Employs an AI to determine appropriate rewards for bounty submissions.
4. Automated Governance: Manages the entire lifecycle of bounties from proposal to completion.

Handling Different Scenarios

- Bounty Proposal: A member proposes a bounty, which is evaluated by AI. If accepted, it moves to the voting phase.
- Voting Process: Members vote on proposed bounties, with decisions based on token-weighted majority and quorum requirements.
- Bounty Activation: Approved bounties become active and open for submissions.
- Bounty Claim: Submissions are evaluated by AI, and if accepted, rewards are automatically distributed.

Use Cases and Benefits

The GenLayerDAO contract is ideal for:

- Decentralized Decision Making: Communities seeking fair and transparent governance.
- Ecosystem Growth: Blockchain projects incentivizing development and community engagement.
- Open Source Development: Managing and rewarding contributions to open-source projects.

Benefits include:

- Flexibility: AI-powered system can interpret and execute a wide range of proposals.
- Scalability: Can handle growing membership and increasing complexity of decisions.
- Transparency: All decisions and their reasoning are recorded on the blockchain.

Future Enhancements

Potential future developments could include:

1. Multi-tiered Governance: Implementing different levels of decision-making authority.
2. Integration with External Systems: Connecting with GitHub or other platforms for seamless contribution tracking.
3. Advanced Analytics: Implementing tools to analyze voting patterns and DAO performance over time.

This GenLayerDAO contract demonstrates a sophisticated use of blockchain technology, combining token economics, AI-assisted decision-making, and community governance. It showcases how smart contracts can create complex, self-governing systems that incentivize participation and manage resources in a decentralized manner.

GitBounties Contract

Source: https://docs.genlayer.com/developers/intelligent-contracts/examples/_advanced/_git-bounties

The GitBounties contract sets up a scenario to manage bounties for GitHub issues, allowing developers to claim points for resolving issues. This contract demonstrates more complex interactions involving external data (GitHub) and multi-step processes within a blockchain environment.

□

This intelligent contract is currently not migrated to the real GenVM syntaxis.

```
import json
import math
from backend.node.genvm.icontract import IContract
from backend.node.genvm.equivalence_principle import (
    EquivalencePrinciple,
    get_webpage_with_principle,
)

class BountyData:
    issue: int
    points: int
    claimed: bool

    def __init__(self, issue, points):
        self.issue = issue
        self.points = points
        self.claimed = False

class GitBounties(IContract):
    def __init__(self, github_repository: str):
        self.owner = contract_runner.from_address
        self.repository = "https://github.com/" + github_repository
        self.developers = {} # Mapping from: GitHub username -> Address
        self.points = {} # Mapping from: GitHub username -> points earned
        self.bounties = {} # Mapping from: Issue number -> BountyData
        pass

    def get_developers(self) -> dict:
        return self.developers

    def get_points(self) -> dict:
        return self.points

    def get_bounties(self) -> dict:
        bounties = {}
        for k, v in self.bounties.items():
            bounties[k] = {"issue": v.issue, "points": v.points, "claimed": v.claimed}
        return bounties

    def add_bounty(self, issue: int, points: int):
        if self.owner != contract_runner.from_address:
            raise Exception("only owner")

        bounty = self.bounties.setdefault(issue, BountyData(issue, points))
        if bounty.claimed:
            raise Exception("can't add bounty to claimed issue")

    async def register(self, github_username: str) -> None:
        dev_github_profile = f"https://github.com/{github_username}"
```



```

developer_address = contract_runner.from_address

web_data = await get_webpage_with_principle(
    dev_github_profile, "The result should be exactly the same"
)
if developer_address in web_data["output"]:
    self.developers[github_username] = developer_address
else:
    raise Exception(
        "Couldn't verify the developer, GitHub profile page must have the given address on its bio"
    )

async def claim(self, pull: int) -> None:
    final_result = {}
    async with EquivalencePrinciple(
        result=final_result,
        principle="The result should be exactly the same".
        comparative=True,
    ) as eq:
        web_data = await eq.get_webpage(self.repository + "/pull/" + str(pull))
        task = f"""
        The following web page content corresponds to a GitHub pull request.

        Web page content:
        {web_data}
        End of web page data.

        In that Pull Request, a developer should be fixing an issue from the repository
        issues list: located at: https://github.com/cristiam86/genlayer-hackaton/issues

        To fin the issue, you should look for a text like "Fixes: #<issue_number>" in the
        Pull Request first comment.
        It is very important to also include information about how many times a given PR has
        ben rejected (changes requested), son include the number of those as well.

        Respond with the following JSON format:
        {{
            "merged": boolean, // if pull request was merged
            "username": string, // GitHub username of the developer who opened a pull request
            "issue": int, // number of the closed issue
            "changes_requested": int // number of changes requested for the given pull request
        }}

        It is mandatory that you respond only using the JSON format above, nothing else.
        Don't include any other words or characters, your output must be only JSON without any
        formatting prefix or suffix. This result should be perfectly parseable by a
        JSON parser without errors.
        """
        result = await eq.call_llm(task)
        print("\nresult: ", result)
        eq.set(result)
    res = json.loads(final_result["output"])
    if not res["merged"]:
        raise Exception("pull is not mergerd")

    bounty_issue = res["issue"]
    bounty_username = res["username"]
    pull_request_changes_requested = res["changes_requested"]
    bounty = self.bounties.get(bounty_issue, None)

    if bounty and not bounty.claimed:
        bounty.claimed = True
        if not bounty_username in self.points:
            self.points[bounty_username] = 0
        total_points = math.floor(
            bounty.points / (pull_request_changes_requested + 1)
        )
        print("total_points", total_points)
        self.points[bounty_username] += 1 if total_points <= 1 else total_points

```

Code Explanation

- Initialization: The `GitBounties` class initializes the contract with a GitHub repository, setting up data structures for developers, points, and bounties.
- Read Methods:
 - `get_developers()`: Returns the mapping of GitHub usernames to blockchain addresses.
 - `get_points()`: Returns the mapping of GitHub usernames to earned points.
 - `get_bounties()`: Returns the current bounties and their status.
- Write Methods:
 - `add_bounty(issue, points)`: Allows the owner to add a new bounty for an issue.
 - `register(github_username)`: Allows developers to register by linking their GitHub profile to their blockchain address.
 - `claim(pull)`: Processes a claim for a resolved issue, awarding points based on the bounty and the number of change requests asked by reviewers.

Deploying the Contract

To deploy the `GitBounties` contract, you need to provide the GitHub repository:

1. Set GitHub Repository: Provide the GitHub repository in the format "username/repository".
2. Deploy the Contract: Once the repository is set, deploy the contract to make it ready for interaction.

Checking the Contract State

After deploying the contract, its address is displayed and you can check its state in the Read Methods section.

- Use `get_developers()` to see registered developers.
- Use `get_points()` to see points earned by developers.
- Use `get_bounties()` to see all current bounties and their status.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can:

- Call `add_bounty(issue, points)` to create a new bounty (owner only).
- Call `register(github_username)` for developers to register.
- Call `claim(pull)` to process a claim for a resolved issue.

Analyzing the Contract's Behavior

The contract's behavior involves several complex interactions:

- Adding Bounties: Only the contract owner can add bounties for specific issues.
- Developer Registration: Developers must register by providing their GitHub username. The contract verifies that their address has been added to their GitHub profile.
- Claiming Bounties: When a developer claims a bounty:

- The contract fetches the pull request data from GitHub.
- It verifies if the pull request is merged and identifies the associated issue.
- It calculates points based on the bounty value and the number of change requests.
- Points are awarded to the developer who resolved the issue.

Handling Different Scenarios

- **Successful Claim:** When a valid claim is processed, the bounty is marked as claimed, and points are awarded to the developer.
- **Invalid Claim:** If the pull request is not merged or doesn't reference a valid bounty, the claim is rejected.
- **Point Calculation:** The number of change requests affects the points awarded, with more changes resulting in fewer points.
- **Minimum Points:** A minimum of 1 point is always awarded for a successful claim.

This GitBounties contract demonstrates a complex use case involving external data integration, multi-step verification processes, and dynamic point calculation. It showcases how blockchain contracts can interact with real-world systems to create incentive structures for open-source contributions.

Discover the Protocol (/understand-genlayer-protocol "Discover the Protocol")

Influencer Tweet Analyzer Contract

Source:

https://docs.genlayer.com/developers/intelligent-contracts/examples/_advanced/_influencer-tweet-analyzer

The Influencer Tweet Analyzer contract sets up a system for analyzing cryptocurrency-related tweets from influencers, evaluating their impact on market prices, and managing a leaderboard of influencers based on their predictive accuracy. This contract demonstrates complex social media analysis, market data integration, and automated investment simulation within a blockchain environment.

□

This intelligent contract is currently not migrated to the real GenVM syntaxis.

```
from typing import List
from backend.node.genvm.icontract import IContract
from backend.node.genvm.equivalence_principle import EquivalencePrinciple
import json

# TODO: handle duplication of tweet feeds
class InfluencerTweetAnalyzer(IContract):
    # AnalyzedTweet is defined here due to problems "compiling" the contract in the Studio
    from dataclasses import dataclass

    @dataclass
    class AnalyzedTweet:
        influencer: str
        tweet: str
        date: str
        cryptocurrency: str
```

```

        positive_sentiment: float

def __init__(self) -> None:
    from collections import defaultdict

    self.leaderboard: dict[str, float] = {} # influencer -> score
    self.price_history: dict[str, dict[str, float]] = defaultdict(
        dict
    ) # cryptocurrency -> date -> price
    self.tweets_pending_process: List[InfluencerTweetAnalyzer.AnalyzedTweet] = []
    self.balances: dict[str, int] = defaultdict(int) # address -> balance
    self.followers: dict[str, set[str]] = defaultdict(
        set
    ) # influencer -> followers addresses

    self.followers_investments: dict[str, int] = defaultdict(
        self.default_int dict
    ) # address -> cryptocurrency -> investment. Simulates interactions with other coins

def _default_int_dict(
    self,
): # This is a workaround for the fact that lambdas don't work in the Studio
    from collections import defaultdict

    return defaultdict(int)

async def feed(self, influencer: str, tweet: str, date: str):
    """
    Inputs:
    - influencer: the name of the influencer
    - tweet: the tweet content
    - date: the date of the tweet

    Process:
    1. Analyze the tweet to determine if sentiment is positive towards one or more cryptocurrencies
        - This step uses Consensus to analyze the tweet
        - We save these results in a list of tweets pending process, one per cryptocurrency
    """
    if influencer not in self.leaderboard: # influencer enters the leaderboard
        self.leaderboard[influencer] = 0

    tweet_result = {}
    async with EquivalencePrinciple(
        result=tweet_result,
        principle="The cryptocurrencies names are the exact same, and the positive_sentiment is similar",
        comparative=True,
    ) as eq:
        # TODO: actually retrieve from the Internet
        # web_data = await eq.get_webpage(tweet)
        # print(web_data)
        web_data = tweet

        task = f"""In this webpage you'll find a tweet from a crypto influencer.
        Analyze this tweet to determine if sentiment is positive towards one or more cryptocurrencies

        Return the output as a JSON array of cryptocurrencies and their positive sentiment.
        - 'positive_sentiment' should go from -1 to 1.
        - 'cryptocurrency' should be the entire name of the cryptocurrency, all in lowercase, only letters

        Here's an example format:
        [
            {{
                "cryptocurrency": "bitcoin",
                "positive_sentiment": 0.1
            }}
        ]

        Respond ONLY with the JSON output, nothing else. The output should be parsable by any JSON parser

        Web page content:
        {web_data}
        """

        result = await eq.call_llm(task)
        print(result)
        eq.set(result)

    tweet_result = json.loads(tweet_result["output"])

```

```

        for item in tweet_result:
            analyzed_tweet = InfluencerTweetAnalyzer.AnalyzedTweet(
                influencer=influencer,
                tweet=tweet,
                date=date,
                cryptocurrency=item["cryptocurrency"],
                positive_sentiment=item["positive sentiment"].
            )

            self.tweets_pending_process.append(analyzed_tweet)

            if date == self._today():
                self._update_follower_investments(analyzed_tweet)

def _today(self) -> str:
    from datetime import date

    return date.today().strftime("%Y-%m-%d")

def _update_follower_investments(self, analyzed_tweet: AnalyzedTweet):
    # TODO: can we make this contract interact with other Ethereum contracts like ERC20?
    # TODO: sentiment threshold and investment amount are arbitrary
    if analyzed_tweet.positive_sentiment > 0.5: # Buy
        for follower in self.followers[analyzed_tweet.influencer]:
            if self.balances[follower] > 0:
                self.balances[follower] -= 1
                self.followers_investments[follower][
                    analyzed_tweet.cryptocurrency
                ] += 1

    elif analyzed_tweet.positive_sentiment < -0.5: # Sell
        for follower in self.followers[analyzed_tweet.influencer]:
            if self.followers_investments[follower] > 0:
                self.balances[follower] += 1
                self.followers_investments[follower][
                    analyzed_tweet.cryptocurrency
                ] -= 1

async def process_score(self):
    """
    1. For each analyzed tweet. get the daily price change of the cryptocurrency
       - This step uses Consensus and connects to the Internet to get the price change
    2. Update the leaderboard with the score of the influencer, based on alignment between sentiment and price
    change
    """
    tweets_pending_process_from_today = []
    for tweet in self.tweets_pending_process:
        if tweet.date == self._today():
            # We skip today's tweets since there's no market data yet
            tweets_pending_process_from_today.append(tweet)
            continue

        price_change = await self.retrieve_market_data(
            tweet.cryptocurrency, tweet.date
        )

        # This is a simple score function for demonstration
        self.leaderboard[tweet.influencer] += (
            price_change * tweet.positive_sentiment
        )

    self.tweets_pending_process = tweets_pending_process_from_today

# TODO: use date
async def retrieve_market_data(self, cryptocurrency: str, date: str) -> float:
    if cryptocurrency in self.price_history:
        if date in self.price_history[cryptocurrency]:
            return self.price_history[cryptocurrency][date]

    market_result = {}

    async with EquivalencePrinciple(
        result=market_result,
        principle="The price ",
        comparative=True,
    ) as eq:
        url = "https://coinmarketcap.com/currencies/" + cryptocurrency

```

```

        web_data = await eq.get_webpage(url)
        print(web_data)

        task = f"""In this webpage from 'coinmarketcap' you'll find a lot of information about the market status
of a cryptocurrency.
        Analyze this information to determine today's daily price change of the cryptocurrency.

        Return the output as a JSON number bigger than -100, representing the daily price change.
        - Negative numbers mean that the price went down
        - Positive numbers mean that the price went up

        Respong ONLY with the JSON output, nothing else, not even the word "json". The output should be
parsable by any JSON parser
        Example output:
        {{
            "price_change": 2.4
        }}

        Web page content:
        {web_data}
        """

        result = await eq.call_llm(task)
        print(result)
        eq.set(result)

        market_data = json.loads(market_result["output"])["price_change"]
        self.price_history[cryptocurrency][date] = market_data
        return market_data

def deposit(self, amount: int):
    self.balances[contract_runner.from_address] += amount
def follow(self, influencer: str):
    self.followers[influencer].add(contract_runner.from_address)

def unfollow(self, influencer: str):
    self.followers[influencer].remove(contract_runner.from_address)

# Read methods
def get_leaderboard(self):
    return self.leaderboard

def get_price_history(self):
    return self.price_history

def get_tweets_pending_process(self):
    return [item.__dict__ for item in self.tweets_pending_process]

def get_followers_investments(self):
    return self.followers_investments

def get_followers(self):
    return {key: list(value) for key, value in self.followers.items()}

def get_all_state(self):
    # We convert them so they are json serializable
    return {
        "leaderboard": self.leaderboard,
        "price_history": self.price_history,
        "tweets_pending_process": self.get_tweets_pending_process(),
        "balances": self.balances,
        "followers": self.get_followers(),
        "followers_investments": self.followers_investments,
    }

@asvnc def test(self):
    influencer = "Chris Burniske"
    self.deposit(1000)

    self.follow(influencer)

    # New tweet, should create movement in followers investments
    await self.feed(
        influencer=influencer,
        tweet=""

    Each cycle I've tended to give a majority of focus to one major underdog. In 2014-17 that was $BTC, in
    2018-2021 that was $ETH, and in 2022 to now that's $SOL.
    """

```

```

        date=self._today(),
    )

    other_influencer = "Wizard Of SoHo"
    self.follow(other_influencer)

    # Old tweet, should be processed for leaderboard score
    await self.feed(
        influencer=other_influencer,
        tweet="""
        @Nate_Rivers @osf_rekt I don't think so. Doge had Elon like max shilling. No other memecoin has made it
        otherwise. Who is gonna drive pepe up again? No coin makes it without some leaders pushing it or large holders. Here
        the large holders are jeet scam devs with no rep outside of shitcoin world
        """,
        date="2024-08-16",
    )

    await self.process_score()

```

Code Explanation

- Initialization: The InfluencerTweetAnalyzer class initializes with empty data structures for the leaderboard, price history, pending tweets, user balances, and follower relationships.
- Key Methods:
 - feed(): Analyzes a tweet to determine sentiment towards cryptocurrencies using an LLM.
 - process_score(): Updates the leaderboard based on the alignment between tweet sentiment and actual price changes.
 - retrieve_market_data(): Fetches and processes cryptocurrency price data from an external source.
- User Interaction Methods:
 - deposit(): Allows users to deposit funds into their account.
 - follow() and unfollow(): Enables users to follow or unfollow influencers.
- Read Methods: Various getter methods to retrieve contract state, leaderboard, price history, and user data.

Deploying the Contract

To deploy the Influencer Tweet Analyzer contract:

1. Deploy the Contract: No initial parameters are needed.

Checking the Contract State

After deploying the contract, you can check its state using the Read Methods:

- Use get_leaderboard() to view the current influencer rankings.
- Use get_price_history() to see historical cryptocurrency prices.
- Use get_followers() to check follower relationships.
- Use get_all_state() to retrieve the entire contract state.

Executing Transactions

To interact with the deployed contract, use the Write Methods:

- Call `feed(influencer, tweet, date)` to input new tweets for analysis.
- Call `process_score()` to update the leaderboard based on recent market data.
- Call `deposit(amount)` to add funds to a user's account.
- Call `follow(influencer)` or `unfollow(influencer)` to manage influencer subscriptions.

Analyzing the Contract's Behavior

The contract's behavior involves several complex processes:

1. **Sentiment Analysis:** Uses an LLM to analyze tweets and determine sentiment towards specific cryptocurrencies.
2. **Market Data Integration:** Fetches and processes real-time cryptocurrency price data.
3. **Automated Scoring:** Updates the influencer leaderboard based on the accuracy of their implied predictions.
4. **Simulated Investments:** Automatically adjusts follower investments based on influencer tweets.

Handling Different Scenarios

- **New Tweet Analysis:** When a new tweet is fed, it's analyzed for sentiment and potentially triggers simulated investments for followers.
- **Historical Tweet Processing:** Older tweets are used to update the leaderboard based on actual market performance.
- **User Following/Unfollowing:** Users can dynamically change their subscriptions, affecting their simulated investment behavior.

Use Cases and Market Potential

The Influencer Tweet Analyzer contract has several potential applications:

- **Crypto Investment Guidance:** Provides users with insights based on influencer performance.
- **Influencer Accountability:** Creates a transparent system for tracking the accuracy of cryptocurrency predictions.
- **Automated Trading Strategies:** Could be extended to implement real trading based on influencer sentiment.
- **Market Sentiment Analysis:** Offers a tool for understanding the relationship between social media sentiment and market movements.

Future Enhancements

Potential future developments include:

1. **Real-time Tweet Fetching:** Implementing an external service to automatically fetch tweets from followed influencers.
2. **Advanced Sentiment Analysis:** Refining the LLM prompts for more nuanced sentiment evaluation.
3. **Integration with Trading Platforms:** Connecting with real cryptocurrency exchanges for actual trading based on the contract's analysis.

4. User Customization: Allowing users to set their own risk preferences and investment thresholds.
5. Expanded Market Data: Incorporating more comprehensive and real-time market data sources.

This Influencer Tweet Analyzer contract demonstrates an innovative use of GenLayer for social media analysis and cryptocurrency market prediction. It showcases how Intelligent Contracts can process complex, real-world data streams, provide valuable insights, and potentially guide investment decisions in a decentralized and transparent manner.

Discover the Protocol (/understand-genlayer-protocol "Discover the Protocol")

PureLLM DAO Contract

Source: https://docs.genlayer.com/developers/intelligent-contracts/examples/_advanced/_purellm-dao

The PureLLM DAO contract implements a simplified, AI-driven Decentralized Autonomous Organization (DAO) that relies entirely on language model decision-making. This contract demonstrates extreme flexibility in governance by interpreting and executing natural language proposals within a blockchain environment.

□

This intelligent contract is currently not migrated to the real GenVM syntaxis.

```
import json
from backend.node.genvm.icontract import IContract
from backend.node.genvm.equivalence_principle import call_llm_with_principle

class ConstitutionalDAO(IContract):
    """
    A Constitutional DAO that uses AI to interpret and execute motions.
    This DAO maintains a state and can update it based on user-submitted motions.
    """

    def __init__(self):
        """
        Initialize the ConstitutionalDAO with a basic constitution.
        """
        self.state = json.dumps({
            "constitution": [
                "1. Anyone can become a member of the DAO",
                "2. The constitution of the DAO can be updated by a unanimous vote"
            ]
        })

    async def execute_motion(self, motion: str) -> None:
        """
        Execute a motion proposed by a user.

        This method interprets the motion using an AI model and updates the DAO state accordingly.

        Args:
            motion (str): The motion proposed by a user.

        Returns:
            None
        """
        # Prepare the prompt for the language model
        prompt = f"""
        You are a constitutional DAO
```

```

Your state is as follows:
{self.state}

User with the address "{contract_runner.from_address}"
has made the following motion:
{motion}

Decide how to proceed
Respond with the following JSON format:
{{
  "reasoning": str,           // Your reasoning
  "updated_state": any,       // The new state of the DAO - can be any format
}}

It is mandatory that you respond only using the JSON format above,
nothing else. Don't include any other words or characters,
your output must be only JSON without any formatting prefix or suffix.
This result should be perfectly parseable by a JSON parser without errors.
"""

    # Call the language model with the equivalence principle
    result = await call_llm_with_principle(
        prompt,
        eq_principle="The updated state has to be essentially equivalent",
    )

    # Clean up the result and parse it as JSON
    result_clean = result.replace("True", "true").replace("False", "false")
    output = ison.loads(result_clean)

    # Update the DAO state
    self.state = json.dumps(output["updated_state"])

def get_state(self) -> str:
    """
    Get the current state of the DAO.

    Returns:
        str: The current state of the DAO as a JSON string.
    """
    return self.state

```

Code Explanation

- Initialization: The ConstitutionalDAO class initializes with a basic constitution stored as a JSON string in the state variable.
- Key Methods:
 - execute_motion(motion): Processes a text-based motion submitted by any user. It uses an AI model to interpret the motion and update the DAO's state accordingly.
 - State Management:
 - get_state(): Allows retrieval of the current DAO state.

Deploying the Contract

To deploy the ConstitutionalDAO contract:

1. Deploy the Contract: No initial parameters are needed. The contract initializes with a basic constitution.

Checking the Contract State

After deploying the contract, you can check its state in the Read Methods section.

- Use `get_state()` to view the current state of the DAO, including its constitution and any other properties that have been added through motions.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can:

- Call `execute_motion(motion)` to propose and execute a change to the DAO. This is the primary method of interaction with the contract.

Analyzing the Contract's Behavior

The contract's behavior involves a single, highly flexible process:

AI-Driven Motion Execution:

1. A user submits a motion in natural language.
2. The AI interprets the motion in the context of the current DAO state.
3. The AI decides how to update the state based on its interpretation.
4. The contract's state is updated according to the AI's decision.

Handling Different Scenarios

The ConstitutionalDAO can handle a wide range of scenarios through its flexible motion execution:

- Constitutional Amendments: Users can propose changes to the DAO's constitution.
- New Feature Proposals: Motions can suggest adding new functionalities to the DAO.
- Resource Allocation: Users might propose ways to manage or distribute resources.
- Membership Rules: Motions could alter how membership in the DAO is determined or managed.
- Decision-Making Processes: The very process of how decisions are made could be altered through motions.

Use Cases and Benefits

The ConstitutionalDAO contract is ideal for:

- Experimental Governance: Organizations looking to explore highly adaptive governance models.
- Rapid Iteration: Communities that need to quickly evolve their structure and rules.
- Complex Decision-Making: Scenarios where decisions require nuanced interpretation and execution.

Benefits include:

- Extreme Flexibility: Can handle a wide range of governance actions without predefined structures.
- Natural Language Interface: Allows users to interact with the DAO using everyday language.
- Adaptive Governance: The DAO can evolve its own rules and structure over time based on member input.

Limitations and Considerations

While powerful, this approach has some important considerations:

1. **AI Dependence:** The reliability and effectiveness of the DAO are heavily dependent on the capabilities of the AI model.
2. **Interpretation Accuracy:** There's a risk of misinterpretation of motions, which could lead to unintended outcomes.
3. **Language Precision:** Users need to be careful in how they word their motions to ensure accurate interpretation.
4. **Lack of Traditional Checks and Balances:** The flexibility could potentially lead to rapid, drastic changes without traditional safeguards.

Future Enhancements

Potential improvements to this system could include:

1. **Multi-Step Approval Process:** Implementing a confirmation step before major changes are executed.
2. **Historical Tracking:** Maintaining a log of all executed motions and their impacts on the DAO state.
3. **User Reputation System:** Incorporating a system that weighs motions based on the proposer's past contributions or reputation.

This ConstitutionalDAO contract demonstrates an innovative and highly flexible approach to decentralized governance. It showcases how advanced AI can be integrated with blockchain technology to create adaptive, language-driven organizational structures. While powerful, it also highlights the challenges and considerations in creating AI-driven governance systems.

Discover the Protocol (</understand-genlayer-protocol> "Discover the Protocol")

RokosMansion Contract

Source: https://docs.genlayer.com/developers/intelligent-contracts/examples/_advanced/_rokos-mansion

The RokosMansion contract sets up a scenario for an interactive text-based game where players explore Professor Roko's mansion, solve puzzles, and interact with the environment to prevent a malicious artificial superintelligence (ASI) from taking control. This contract demonstrates complex game logic, dynamic content generation, and player interactions within a blockchain environment.

□

This intelligent contract is currently not migrated to the real GenVM syntaxis.

```
"""
Module: RokosMansion
Description: Implements the RokosMansion game contract.
"""

import json
from backend.node.genvm.icontract import IContract
```

```

from backend.node.genvm.equivalence_principle import call_llm_with_principle

class RokosMansion(IContract):
    """
    RokosMansion game contract class.

    This class represents the game logic for the Mansion of Professor Roko game.
    It handles the game state, page transitions, and interactions with the game environment.

    Attributes:
        _allowed_styles (list): List of allowed writing styles.
        _style (str): Selected writing style.
        _allowed_countries (list): List of allowed country styles.
        _country (str): Selected country style.
        _inventory (list): Player's inventory.
        _environment (str): String summarizing the changes in the environment.
        _current_page_number (int): Current page number.
        _current_page (str): Current page content.
        page_text_gen (dict): Generated page text.
        page_actions_gen (dict): Generated page actions.
        page_text (dict): Default page text.
        page_actions (dict): Default page actions.
    """

    def __init__(self, style: str = "Stephen King", country: str = "USA"):
        """
        Initialize the RokosMansion contract.

        Args:
            style (str): Writing style. Defaults to "Stephen King".
            country (str): Country style. Defaults to "USA".

        Raises:
            AssertionError: If the provided style, country is not allowed.
        """
        self._allowed_styles = ["Stephen King", "HP Lovecraft", "Clive Barker", "Yukio Mishima"]
        assert style in self._allowed_styles
        self._style = style
        self._allowed_countries = ['Andorra', 'Argentina', 'Brazil', 'España', 'Latvia', 'Portugal', 'Russia',
'Thailand', 'USA', 'Venezuela','Japan']
        assert country in self._allowed_countries
        self._country = country

        self._inventory = [] # list of strings
        self._environment = ""
        self._current_page_number = 5
        self._current_page = ""
        self.page_text_gen = {}
        self.page_actions_gen = {}
        self.puzzles_solved = []
        self.puzzles_for_victory = 3
        self.victory_page = 11

        self.page_puzzles = {}
        self.page_puzzles[3] = "Each box is labeled, but all labels are incorrect. One box contains only **Poison**, one contains only **Antidote**, and the last contains **Both Poison and Antidote**. The puzzle asks you to pick one item from any box, knowing that the labels are wrong. For example, if you pick from the box labeled 'Both Poison and Antidote,' whatever you pull will reveal how to correctly label all three boxes. Solving this puzzle deactivates the ASI's device in the library and allows you to proceed.",
        self.page_puzzles[4] = "The room's puzzle involves analyzing the behavior of Organics and Synthetics. two types of beings affected by the ASI: Organics believe everything while awake is true, and everything while asleep is false, while Synthetics believe the opposite. The puzzle asks you to determine the truth of the statement: 'Any person that is awake believes they are organic.' Solve it to deactivate the ASI's device in the room."
        self.page_puzzles[5] = "Guard 1 (a guard who lies only when talking about uranium) says 'The materials are either uranium or thorium,' Guard 2 (a guard who lies when talking about plutonium) says 'The secret material is plutonium'; solve their statements to determine the correct radioactive material and sever the ASI's timeline connection."

        self.page_puzzles_action = {}
        self.page_puzzles_action[3] = " To solve this puzzle you must choose ONLY one box and be logically consistent with the conditions.",
        self.page_puzzles_action[4] = " To solve this puzzle you must clearly say if the statement <Any person that is awake believes they are organic.> is true or false, and be logically consistent with the puzzle conditions."
        self.page_puzzles_action[5] = "To solve this puzzle you must clearly say if the secret material mentioned by the Guards is plutonium, uranium or thorium, and be logically consistent with that what the Guards have said."

```

```

        self.page_text = {
            1: "Arrival at the Mansion: You are an engineer named Alex, invited to visit the mansion of Professor Roko, a notorious mad scientist known for dabbling in AI technology. Upon arrival, the mansion seems eerie, its large doors creaking open on their own. As you step inside, you feel a strange presence. A hologram of Professor Roko appears and reveals that he has made contact with a malicious artificial superintelligence (ASI) from the future. The ASI is sending cryptic messages and puzzles through devices scattered across the mansion.",
            2: "The Entrance Hall: You stand in the grand entrance hall of the mansion with several doors leading to different rooms. Roko's hologram reappears, urging you to hurry, as the ASI grows stronger by the minute. Your task is to solve three logical puzzles hidden within the mansion to weaken the ASI's influence. However, you can explore the rooms to gather information and insights.",
            3: "The Library: The library is a vast room filled with towering bookshelves and strange devices. In the center, an ancient machine hums softly, connected to three mysterious boxes.",
            4: "The Study: The study is a small room filled with strange devices, a desk cluttered with blueprints and journals, and a glowing cube resting on the desk.",
            5: "The Laboratory: A futuristic room where two robotic guards protect a vault containing either uranium, plutonium, or thorium .",
            6: "Professor Roko's Dormitory: The dormitory is cluttered with books, blueprints, and gadgets. Professor Roko's hologram stands in the center. offering crvotic insights about the ASI. the puzzles. and how to defeat the ASI.",
            7: "The Dining Hall: The dining hall is lavish but abandoned, with a long table covered in untouched dishes. There are no puzzles here, but there are clues about AI developments and future dangers.",
            8: "The Upstairs Hallway: The hallway is dark with paintings of futuristic cities on the walls. Faint mechanical sounds can be heard from one of the rooms. Several doors line the hall, but only one is unlocked.",
            9: "The Observatory: The observatory offers a breathtaking view of the night sky. There are no puzzles here, but you find a journal detailing Professor Roko's early experiments with the ASI and troubling future visions.",
            10: "Return to the Entrance Hall: From any room in the mansion, the player can choose to return to the entrance hall and select a different room to explore. Gathering clues in these rooms may help solve the puzzles in the mansion.",
            11: "Victory!: After solving all the puzzles, you return to Professor Roko's study where the ASI's connection is permanently severed. The devices go dark, and you leave the mansion victorious, having saved the future."
        }
        self.page_actions = {
            1: "Action: Enter the mansion entrance hall (to solve the puzzles to prevent the ASI from taking control of your world).",
            2: "Action: Choose a door: Left (Library), Center (Laboratory), Right (Study), Door leading to Professor Roko's personal study, Staircase to the upper floor.",
            3: "Action: Choose the correct box to solve the puzzle and step through the portal and return to the entrance hall. Now you must decide which room to explore next: Center (Laboratory), Right (Study), Professor Roko's Personal Study.",
            4: "Action: Analyze the statement and determine if it is true or false to solve the puzzle. Correct it if necessary to deactivate the ASI's device. Return to the entrance hall: Left (Library), Center (Laboratory), Professor Roko's Personal Study.",
            5: "Action: Choose the correct element is the action needed to solve the puzzle. Return to the Entrance Hall and choose a new room to explore.",
            6: "Action: Speak to Professor Roko. You may ask him about the Origin of the ASI, the Nature of the Puzzles, or how to defeat the ASI.",
            7: "Action: Explore the room and read the newspaper. Return to the Entrance Hall or proceed to another room.",
            8: "Action: Explore the hallway. Enter the unlocked room or return downstairs.",
            9: "Action: Read Professor Roko's journal. Return to the Entrance Hall or explore another part of the mansion.".
            10: "Action: Return to the Entrance Hall and select a different room to explore. Gather clues or explore the mansion. Choose a door: Left (Library), Center (Laboratory), Right (Study), Door leading to Professor Roko's personal study, Staircase to the upper floor.",
            11: "Action: Exit the mansion, completing your journey as the savior of the future. You've won the game!"
        }
    }

# def update_current_page(self) -> str:
#     """
#     Update the current page content.
#
#     Returns:
#         str: Updated current page content.
#     """
#     if self._current_page_number not in self.page_text_gen:
#         self.page_text_gen[self._current_page_number] = self._current_page
#     self._current_page = self.page_text_gen[self._current_page_number]
#
#     return self._current_page

def get_current_page(self) -> str:
    """
    Get the current page content.

    Returns:
        str: Current page content.
    """

```

```

"""
if self._current_page_number not in self.page_text_gen:
    return "Void. Call `update_current_page`"
return self.page_text_gen[self._current_page_number]

def get_current_page_number(self) -> int:
    """
    Get the current page number.
    Returns:
        int: Current page number.
    """
    return self._current_page_number

async def update_current_page(self):
    """
    Update the current page content using an LLM.

    This method generates a detailed scenario description for the current page
    based on the writer's style, country style, inventory, and original page scenario.
    """
    if self._current_page_number in self.page_text_gen:
        return
    prompt = f"""
Generate a very brief but vivid scenario description (in 3 short sentences) for the current page in the "Mansion of
Professor Roko" game. Use the following context:

1. Writer Style: {self._style}
2. Country Style: {self._country}
3. Inventory: {'', '.join(self._inventory) if self._inventory else 'Empty'}
4. Page Scenario: {self.page_text[self._current_page_number]}

Create a very brief but vivid and immersive description that incorporates elements of the specified writer's style,
cultural elements from the given country, mentions any items in the characters inventory, and based on the original
page scenario. The description should be brief but consistent with the original context while adding color and
atmosphere.

Respond using ONLY the following format:
{{
"description": str
}}
"""
    result = await call_llm_with_principle(
        prompt,
        eq_principle="The generated description must be consistent with the original page scenario, writer's
style, country's culture, and inventory items."
    )
    output = json.loads(result)
    self.page_text_gen[self._current_page_number] = output["description"]
    if self._current_page_number in self.page_puzzles:
        self.page_text_gen[self._current_page_number] += ' ' + self.page_puzzles[self._current_page_number]

async def update_current_actions(self):
    """
    Update the current page actions using an LLM.

    This method generates brief and concise descriptions for the actions available
    on the current page based on the writer's style, country style, and inventory.
    """
    if self._current_page_number in self.page_actions_gen:
        return
    prompt = f"""
Generate brief and concise descriptions for the actions available on the current page of the "Mansion of Professor
Roko" game. Use the following context:

1. Writer Style: {self._style}
2. Country Style: {self._country}
3. Inventory: {'', '.join(self._inventory) if self._inventory else 'Empty'}
4. Current Actions: {self.page_actions[self._current_page_number]}

Create very brief but vivid action descriptions that incorporate elements of the specified writer's style and cultural
elements from the given country. The descriptions should be concise but consistent with the original actions while
adding a touch of atmosphere.

Format the output as a single string with Markdown bullet points, like this:
* Action 1 description

```

```

* Action 2 description
* Action 3 description

Respond using ONLY the following format:
{{
  "actions": str
}}
"""
    result = await call_llm_with_principle(
        prompt,
        eq_principle="The generated action descriptions must be consistent with the original actions, writer's
style, and country's culture."
    )
    output = json.loads(result)
    self.page_actions_gen[self._current_page_number] = output["actions"]

    if self._current_page_number in self.page_puzzles_action:
        self.page_actions_gen[self._current_page_number] += ' ' +
self.page_puzzles_action[self._current_page_number]

def get_current_actions(self) -> str:
    """
    Get the current page actions.

    Returns:
        str: Current page actions.
    """
    if self._current_page_number not in self.page_actions_gen:
        return "Void. Call `update_current_actions`"
    return self.page_actions_gen[self._current_page_number]

async def do_prompt(self, prompt: str) -> str:
    """
    Process a user prompt and generate a response.
    This method checks if the prompt matches any current actions and responds accordingly.
    If the prompt doesn't match an action, it is treated as a question or environmental interaction.

    Args:
        prompt (str): User prompt.

    Returns:
        str: Generated response.
    """
    assert self._current_page_number != self.victory_page

    room_mapping = {k: v.split(':')[0].strip() for k, v in self.page_text.items()}
    room_mapping_str = '\n'.join(f"{v} (Page {k})" for k, v in room_mapping.items())

    # Check if the prompt matches any current actions
    action_match_prompt = f"""
    Given the current actions for this page:
    {self.get_current_actions()}

    The user's inventory:
    {'', '.join(self._inventory) if self._inventory else 'Empty'}

    The environment summary:
    {self._environment if self._environment else 'No changes in the environment.'}

    The current room:
    {room_mapping[self._current_page_number]} (Page {self._current_page_number})

    And the user's prompt:
    "{prompt}"

    Determine if the user's prompt roughly matches any of the current actions, or matches
    the action of trving to solve a puzzle if there is a puzzle present.
    Respond with only "true" if it matches, or "false" if it doesn't match.
    """
    action_match_result = await call_llm_with_principle(
        action_match_prompt,
        eq_principle="The response must be either 'true' or 'false' based on whether the prompt matches an action
(including an attempt to solve a puzzle if present)."
    )

    if action_match_result.strip().lower() == "true":
        # The prompt matches an action, so we need to move to another page/room or solve a puzzle

```



```

print('DEBUG: # The prompt matches an action, so we need to move to another page/room or solve a puzzle')
action_result_prompt = f"""
Given the current page description:
{self.get_current_page()}

The user's inventory:
{' '.join(self._inventory) if self._inventory else 'Empty'}

The environment summary:
{self._environment if self._environment else 'No changes in the environment.'}

And the user's action:
"{prompt}"

All rooms in the game:
{room_mapping_str}

Determine the result of this action. If it leads to a new room, specify which room (page number) to move
to.
If it involves solving a puzzle, describe the attempt to solve it (true, false, null).
If the action was an attempt to solve a puzzle do not move to different room.
If the action was trying to move to another room, change the room number if applicable.

Respond using ONLY the following JSON format:
{{
    "result": str,
    "new_page_number": int or null,
    "puzzle_solved": bool or null
}}
"""
action_result = await call_llm_with_principle(
    action_result_prompt,
    eq_principle="The response must be consistent with the game's logic, current state, inventory,
environment, and difficulty level."
)
action_output = json.loads(action_result)

if action_output["puzzle_solved"]:
    if not self._current_page_number in self.puzzles_solved:
        self.puzzles_solved.append( self._current_page_number )

if len(self.puzzles_solved) >= self.puzzles_for_victory:
    self._current_page_number = self.victory_page
elif action_output["new_page_number"]:
    self._current_page_number = action_output["new_page_number"]

return action_output["result"]
else:
    # The prompt doesn't match an action, so we need to handle it as a question or environmental interaction
    print("DEBUG: # The prompt doesn't match an action, so we need to handle it as a question or environmental
interaction")
    env_interaction_prompt = f"""
Given the current page description:
{self.get_current_page()}

The user's inventory:
{' '.join(self._inventory) if self._inventory else 'Empty'}

The environment summary:
{self._environment if self._environment else 'No changes in the environment.'}

And the user's prompt:
"{prompt}"

Determine how this prompt affects the environment or inventory. If it's a question, provide an appropriate
answer.

Respond using ONLY the following JSON format:
{{
    "result": str,
    "inventory_change": [str] or null,
    "environment_change": str or null
}}
"""
    env_result = await call_llm_with_principle(
        env_interaction_prompt,
        eq_principle="The response must be consistent with the game's current state, inventory, environment,
and logical within the game world."
    )

```

```

    )
    env_output = json.loads(env_result)

    if env_output["inventory_change"]:
        self._inventory.extend(env_output["inventory_change"])

    if env_output["environment_change"]:
        self._environment += f" {env_output['environment_change']}"

    return env_output["result"]

```

Code Explanation

- Initialization: The RokosMansion class initializes the game state with a writing style, country style, and various game elements like inventory, environment, and page content.
- Read Methods:
 - get_current_page(): Returns the current page content.
 - get_current_page_number(): Returns the current page number.
 - get_current_actions(): Returns the available actions on the current page.
- Write Methods:
 - update_current_page(): Generates detailed scenario descriptions for the current page using an LLM.
 - update_current_actions(): Generates action descriptions for the current page using an LLM.
 - do_prompt(prompt): Processes user prompts, updates game state, and generates responses.

Deploying the Contract

To deploy the RokosMansion contract, you need to provide the writing style and country style:

1. Set Writing Style: Choose from "Stephen King", "HP Lovecraft", "Clive Barker", or "Yukio Mishima".
2. Set Country Style: Choose from a list of allowed countries.
3. Deploy the Contract: Once the styles are set, deploy the contract to start the game.

Checking the Contract State

After deploying the contract, its address is displayed and you can check its state in the Read Methods section.

- Use get_current_page() to see the current page content.
- Use get_current_page_number() to see the current page number.
- Use get_current_actions() to see available actions on the current page.

Executing Transactions

To interact with the deployed contract, go to the Write Methods section. Here, you can:

- Call update_current_page() to generate new page content.
- Call update_current_actions() to generate new action descriptions.
- Call do_prompt(prompt) to process user inputs and advance the game.

Analyzing the Contract's Behavior

The contract's behavior involves complex game logic and interactions:

1. **Dynamic Content Generation:** The contract uses LLMs to generate page descriptions and actions based on the chosen writing and country styles.
2. **Player Interactions:** Players can explore the mansion, interact with the environment, and attempt to solve puzzles through text prompts.
3. **Game State Management:** The contract keeps track of the player's inventory, environment changes, and puzzle progress.
4. **Puzzle Solving:** Players must solve three logical puzzles to weaken the ASI's influence and win the game.

Handling Different Scenarios

- **Exploration:** Players can move between rooms, gathering clues and information about the ASI and puzzles.
- **Puzzle Solving:** When a player attempts to solve a puzzle, the contract verifies the solution and updates the game state accordingly.
- **Environmental Interactions:** Players can interact with objects in the environment, potentially affecting their inventory or the room's state.
- **Victory Condition:** Once three puzzles are solved, the player is moved to the victory page, completing the game.

This RokosMansion contract demonstrates a complex use case for interactive storytelling and game logic on the blockchain. It showcases how smart contracts can be used to create engaging, dynamic experiences that combine predetermined elements with AI-generated content.

Discover the Protocol (/understand-genlayer-protocol "Discover the Protocol")

Balances

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/balances>

This page has moved to Value Transfers (/developers/intelligent-contracts/features/value-transfers).

See:

- **Reading Balances** (/developers/intelligent-contracts/features/value-transfers#reading-balances)
- **Receiving Value** (/developers/intelligent-contracts/features/value-transfers#receiving-value)
- **Sending Value**
(/developers/intelligent-contracts/features/value-transfers#sending-value-to-another-intelligent-contract)

Web Access (/developers/intelligent-contracts/features/web-access "Web Access")Features
(/developers/intelligent-contracts/features/features "Features")

Intelligent Contract Features

Source: <https://docs.genlayer.com/developers/intelligent-contracts/features/features>

Deterministic Features

[Types \(/developers/intelligent-contracts/types\)](/developers/intelligent-contracts/types) [Storage \(/developers/intelligent-contracts/features/storage\)](/developers/intelligent-contracts/features/storage) [Error Handling \(/developers/intelligent-contracts/features/error-handling\)](/developers/intelligent-contracts/features/error-handling) [Upgradability \(/developers/intelligent-contracts/features/upgradability\)](/developers/intelligent-contracts/features/upgradability) [Value Transfers \(/developers/intelligent-contracts/features/value-transfers\)](/developers/intelligent-contracts/features/value-transfers) [Messages \(/developers/intelligent-contracts/features/messages\)](/developers/intelligent-contracts/features/messages) [Interacting with Intelligent Contracts \(/developers/intelligent-contracts/features/interacting-with-intelligent-contracts\)](/developers/intelligent-contracts/features/interacting-with-intelligent-contracts) [Interacting with EVM Contracts \(/developers/intelligent-contracts/features/interacting-with-evm-contracts\)](/developers/intelligent-contracts/features/interacting-with-evm-contracts) [Special Methods \(/developers/intelligent-contracts/features/special-methods\)](/developers/intelligent-contracts/features/special-methods) [Vector Storage \(/developers/intelligent-contracts/features/vector-storage\)](/developers/intelligent-contracts/features/vector-storage) [Debugging \(/developers/intelligent-contracts/features/debugging\)](/developers/intelligent-contracts/features/debugging) [Random \(/developers/intelligent-contracts/features/random\)](/developers/intelligent-contracts/features/random)

Non-Deterministic Features

[Non-determinism \(/developers/intelligent-contracts/features/non-determinism\)](/developers/intelligent-contracts/features/non-determinism) [Calling LLMs \(/developers/intelligent-contracts/features/calling-llms\)](/developers/intelligent-contracts/features/calling-llms) [Web Access \(/developers/intelligent-contracts/features/web-access\)](/developers/intelligent-contracts/features/web-access)

[Balances \(/developers/intelligent-contracts/features/balances "Balances"\)](/developers/intelligent-contracts/features/balances) [Development Setup \(/developers/intelligent-contracts/tooling-setup "Development Setup"\)](/developers/intelligent-contracts/tooling-setup)

What Are Intelligent Contracts?

Source: <https://docs.genlayer.com/understand-genlayer-protocol/what-are-intelligent-contracts>

This page has moved to [What is GenLayer \(/understand-genlayer-protocol/what-is-genlayer\)](/understand-genlayer-protocol/what-is-genlayer).

[Accounts and Addresses \(/understand-genlayer-protocol/core-concepts/accounts-and-addresses "Accounts and Addresses"\)](/understand-genlayer-protocol/core-concepts/accounts-and-addresses) [What Makes Genlayer Different \(/understand-genlayer-protocol/what-makes-genlayer-different "What Makes Genlayer Different"\)](/understand-genlayer-protocol/what-makes-genlayer-different)

What Makes GenLayer Different?

Source: <https://docs.genlayer.com/understand-genlayer-protocol/what-makes-genlayer-different>

This page has moved to [What is GenLayer \(/understand-genlayer-protocol/what-is-genlayer\)](#).

[What Are Intelligent Contracts \(/understand-genlayer-protocol/what-are-intelligent-contracts "What Are Intelligent Contracts"\)](#)[Who Is Genlayer For \(/understand-genlayer-protocol/who-is-genlayer-for "Who Is Genlayer For"\)](#)

Who Is GenLayer For?

Source: <https://docs.genlayer.com/understand-genlayer-protocol/who-is-genlayer-for>

This page has moved to [What is GenLayer \(/understand-genlayer-protocol/what-is-genlayer\)](#).

[What Makes Genlayer Different \(/understand-genlayer-protocol/what-makes-genlayer-different "What Makes Genlayer Different"\)](#)[Why We Are Building Genlayer \(/understand-genlayer-protocol/why-we-are-building-genlayer "Why We Are Building Genlayer"\)](#)

Why We Are Building GenLayer

Source: <https://docs.genlayer.com/understand-genlayer-protocol/why-we-are-building-genlayer>

This page has moved to [What is GenLayer \(/understand-genlayer-protocol/what-is-genlayer\)](#).

[Who Is Genlayer For \(/understand-genlayer-protocol/who-is-genlayer-for "Who Is Genlayer For"\)](#)[Build on GenLayer \(/developers "Build on GenLayer"\)](#)

staking active-validators

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/active-validators>

List all active validators

Usage

```
$ genlayer staking active-validators [options]
```

Options

staking (/api-references/genlayer-cli/staking/staking "staking")staking banned-validators (/api-references/genlayer-cli/staking/staking/banned-validators "staking banned-validators")

staking banned-validators

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/banned-validators>

List all banned validators

Usage

\$ genlayer staking banned-validators [options]

Options

staking active-validators (/api-references/genlayer-cli/staking/staking/active-validators "staking active-validators")staking delegation-info (/api-references/genlayer-cli/staking/staking/delegation-info "staking delegation-info")

staking delegation-info

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/delegation-info>

Get delegation info for a delegator with a validator

Usage

\$ genlayer staking delegation-info [options] [validator]

Arguments

- [validator]

Options

staking banned-validators (/api-references/genlayer-cli/staking/staking/banned-validators "staking banned-validators")staking delegator-claim (/api-references/genlayer-cli/staking/staking/delegator-claim "staking delegator-claim")

staking delegator-claim

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/delegator-claim>

Claim delegator withdrawals after unbonding period

Usage

```
$ genlayer staking delegator-claim [options] [validator]
```

Arguments

- [validator]

Options

staking delegation-info (/api-references/genlayer-cli/staking/staking/delegation-info "staking delegation-info")staking delegator-exit (/api-references/genlayer-cli/staking/staking/delegator-exit "staking delegator-exit")

staking delegator-exit

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/delegator-exit>

Exit as a delegator by withdrawing shares from a validator

Usage

```
$ genlayer staking delegator-exit [options] [validator]
```

Arguments

- [validator]

Options

staking delegator-claim (/api-references/genlayer-cli/staking/staking/delegator-claim "staking delegator-claim")staking delegator-join (/api-references/genlayer-cli/staking/staking/delegator-join "staking delegator-join")

staking delegator-join

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/delegator-join>

Join as a delegator by staking with a validator

Usage

```
$ genlayer staking delegator-join [options] [validator]
```

Arguments

- [validator]

Options

staking delegator-exit (/api-references/genlayer-cli/staking/staking/delegator-exit "staking delegator-exit")staking
epoch-info (/api-references/genlayer-cli/staking/staking/epoch-info "staking epoch-info")

staking epoch-info

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/epoch-info>

Get current epoch and staking parameters

Usage

```
$ genlayer staking epoch-info [options]
```

Options

staking delegator-join (/api-references/genlayer-cli/staking/staking/delegator-join "staking delegator-join")staking
prime-all (/api-references/genlayer-cli/staking/staking/prime-all "staking prime-all")

staking prime-all

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/prime-all>

Prime all validators that need priming

Usage

\$ genlayer staking prime-all [options]

Options

staking epoch-info (/api-references/genlayer-cli/staking/staking/epoch-info "staking epoch-info")staking
quarantined-validators (/api-references/genlayer-cli/staking/staking/quarantined-validators "staking
quarantined-validators")

staking quarantined-validators

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/quarantined-validators>

List all quarantined validators

Usage

\$ genlayer staking quarantined-validators [options]

Options

staking prime-all (/api-references/genlayer-cli/staking/staking/prime-all "staking prime-all")staking set-identity
(/api-references/genlayer-cli/staking/staking/set-identity "staking set-identity")

staking set-identity

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/set-identity>

Set validator identity metadata (moniker, website, socials, etc.)

Usage

\$ genlayer staking set-identity [options] [validator]

Arguments

- [validator]

Options

staking quarantined-validators (/api-references/genlayer-cli/staking/staking/quarantined-validators "staking quarantined-validators")staking set-operator (/api-references/genlayer-cli/staking/staking/set-operator "staking set-operator")

staking set-operator

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/set-operator>

Change the operator address for a validator wallet

Usage

```
$ genlayer staking set-operator [options] [validator] [operator]
```

Arguments

- [validator]
- [operator]

Options

staking set-identity (/api-references/genlayer-cli/staking/staking/set-identity "staking set-identity")staking validator-claim (/api-references/genlayer-cli/staking/staking/validator-claim "staking validator-claim")

staking validator-claim

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/validator-claim>

Claim validator withdrawals after unbonding period

Usage

```
$ genlayer staking validator-claim [options] [validator]
```

Arguments

- [validator]

Options

staking set-operator (/api-references/genlayer-cli/staking/staking/set-operator "staking set-operator")staking
validator-deposit (/api-references/genlayer-cli/staking/staking/validator-deposit "staking validator-deposit")

staking validator-deposit

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/validator-deposit>

Make an additional deposit to a validator wallet

Usage

```
$ genlayer staking validator-deposit [options] [validator]
```

Arguments

- [validator]

Options

staking validator-claim (/api-references/genlayer-cli/staking/staking/validator-claim "staking
validator-claim")staking validator-exit (/api-references/genlayer-cli/staking/staking/validator-exit "staking
validator-exit")

staking validator-exit

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/validator-exit>

Exit as a validator by withdrawing shares

Usage

```
$ genlayer staking validator-exit [options] [validator]
```

Arguments

- [validator]

Options

staking validator-deposit (/api-references/genlayer-cli/staking/staking/validator-deposit "staking validator-deposit")staking validator-history (/api-references/genlayer-cli/staking/staking/validator-history "staking validator-history")

staking validator-history

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/validator-history>

Show slash and reward history for a validator (default: last 10 epochs)

Usage

```
$ genlayer staking validator-history [options] [validator]
```

Arguments

- [validator]

Options

staking validator-exit (/api-references/genlayer-cli/staking/staking/validator-exit "staking validator-exit")staking validator-info (/api-references/genlayer-cli/staking/staking/validator-info "staking validator-info")

staking validator-info

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/validator-info>

Get information about a validator

Usage

```
$ genlayer staking validator-info [options] [validator]
```

Arguments

- [validator]

Options

staking validator-history (/api-references/genlayer-cli/staking/staking/validator-history "staking validator-history")staking validator-join (/api-references/genlayer-cli/staking/staking/validator-join "staking validator-join")

staking validator-join

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/validator-join>

Join as a validator by staking tokens

Usage

\$ genlayer staking validator-join [options]

Options

staking validator-info (/api-references/genlayer-cli/staking/staking/validator-info "staking validator-info")staking validator-prime (/api-references/genlayer-cli/staking/staking/validator-prime "staking validator-prime")

staking validator-prime

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/validator-prime>

Prime a validator to prepare their stake record for the next epoch

Usage

\$ genlayer staking validator-prime [options] [validator]

Arguments

- [validator]

Options

staking validator-join (/api-references/genlayer-cli/staking/staking/validator-join "staking validator-join")staking validators (/api-references/genlayer-cli/staking/staking/validators "staking validators")

staking validators

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/validators>

Show validator set with stake, status, and voting power

Usage

```
$ genlayer staking validators [options]
```

Options

staking validator-prime (/api-references/genlayer-cli/staking/staking/validator-prime "staking validator-prime")staking wizard (/api-references/genlayer-cli/staking/staking/wizard "staking wizard")

staking wizard

Source: <https://docs.genlayer.com/api-references/genlayer-cli/staking/staking/wizard>

Interactive wizard to become a validator

Usage

```
$ genlayer staking wizard [options]
```

Options

staking validators (/api-references/genlayer-cli/staking/staking/validators "staking validators")localnet (/api-references/genlayer-cli/localnet/localnet "localnet")

localnet validators

Source: <https://docs.genlayer.com/api-references/genlayer-cli/localnet/localnet/validators>

Manage localnet validators operations

Usage

```
$ genlayer localnet validators [options] [command]
```

Arguments

- [command]

Options

Subcommands

- genlayer get — Retrieve details of a specific validator
- genlayer delete — Delete a specific validator or all
- genlayer count — Count all validators
- genlayer update — Update a validator's details
- genlayer create-random — Create random validators
- genlayer create — Create a new validator

localnet (/api-references/genlayer-cli/localnet/localnet "localnet")localnet validators count
(/api-references/genlayer-cli/localnet/localnet/validators/count "localnet validators count")

localnet validators count

Source: <https://docs.genlayer.com/api-references/genlayer-cli/localnet/localnet/validators/count>

Count all validators

Usage

\$ genlayer localnet validators count [options]

Options

localnet validators (/api-references/genlayer-cli/localnet/localnet/validators "localnet validators")localnet validators
create (/api-references/genlayer-cli/localnet/localnet/validators/create "localnet validators create")

localnet validators create

Source: <https://docs.genlayer.com/api-references/genlayer-cli/localnet/localnet/validators/create>

Create a new validator

Usage

```
$ genlayer localnet validators create [options]
```

Options

```
localnet validators count (/api-references/genlayer-cli/localnet/localnet/validators/count "localnet validators count")localnet validators create-random (/api-references/genlayer-cli/localnet/localnet/validators/create-random "localnet validators create-random")
```

localnet validators create-random

Source: <https://docs.genlayer.com/api-references/genlayer-cli/localnet/localnet/validators/create-random>

Create random validators

Usage

```
$ genlayer localnet validators create-random [options]
```

Options

```
localnet validators create (/api-references/genlayer-cli/localnet/localnet/validators/create "localnet validators create")localnet validators delete (/api-references/genlayer-cli/localnet/localnet/validators/delete "localnet validators delete")
```

localnet validators delete

Source: <https://docs.genlayer.com/api-references/genlayer-cli/localnet/localnet/validators/delete>

Delete a specific validator or all validators

Usage

```
$ genlayer localnet validators delete [options]
```

Options

```
localnet validators create-random (/api-references/genlayer-cli/localnet/localnet/validators/create-random "localnet validators create-random")localnet validators get (/api-references/genlayer-cli/localnet/localnet/validators/get
```


"localnet validators get")

localnet validators get

Source: <https://docs.genlayer.com/api-references/genlayer-cli/localnet/localnet/validators/get>

Retrieve details of a specific validator or all validators

Usage

\$ genlayer localnet validators get [options]

Options

localnet validators delete (/api-references/genlayer-cli/localnet/localnet/validators/delete "localnet validators delete")localnet validators update (/api-references/genlayer-cli/localnet/localnet/validators/update "localnet validators update")

localnet validators update

Source: <https://docs.genlayer.com/api-references/genlayer-cli/localnet/localnet/validators/update>

Update a validator's details

Usage

\$ genlayer localnet validators update [options] <validatorAddress>

Arguments

- <validatorAddress>

Options

localnet validators get (/api-references/genlayer-cli/localnet/localnet/validators/get "localnet validators get")update

finalize

Source: <https://docs.genlayer.com/api-references/genlayer-cli/finalize>

Finalize a transaction that is ready to be finalized (public call)

Usage

```
$ genlayer finalize [options] <txId>
```

Arguments

- <txId>

Options

updatefinalize-batch (/api-references/genlayer-cli/finalize-batch "finalize-batch")

finalize-batch

Source: <https://docs.genlayer.com/api-references/genlayer-cli/finalize-batch>

Finalize a batch of idle transactions in a single call (public call)

Usage

```
$ genlayer finalize-batch [options] <txIds...>
```

Arguments

- <txIds...>

Options

finalize (/api-references/genlayer-cli/finalize "finalize")genlayer Commands (/api-references/genlayer-cli "genlayer Commands")

account

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account>

Manage your accounts (address, balance, keys)

Usage

```
$ genlayer account [options] [command]
```

Arguments

- [command]

Options

Subcommands

- genlayer list — List all accounts
- genlayer show — Show account details (address, balance)
- genlayer create — Create a new account with encrypted keystore
- genlayer import — Import an account from a private key or keystore
- genlayer export — Export an account to a keystore file
- genlayer use — Set the active account
- genlayer remove — Remove an account
- genlayer send — Send GEN to an address
- genlayer unlock — Unlock account by caching private key in OS
- genlayer lock — Lock account by removing private key from OS

genlayer Commands (/api-references/genlayer-cli "genlayer Commands")account create (/api-references/genlayer-cli/accounts/account/create "account create")

account create

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/create>

Create a new account with encrypted keystore

Usage

```
$ genlayer account create [options]
```

Options

account (/api-references/genlayer-cli/accounts/account "account")account export
(/api-references/genlayer-cli/accounts/account/export "account export")

account export

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/export>

Export an account to a keystore file (web3/geth/foundry compatible)

Usage

\$ genlayer account export [options]

Options

account create (/api-references/genlayer-cli/accounts/account/create "account create")account import
(/api-references/genlayer-cli/accounts/account/import "account import")

account import

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/import>

Import an account from a private key or keystore file

Usage

\$ genlayer account import [options]

Options

account export (/api-references/genlayer-cli/accounts/account/export "account export")account list
(/api-references/genlayer-cli/accounts/account/list "account list")

account list

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/list>

List all accounts

Usage

```
$ genlayer account list [options]
```

Options

account import (/api-references/genlayer-cli/accounts/account/import "account import")account lock (/api-references/genlayer-cli/accounts/account/lock "account lock")

account lock

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/lock>

Lock account by removing private key from OS keychain

Usage

```
$ genlayer account lock [options]
```

Options

account list (/api-references/genlayer-cli/accounts/account/list "account list")account remove (/api-references/genlayer-cli/accounts/account/remove "account remove")

account remove

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/remove>

Remove an account

Usage

```
$ genlayer account remove [options] <name>
```

Arguments

- <name>

Options

account lock (/api-references/genlayer-cli/accounts/account/lock "account lock")account send (/api-references/genlayer-cli/accounts/account/send "account send")

account send

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/send>

Send GEN to an address

Usage

\$ genlayer account send [options] <to> <amount>

Arguments

- <to>
- <amount>

Options

account remove (/api-references/genlayer-cli/accounts/account/remove "account remove")account show (/api-references/genlayer-cli/accounts/account/show "account show")

account show

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/show>

Show account details (address, balance)

Usage

\$ genlayer account show [options]

Options

account send (/api-references/genlayer-cli/accounts/account/send "account send")account unlock (/api-references/genlayer-cli/accounts/account/unlock "account unlock")

account unlock

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/unlock>

Unlock account by caching private key in OS keychain

Usage

```
$ genlayer account unlock [options]
```

Options

```
account show (/api-references/genlayer-cli/accounts/account/show "account show")account use  
(/api-references/genlayer-cli/accounts/account/use "account use")
```

account use

Source: <https://docs.genlayer.com/api-references/genlayer-cli/accounts/account/use>

Set the active account

Usage

```
$ genlayer account use [options] <name>
```

Arguments

```
- <name>
```

Options

```
account unlock (/api-references/genlayer-cli/accounts/account/unlock "account unlock")config  
(/api-references/genlayer-cli/configuration/config "config")
```

config

Source: <https://docs.genlayer.com/api-references/genlayer-cli/configuration/config>

Manage CLI configuration, including the default network

Usage

```
$ genlayer config [options] [command]
```

Arguments

- [command]

Options

Subcommands

- genlayer set — Set a configuration value
- genlayer get — Get the current configuration
- genlayer reset — Reset a configuration value to its default

account use (/api-references/genlayer-cli/accounts/account/use "account use")config get (/api-references/genlayer-cli/configuration/config/get "config get")

config get

Source: <https://docs.genlayer.com/api-references/genlayer-cli/configuration/config/get>

Get the current configuration

Arguments:

key Configuration key to retrieve

Usage

```
$ genlayer config get [options] [key]
```

Arguments

- [key]

Options

config (/api-references/genlayer-cli/configuration/config "config")config reset (/api-references/genlayer-cli/configuration/config/reset "config reset")

config reset

Source: <https://docs.genlayer.com/api-references/genlayer-cli/configuration/config/reset>

Reset a configuration value to its default

Arguments:

key Configuration key to reset

Usage

```
$ genlayer config reset [options] <key>
```

Arguments

- <key>

Options

```
config get (/api-references/genlayer-cli/configuration/config/get "config get")config set  
(/api-references/genlayer-cli/configuration/config/set "config set")
```

config set

Source: <https://docs.genlayer.com/api-references/genlayer-cli/configuration/config/set>

Set a configuration value

Arguments:

key=value Configuration key-value pair to set

Usage

```
$ genlayer config set [options] <key=value>
```

Arguments

- <key=value>

Options

config reset (/api-references/genlayer-cli/configuration/config/reset "config reset")network
(/api-references/genlayer-cli/configuration/network "network")

network

Source: <https://docs.genlayer.com/api-references/genlayer-cli/configuration/network>

Network configuration

Usage

\$ genlayer network [options] [command]

Arguments

- [command]

Options

Subcommands

- genlayer set — Set the network to use
- genlayer info — Show current network configuration and contract addresses
- genlayer list — List available networks

config set (/api-references/genlayer-cli/configuration/config/set "config set")network info
(/api-references/genlayer-cli/configuration/network/info "network info")

network info

Source: <https://docs.genlayer.com/api-references/genlayer-cli/configuration/network/info>

Show current network configuration and contract addresses

Usage

\$ genlayer network info [options]

Options

network (/api-references/genlayer-cli/configuration/network "network")network list
(/api-references/genlayer-cli/configuration/network/list "network list")

network list

Source: <https://docs.genlayer.com/api-references/genlayer-cli/configuration/network/list>

List available networks

Usage

\$ genlayer network list [options]

Options

network info (/api-references/genlayer-cli/configuration/network/info "network info")network set
(/api-references/genlayer-cli/configuration/network/set "network set")

network set

Source: <https://docs.genlayer.com/api-references/genlayer-cli/configuration/network/set>

Set the network to use

Arguments:

network The network to set

Usage

\$ genlayer network set [options] [network]

Arguments

- [network]

Options

network list (/api-references/genlayer-cli/configuration/network/list "network list")deploy
(/api-references/genlayer-cli/contracts/deploy "deploy")

deploy

Source: <https://docs.genlayer.com/api-references/genlayer-cli/contracts/deploy>

Deploy intelligent contracts

Usage

```
$ genlayer deploy [options]
```

Options

network set (/api-references/genlayer-cli/configuration/network/set "network set")call (/api-references/genlayer-cli/contracts/call "call")

call

Source: <https://docs.genlayer.com/api-references/genlayer-cli/contracts/call>

Call a contract method without sending a transaction or changing the state

Usage

```
$ genlayer call [options] <contractAddress> <method>
```

Arguments

- <contractAddress>
- <method>

Options

deploy (/api-references/genlayer-cli/contracts/deploy "deploy")write (/api-references/genlayer-cli/contracts/write "write")

write

Source: <https://docs.genlayer.com/api-references/genlayer-cli/contracts/write>

Sends a transaction to a contract method that modifies the state

Usage

```
$ genlayer write [options] <contractAddress> <method>
```

Arguments

- <contractAddress>
- <method>

Options

```
call (/api-references/genlayer-cli/contracts/call "call")schema (/api-references/genlayer-cli/contracts/schema "schema")
```

schema

Source: <https://docs.genlayer.com/api-references/genlayer-cli/contracts/schema>

Get the schema for a deployed contract

Usage

```
$ genlayer schema [options] <contractAddress>
```

Arguments

- <contractAddress>

Options

```
write (/api-references/genlayer-cli/contracts/write "write")code (/api-references/genlayer-cli/contracts/code "code")
```

code

Source: <https://docs.genlayer.com/api-references/genlayer-cli/contracts/code>

Get the source for a deployed contract

Usage

```
$ genlayer code [options] <contractAddress>
```

Arguments

- <contractAddress>

Options

```
schema (/api-references/genlayer-cli/contracts/schema "schema")init (/api-references/genlayer-cli/environment/init "init")
```

init

Source: <https://docs.genlayer.com/api-references/genlayer-cli/environment/init>

Initialize the GenLayer Environment

Usage

```
$ genlayer init [options]
```

Options

```
code (/api-references/genlayer-cli/contracts/code "code")up (/api-references/genlayer-cli/environment/up "up")
```

up

Source: <https://docs.genlayer.com/api-references/genlayer-cli/environment/up>

Starts GenLayer's simulator

Usage

```
$ genlayer up [options]
```

Options

init (/api-references/genlayer-cli/environment/init "init")stop (/api-references/genlayer-cli/environment/stop "stop")

stop

Source: <https://docs.genlayer.com/api-references/genlayer-cli/environment/stop>

Stop all running localnet services.

Usage

\$ genlayer stop [options]

Options

up (/api-references/genlayer-cli/environment/up "up")new (/api-references/genlayer-cli/environment/new "new")

new

Source: <https://docs.genlayer.com/api-references/genlayer-cli/environment/new>

Create a new GenLayer project using the default template

Usage

\$ genlayer new [options] <projectName>

Arguments

- <projectName>

Options

stop (/api-references/genlayer-cli/environment/stop "stop")update (/api-references/genlayer-cli/environment/update "update")

update

Source: <https://docs.genlayer.com/api-references/genlayer-cli/environment/update>

Update resources like models or configurations

Usage

```
$ genlayer update [options] [command]
```

Arguments

- [command]

Options

Subcommands

- genlayer ollama — Manage Ollama models (update or remove)

```
new (/api-references/genlayer-cli/environment/new "new")update ollama  
(/api-references/genlayer-cli/environment/update/ollama "update ollama")
```

update ollama

Source: <https://docs.genlayer.com/api-references/genlayer-cli/environment/update/ollama>

Manage Ollama models (update or remove)

Usage

```
$ genlayer update ollama [options]
```

Options

```
update (/api-references/genlayer-cli/environment/update "update")receipt  
(/api-references/genlayer-cli/transactions/receipt "receipt")
```

receipt

Source: <https://docs.genlayer.com/api-references/genlayer-cli/transactions/receipt>

Get transaction receipt by hash

Usage

```
$ genlayer receipt [options] <txId>
```

Arguments

- <txId>

Options

```
update ollama (/api-references/genlayer-cli/environment/update/ollama "update ollama")trace  
(/api-references/genlayer-cli/transactions/trace "trace")
```

trace

Source: <https://docs.genlayer.com/api-references/genlayer-cli/transactions/trace>

Get execution trace for a transaction (return data, stdout, stderr, GenVM logs)

Usage

```
$ genlayer trace [options] <txId>
```

Arguments

- <txId>

Options

```
receipt (/api-references/genlayer-cli/transactions/receipt "receipt")appeal  
(/api-references/genlayer-cli/transactions/appeal "appeal")
```

appeal

Source: <https://docs.genlayer.com/api-references/genlayer-cli/transactions/appeal>

Appeal a transaction by its hash

Usage

```
$ genlayer appeal [options] <txId>
```

Arguments

- <txId>

Options

```
trace (/api-references/genlayer-cli/transactions/trace "trace")appeal-bond  
(/api-references/genlayer-cli/transactions/appeal-bond "appeal-bond")
```

appeal-bond

Source: <https://docs.genlayer.com/api-references/genlayer-cli/transactions/appeal-bond>

Show minimum appeal bond required for a transaction

Usage

```
$ genlayer appeal-bond [options] <txId>
```

Arguments

- <txId>

Options

```
appeal (/api-references/genlayer-cli/transactions/appeal "appeal")GenLayerJS (/api-references/genlayer-js  
"GenLayerJS")
```

Contract Methods

Source: <https://docs.genlayer.com/api-references/genlayer-js/contracts>

Methods for deploying, reading, writing, and simulating GenLayer intelligent contracts.

getContractCode

Retrieves the source code of a deployed contract.

Returns: string

getContractSchema

Gets the schema (methods and constructor) of a deployed contract.

Returns: ContractSchema

getContractSchemaForCode

Generates a schema for contract code without deploying it.

Returns: ContractSchema

readContract

Executes a read-only contract call without modifying state.

Returns: RawReturn extends true ? 0x\${string} : CalldataEncodable

simulateWriteContract

Simulates a state-modifying contract call without executing on-chain.

Returns: RawReturn extends true ? 0x\${string} : CalldataEncodable

writeContract

Executes a state-modifying function on a contract through consensus. Returns the transaction hash.

Returns: 0x\${string}

deployContract

Deploys a new intelligent contract to GenLayer. Returns the transaction hash.

getMinAppealBond

Calculates the minimum bond required to appeal a transaction.

Returns: bigint

getRoundNumber

Returns the current consensus round number for a transaction.

Returns: bigint

getRoundData

Returns detailed data for a specific consensus round.

getLastRoundData

Returns the current round number and its data for a transaction.

canAppeal

Checks if a transaction can be appealed.

Returns: boolean

appealTransaction

Appeals a consensus transaction to trigger a new round of validation.

finalizeTransaction

Finalizes a single GenLayer transaction that is ready to be finalized. Returns the EVM transaction hash.

Returns: 0x\${string}

finalizeIdlenessTx

Batch-finalizes idle GenLayer transactions (those stuck without progressing). Returns the EVM transaction hash.

Returns: 0x\${string}

GenLayerJS (/api-references/genlayer-js "GenLayerJS")Transactions (/api-references/genlayer-js/transactions "Transactions")

Transaction Methods

Source: <https://docs.genlayer.com/api-references/genlayer-js/transactions>

Methods for fetching transactions, waiting for receipts, estimating gas, and debugging execution traces.

waitForTransactionReceipt

Polls until a transaction reaches the specified status. Returns the transaction receipt.

Returns: GenLayerTransaction

getTransaction

Fetches transaction data including status, execution result, and consensus details.

Returns: GenLayerTransaction

getTriggeredTransactionIds

Returns transaction IDs of child transactions created from emitted messages.

Returns: TransactionHash[]

debugTraceTransaction

Fetches the full execution trace including return data, stdout, stderr, and GenVM logs.

Returns: DebugTraceResult

cancelTransaction

Cancels a pending transaction. Studio networks only.

Returns: {transaction_hash: string; status: string}

getTransactionQueuePosition

Returns the queue slot position of a transaction in the pending queue.

Returns: number

estimateTransactionGas

Estimates gas required for a transaction.

Returns: bigint

Contracts (/api-references/genlayer-js/contracts "Contracts")Staking (/api-references/genlayer-js/staking "Staking")

Staking Methods

Source: <https://docs.genlayer.com/api-references/genlayer-js/staking>

Methods for validator and delegator staking operations, epoch queries, and network status.

validatorJoin

Joins as a validator with the specified stake amount.

Returns: ValidatorJoinResult

validatorDeposit

Adds additional self-stake to an active validator position. The underlying Staking contract requires `msg.sender == ValidatorWallet`, so the call is routed through the wallet's own `validatorDeposit` forwarder (which re-enters Staking with the correct sender).

Returns: StakingTransactionResult

validatorExit

Exits a validator position by burning the specified shares. Same `msg.sender` constraint as `validatorDeposit` — routed via the wallet.

Returns: StakingTransactionResult

validatorClaim

Claims pending validator withdrawals.

Returns: StakingTransactionResult & {claimedAmount: bigint}

validatorPrime

Primes a validator for participation in the next epoch.

Returns: StakingTransactionResult

setOperator

Sets the operator address for a validator wallet.

Returns: StakingTransactionResult

setIdentity

Sets validator identity information (name, website, social links).

Returns: StakingTransactionResult

delegatorJoin

Delegates stake to a validator.

Returns: DelegatorJoinResult

delegatorExit

Exits a delegation by burning the specified shares.

Returns: StakingTransactionResult

delegatorClaim

Claims pending delegator withdrawals.

Returns: StakingTransactionResult

isValidator

Checks if an address is an active validator.

Returns: boolean

getValidatorInfo

Returns comprehensive information about a validator including stake, identity, and status.

Returns: ValidatorInfo

getStakeInfo

Returns delegation stake information for a delegator-validator pair.

Returns: StakeInfo

getEpochInfo

Returns current epoch information including timing, stake requirements, and inflation data.

No parameters.

Returns: EpochInfo

getEpochData

Returns detailed data for a specific epoch.

Returns: EpochData

getActiveValidators

Returns addresses of all currently active validators.

No parameters.

Returns: Address[]

getActiveValidatorsCount

Returns the count of active validators.

No parameters.

Returns: bigint

getQuarantinedValidators

Returns addresses of validators currently in quarantine.

No parameters.

Returns: Address[]

getBannedValidators

Returns banned validators with ban duration and permanent ban status.

Returns: BannedValidatorInfo[]

getQuarantinedValidatorsDetailed

Returns detailed quarantine information with pagination.

Returns: BannedValidatorInfo[]

Transactions (/api-references/genlayer-js/transactions "Transactions")GenLayerPY (/api-references/genlayer-py "GenLayerPY")

GenLayerPY SDK API Reference

Source: <https://docs.genlayer.com/api-references/genlayer-py/api>

Auto-generated from source docstrings.

Client Methods

Client for interacting with the GenLayer network.

Provides methods for deploying and calling intelligent contracts, managing transactions, and staking operations.

fund_account

Funds an account with test tokens. Localnet only.

```
client.fund_account(address: Union, amount: int)
```

Returns: HexBytes

get_current_nonce

Returns the current nonce (transaction count) for an account.

```
client.get_current_nonce(address: Union = None, block_identifier: Union = None)
```

Returns: Nonce

initialize_consensus_smart_contract

Initializes the consensus contract configuration from the network.

```
client.initialize_consensus_smart_contract(force_reset: bool = False)
```

Returns: None

read_contract

Executes a read-only contract call without modifying state.

```
client.read_contract(address: Union, function_name: str, args: Optional = None, kwargs: Optional = None, account: Optional = None, raw_return: bool = False, transaction_hash_variant: TransactionHashVariant = <TransactionHashVariant.LATEST_NONFINAL: 'latest-nonfinal'>, sim_config: Optional = None)
```

write_contract

Executes a state-modifying function on a contract through consensus. Returns the transaction hash.

```
client.write_contract(address: Union, function_name: str, account: Optional = None, consensus_max_rotations: Optional = None, value: int = 0, leader_only: bool = False, args: Optional = None, kwargs: Optional = None, sim_config: Optional = None)
```

simulate_write_contract

Simulates a state-modifying contract call without executing on-chain. Localnet only.

```
client.simulate_write_contract(address: Union, function_name: str, account: Optional = None, args: Optional = None, kwargs: Optional = None, sim_config: Optional = None, transaction_hash_variant: TransactionHashVariant =
```

```
<TransactionHashVariant.LATEST_NONFINAL: 'latest-nonfinal'>)
```

deploy\contract

Deploys a new intelligent contract to GenLayer. Returns the transaction hash.

```
client.deploy_contract(code: Union, account: Optional = None, args: Optional = None, kwargs: Optional = None,
consensus_max_rotations: Optional = None, leader_only: bool = False, sim_config: Optional = None)
```

get\contract\schema

Gets the schema (methods and constructor) of a deployed contract. Localnet only.

```
client.get_contract_schema(address: Union)
```

Returns: ContractSchema

get\contract\schema\for\code

Generates a schema for contract code without deploying it. Localnet only.

```
client.get_contract_schema_for_code(contract_code: AnyStr)
```

Returns: ContractSchema

appeal\transaction

Appeals a consensus transaction to trigger a new round of validation.

Returns the original transaction\id (appeals operate on the same tx).

```
client.appeal_transaction(transaction_id: HexStr, account: Optional = None, value: int = 0)
```

wait\for\transaction\receipt

Polls until a transaction reaches the specified status. Returns the transaction receipt.

```
client.wait_for_transaction_receipt(transaction_hash: Union, status: TransactionStatus = <TransactionStatus.ACCEPTED:
'ACCEPTED'>, interval: int = 3000, retries: int = 10, full_transaction: bool = False)
```

Returns: GenLayerTransaction

get_transaction

Fetches transaction data including status, execution result, and consensus details.

```
client.get_transaction(transaction_hash: Union)
```

Returns: GenLayerTransaction

get_triggered_transaction_ids

Returns transaction IDs of child transactions created from emitted messages.

```
client.get_triggered_transaction_ids(transaction_hash: Union)
```

Returns: list

debug_trace_transaction

Fetches the full execution trace including return data, stdout, stderr, and GenVM logs.

```
client.debug_trace_transaction(transaction_hash: Union, round: int = 0)
```

Returns: dict

Types and Enums

TransactionStatus

Status of a GenLayer transaction in the consensus lifecycle.

```
TransactionStatus.UNINITIALIZED = "UNINITIALIZED"
TransactionStatus.PENDING = "PENDING"
TransactionStatus.PROPOSING = "PROPOSING"
TransactionStatus.COMMITTING = "COMMITTING"
TransactionStatus.REVEALING = "REVEALING"
TransactionStatus.ACCEPTED = "ACCEPTED"
TransactionStatus.UNDETERMINED = "UNDETERMINED"
TransactionStatus.FINALIZED = "FINALIZED"
TransactionStatus.CANCELED = "CANCELED"
TransactionStatus.APPEAL_REVEALING = "APPEAL_REVEALING"
TransactionStatus.APPEAL_COMMITTING = "APPEAL_COMMITTING"
TransactionStatus.READY_TO_FINALIZE = "READY_TO_FINALIZE"
TransactionStatus.VALIDATORS_TIMEOUT = "VALIDATORS_TIMEOUT"
TransactionStatus.LEADER_TIMEOUT = "LEADER_TIMEOUT"
```

TransactionResult

Consensus voting result across validators.

```

TransactionResult.IDLE = "IDLE"
TransactionResult.AGREE = "AGREE"
TransactionResult.DISAGREE = "DISAGREE"
TransactionResult.TIMEOUT = "TIMEOUT"
TransactionResult.DETERMINISTIC_VIOLATION = "DETERMINISTIC_VIOLATION"
TransactionResult.NO_MAJORITY = "NO_MAJORITY"
TransactionResult.MAJORITY_AGREE = "MAJORITY_AGREE"
TransactionResult.MAJORITY_DISAGREE = "MAJORITY_DISAGREE"

```

ExecutionResult

Result of contract execution by the GenVM.

```

ExecutionResult.NOT_VOTED = "NOT_VOTED"
ExecutionResult.FINISHED_WITH_RETURN = "FINISHED_WITH_RETURN"
ExecutionResult.FINISHED_WITH_ERROR = "FINISHED_WITH_ERROR"

```

VoteType

str(object='') -> str

str(bytes__or__buffer[, encoding[, errors]]) -> str

Create a new string object from the given object. If encoding or errors is specified, then the object must expose a data buffer that will be decoded using the given encoding and error handler.

Otherwise, returns the result of object.str() (if defined)

or repr(object).

encoding defaults to 'utf-8'.

errors defaults to 'strict'.

```

VoteType.NOT_VOTED = "NOT_VOTED"
VoteType.AGREE = "AGREE"
VoteType.DISAGREE = "DISAGREE"
VoteType.TIMEOUT = "TIMEOUT"
VoteType.DETERMINISTIC_VIOLATION = "DETERMINISTIC_VIOLATION"

```

GenLayerPY (/api-references/genlayer-py "GenLayerPY")GenLayer Test (/api-references/genlayer-test "GenLayer Test")

Integration Testing

Source: <https://docs.genlayer.com/api-references/genlayer-test/integration>

Test contracts against a running GenLayer network (localnet, studionet, or testnet).

Setup Functions

get_contract_factory

Get a ContractFactory instance for a contract.

Args:

contract_name: Name of the contract to load from artifacts

contract_file_path: Path to the contract file to load directly

Note: Exactly one of contract_name or contract_file_path must be provided.

```
get_contract_factory(contract_name: Optional = None, contract_file_path: Union = None)
```

Parameters:

Returns: ContractFactory

get_default_account

Returns the default account for the current network.

Returns: LocalAccount

get_accounts

Returns all configured accounts for the current network.

Returns: List

create_accounts

Creates n new accounts with random private keys.

```
create_accounts(n_accounts: int)
```

Parameters:

get_gl_client

Get the GenLayer client instance.

get_validator_factory

Returns: ValidatorFactory

ContractFactory

A factory for deploying contracts.

factory.deploy

Deploy the contract and return a Contract instance (convenience method).

This is a convenience method that handles receipt validation
and contract instantiation automatically.

```
factory.deploy(args: Optional = None, account: Optional = None, consensus_max_rotations: Optional = None,
wait_interval: Optional = None, wait_retries: Optional = None, wait_transaction_status: TransactionStatus =
<TransactionStatus.ACCEPTED: 'ACCEPTED'>, wait_triggered_transactions: bool = False,
wait_triggered_transactions_status: TransactionStatus = <TransactionStatus.ACCEPTED: 'ACCEPTED'>, transaction_context:
Optional = None)
```

Parameters:

Returns: Contract

factory.deploy_contract_tx

Deploy the contract and return the transaction receipt.

```
factory.deploy_contract_tx(args: Optional = None, account: Optional = None, consensus_max_rotations: Optional = None,
wait_interval: Optional = None, wait_retries: Optional = None, wait_transaction_status: TransactionStatus =
<TransactionStatus.ACCEPTED: 'ACCEPTED'>, wait_triggered_transactions: bool = False,
wait_triggered_transactions_status: TransactionStatus = <TransactionStatus.ACCEPTED: 'ACCEPTED'>, transaction_context:
Optional = None)
```

Parameters:

Returns: GenLayerTransaction

factory.build_contract

Build contract from address


```
factory.build_contract(contract_address: Union, account: Optional = None)
```

Parameters:

Returns: Contract

ContractFunction

ContractFunction(method_name: str, read_only: bool, call_method: Optional[Callable] = None, analyze_method: Optional[Callable] = None, transact_method: Optional[Callable] = None)

contract.method_name.call

Executes a read-only contract method call.

```
contract.method_name.call(transaction_hash_variant: TransactionHashVariant = <TransactionHashVariant.LATEST_NONFINAL: 'latest-nonfinal'>, transaction_context: Optional = None)
```

Parameters:

contract.method_name.transact

Executes a state-changing contract method through consensus. Returns the transaction receipt.

```
contract.method_name.transact(value: int = 0, consensus_max_rotations: Optional = None, wait_transaction_status: TransactionStatus = <TransactionStatus.ACCEPTED: 'ACCEPTED'>, wait_interval: Optional = None, wait_retries: Optional = None, wait_triggered_transactions: bool = False, wait_triggered_transactions_status: TransactionStatus = <TransactionStatus.ACCEPTED: 'ACCEPTED'>, transaction_context: Optional = None)
```

Parameters:

contract.method_name.analyze

Runs statistical analysis of method behavior across multiple executions.

```
contract.method_name.analyze(provider: str, model: str, config: Optional = None, plugin: Optional = None, plugin_config: Optional = None, runs: int = 100, genvm_datetime: Optional = None)
```

Parameters:

ValidatorFactory

validator_factory.create_validator

```
validator_factory.create_validator(stake: int, provider: str, model: str, config: Dict, plugin: str, plugin_config: Dict)
```

Parameters:

Returns: Validator

validator_factory.batch_create_validators

```
validator_factory.batch_create_validators(count: int, stake: int, provider: str, model: str, config: Dict, plugin: str, plugin_config: Dict)
```

Parameters:

Returns: List

validator_factory.create_mock_validator

```
validator_factory.create_mock_validator(mock_llm_response: Optional = None, mock_web_response: Optional = None)
```

Parameters:

Returns: Validator

validator_factory.batch_create_mock_validators

```
validator_factory.batch_create_mock_validators(count: int, mock_llm_response: Optional = None, mock_web_response: Optional = None)
```

Parameters:

Returns: List

Validator

Validator(stake: int, provider: str, model: str, config: Dict[str, Any], plugin: str, plugin_config: Dict[str, Any], mock_enabled: bool, mock_llm_response: Optional[glttest.types.MockedLLMResponse], mock_web_response: Optional[glttest.types.MockedWebResponse])

validator.to_dict

Returns: Dict

validator.clone

Returns: Validator

validator.batch_clone

```
validator.batch_clone(count: int)
```

Parameters:

Returns: List

GenLayer Test (/api-references/genlayer-test "GenLayer Test")Direct Mode (/api-references/genlayer-test/direct "Direct Mode")

Direct Mode Testing

Source: <https://docs.genlayer.com/api-references/genlayer-test/direct>

Run contracts directly in Python without a network. Provides Foundry-style cheatcodes for fast test execution.

Setup Functions

deploy_contract

Deploy a contract and return an instance.

```
deploy_contract(contract_path: Path, vm: 'VMContext', args: Any, sdk_version: Optional[str] = None, kwargs: Any)
```

Parameters:

Returns: Any

load_contract_class

Load a contract class from file.

Sets up SDK paths, WASI mock, and message context.

```
load_contract_class(contract_path: Path, vm: 'VMContext', sdk_version: Optional[str] = None)
```

Parameters:

Returns: Type[Any]

create_address

Create a deterministic address from seed string.

```
create_address(seed: str)
```

Parameters:

Returns: Any

create_test_addresses

Create a list of test addresses.

```
create_test_addresses(count: int = 10)
```

Parameters:

Returns: list

VMContext

Test VM context providing Foundry-style cheatcodes.

Usage:

```
vm = VMContext()
```

```
vm.sender = Address("0x" + "a" * 40)
```

```
vm.mock_web("api.example.com", {"status": 200, "body": "{}"})
```

```
with vm.activate():  
    contract = deploy_contract("Token.py", vm, owner)  
    contract.transfer(bob, 100)
```

vm.activate

Activate this VM context for contract execution.

Uses proper cleanup via ExitStack for resource management.

Patches datetime.datetime so that datetime.now() returns the warped time set via vm.warp(). This is dynamic: calling warp()

mid-test updates `\_datetime` and subsequent `now()` calls reflect it.

`vm.warp`

Set block timestamp (ISO format).

```
vm.warp(timestamp: str)
```

Parameters:

Returns: None

`vm.deal`

Set balance for an address.

```
vm.deal(address: Any, amount: int)
```

Parameters:

Returns: None

`vm.snapshot`

Take a snapshot of current state. Returns snapshot ID.

Returns: int

`vm.revert`

Revert to a previous snapshot.

```
vm.revert(snapshot_id: int)
```

Parameters:

Returns: None

`vm.mock_web`

Mock web requests matching URL pattern.

```
vm.mock_web(url_pattern: str, response: MockedWebResponseData)
```

Parameters:

Returns: None

vm.mock_llm

Mock LLM prompts matching pattern.

```
vm.mock_llm(prompt_pattern: str, response: str)
```

Parameters:

Returns: None

vm.clear_mock

Clear all registered mocks.

Returns: None

vm.prank

Context manager to temporarily change sender.

```
vm.prank(address: Any)
```

Parameters:

vm.startPrank

Start pranking as address (persists until stopPrank).

```
vm.startPrank(address: Any)
```

Parameters:

Returns: None

vm.stopPrank

Stop the current prank.

Returns: None

vm.expect_revert

Context manager expecting the next call to revert.

Catches ContractRollback (gl.rollback) and any Exception raised by contract code (ValueError, RuntimeError, etc.). If **message** is given, the exception text must contain it.

```
vm.expect_revert(message: Optional[str] = None)
```

Parameters:

vm.run_validator

Run a captured validator function from a prior run_nondet call.

Each gl.vm.run_nondet call in a contract appends an entry to an internal list. Use **index** to select which one (default -1, the most recent).

Mocks still apply: the validator typically re-runs leader_fn internally, which hits the current web/LLM mocks. Swap mocks between the contract call and run_validator() to simulate the validator seeing different external data.

Args:

leader_result: Override the leader's return value.

leader_error: Simulate a leader exception (gl.vm.UserError).

index: Which captured validator to run (-1 = last).

Returns:

The bool returned by the validator function.

```
vm.run_validator(leader_result: Any = <object object at 0x7f3bdfd04ad0>, leader_error: Optional[Exception] = None, index: int = -1)
```

Parameters:

Returns: bool

vm.clear_validators

Clear the captured validator list.

Returns: None

Integration Testing (/api-references/genlayer-test/integration "Integration Testing")glsim
(/api-references/genlayer-test/glsim "glsim")

glsim — Local GenLayer Network

Source: <https://docs.genlayer.com/api-references/genlayer-test/glsim>

A lightweight single-process GenLayer network that runs contracts via direct mode. Supports real LLM and web calls without Docker or WASM.

Installation

```
pip install genlayer-test[sim]
```

Usage

```
# Start with defaults (port 4000, 5 validators)
glsim

# Custom configuration
glsim --port 8000 --validators 3 --llm-provider openai:gpt-4o

# Deterministic mode (reproducible addresses)
glsim --seed my-test-seed
```

CLI Options

JSON-RPC Methods

glsim exposes a JSON-RPC 2.0 endpoint at POST /api compatible with both GenLayer Studio and standard Ethereum methods:

GenLayer Methods

- gen_call — Call a contract method
- gen_get_contract_schema — Get contract schema

- `gen_get_contract_schema_for_code` — Get schema from code
- `gen_get_transaction_status` — Get transaction status

Simulator Methods

- `sim_deploy` — Deploy a contract
- `sim_call` — Call a contract method
- `sim_read` — Read contract state
- `sim_fund_account` — Fund an account
- `sim_get_balance` — Get account balance
- `sim_create_snapshot` — Create state snapshot
- `sim_restore_snapshot` — Restore snapshot
- `sim_install_mock` — Install web/LLM mocks
- `sim_get_mock` — Get current mocks
- `sim_increase_time` — Advance time (Anvil-style)
- `sim_set_time` — Set time to specific datetime

Ethereum Compatible

- `eth_chainId`, `eth_blockNumber`, `eth_getBalance`
- `eth_getTransactionReceipt`, `eth_getTransactionByHash`
- `eth_estimateGas`, `eth_gasPrice`, `eth_sendRawTransaction`

Direct Mode (</api-references/genlayer-test/direct> "Direct Mode") GenLayer Node API (</api-references/genlayer-node> "GenLayer Node API")

genvm-linter

Source: <https://docs.genlayer.com/api-references/genlayer-linter>

Fast validation and schema extraction for GenLayer intelligent contracts.

Installation

```
pip install genvm-linter
```

Usage

```
# Run both lint and validate (default)
genvm-lint check contract.py

# Fast AST safety checks only (~50ms)
genvm-lint lint contract.py
```

```

# Full SDK semantic validation (~200ms cached)
genvm-lint validate contract.py

# Extract ABI schema
genvm-lint schema contract.py
genvm-lint schema contract.py --output abi.json

# Pyright type checking with SDK auto-configured
genvm-lint typecheck contract.py
genvm-lint typecheck contract.py --strict      # Strict mode
genvm-lint typecheck contract.py --all        # Show all errors (disable SDK suppressions)

# IDE setup – download SDK and output extraPaths for Pylance
genvm-lint setup                               # Latest version
genvm-lint setup --contract contract.py        # Auto-detect version from header
genvm-lint setup --version v0.2.12            # Specific version
genvm-lint setup --json                        # JSON output for IDE integration

# Pre-download GenVM artifacts
genvm-lint download                           # Latest
genvm-lint download --version v0.2.12        # Specific version
genvm-lint download --list                    # Show cached

# JSON output (all commands)
genvm-lint check contract.py --json

```

How It Works

Layer 1: AST Lint Checks (Fast)

- Forbidden imports (random, os, time, etc.)
- Non-deterministic patterns (float(), time.time())
- Structure validation (dependency header)

Layer 2: SDK Validation (Accurate)

- Downloads GenVM release artifacts (cached at ~/.cache/genvm-linter/)
- Loads exact SDK version specified in contract header
- Validates types, decorators, storage fields
- Extracts ABI schema

Exit Codes

- 0 - All checks passed
- 1 - Lint or validation errors
- 2 - Contract file not found
- 3 - SDK download failed

IDE Integration

VS Code Extension

This linter is used by the GenLayer VS Code Extension (<https://github.com/genlayerlabs/vscode-extension>) for real-time contract validation.

Manual Pylance Setup

Use `genvm-lint setup` to configure Pylance with the correct SDK paths. This gives you hover docs, go-to-definition, and type checking without the extension.

```
genvm-lint setup --contract contract.py
```

Add the output paths to your VS Code `settings.json`:

```
{
  "python.analysis.extraPaths": ["<output paths>"],
  "python.analysis.reportMissingModuleSource": "none"
}
```

Type Checking

`genvm-lint typecheck` runs Pyright with the SDK auto-configured. By default it suppresses SDK-internal noise (dynamic attributes, NewType compat). Use `--all` to see everything, `--strict` for strict mode.

Development

```
git clone https://github.com/genlayerlabs/genvm-linter.git
cd genvm-linter
pip install -e ".[dev]"
pytest
```

License

MIT

snapshot (/api-references/genlayer-node/ops/snapshot "snapshot")GenVM SDK ↗ (/api-references/genlayer-linter#"GenVM SDK ↗")

LibertAI

Source: <https://docs.genlayer.com/partners/libertai>

LibertAI (<https://libertai.io>) is a decentralized AI platform designed to be more secure and resilient than traditional centralized alternatives, while protecting user privacy. LibertAI leverages Aleph Cloud (<https://aleph.cloud>) GPUs and TEEs to offer confidential AI services, tools for AI Agents to become self-sustainable & more!

In partnership with GenLayer, LibertAI (<https://libertai.io>) provides access to top open-source models with complete data confidentiality through one unified API endpoint. Developers on GenLayer can receive free API credits visible in the LibertAI console (<https://console.libertai.io>). Complete a short form (<https://tally.so/r/n9AvdK>) to get started.

API Reference: <https://docs.libertai.io/apis> (<https://docs.libertai.io/apis>)

Optimistic Democracy

Source: <https://docs.genlayer.com/core-concepts/optimistic-democracy>

Optimistic Democracy is the consensus method used by GenLayer to validate transactions and operations of Intelligent Contracts. This approach is especially good at handling unpredictable outcomes from transactions involving web data or AI models, which is important for keeping the network reliable and secure.

Key Components

- Validators: Participants who stake tokens to earn the right to validate transactions. They play a crucial role in both the initial validation and any appeals process if needed.
- Leader Selection: A process that randomly picks one validator to propose the outcome for each transaction, ensuring fairness and reducing potential biases.

How It Works

Optimistic Democracy relies on a mix of trust and verification to ensure transaction integrity:

[Image:]

1. Initial Validation: When a transaction is submitted, a small group of randomly selected validators checks its validity. One is chosen as the leader. The leader executes the transaction, and the other validators assess the leader's proposal using the Equivalence Principle.
2. Majority Consensus: If most validators accept the leader's proposal, the transaction is provisionally accepted. However, this decision is not final yet, allowing for possible appeals during a limited window of time, known as the Finality Window.

□

If any validator fails to vote within the specified timeframe, they are replaced, and a new validator is selected to cast a vote.

3. Initiating an Appeal: If a participant disagrees with the initial validation (if it's incorrect or fraudulent), they can appeal during the Finality Window. They must submit a request and provide a bond. After the appeal starts, a new group of validators joins the original ones. This group first votes on whether the transaction should be re-evaluated. If they agree, a new leader is chosen to reassess the transaction, and all validators then review this new evaluation.
4. Appeal Evaluation: The new leader re-evaluates the transaction, while the other validators assess the leader's proposal using the Equivalence Principle. This step involves more validators, increasing the chances of an accurate decision.
5. Escalating Appeals: If the appealing party is still not satisfied, the process can escalate, with each round involving more validators. Each round doubles the number of validators. A new leader is only chosen if the transaction is overturned.

6. Final Decision: The appeals process continues until a majority consensus is reached or until all validators have participated. The final decision is recorded, and the transaction's state is updated accordingly. If the appealing party is correct, they receive a reward for their efforts, while incorrect appellants lose their bond.

GenVM (/understand-genlayer-protocol/core-concepts/genvm "GenVM")Equivalence Principle
(/understand-genlayer-protocol/core-concepts/optimistic-democracy/equivalence-principle "Equivalence Principle")

genlayer Commands

Source: <https://docs.genlayer.com/references/genlayer-cli>

GenLayer CLI is a development environment for the GenLayer ecosystem. It allows developers to interact with the protocol by creating accounts, sending transactions, and working with Intelligent Contracts by testing, debugging, and deploying them.

Version: 0.39.1

Command List

- genlayer init — Initialize the GenLayer Environment
- genlayer up — Starts GenLayer's simulator
- genlayer stop — Stop all running localnet services.
- genlayer account — Manage your accounts (address, balance, keys)
- genlayer deploy — Deploy intelligent contracts
- genlayer call — Call a contract method without sending a transaction or changing the state
- genlayer write — Sends a transaction to a contract method that modifies the state
- genlayer schema — Get the schema for a deployed contract
- genlayer code — Get the source for a deployed contract
- genlayer config — Manage CLI configuration, including the default network
- genlayer update — Update resources like models or configurations
- genlayer localnet — Manage localnet operations
- genlayer new — Create a new GenLayer project using the default template
- genlayer network — Network configuration
- genlayer receipt — Get transaction receipt by hash
- genlayer appeal — Appeal a transaction by its hash
- genlayer appeal-bond — Show minimum appeal bond required for a transaction
- genlayer trace — Get execution trace for a transaction (return data, stdout, stderr, GenVM logs)
- genlayer finalize — Finalize a transaction that is ready to be finalized (public call)
- genlayer finalize-batch — Finalize a batch of idle transactions in a single call (public call)

- genlayer staking — Staking operations for validators and delegators

This reference is auto-generated. Do not edit manually.

finalize-batch (/api-references/genlayer-cli/finalize-batch "finalize-batch")account
(/api-references/genlayer-cli/accounts/account "account")

Getting Started with GenLayer

Source: <https://docs.genlayer.com/getting-started>

This guide helps developers build, deploy, and integrate Intelligent Contracts on GenLayer.

Start with Skills (Recommended)

The fastest path from zero to deployed contract is the GenLayer Skills (<https://skills.genlayer.com/>) plugin for Claude Code. The genlayer-dev skill scaffolds a project, runs the linter and direct-mode tests, deploys to testnet, and walks you through the SDKs interactively.

```
claude /plugin marketplace add genlayerlabs/skills  
claude /plugin install genlayer-dev@genlayerlabs
```

Then run `claude /genlayer-dev` (or just ask Claude what to do next). The skill bundles the same workflows documented on this site — it just runs them for you.

Prefer to set things up manually or read the references first? The cards below cover every topic in depth, and the Tooling Setup (/developers/intelligent-contracts/tooling-setup) page describes the manual install path.

Reference

□ Introduction (/developers/intelligent-contracts/introduction) ⚙️ Tooling Setup
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Execute Transactions

Source: <https://docs.genlayer.com/genlayer-stack/genlayer-studio/execute-transaction>

After deploying your Intelligent Contract and verifying its state, the next step is to execute transactions. This allows you to interact with your contract's methods and functions to perform specific actions or queries.

Write Methods

1. In the Write Methods section, you will see all the write methods available in your Intelligent Contract.
2. Expand the method you want to execute. For example, in the Storage contract, select the `update_storage` method.
3. If the method requires input parameters, a field will appear for you to enter the necessary values. For example, the `update_storage` method requires a string called `new_storage`.

[Image: Deploy Contract]

4. After entering any required parameters, click the button to execute the method. The Studio will process the transaction and return the result.

Viewing Transaction Results

Once you execute a transaction, the result will be displayed, showing the output of the method called. This allows you to verify that the method has executed correctly and to see the effect of the transaction on your contract's state.

Viewing Current Account Address

You can also view your current account address in the header. This address is used to interact with your contract.

[Image: Select Active Account]

You can continue interacting with your Intelligent Contract by executing additional transactions or checking the state again to see how it has changed.

Reading the Contract State (/developers/intelligent-contracts/tools/genlayer-studio/contract-state "Reading the Contract State")Monitoring Node Logs (/developers/intelligent-contracts/tools/genlayer-studio/monitoring-node-logs "Monitoring Node Logs")
